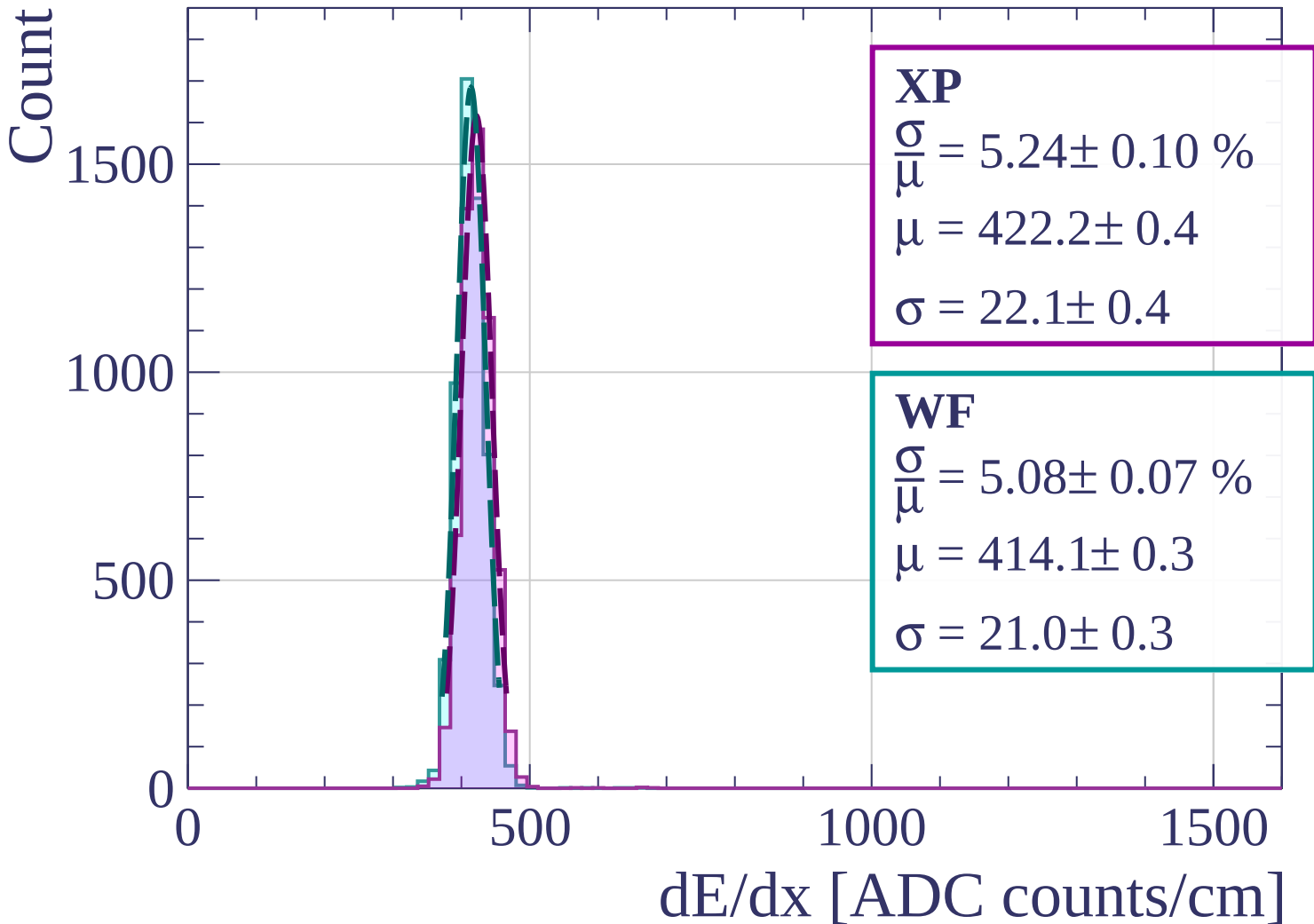
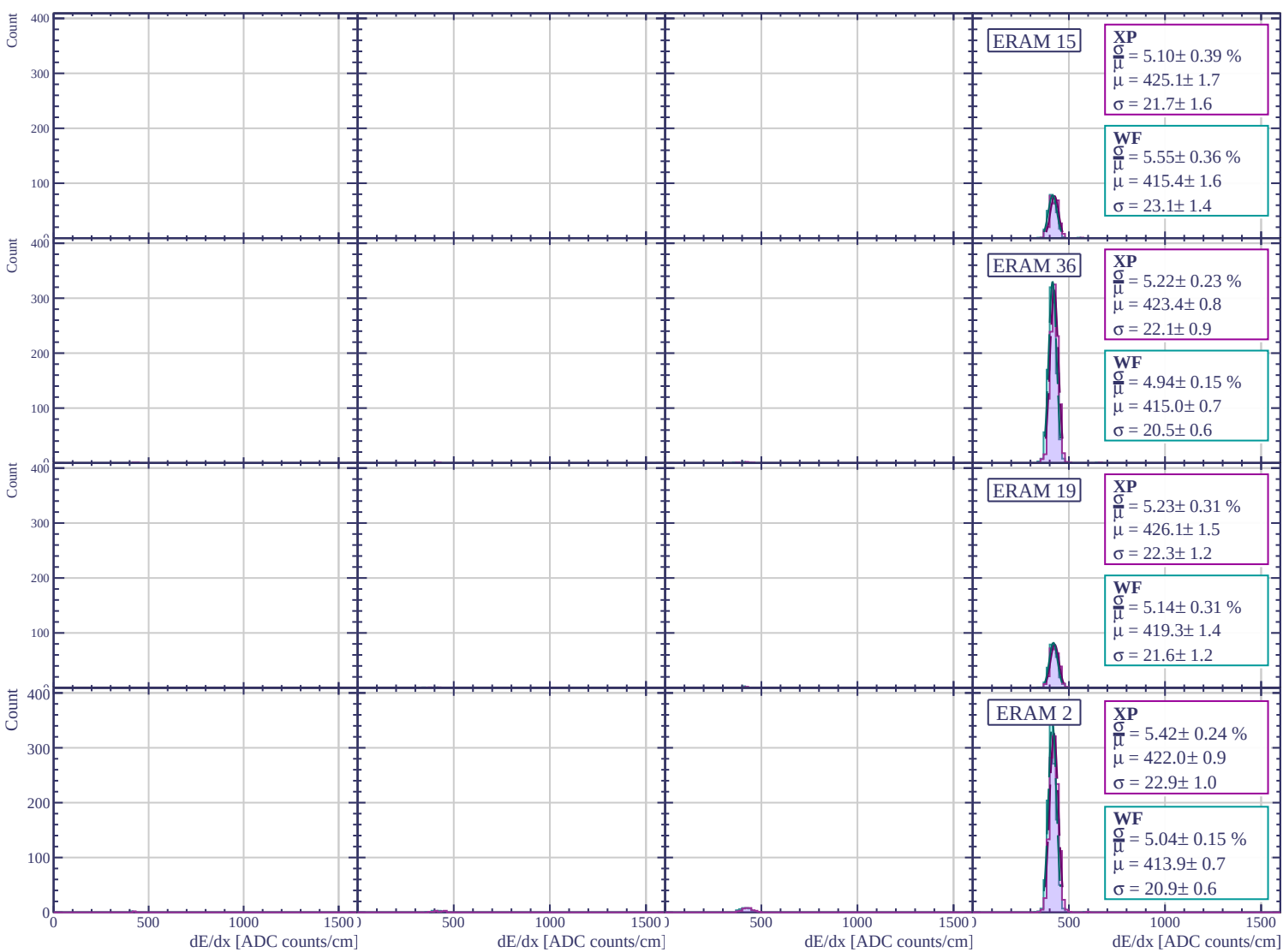
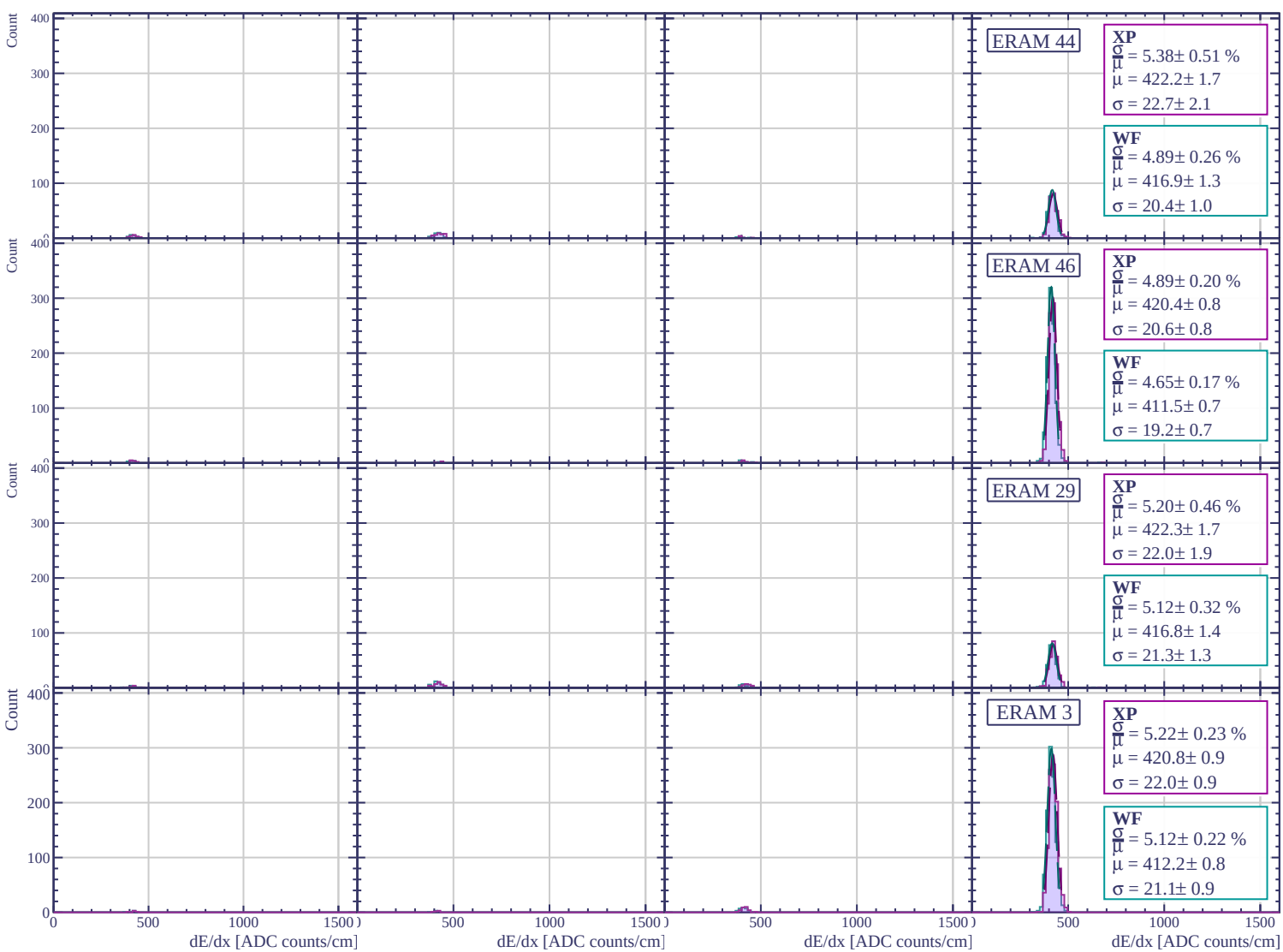
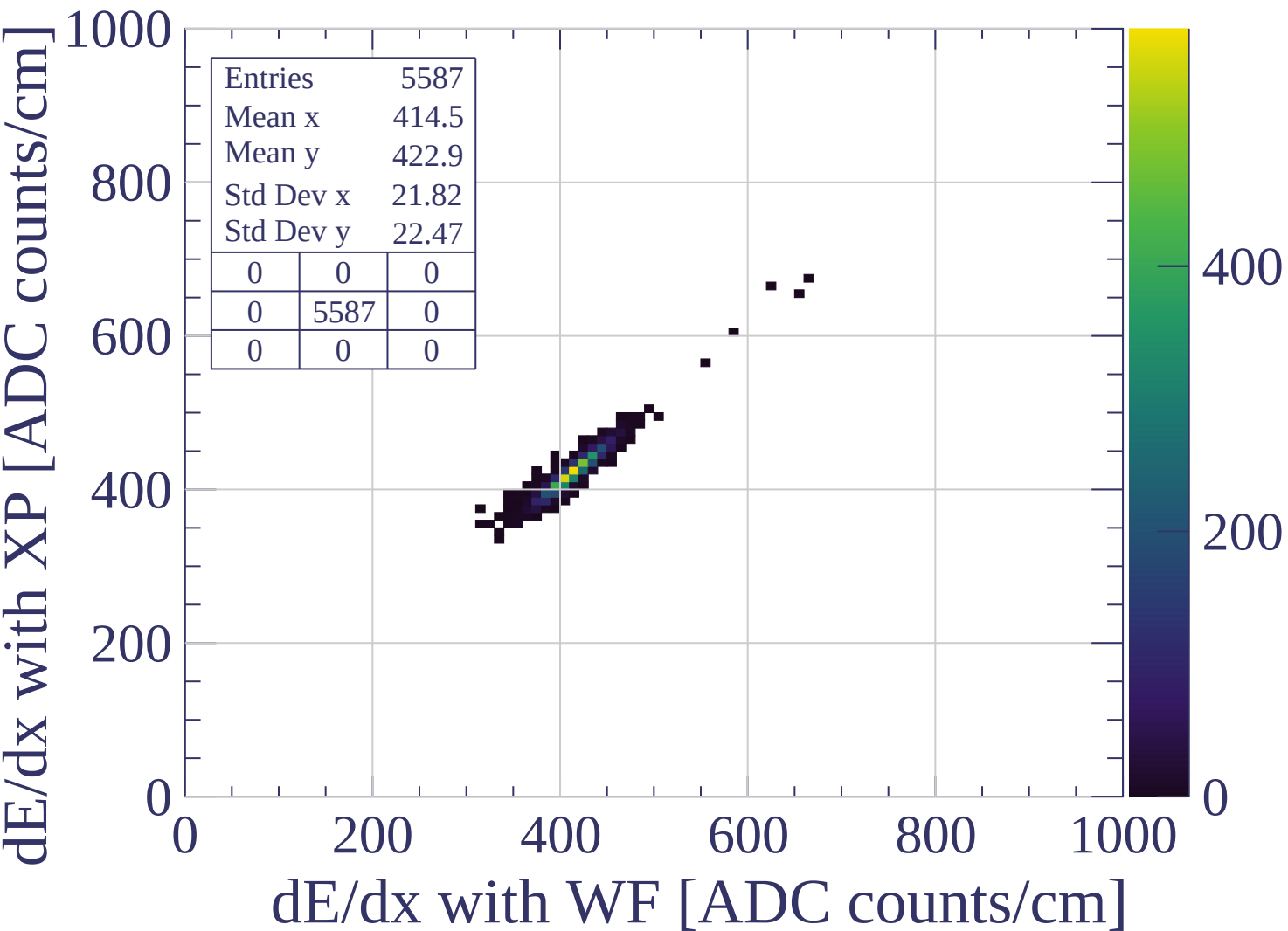
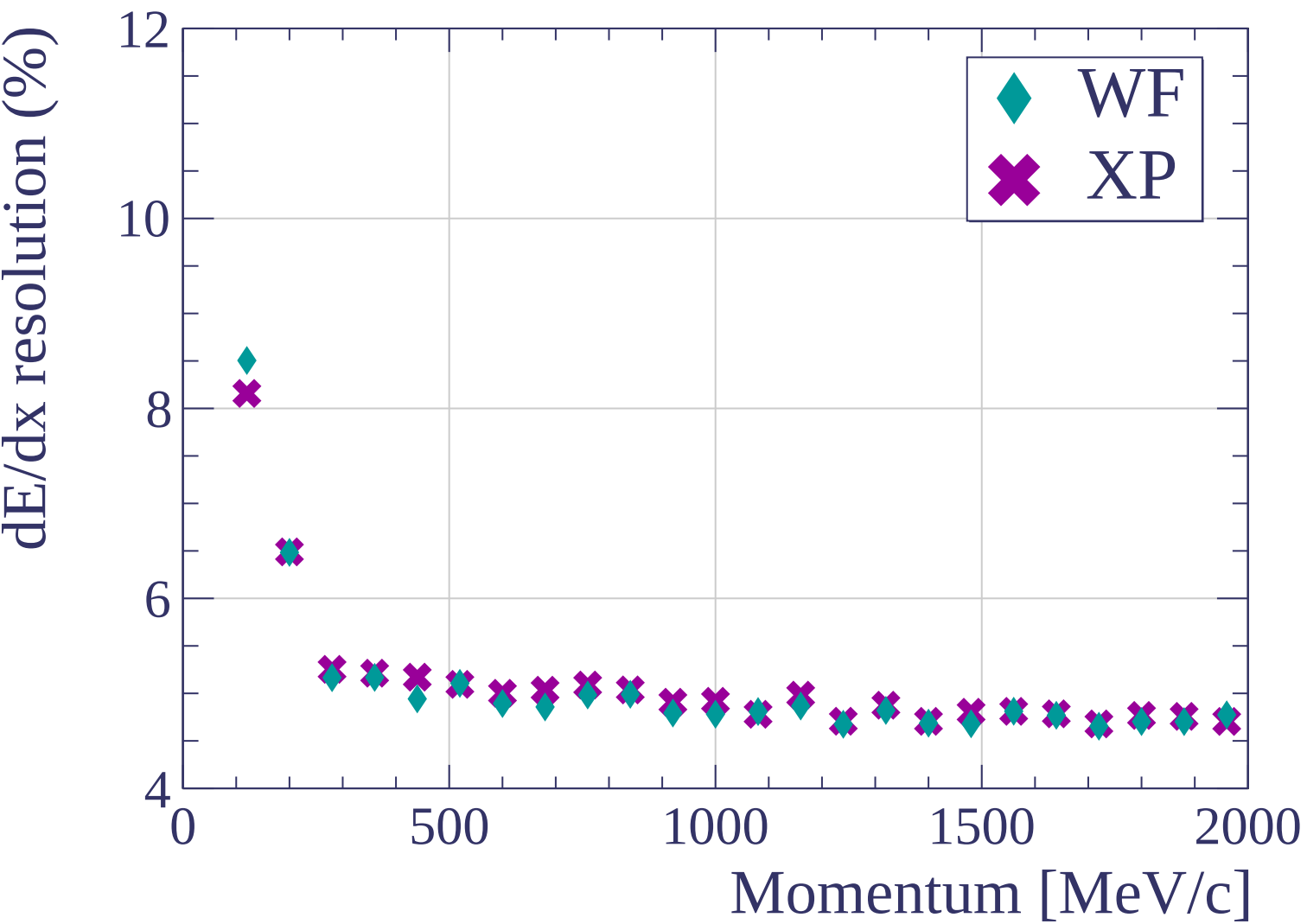


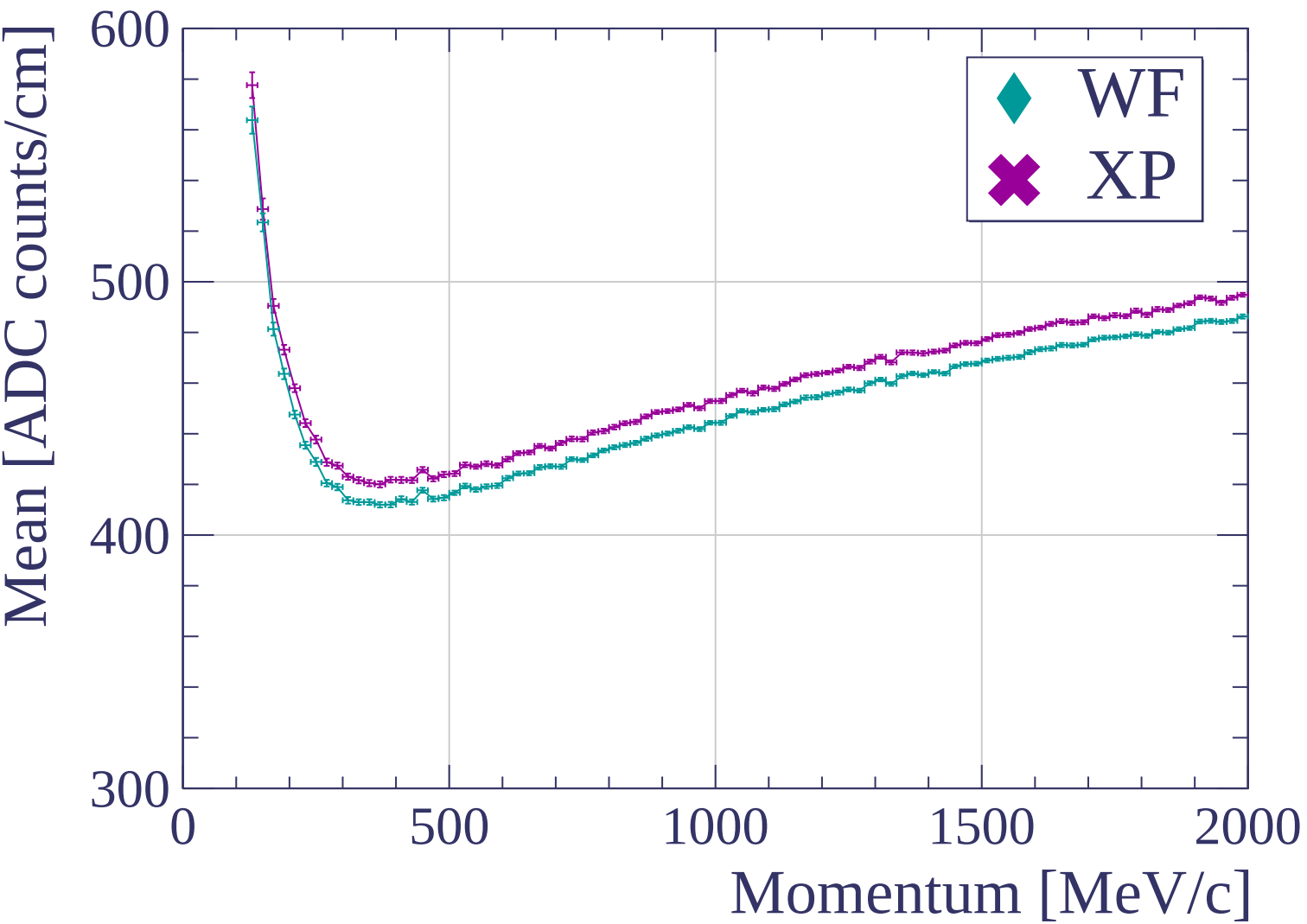


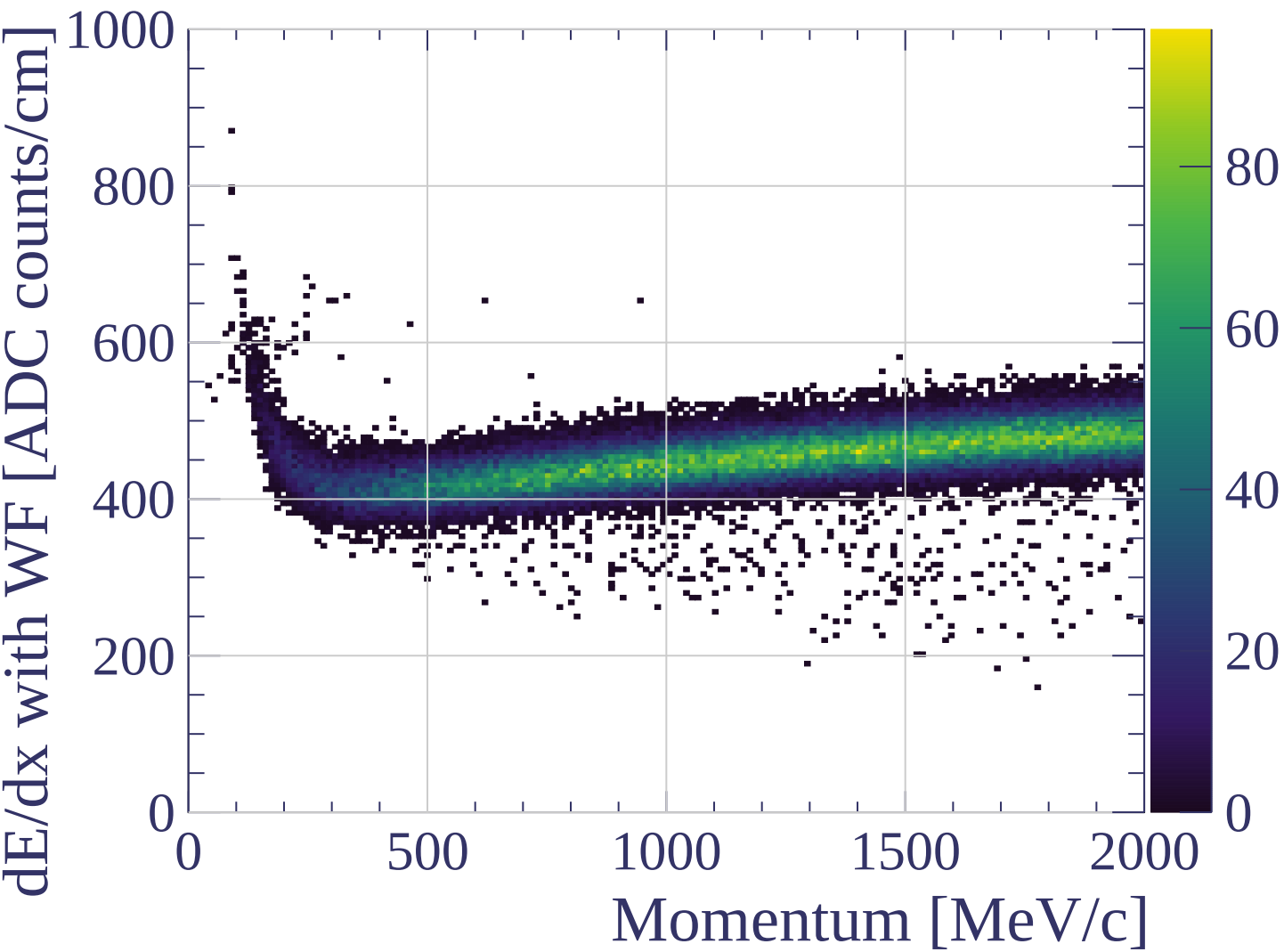


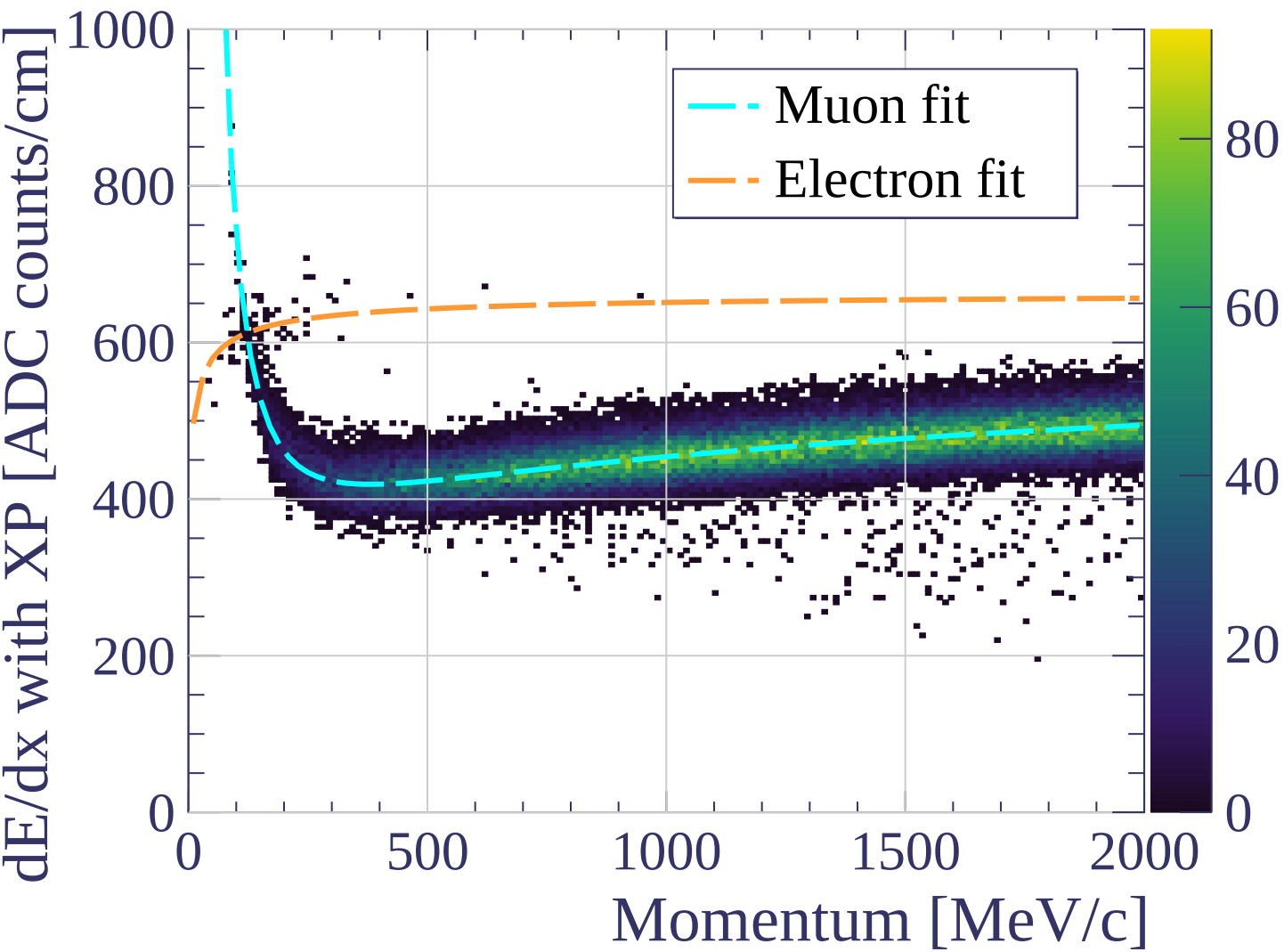


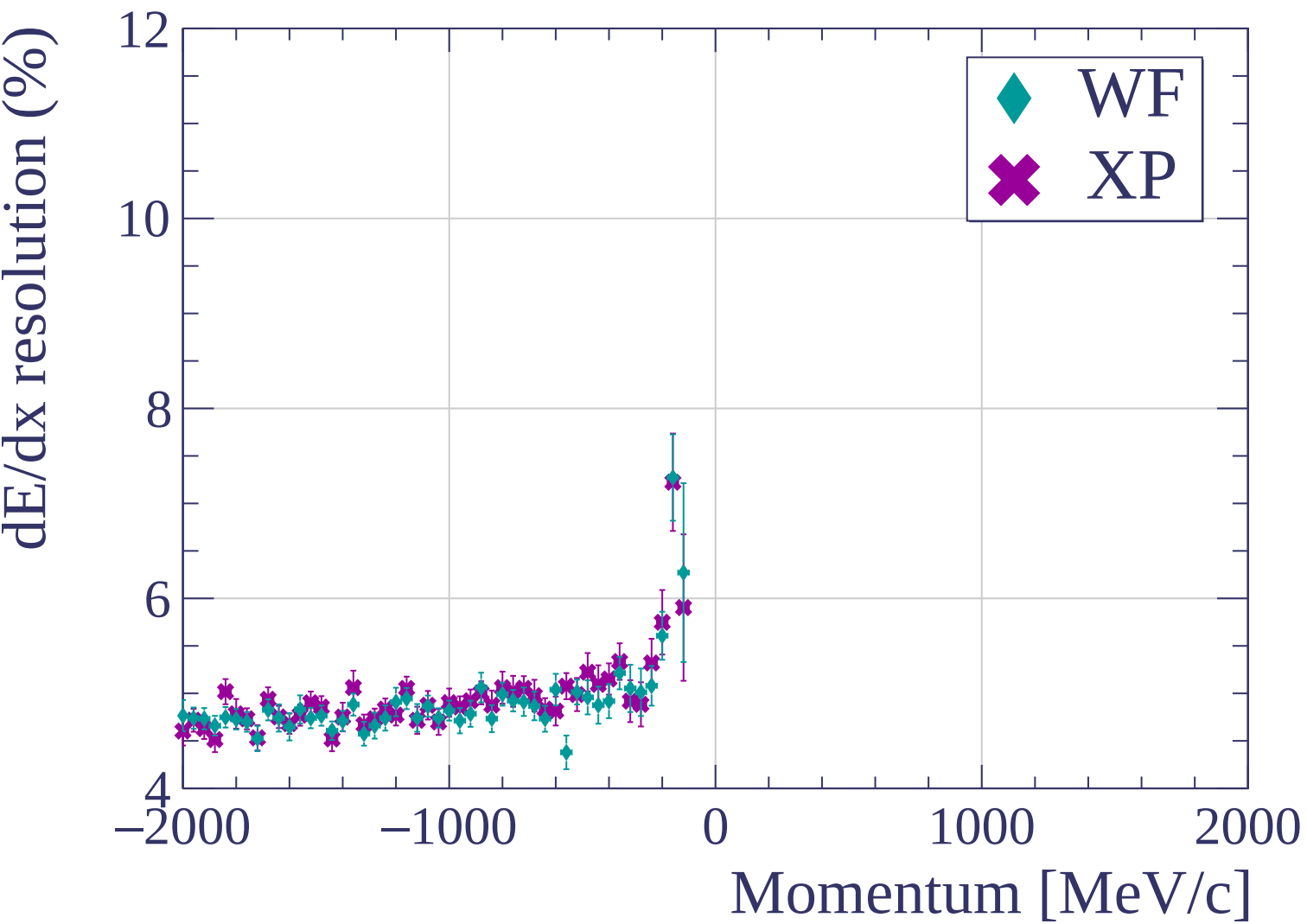


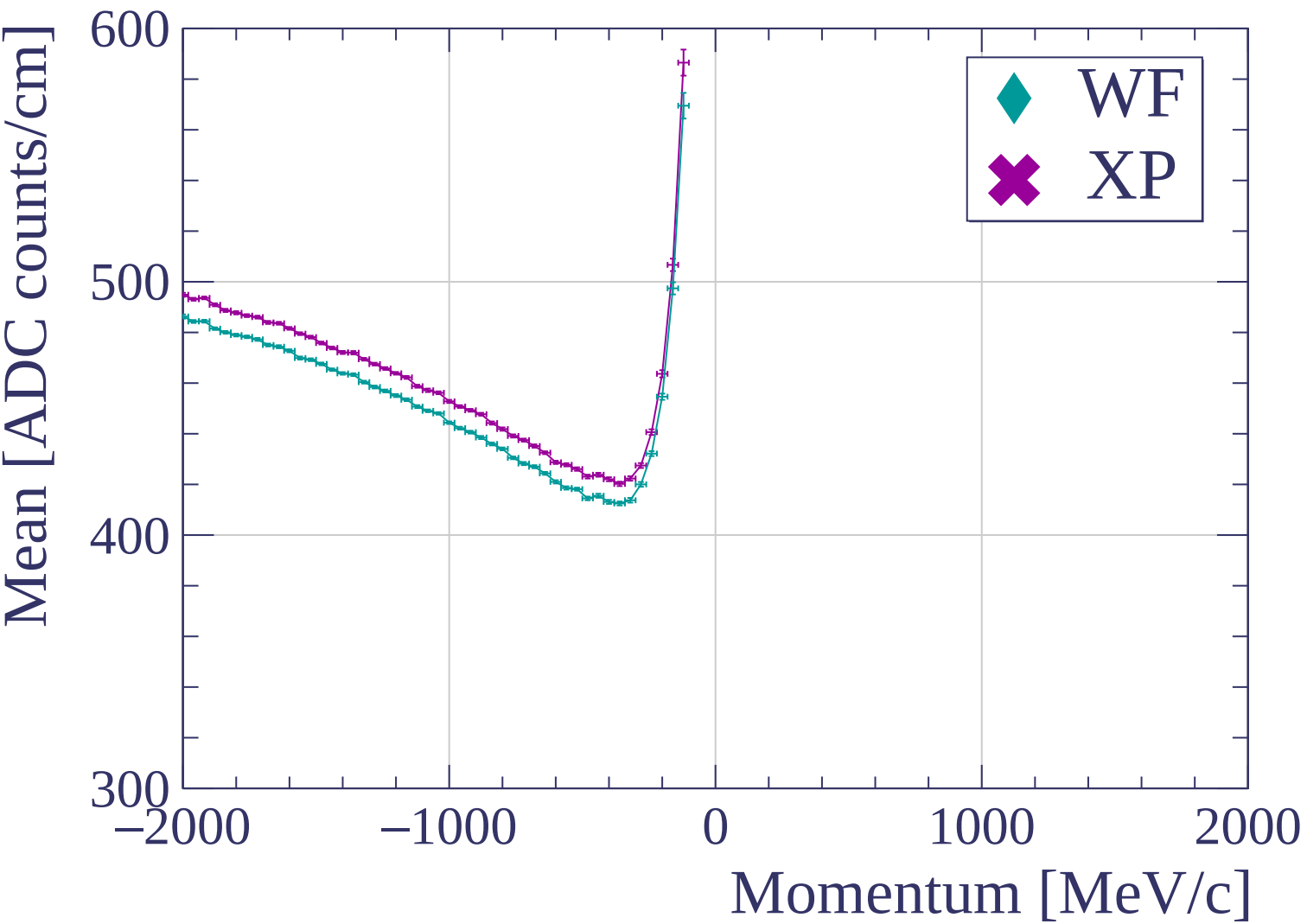


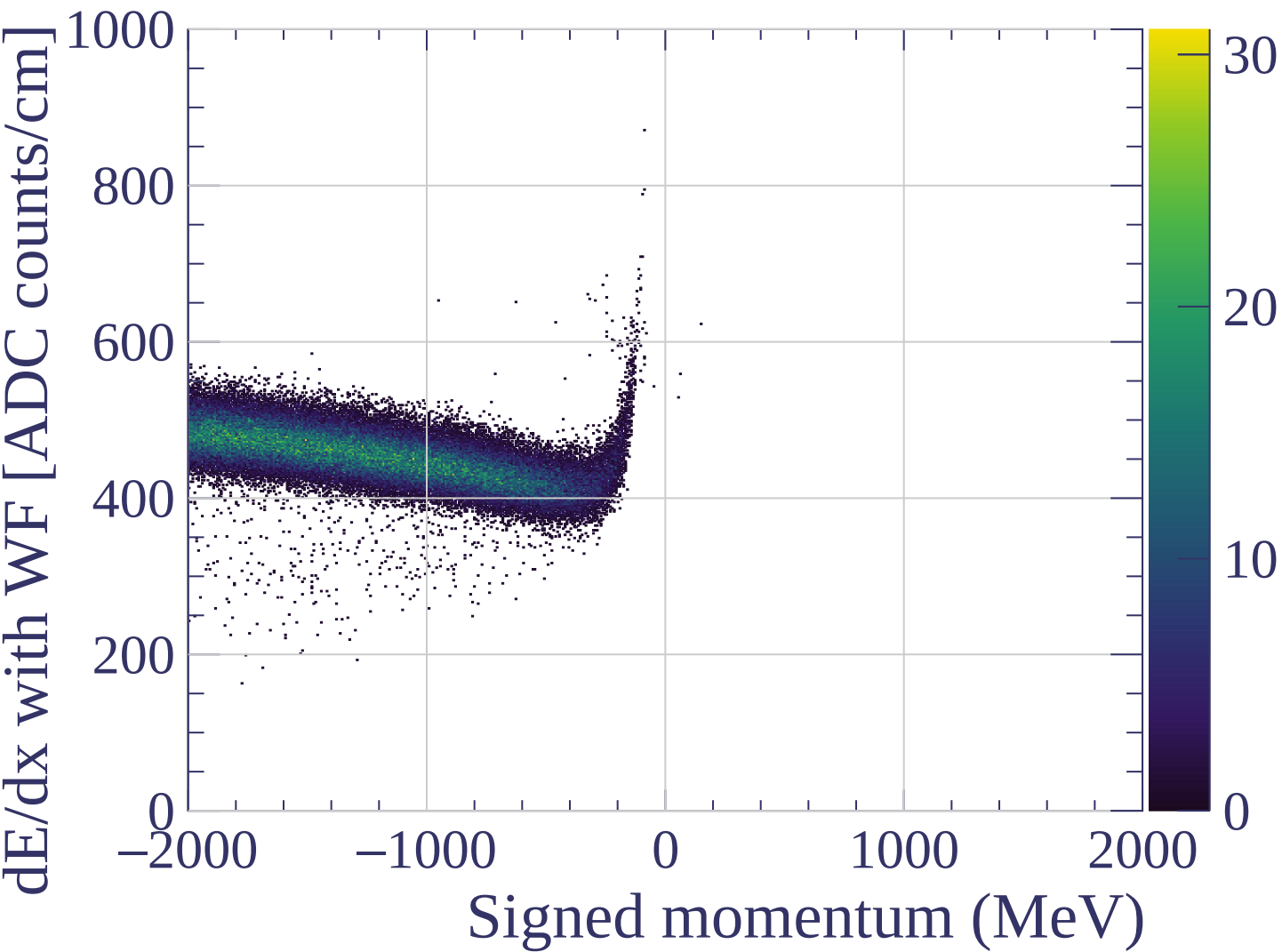


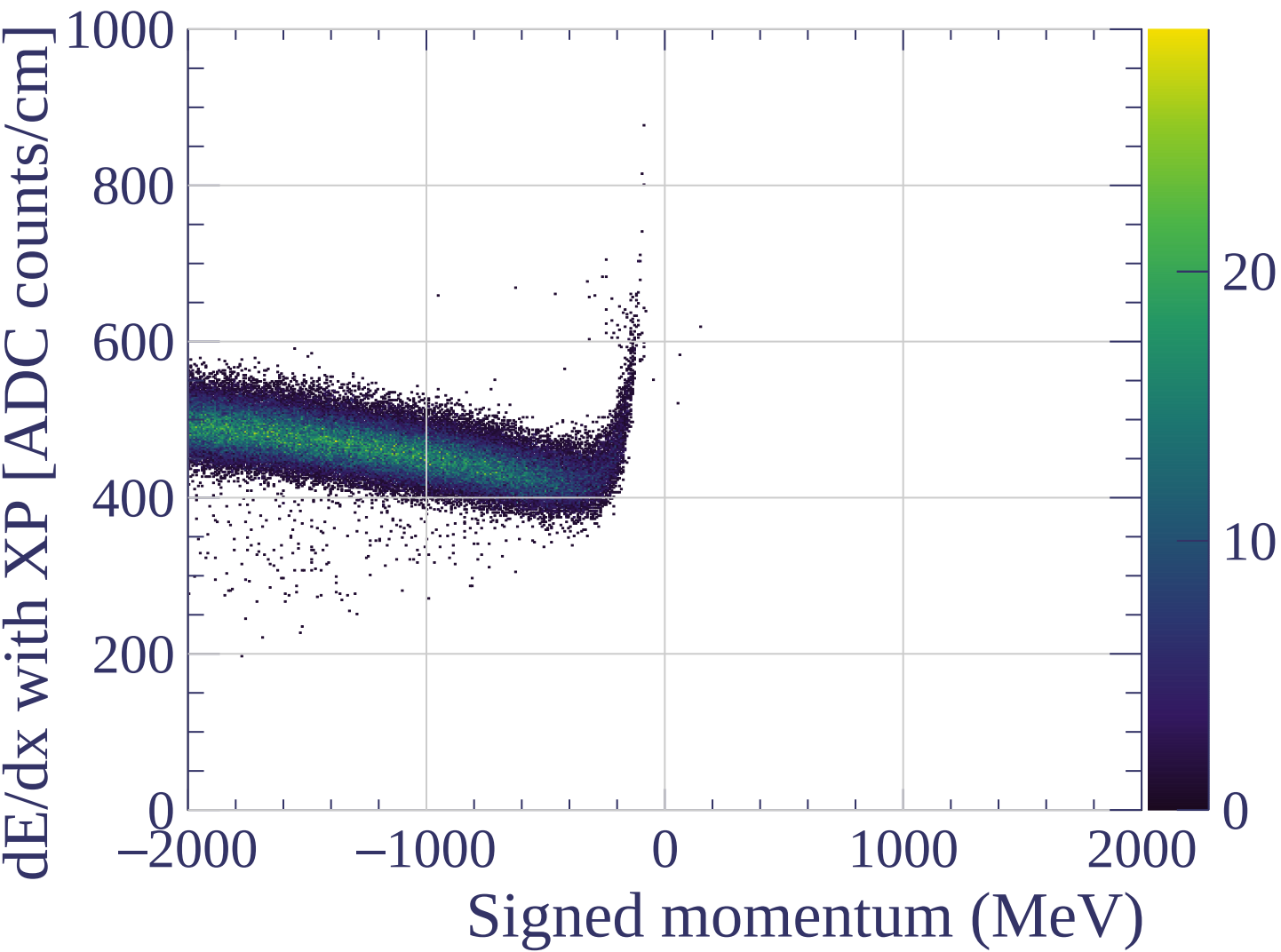




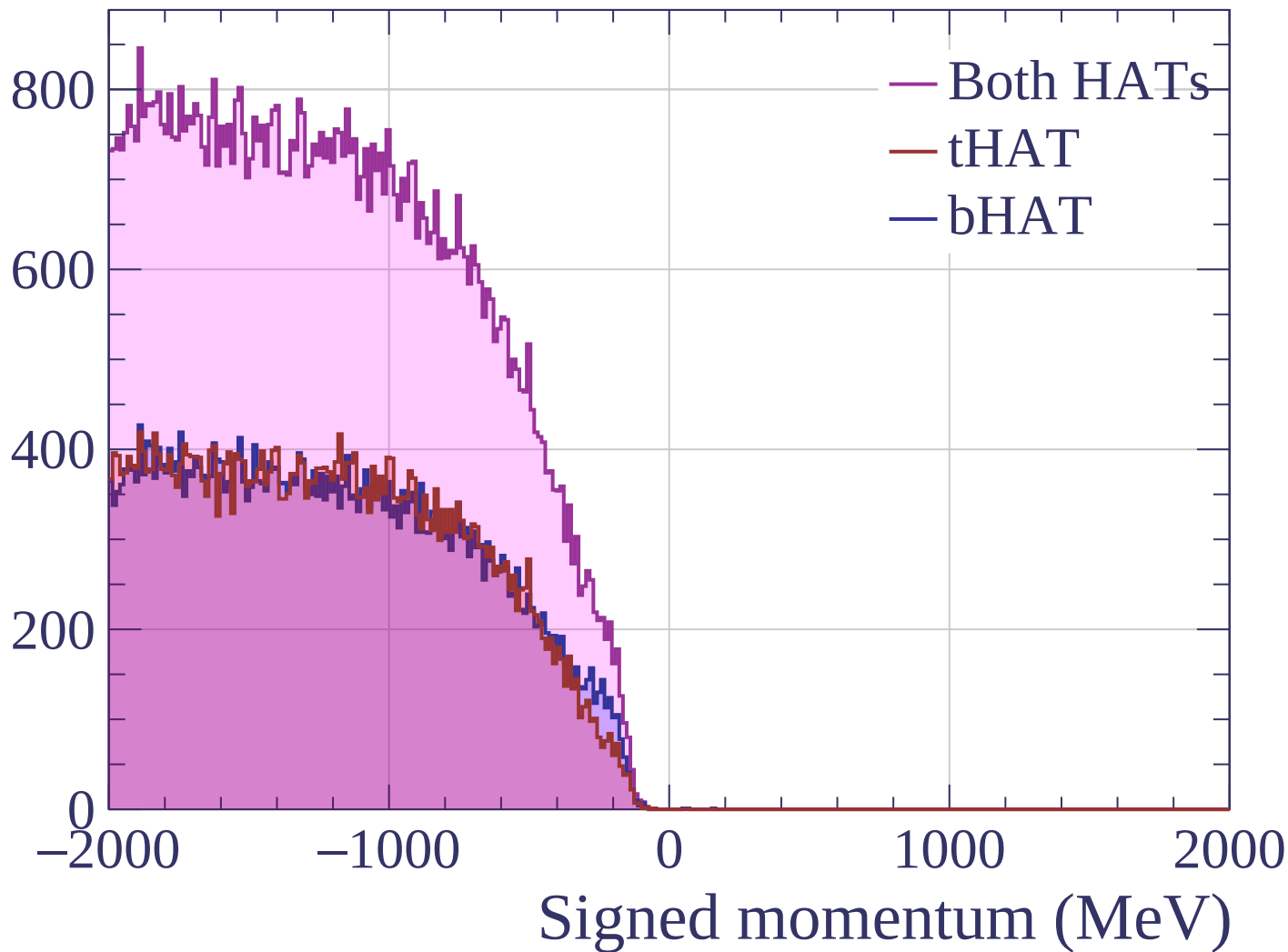


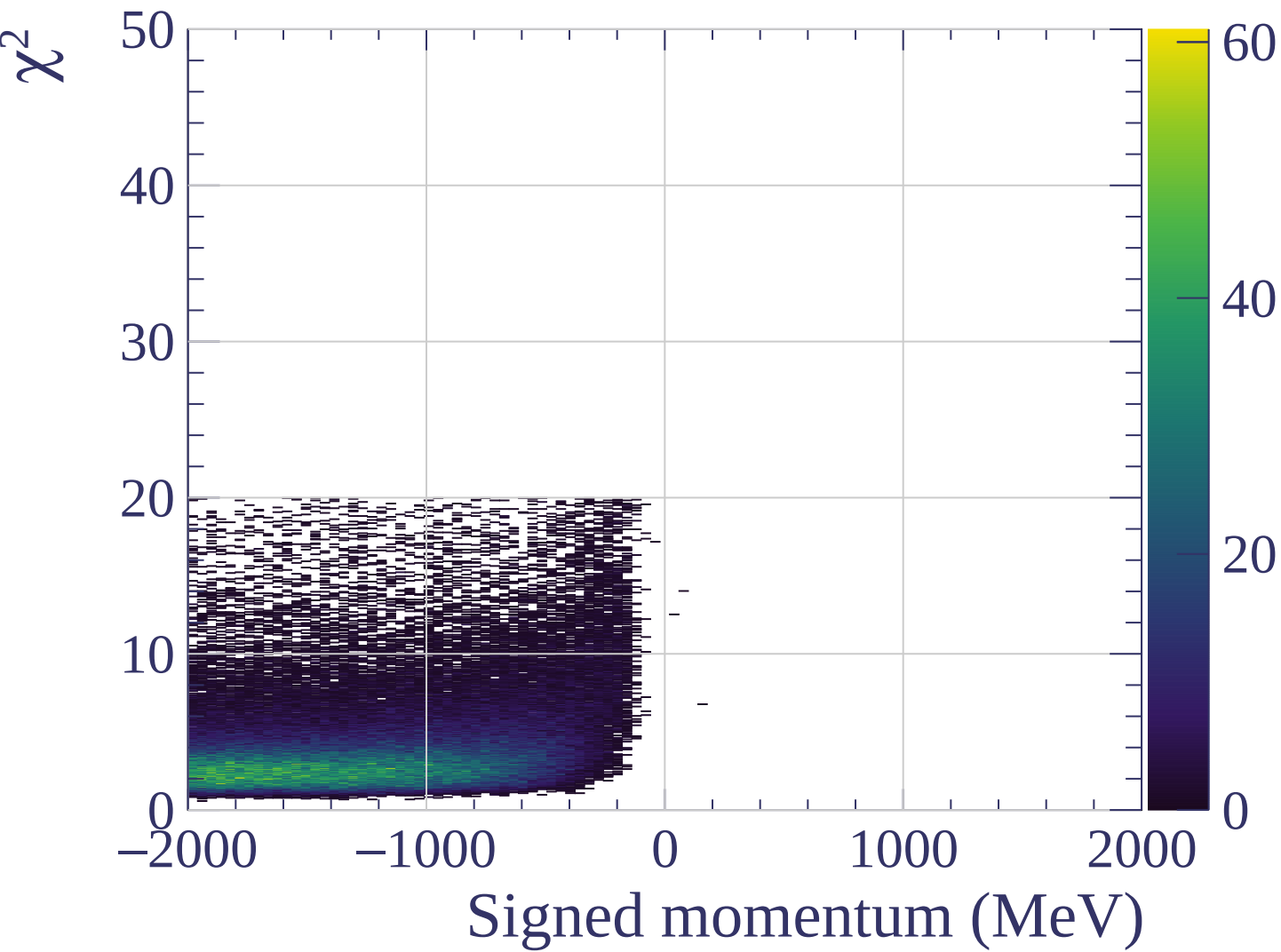


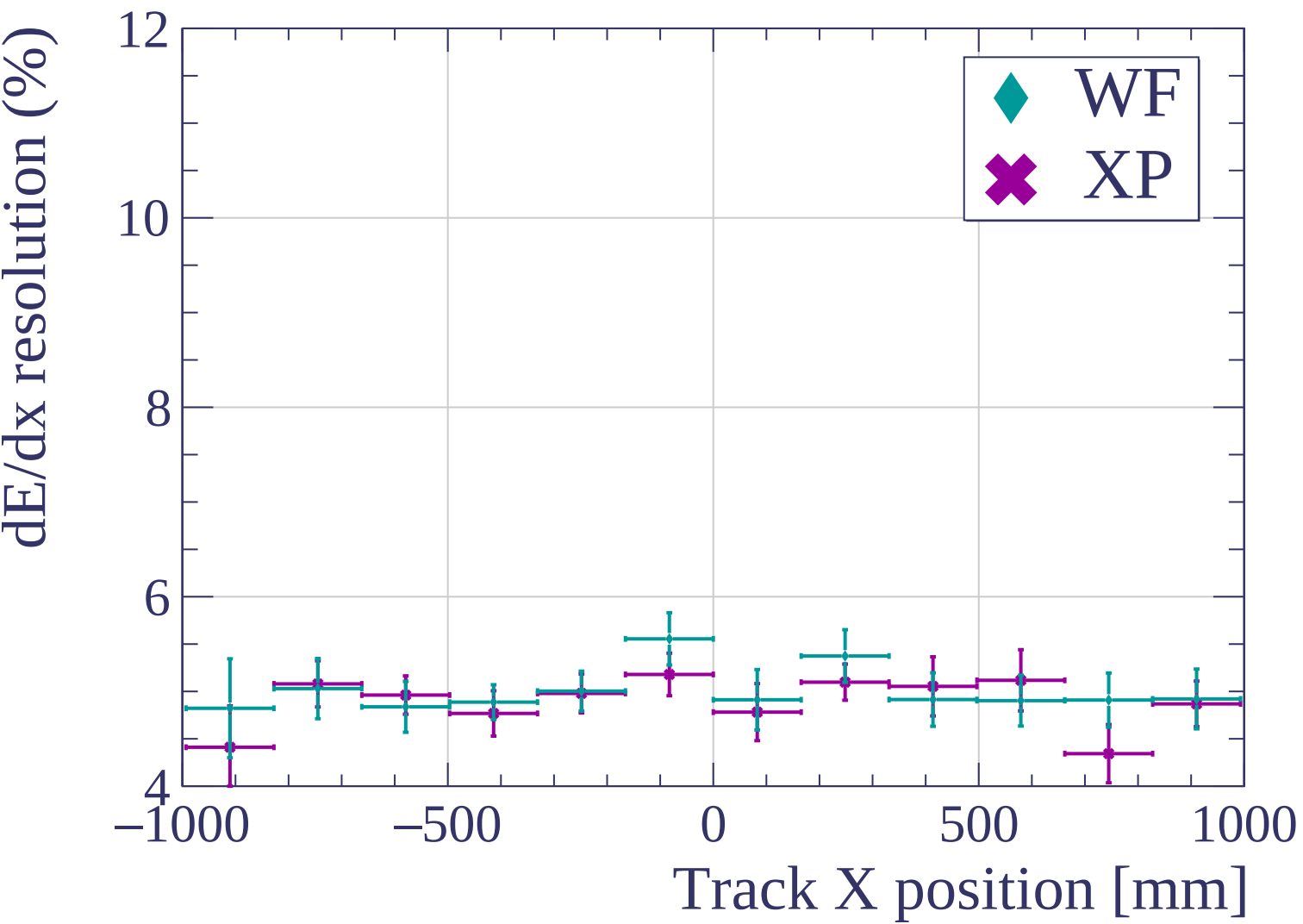


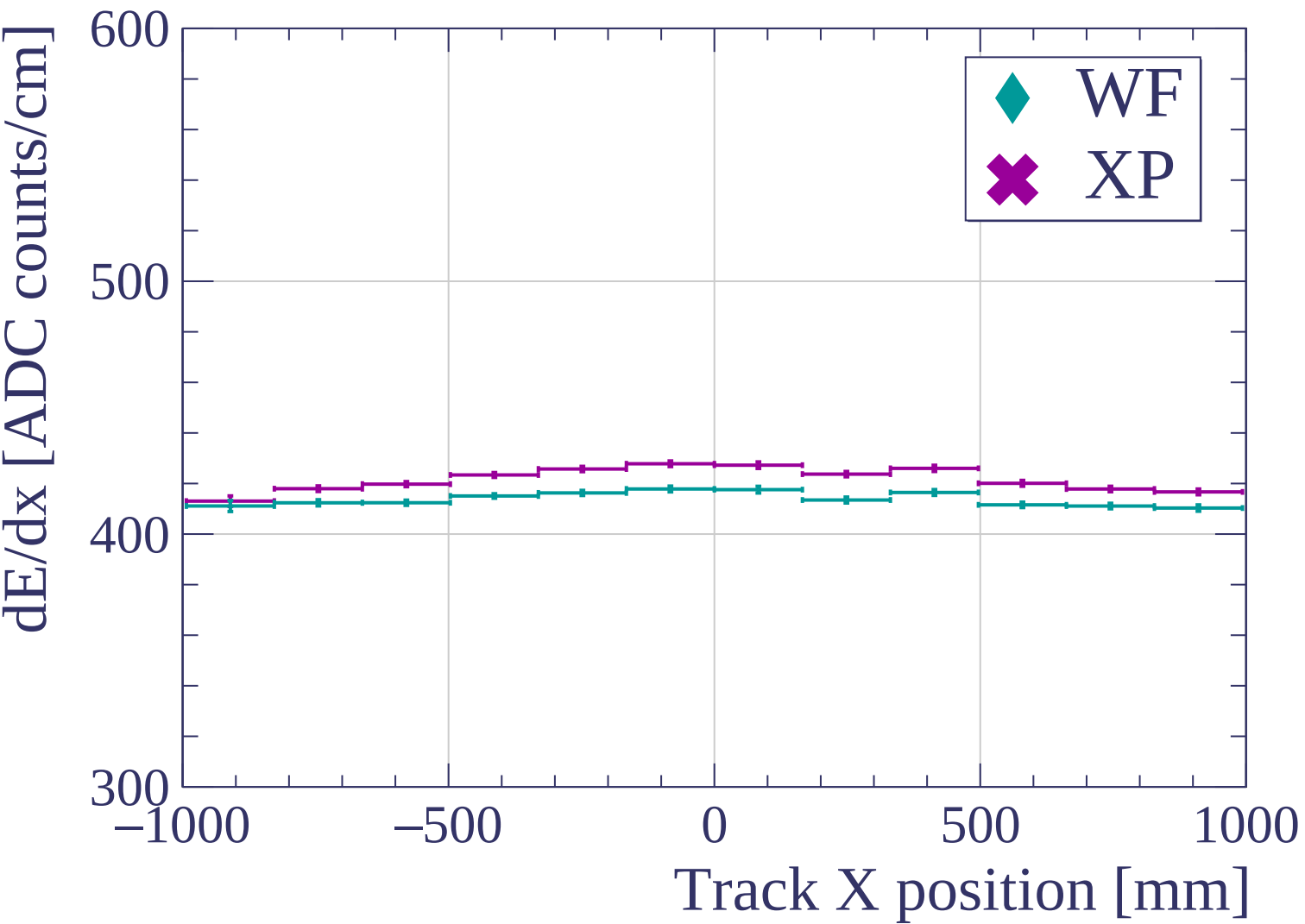


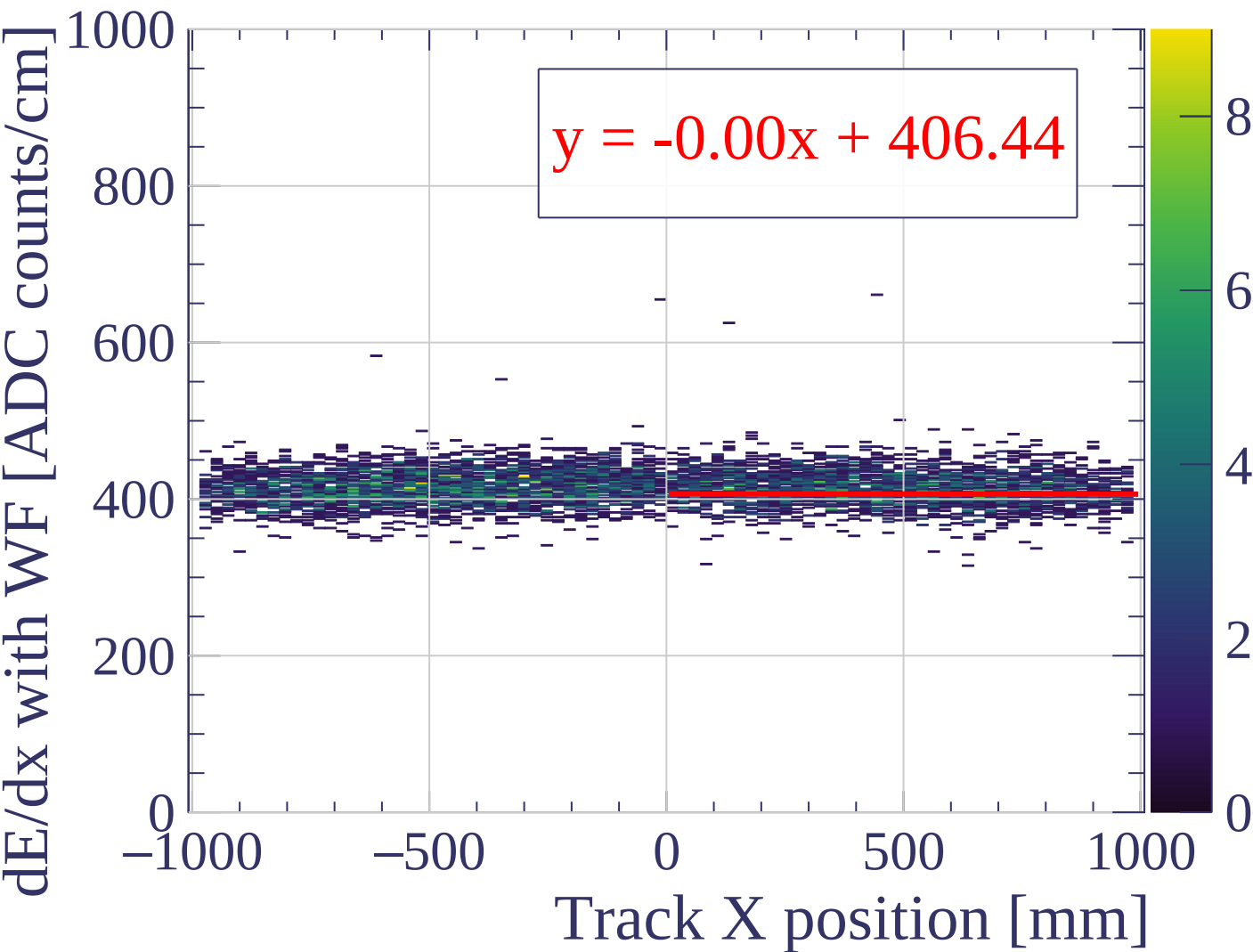
Count

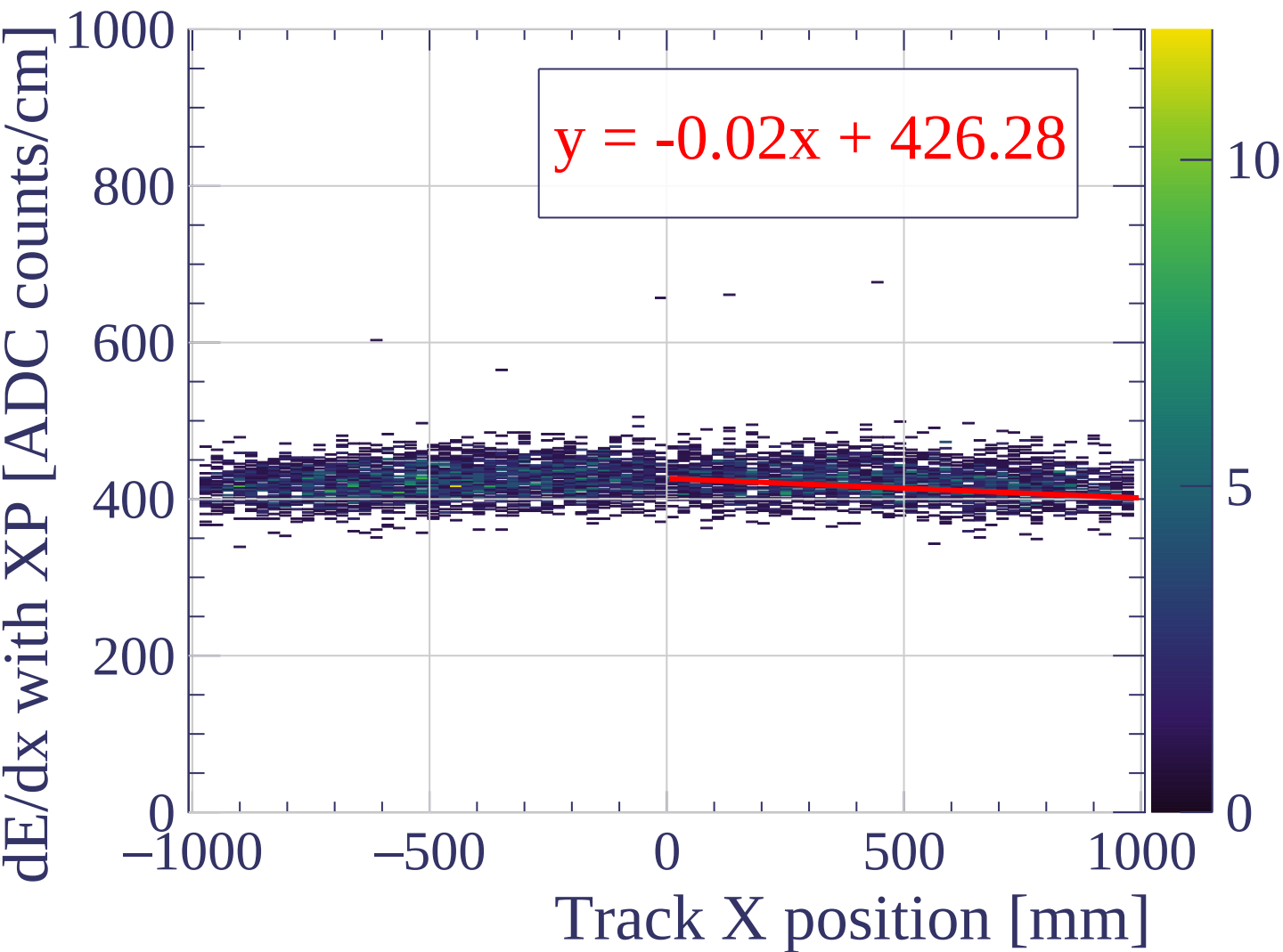


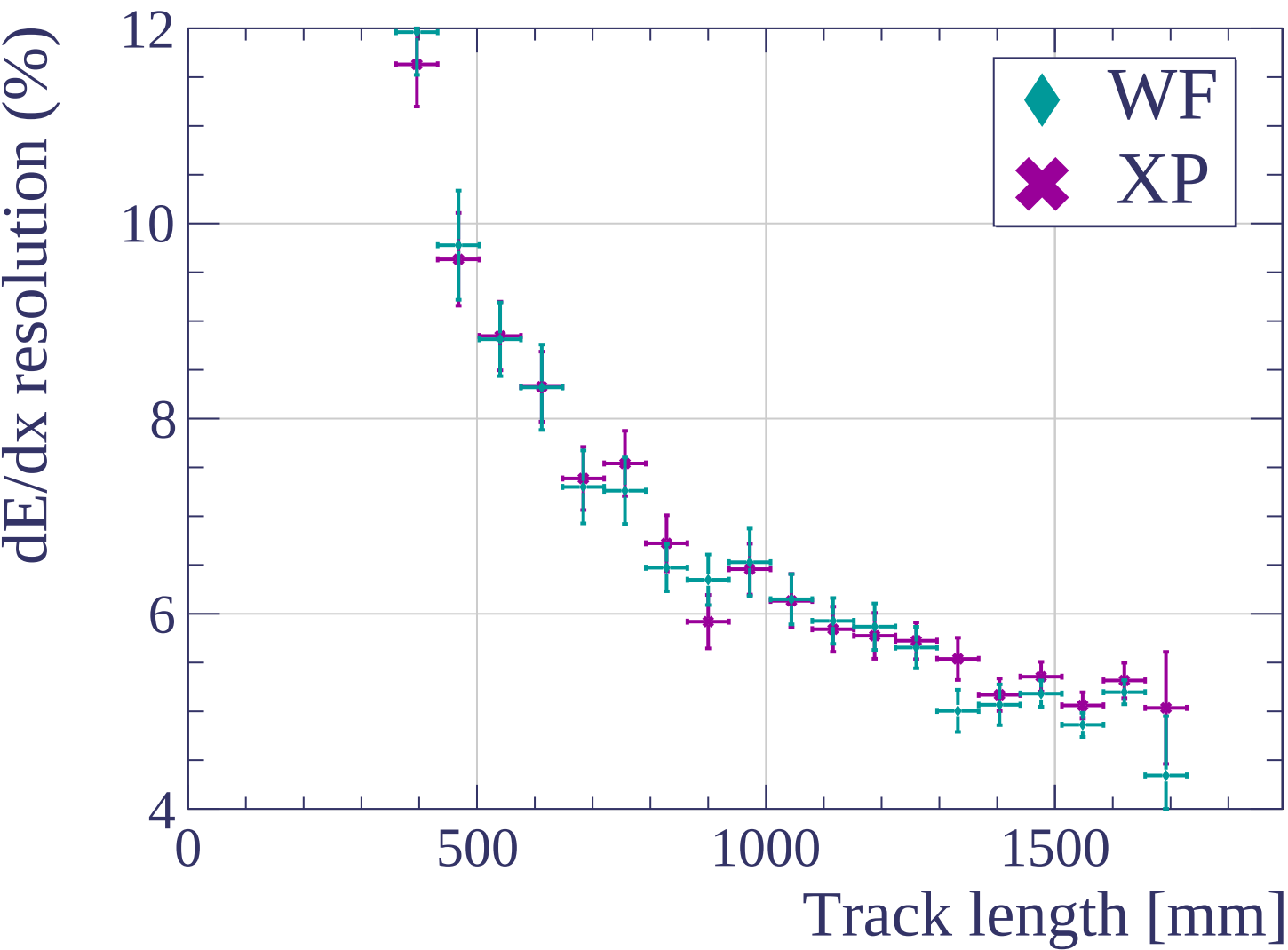


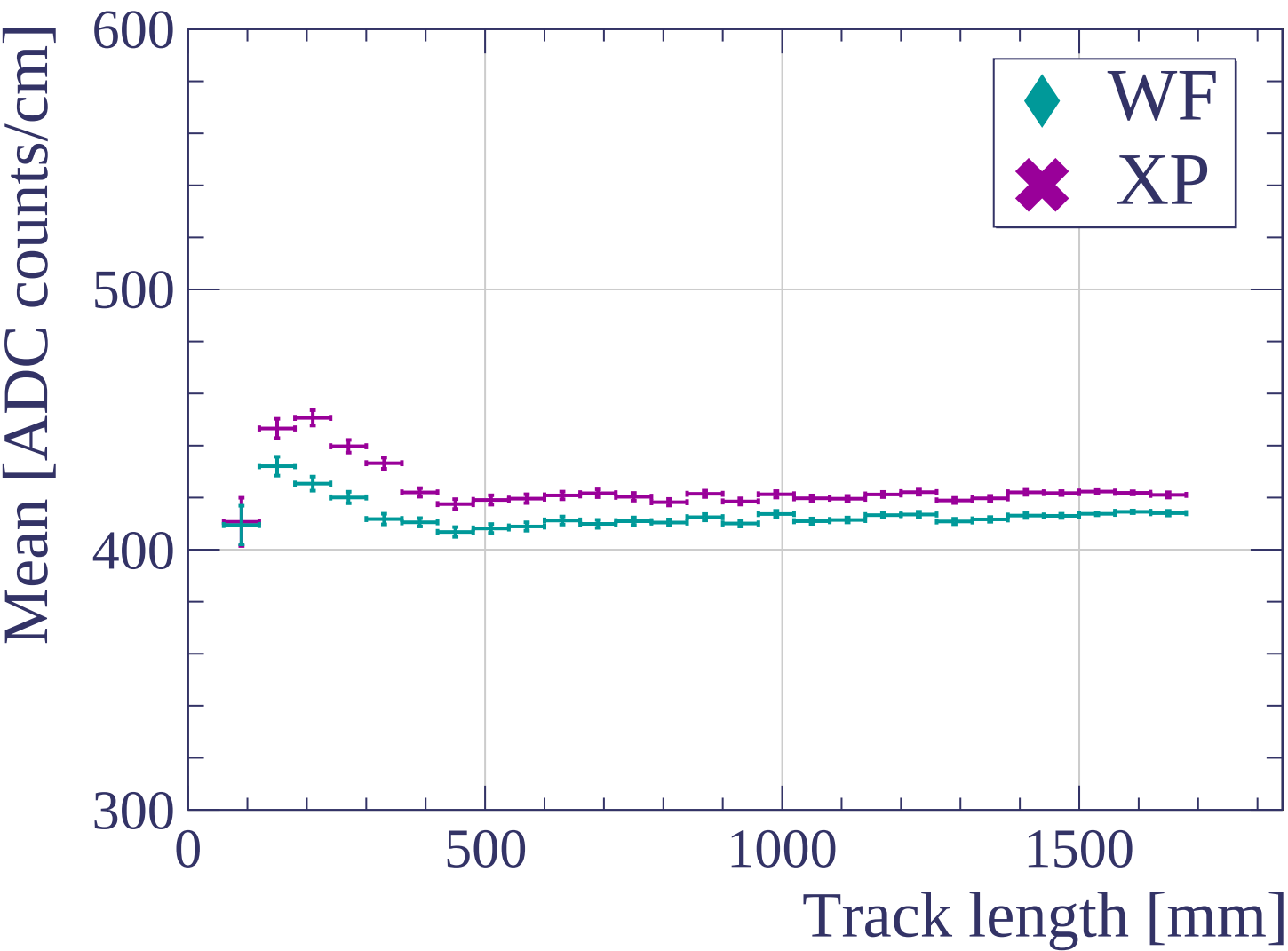


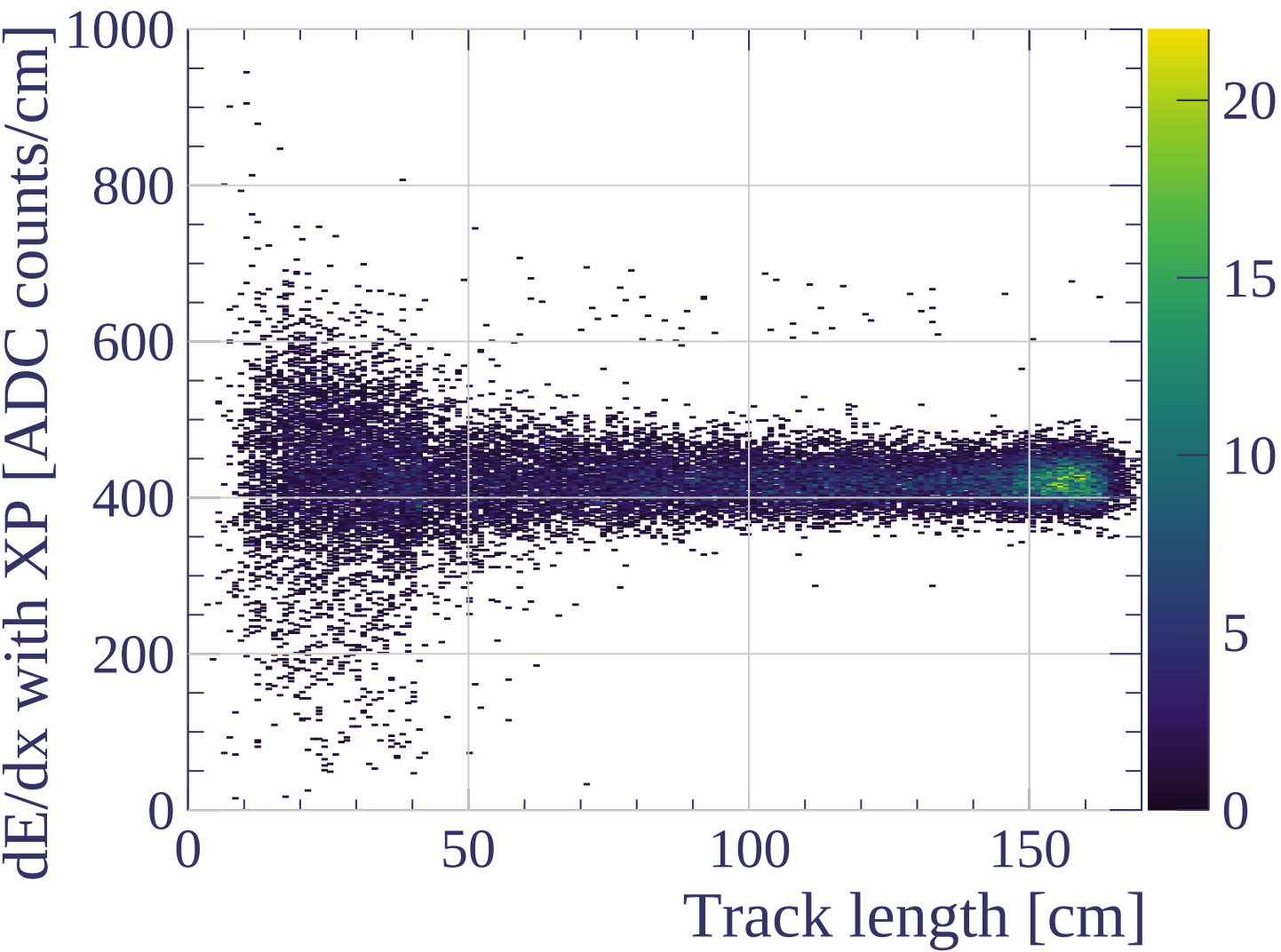


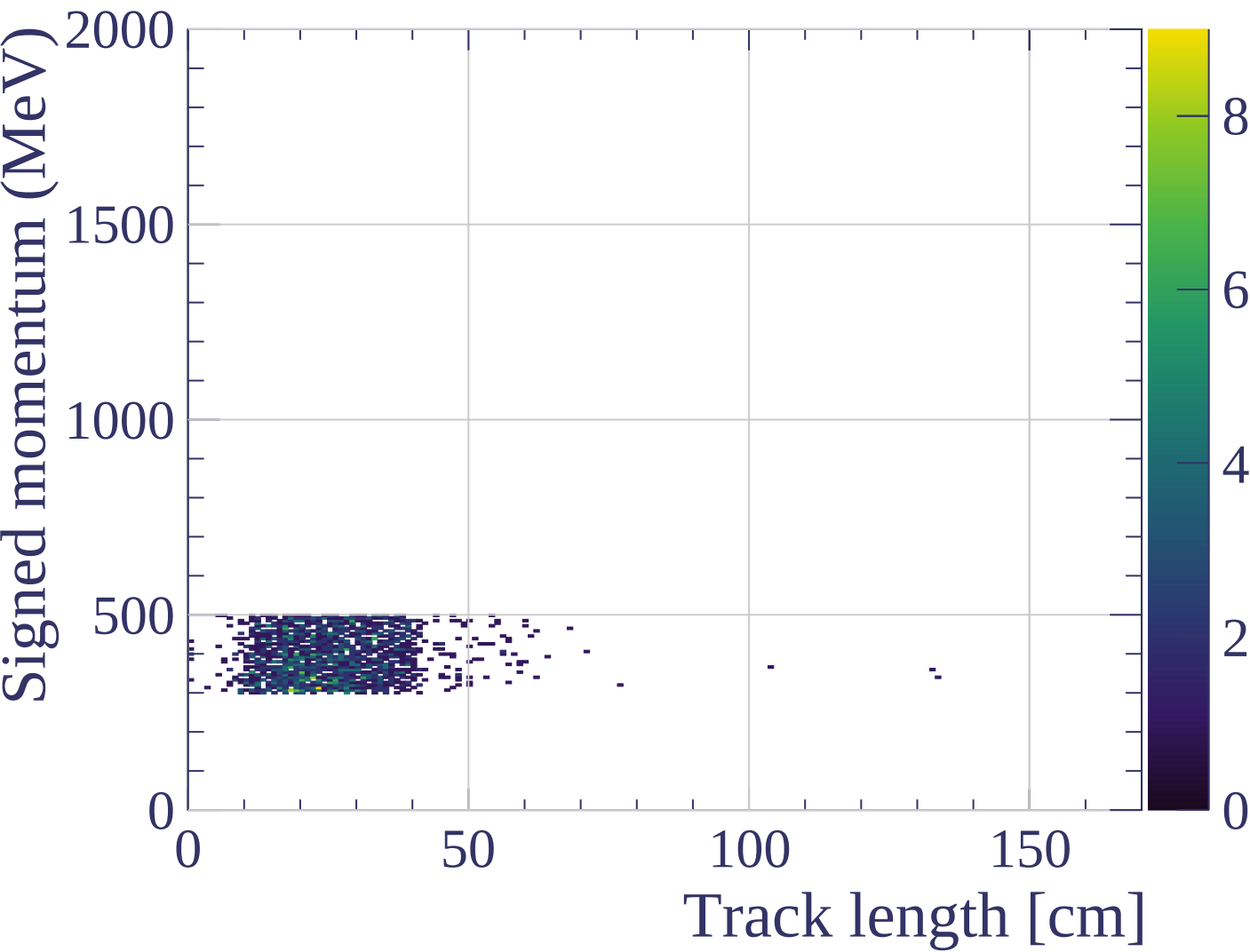


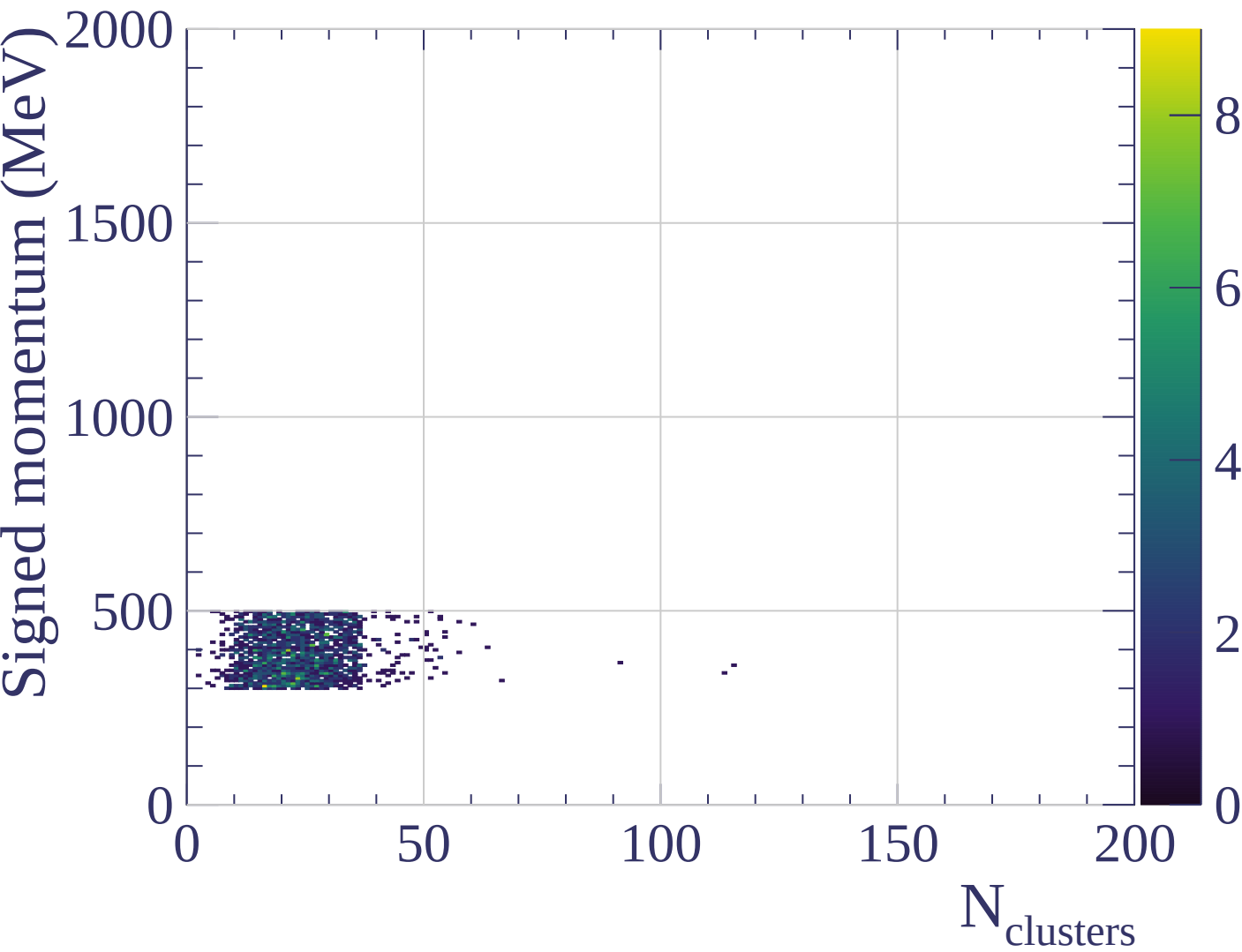


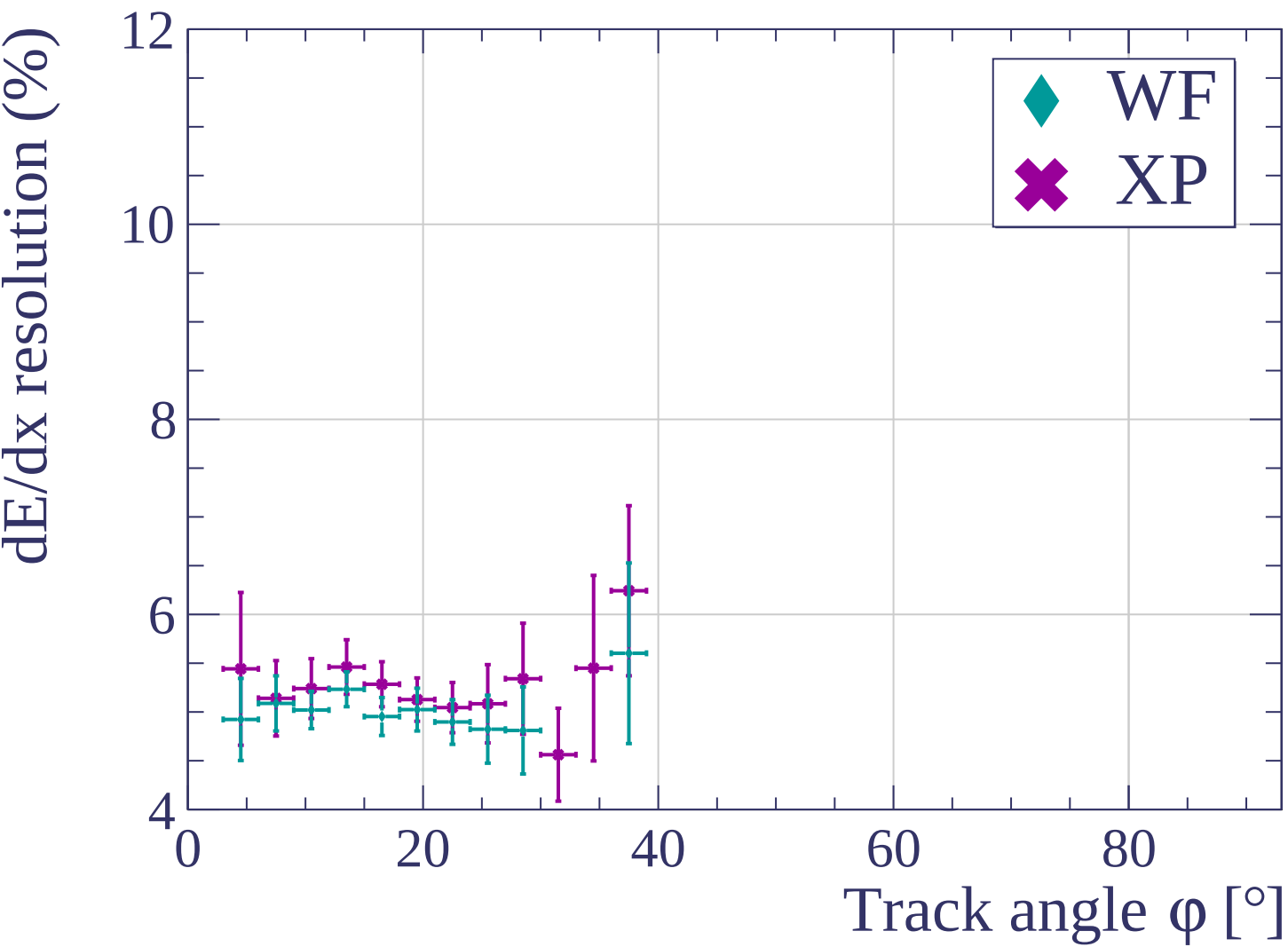


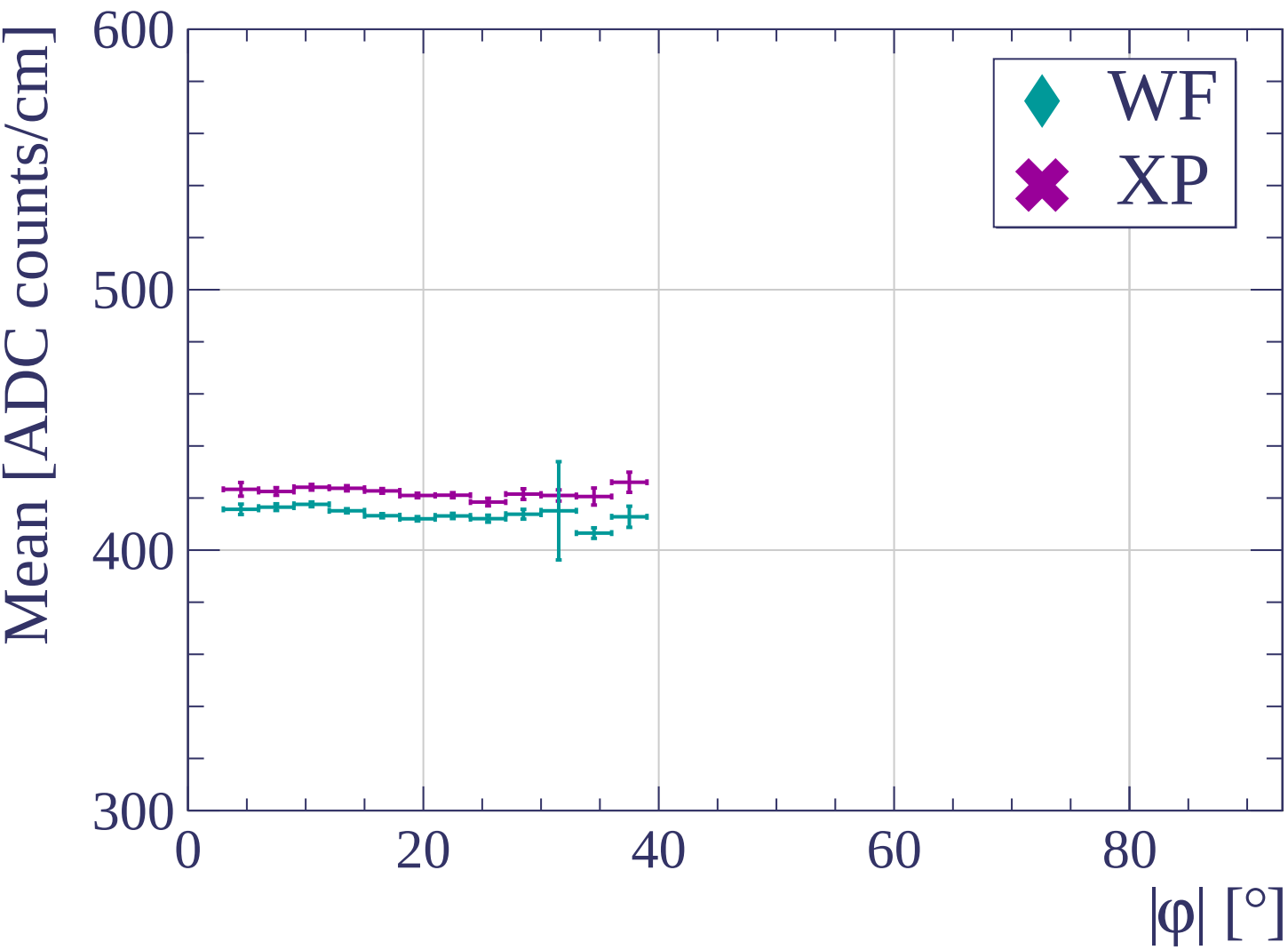


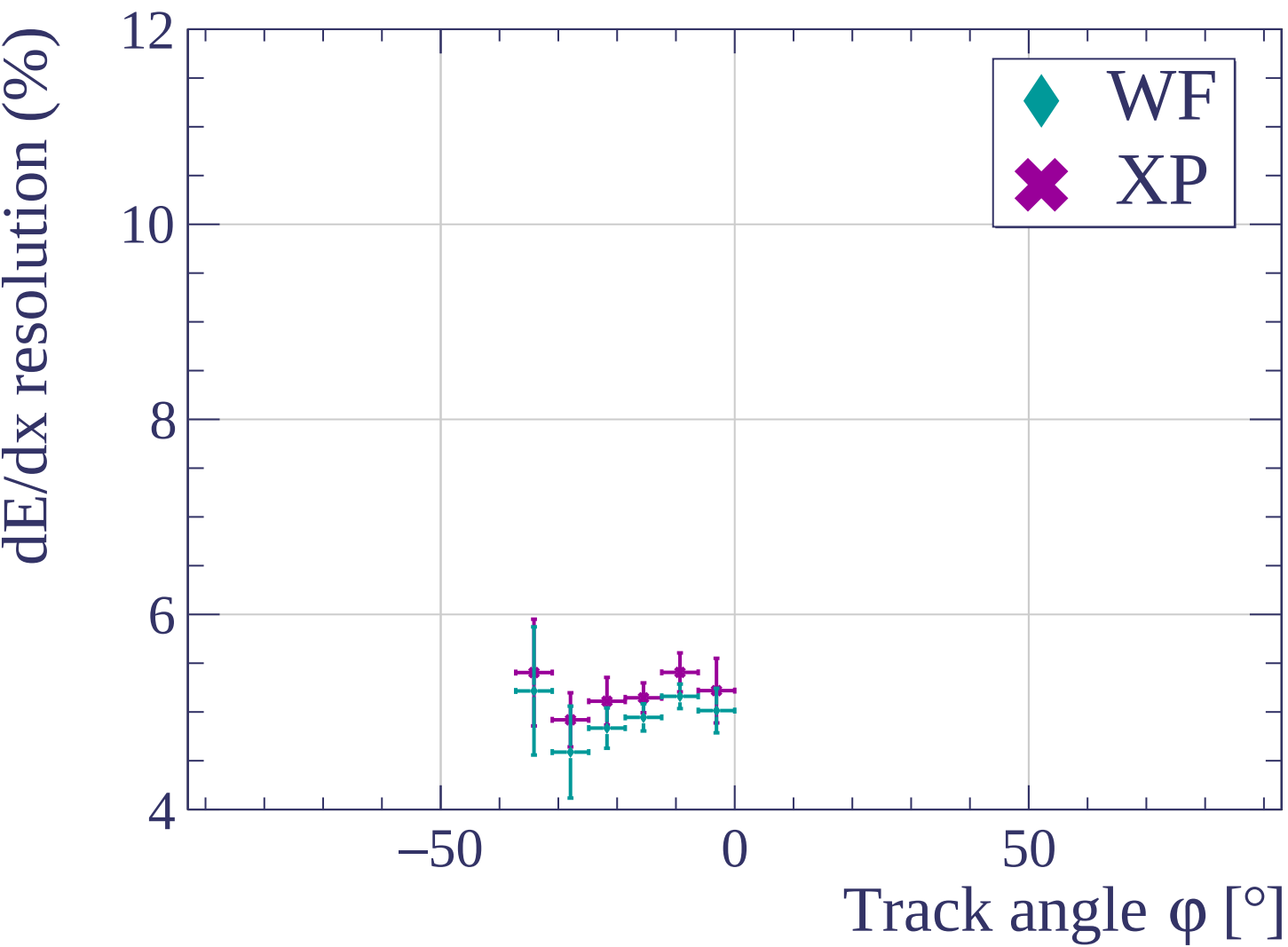


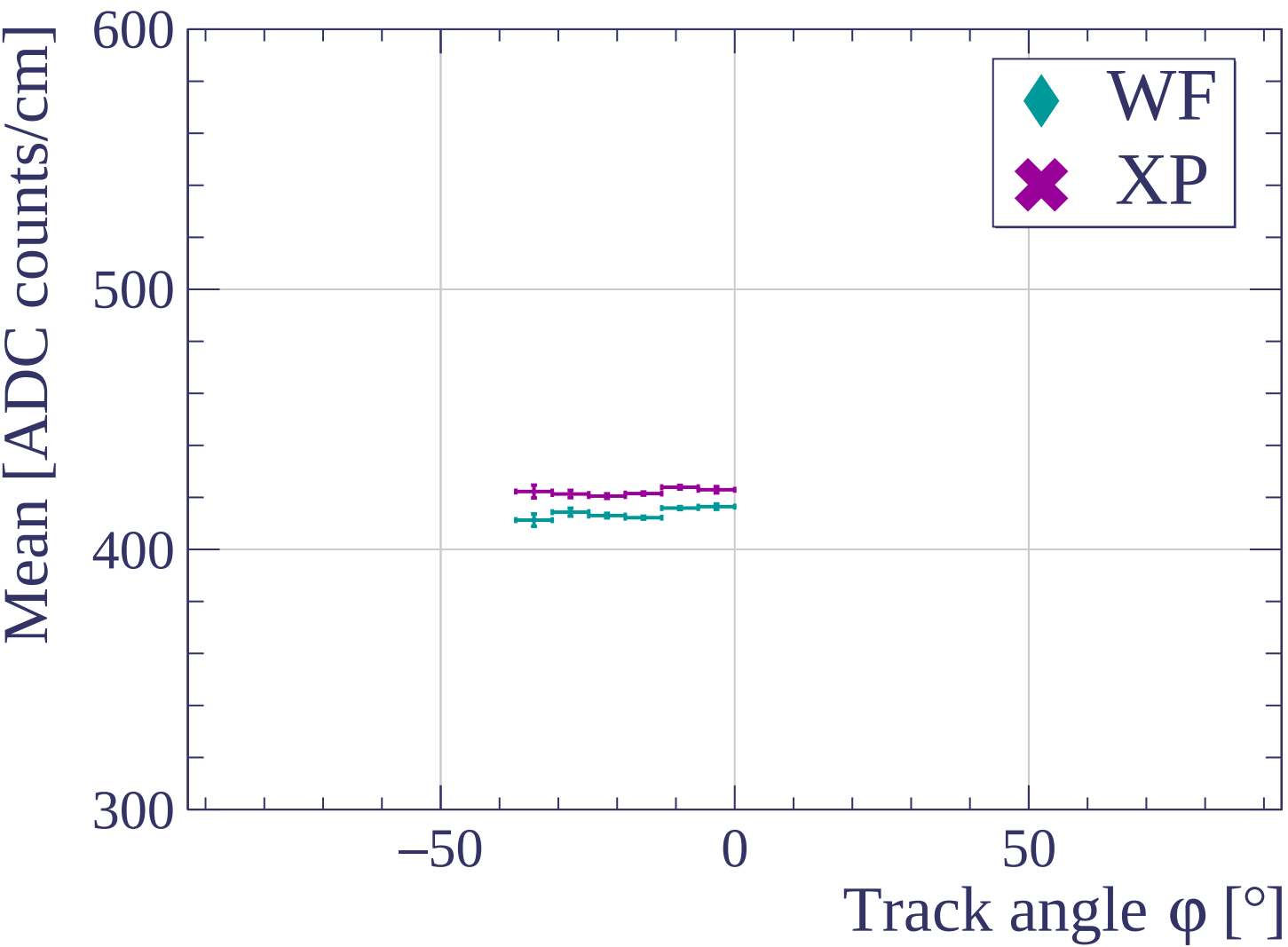


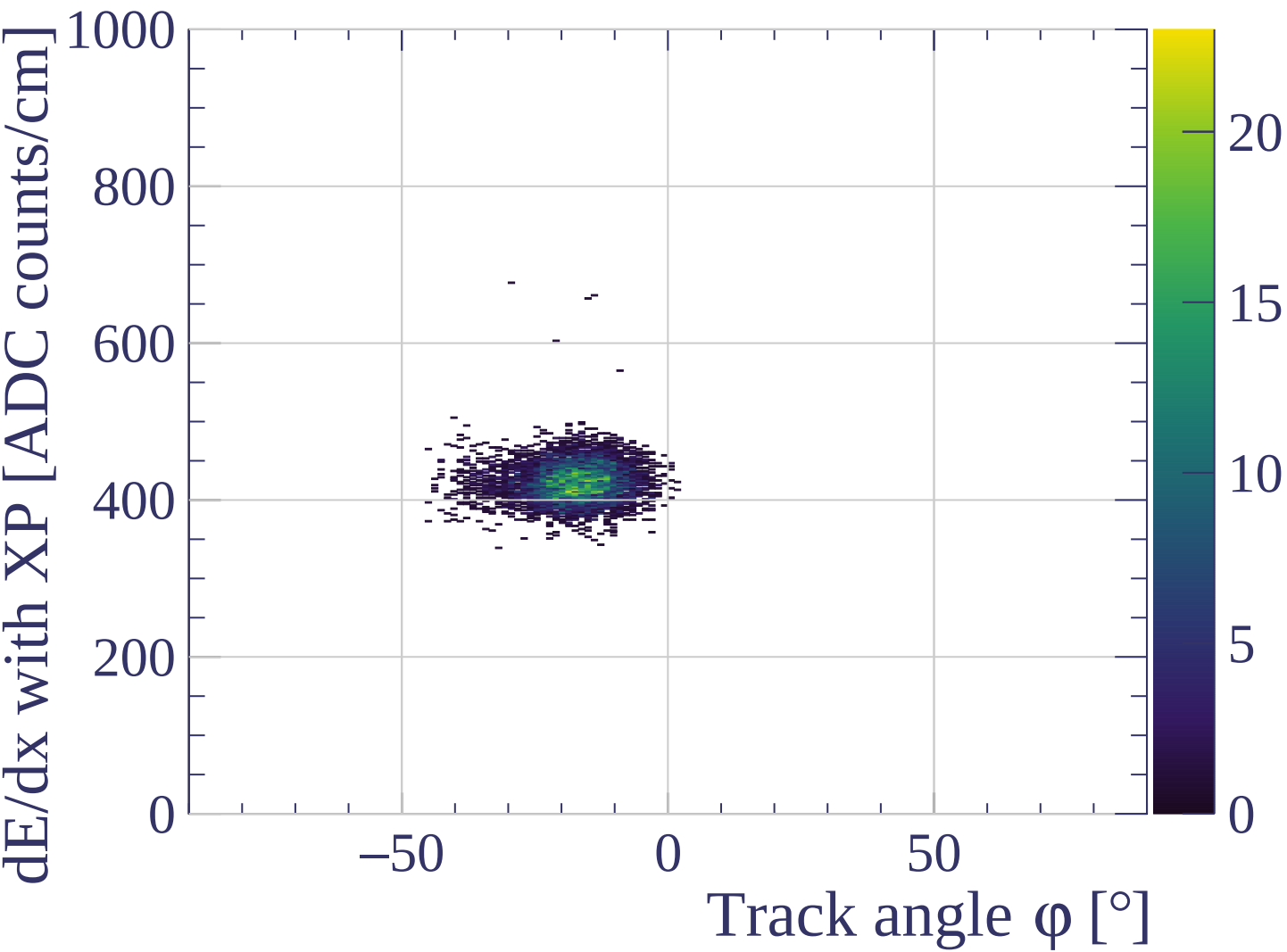




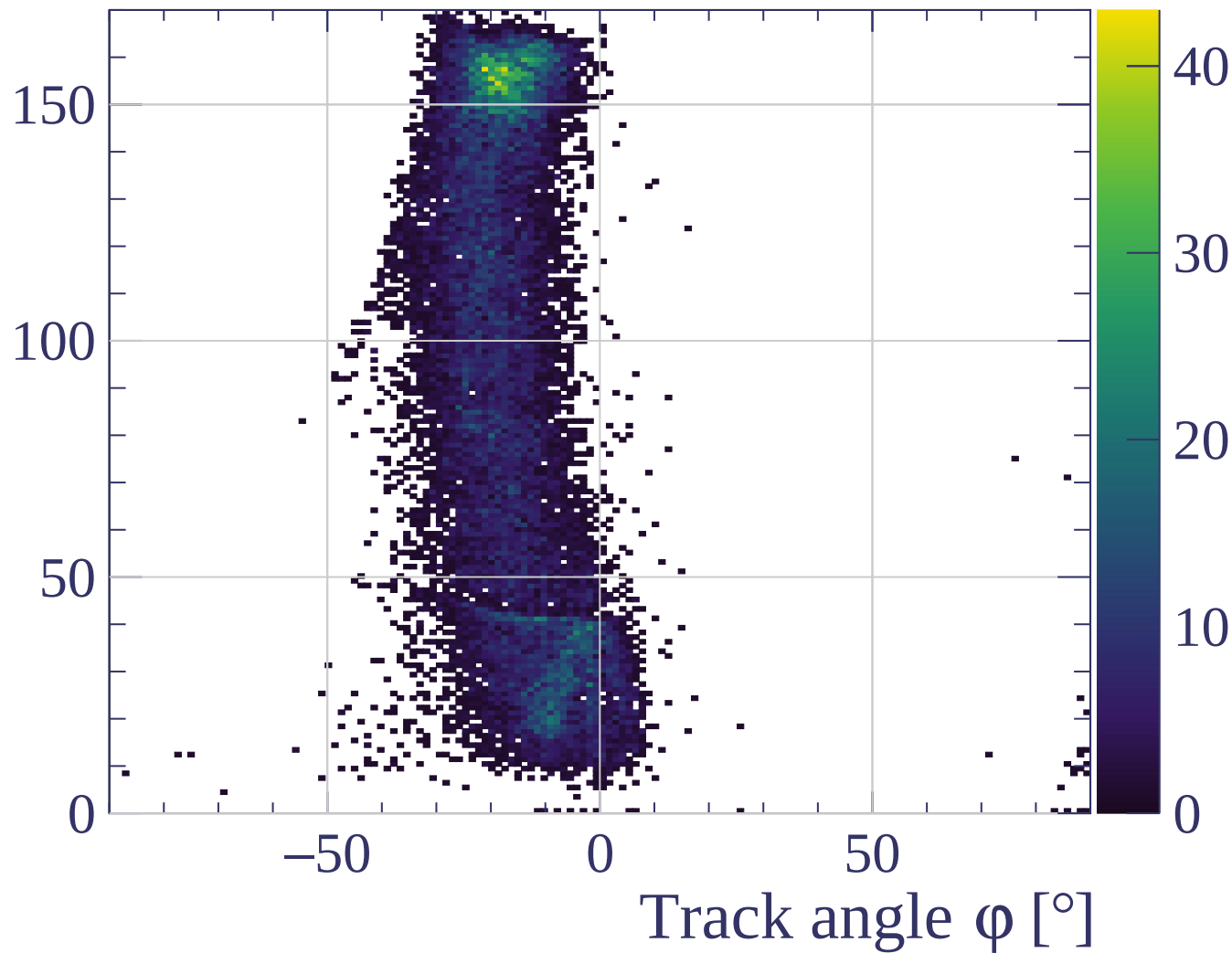


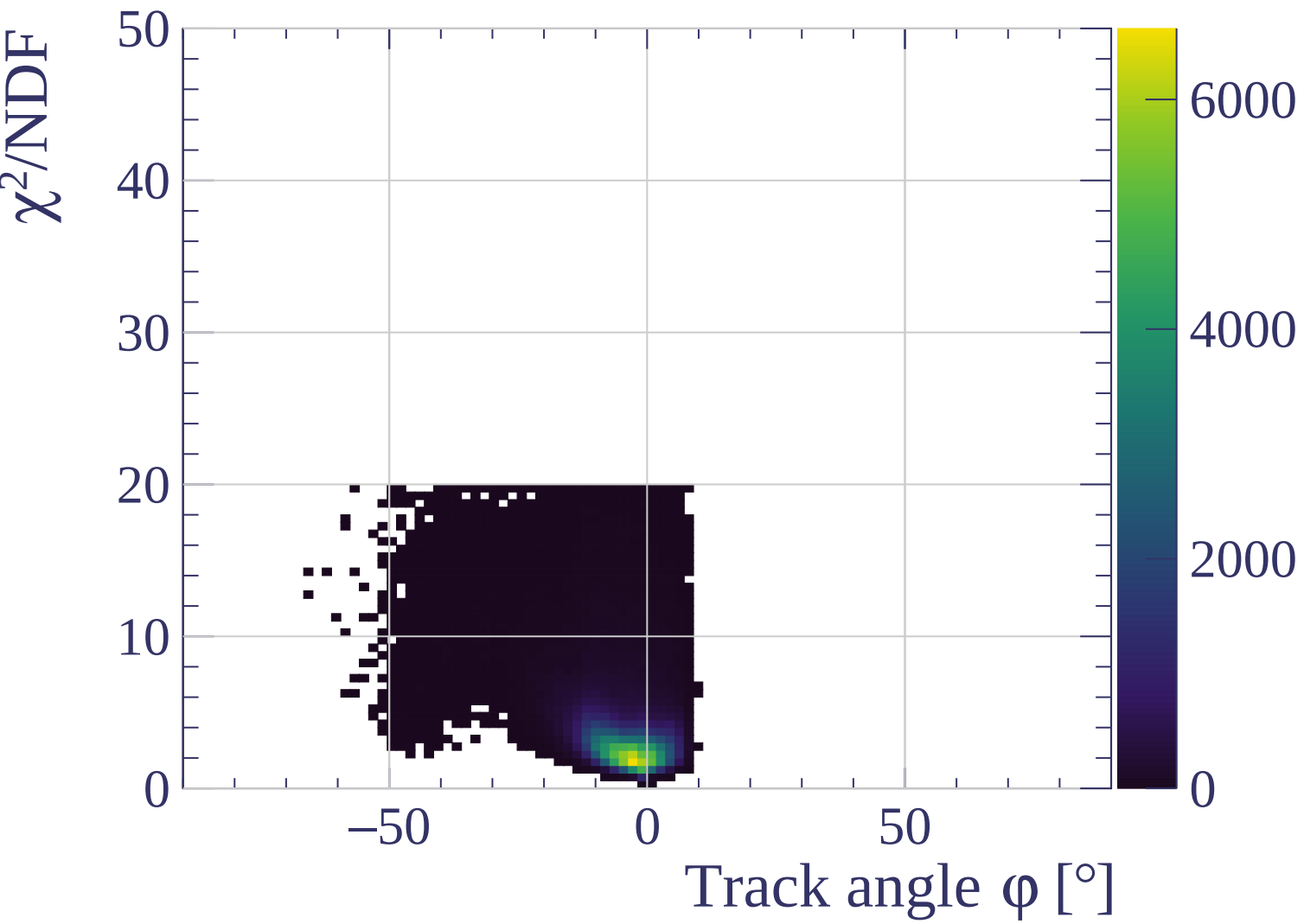


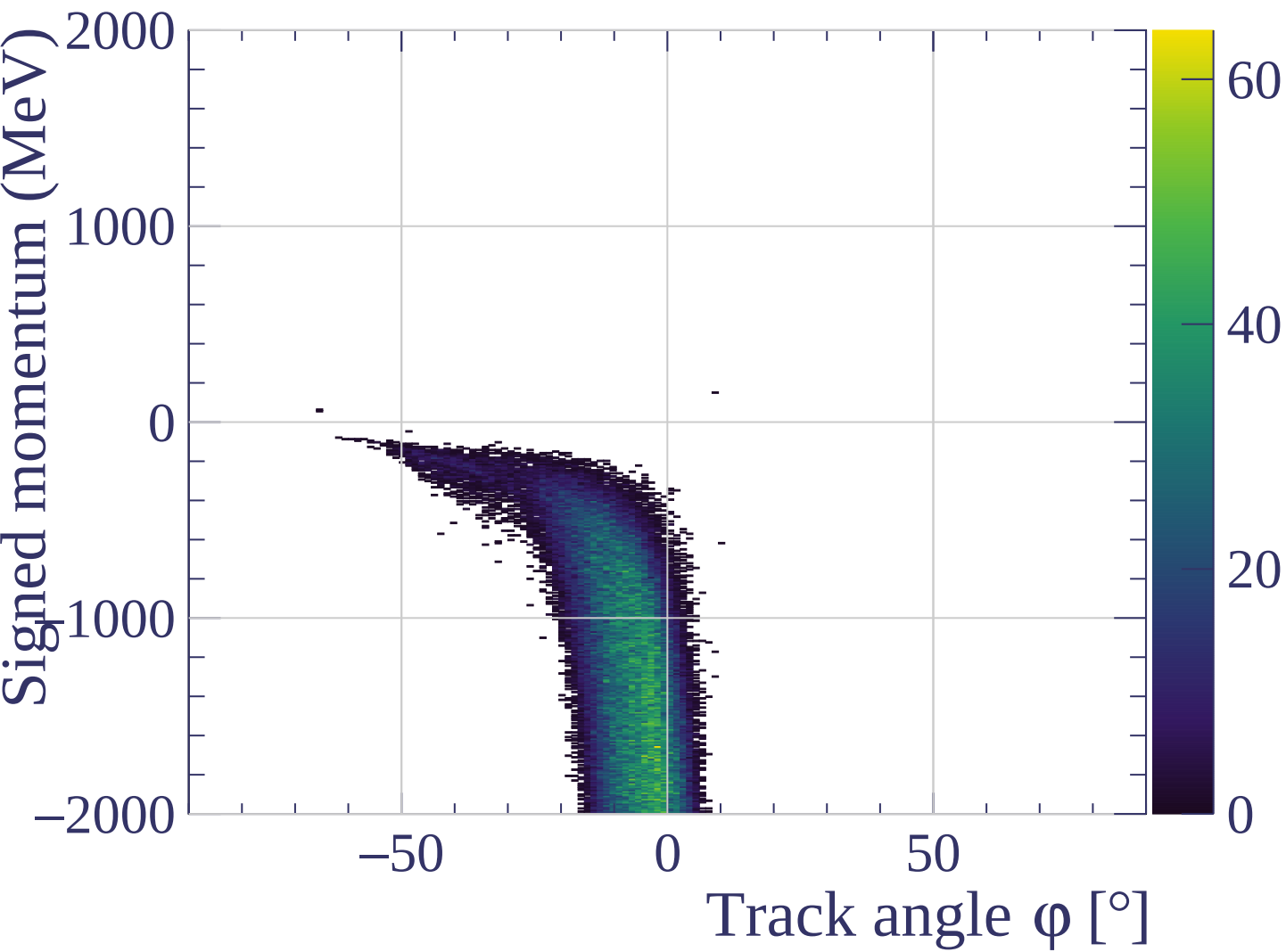




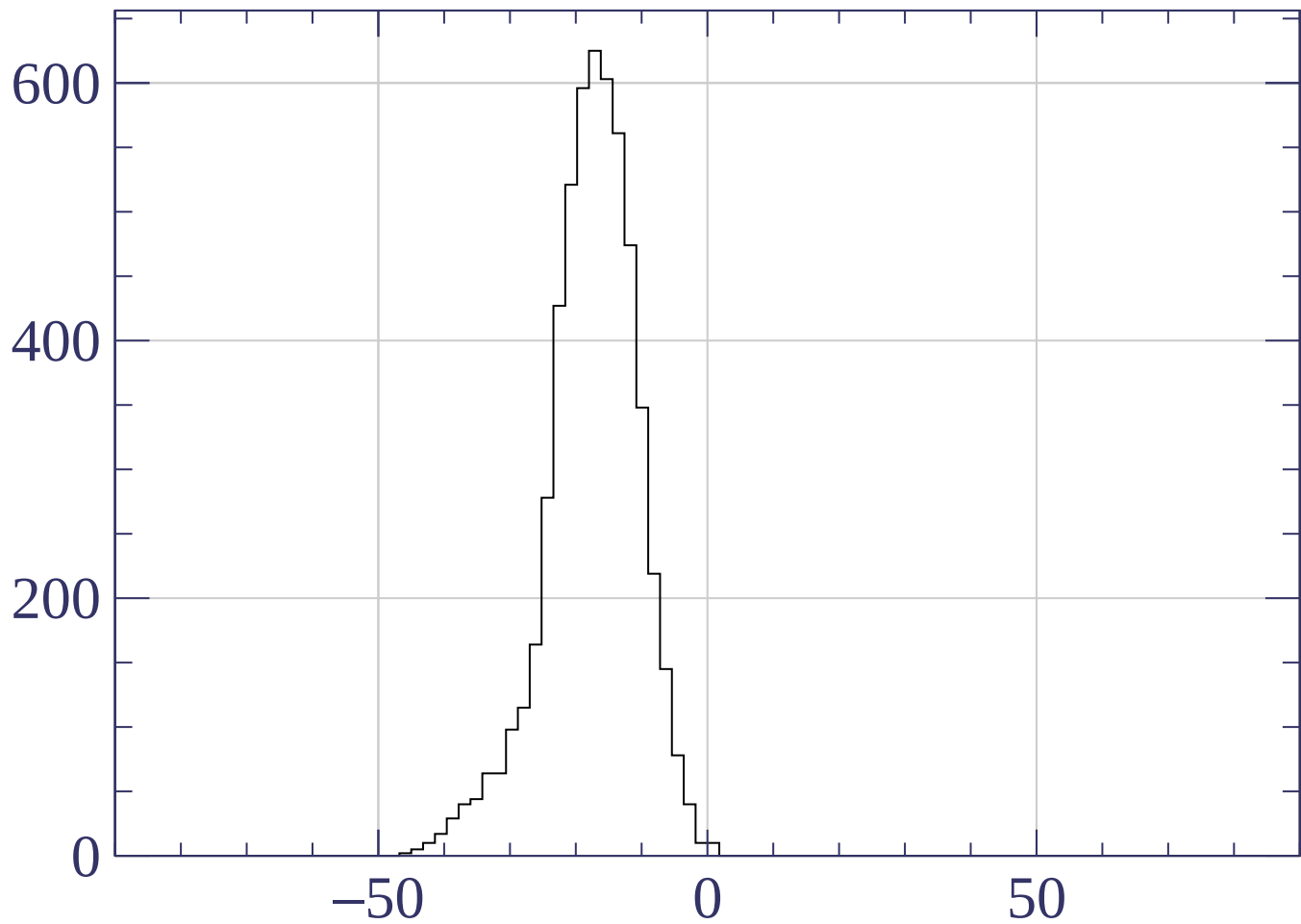
Track length [cm]



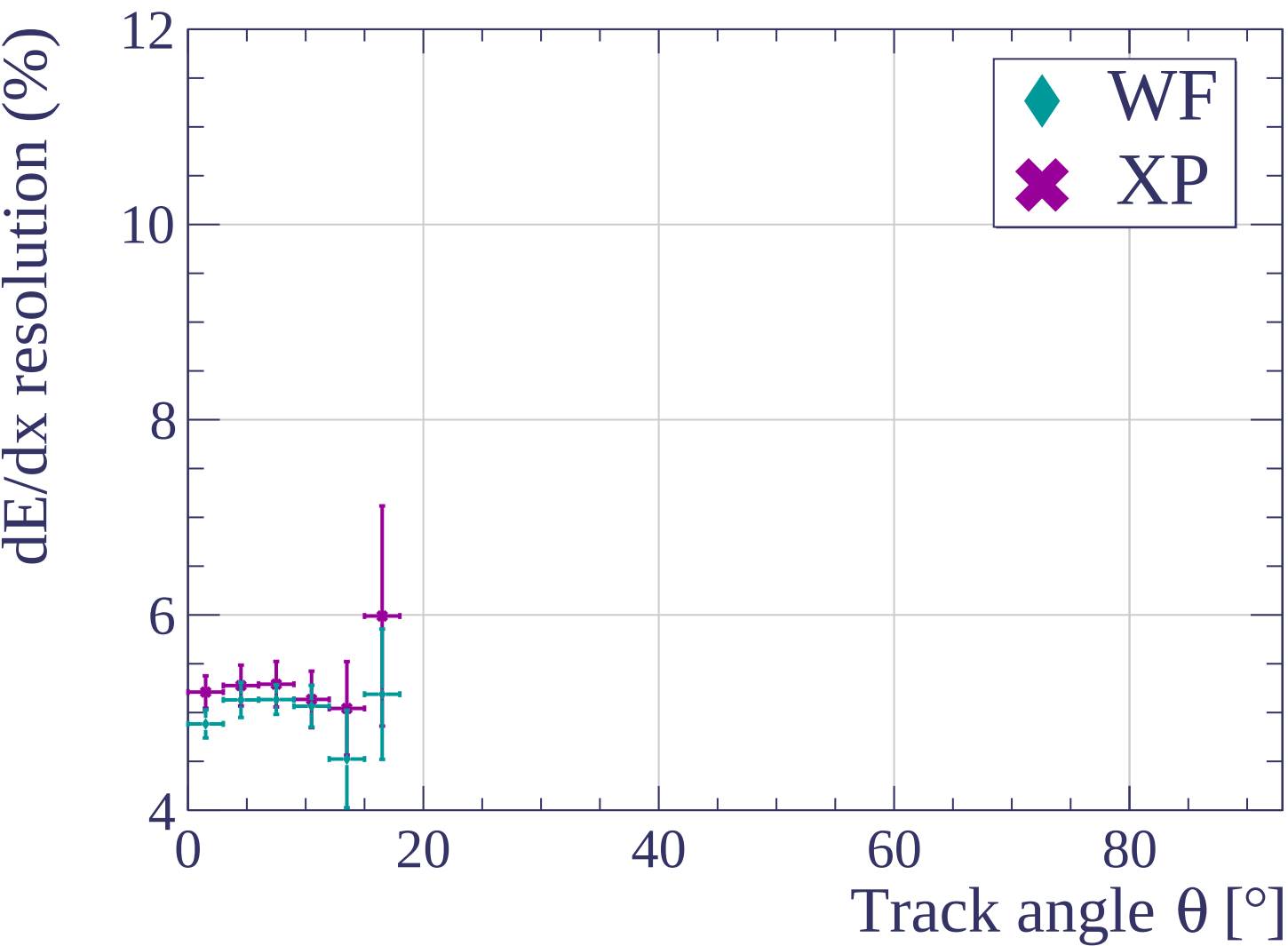


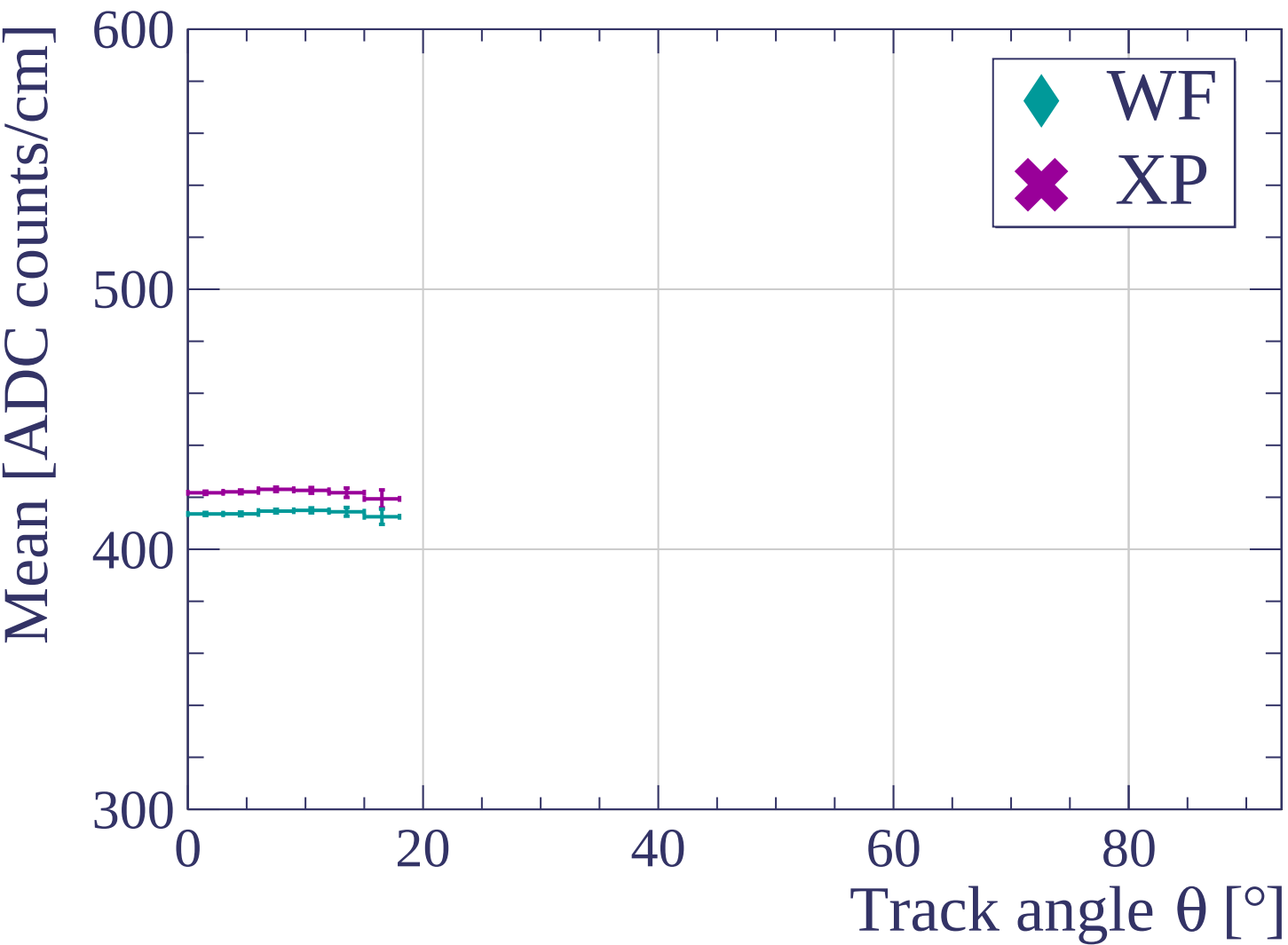


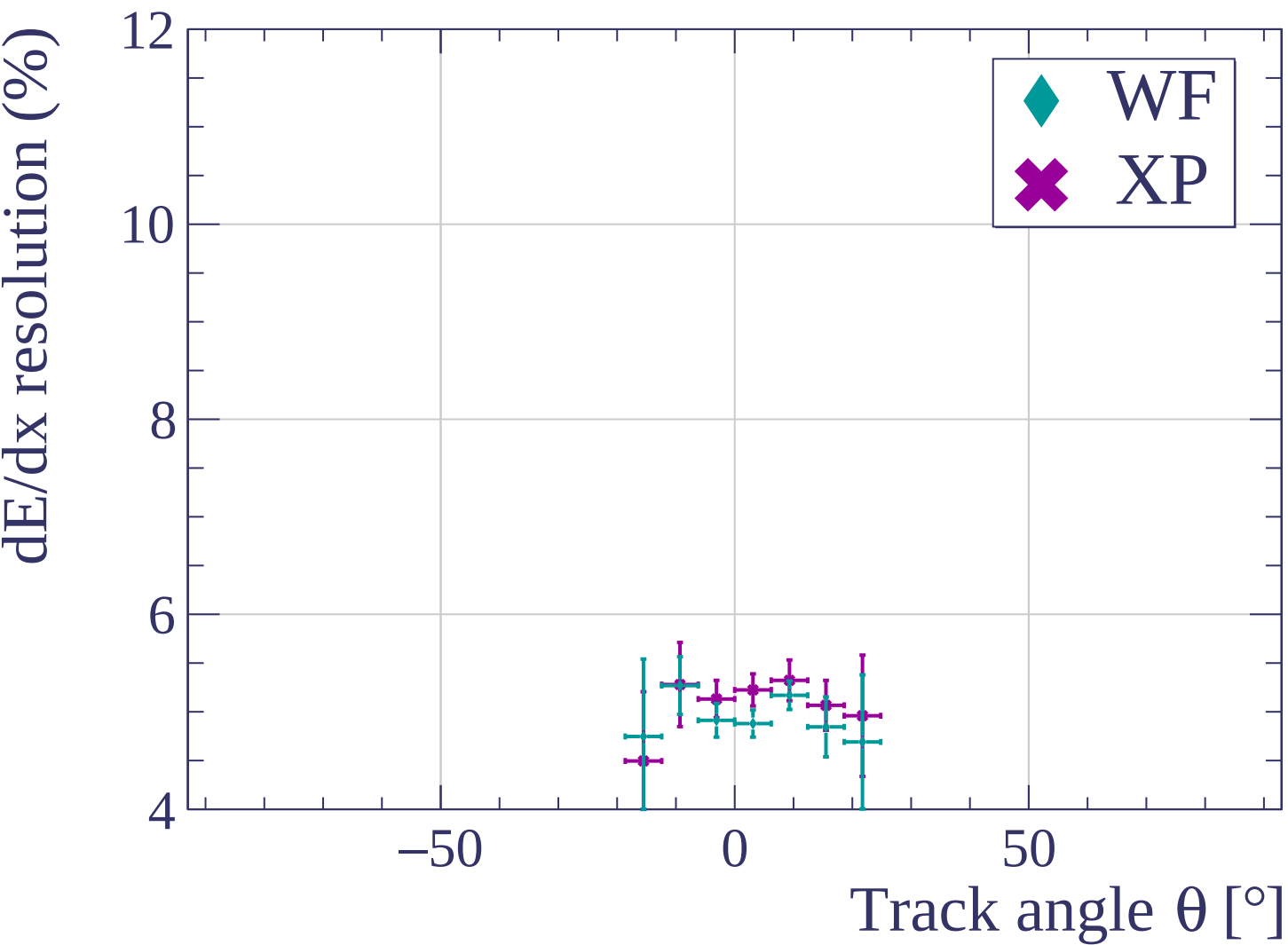
Count

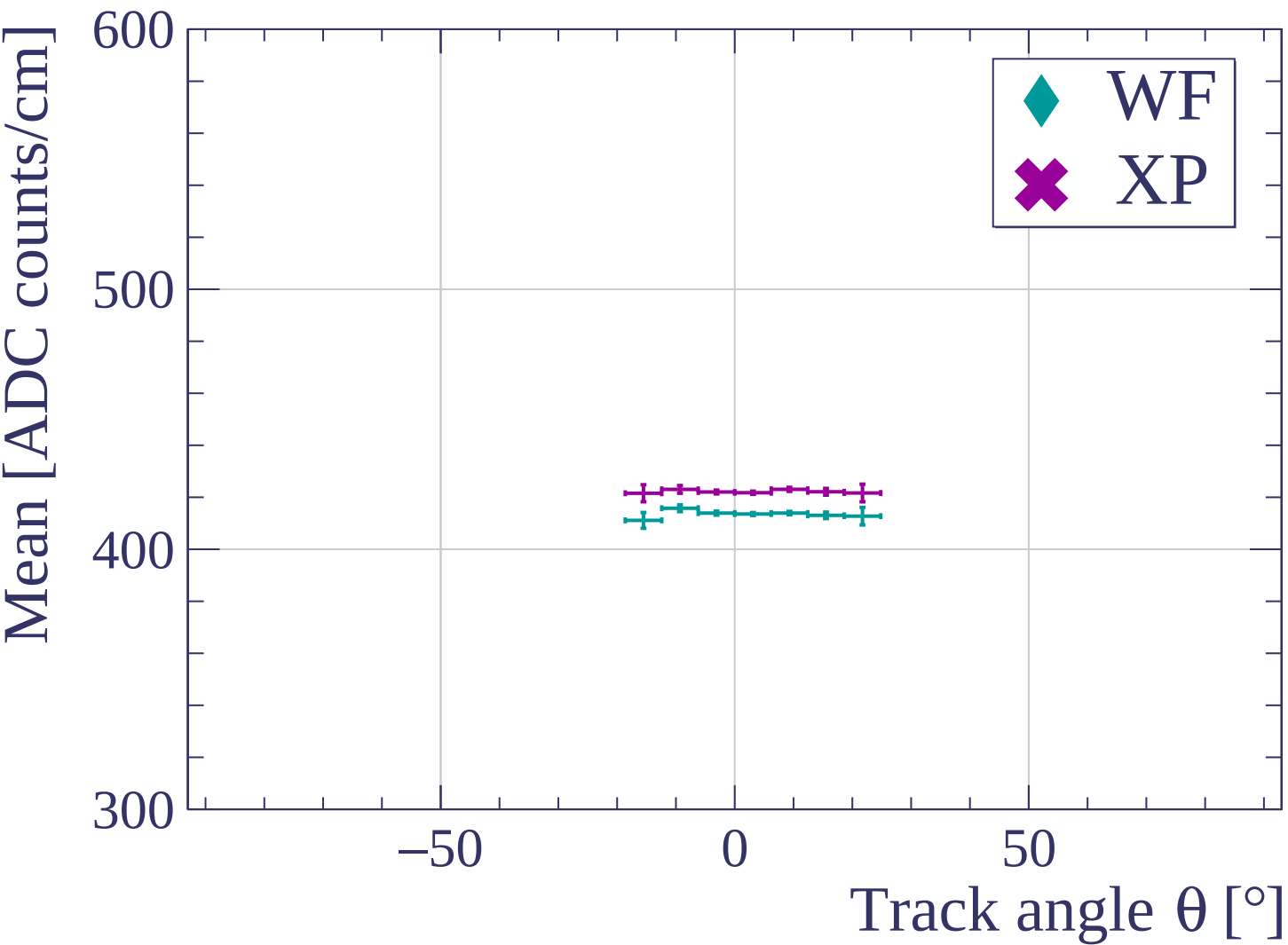


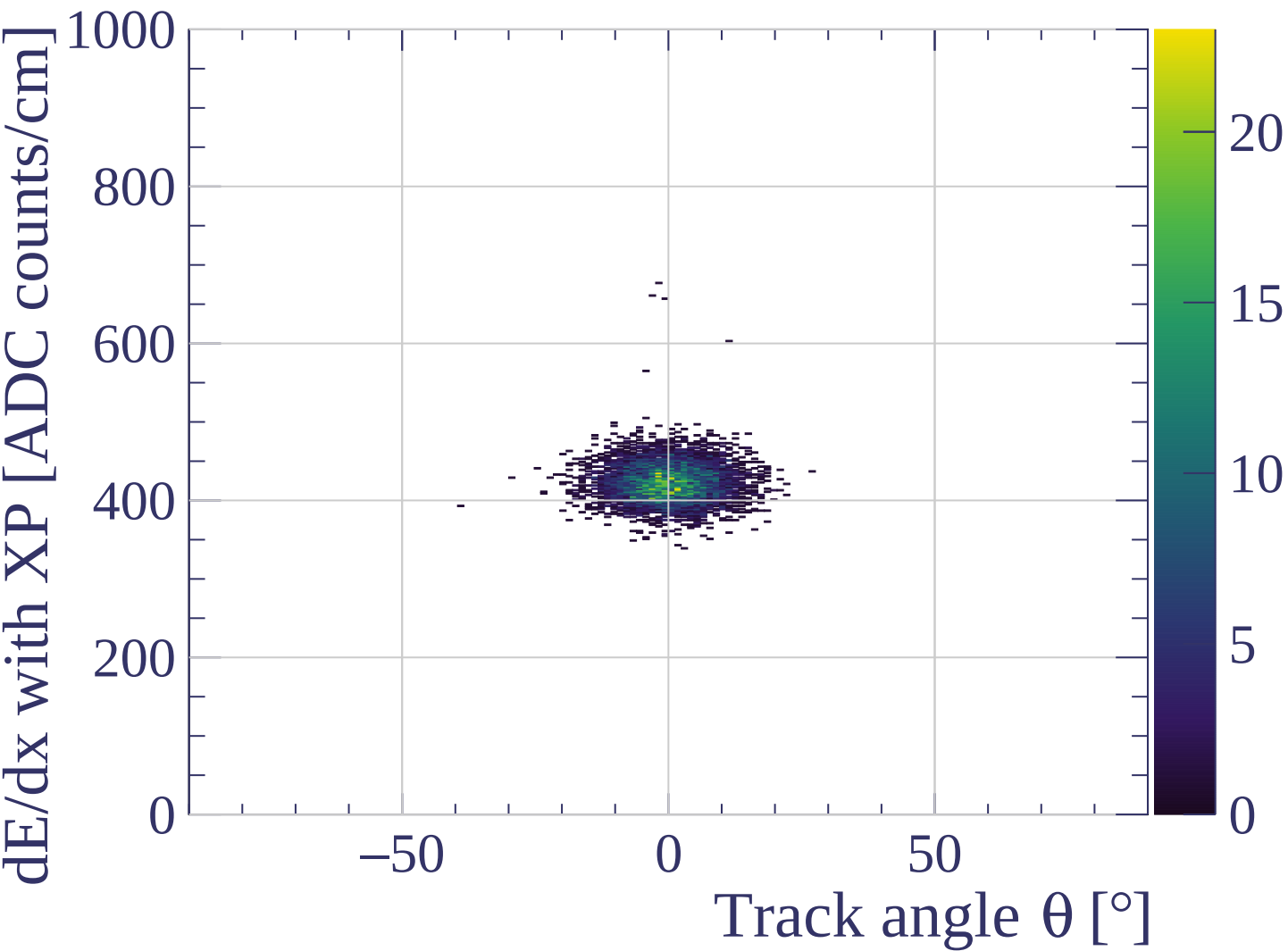
Track angle ϕ [°] angle



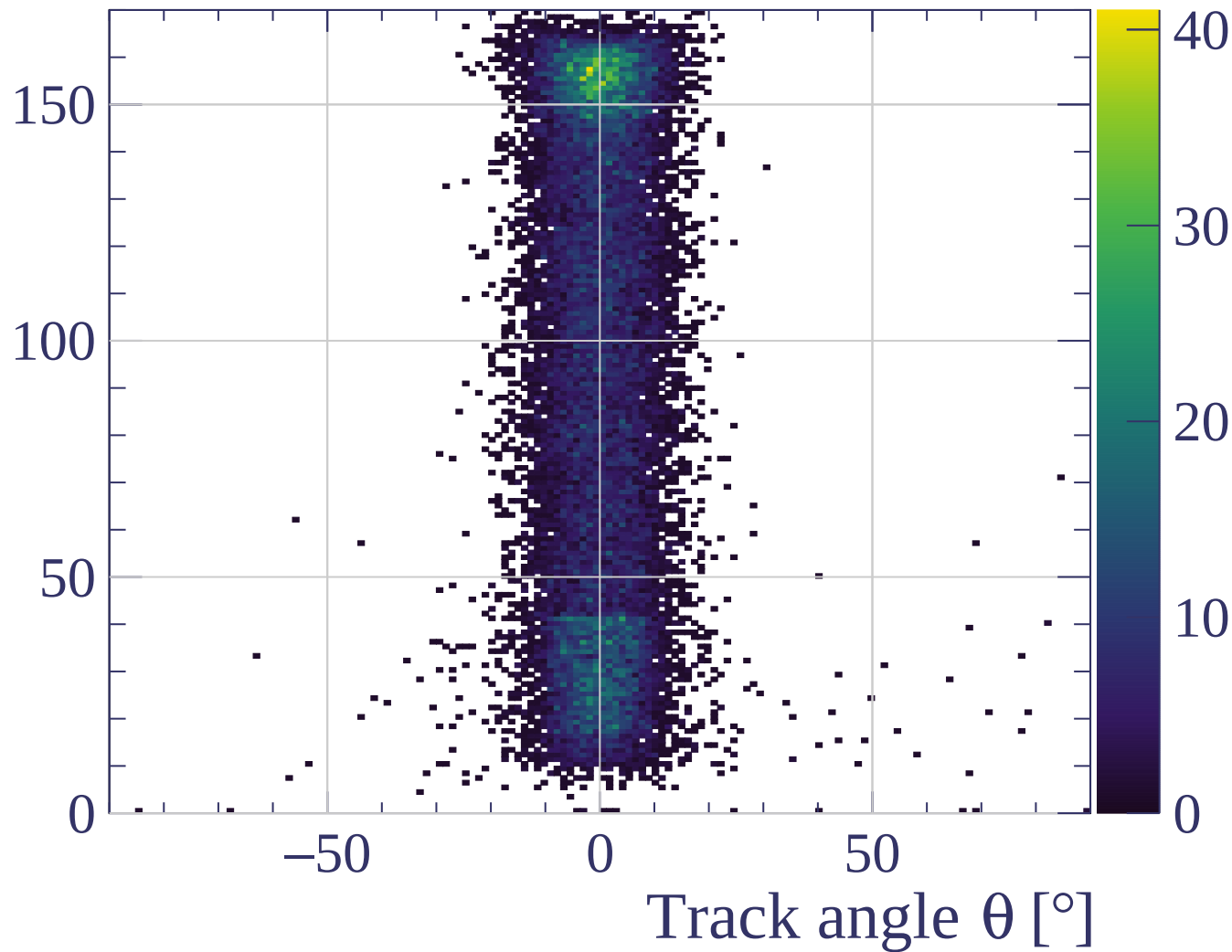


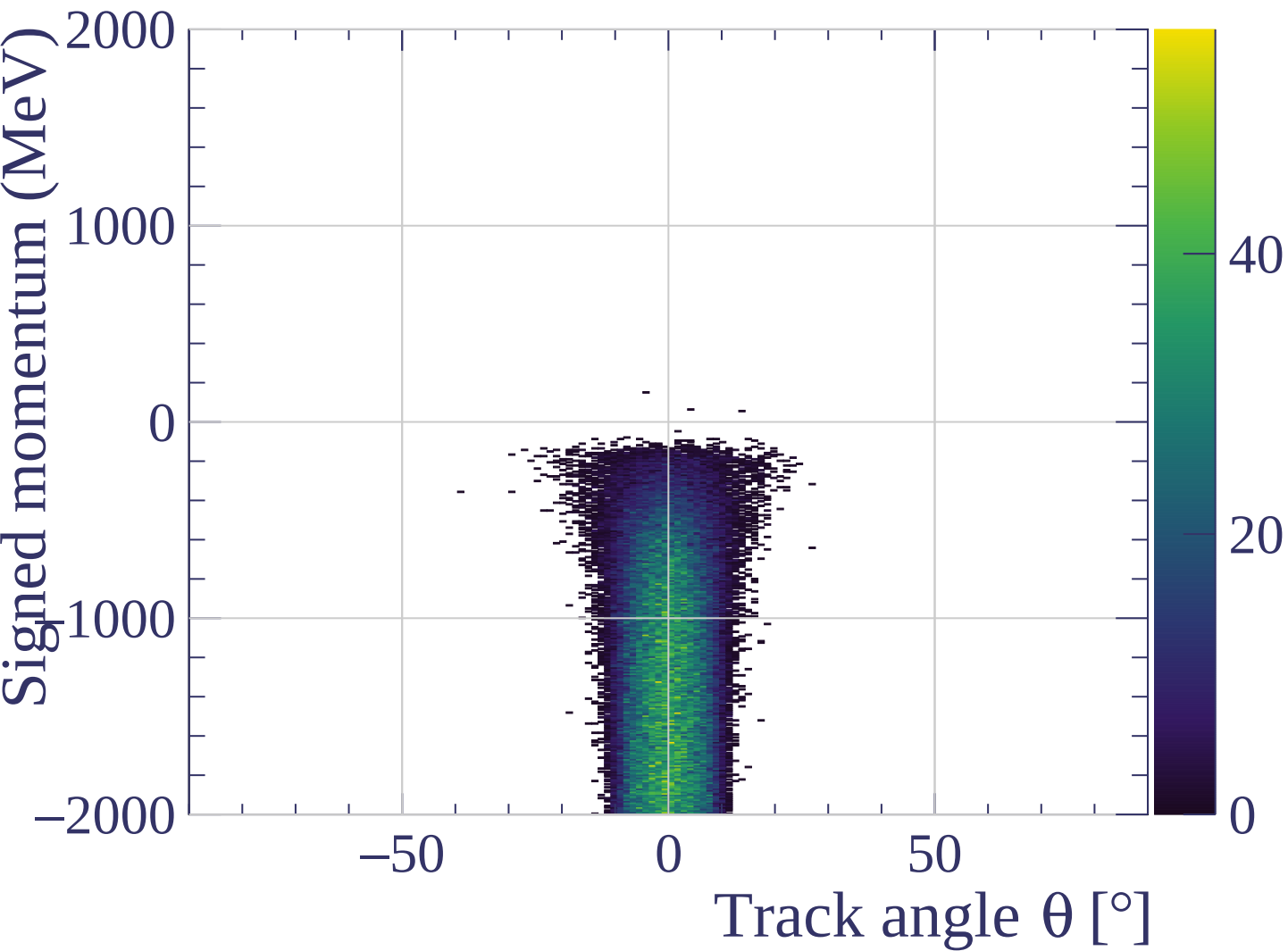




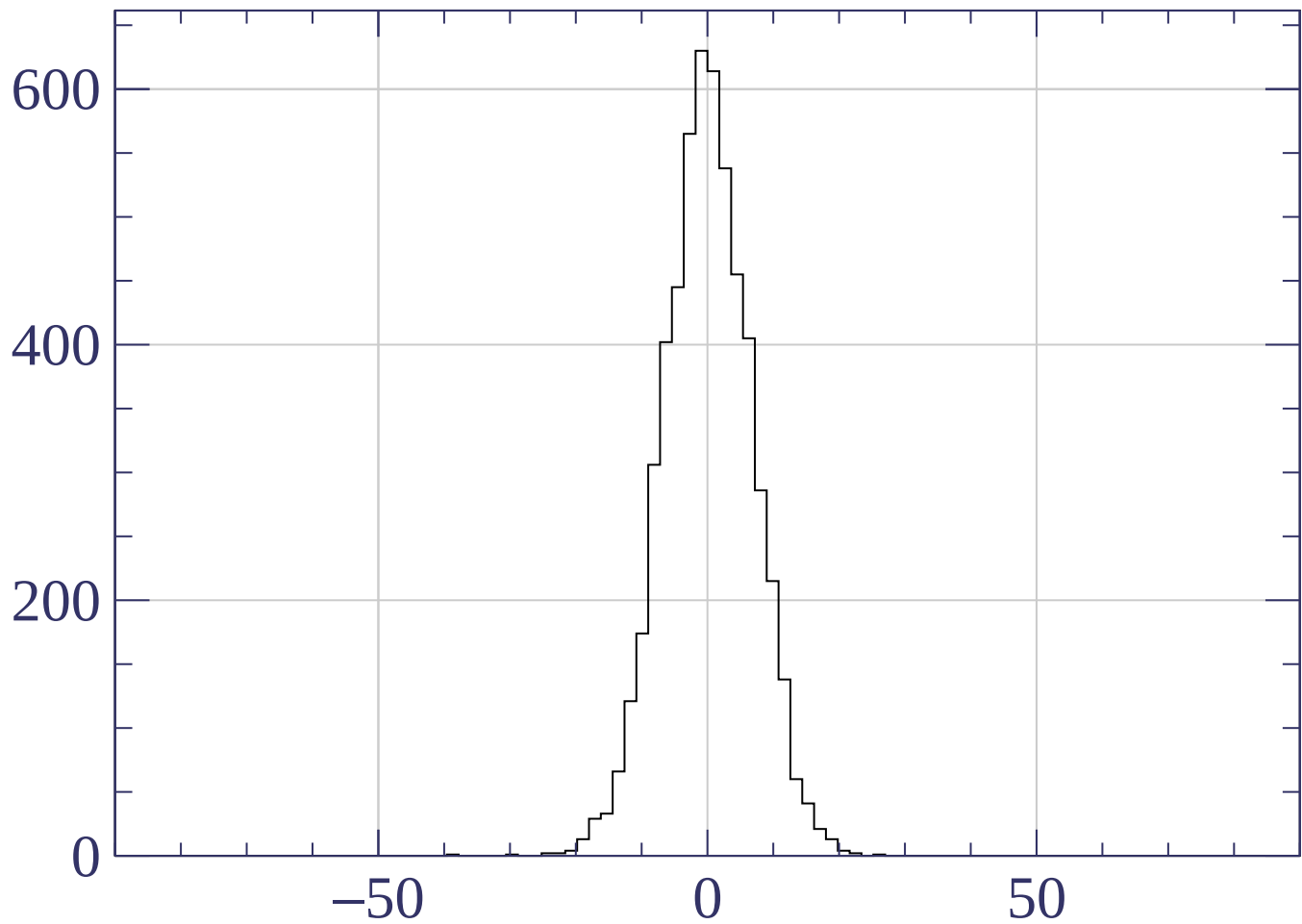


Track length [cm]



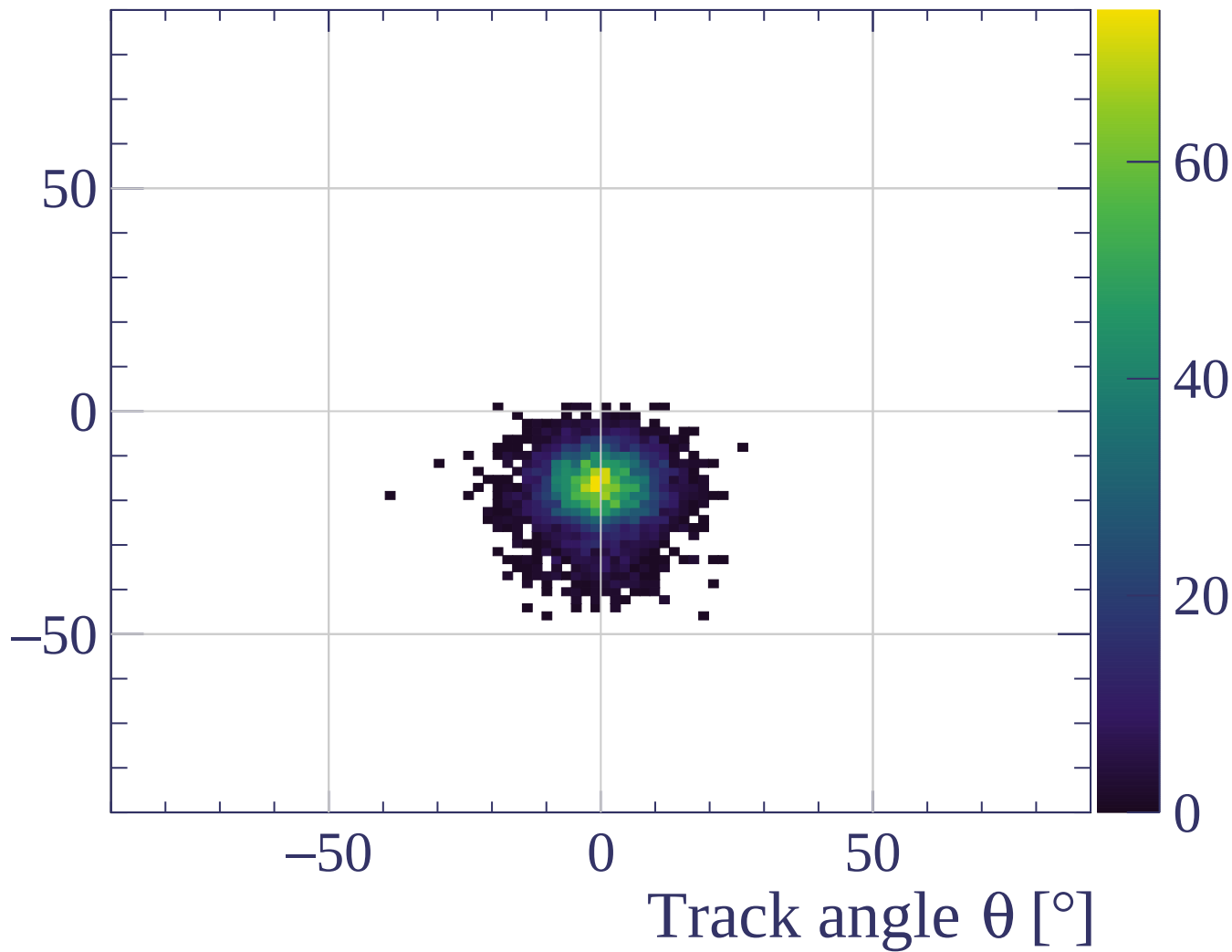


Count

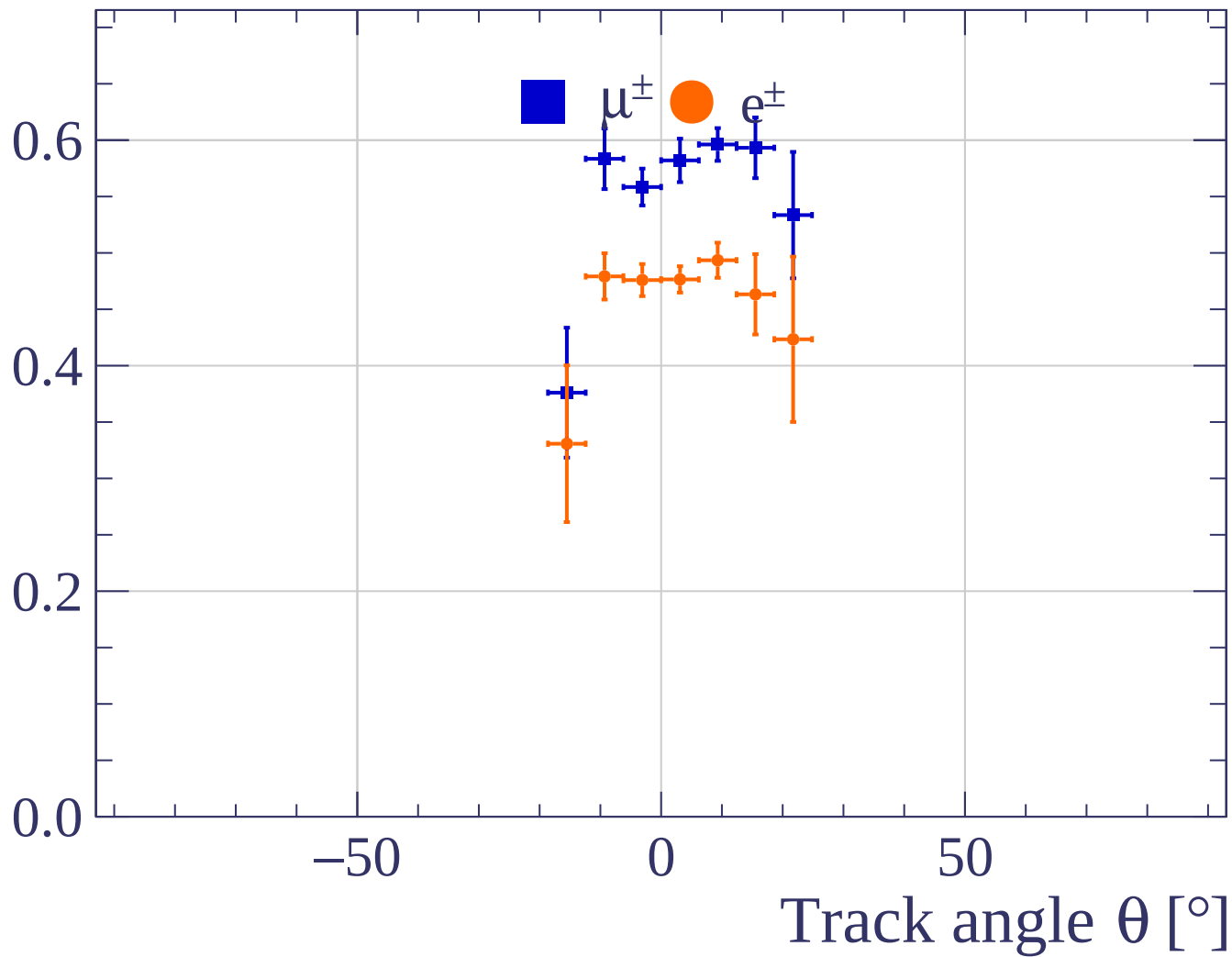


Track angle θ [°]

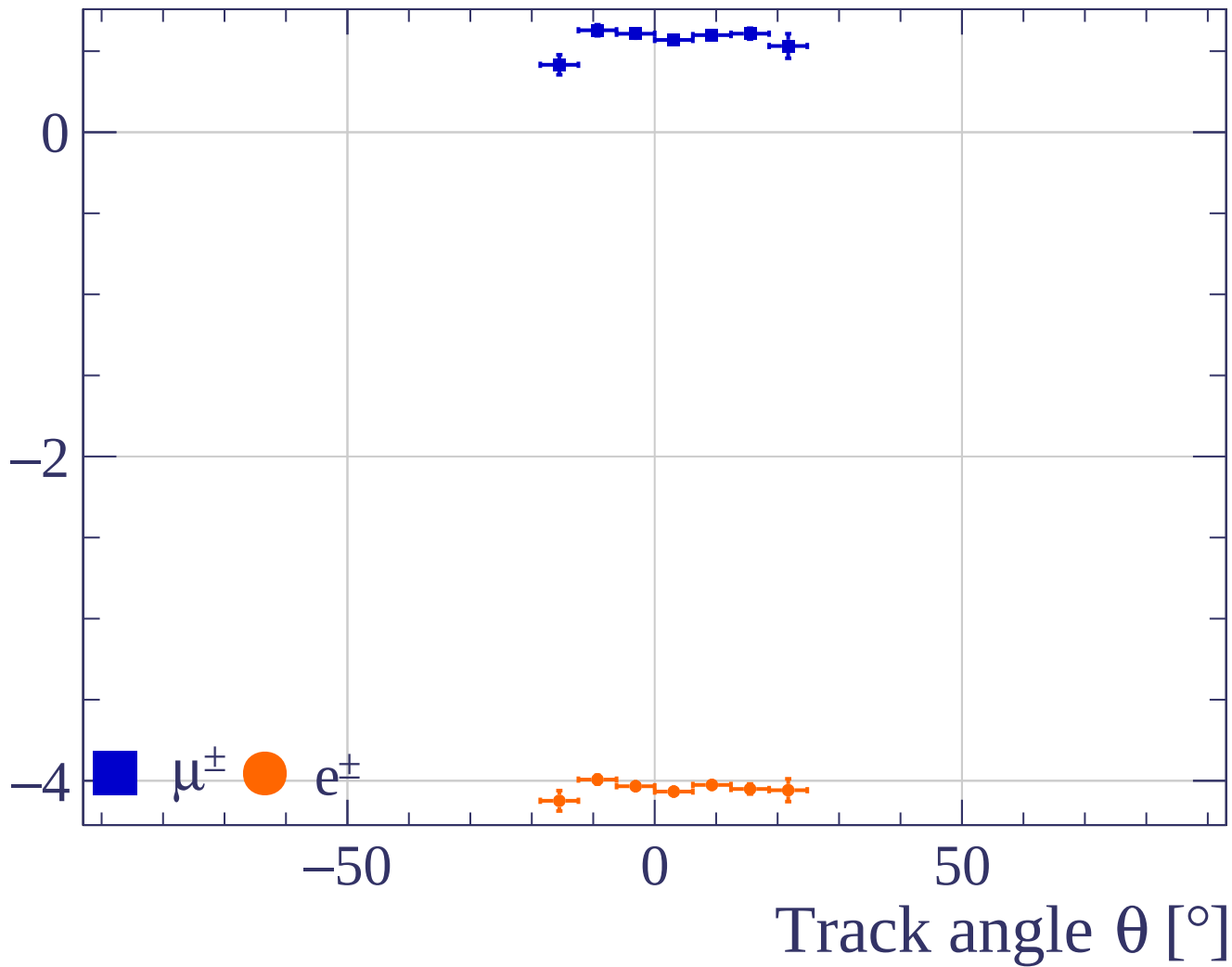
Track angle ϕ [°]



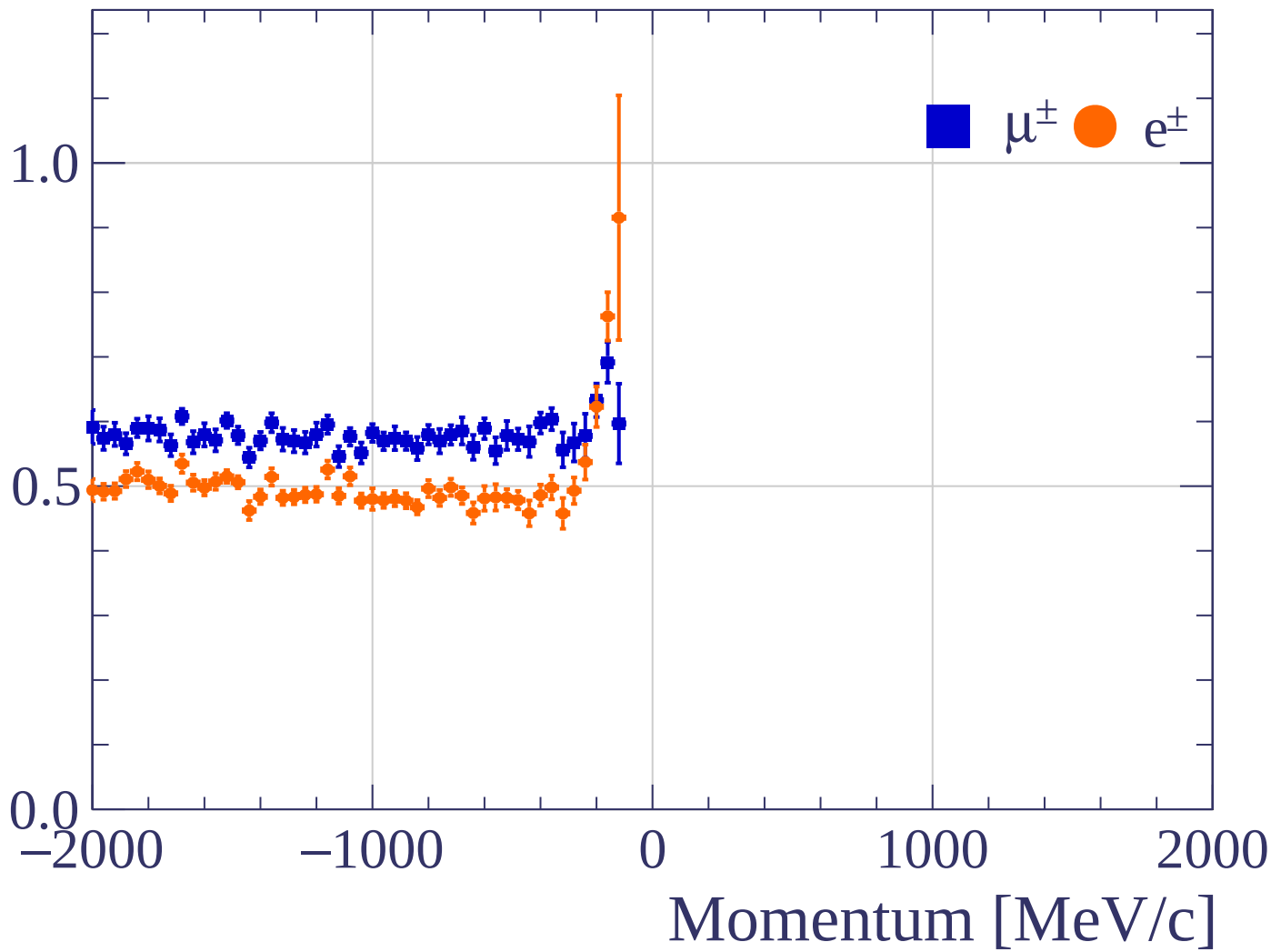
Pulls standard deviation



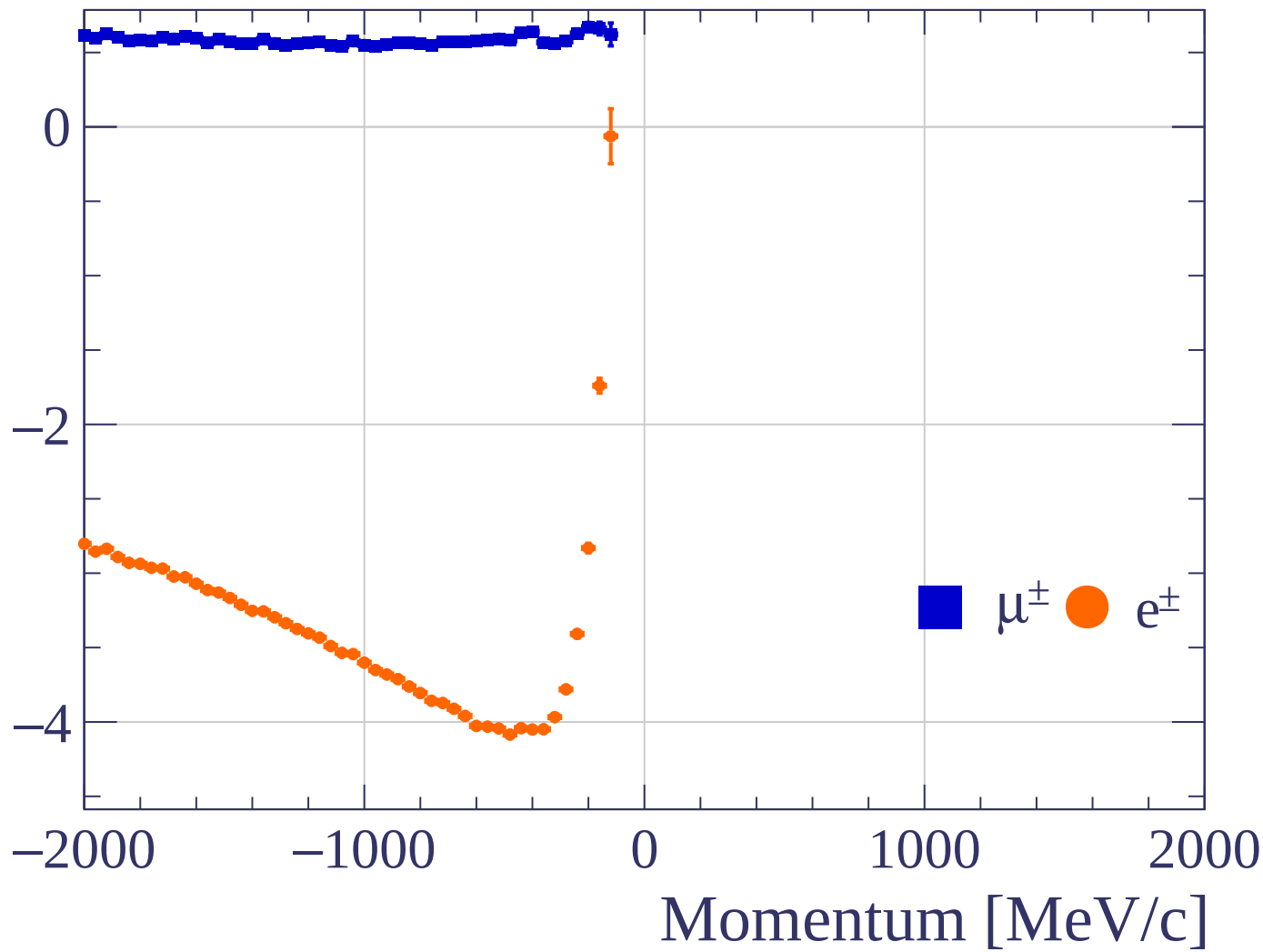
Pulls mean

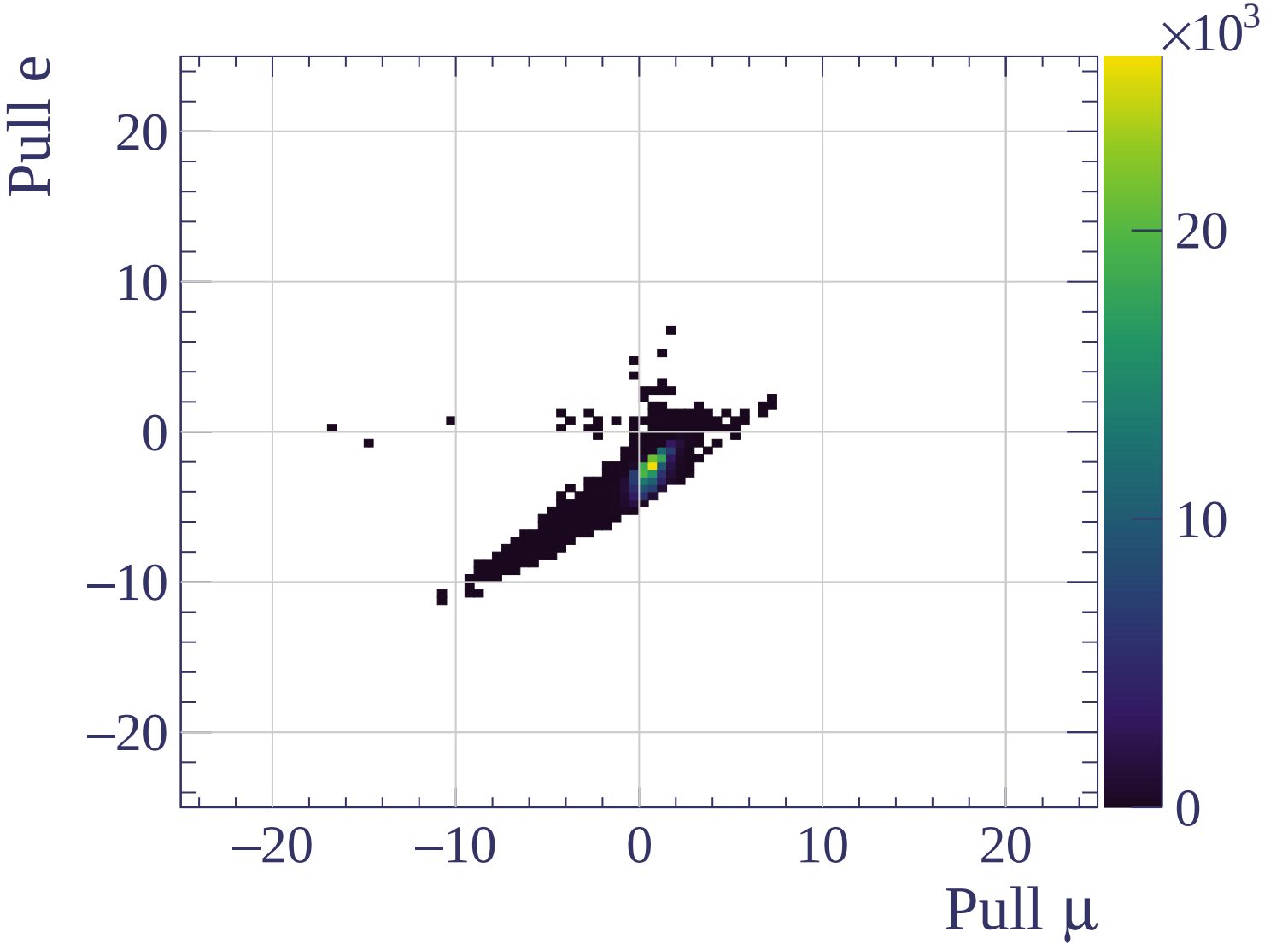


Pulls standard deviation

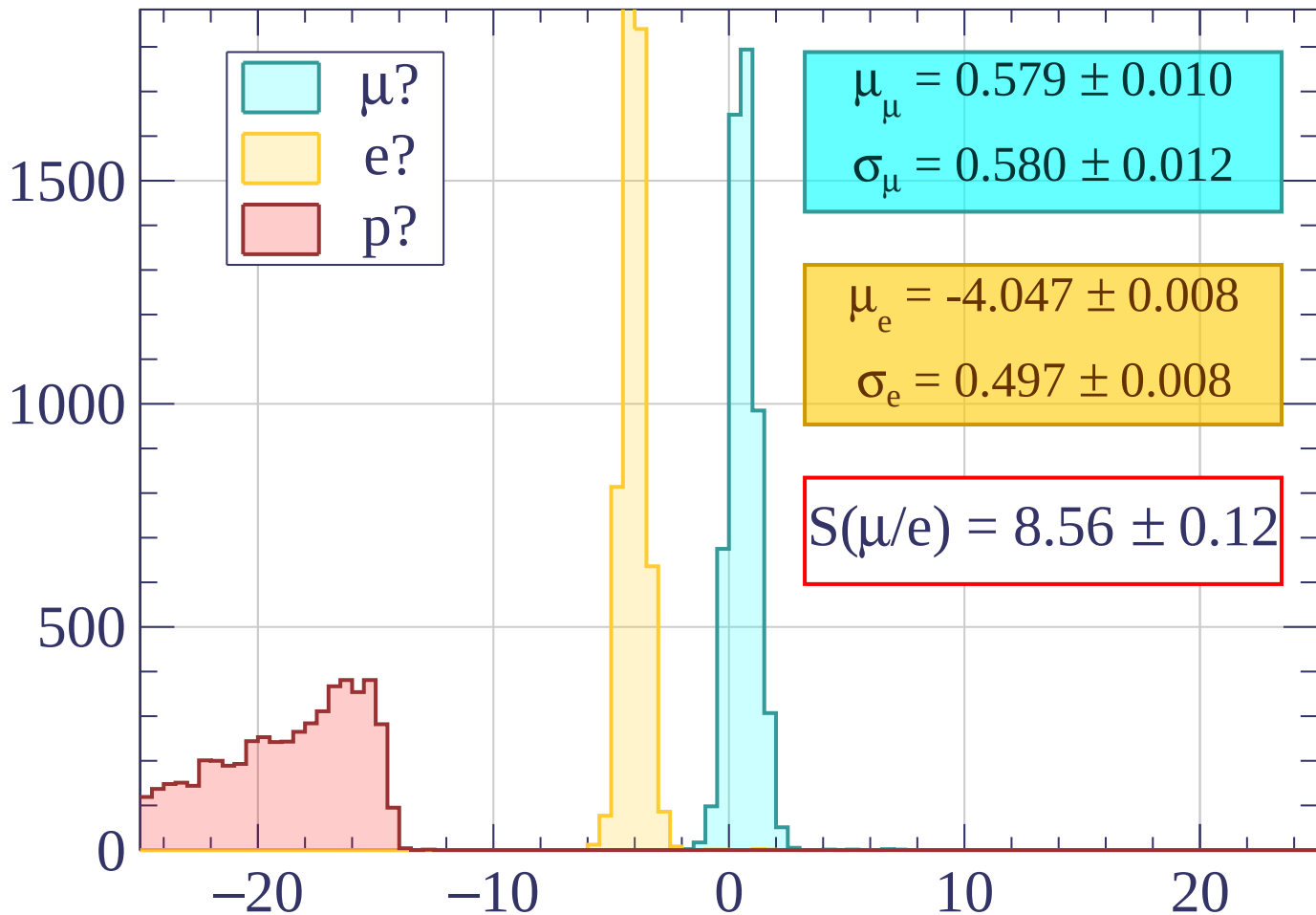


Pulls mean



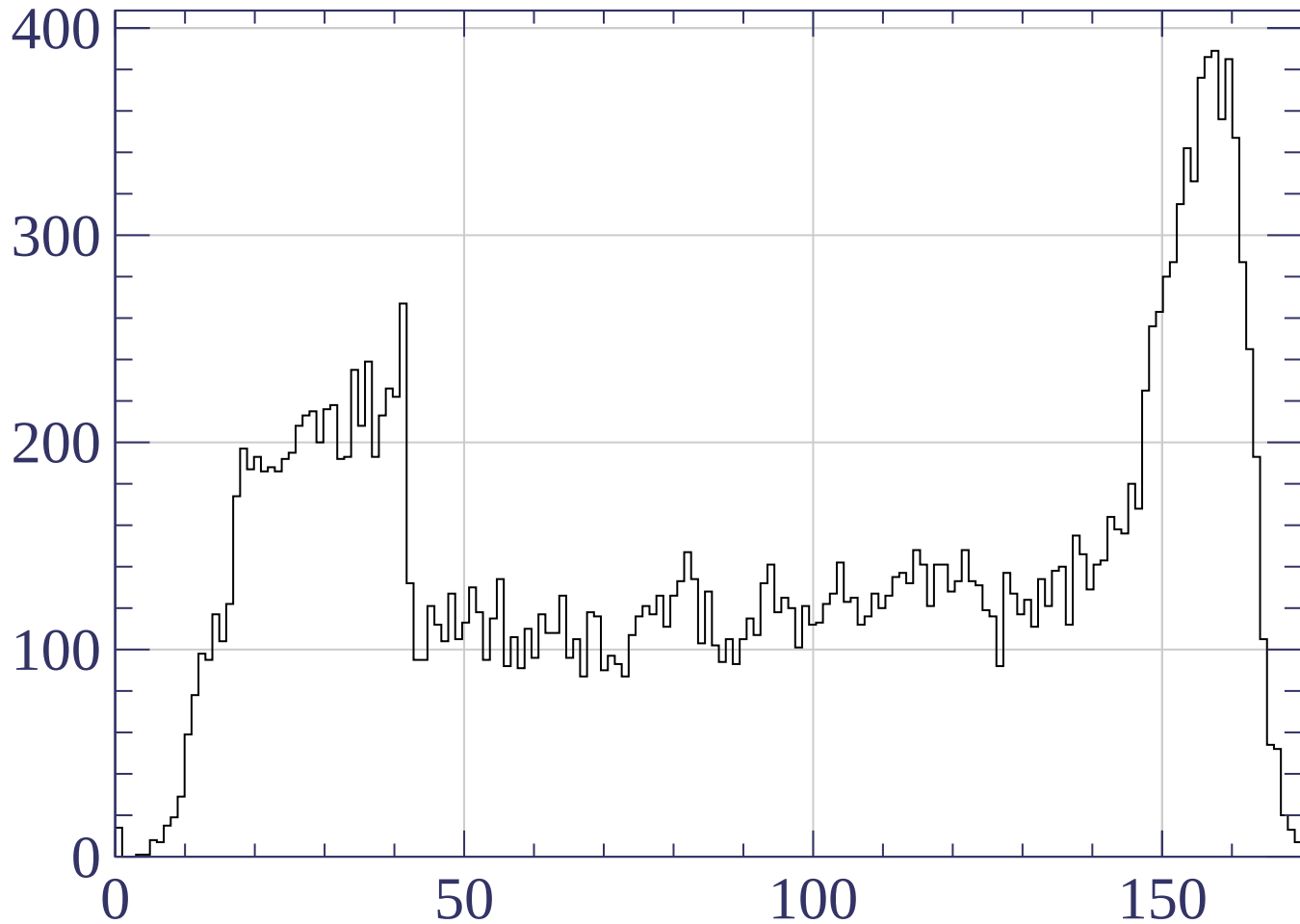


Count



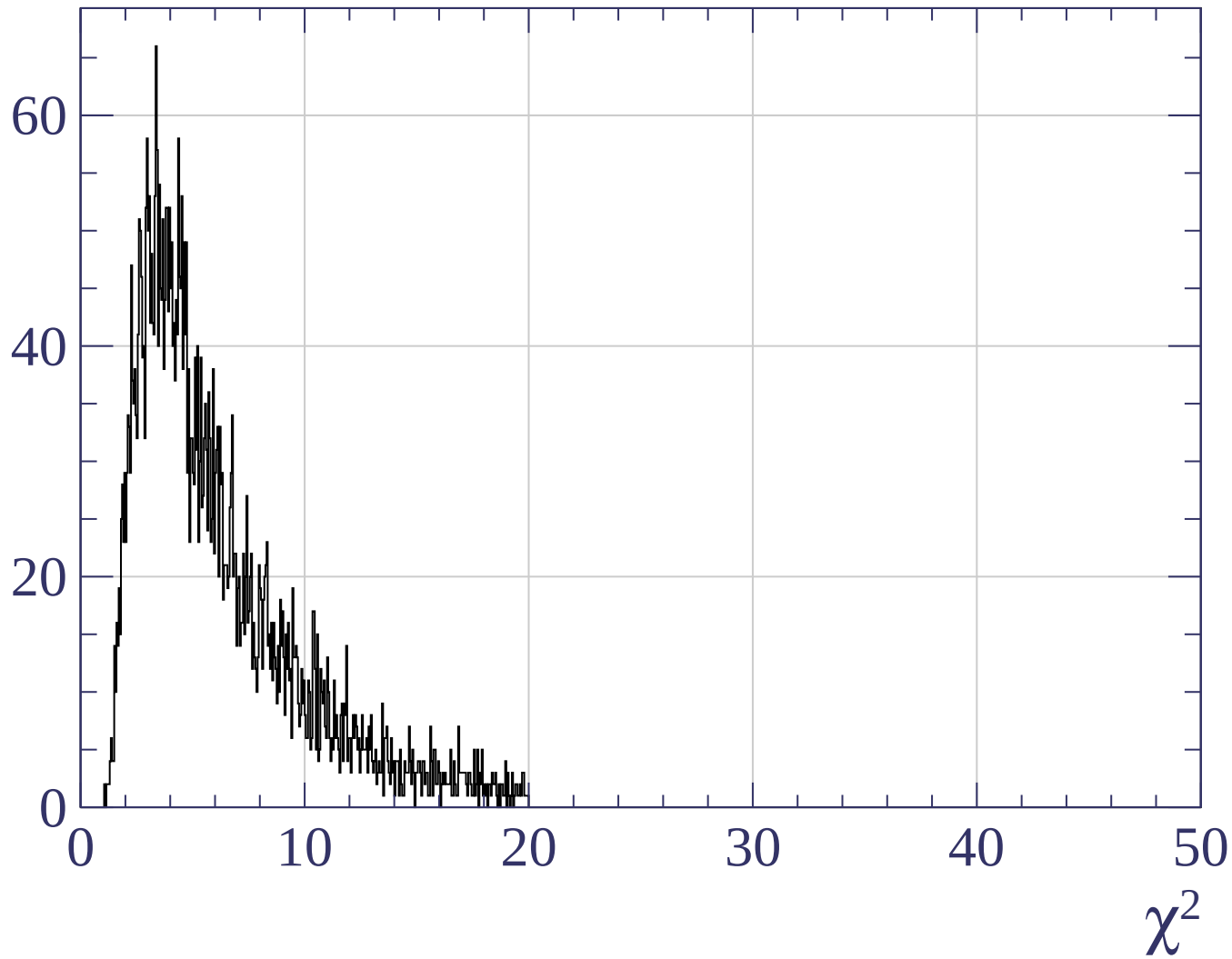
Pull

Count

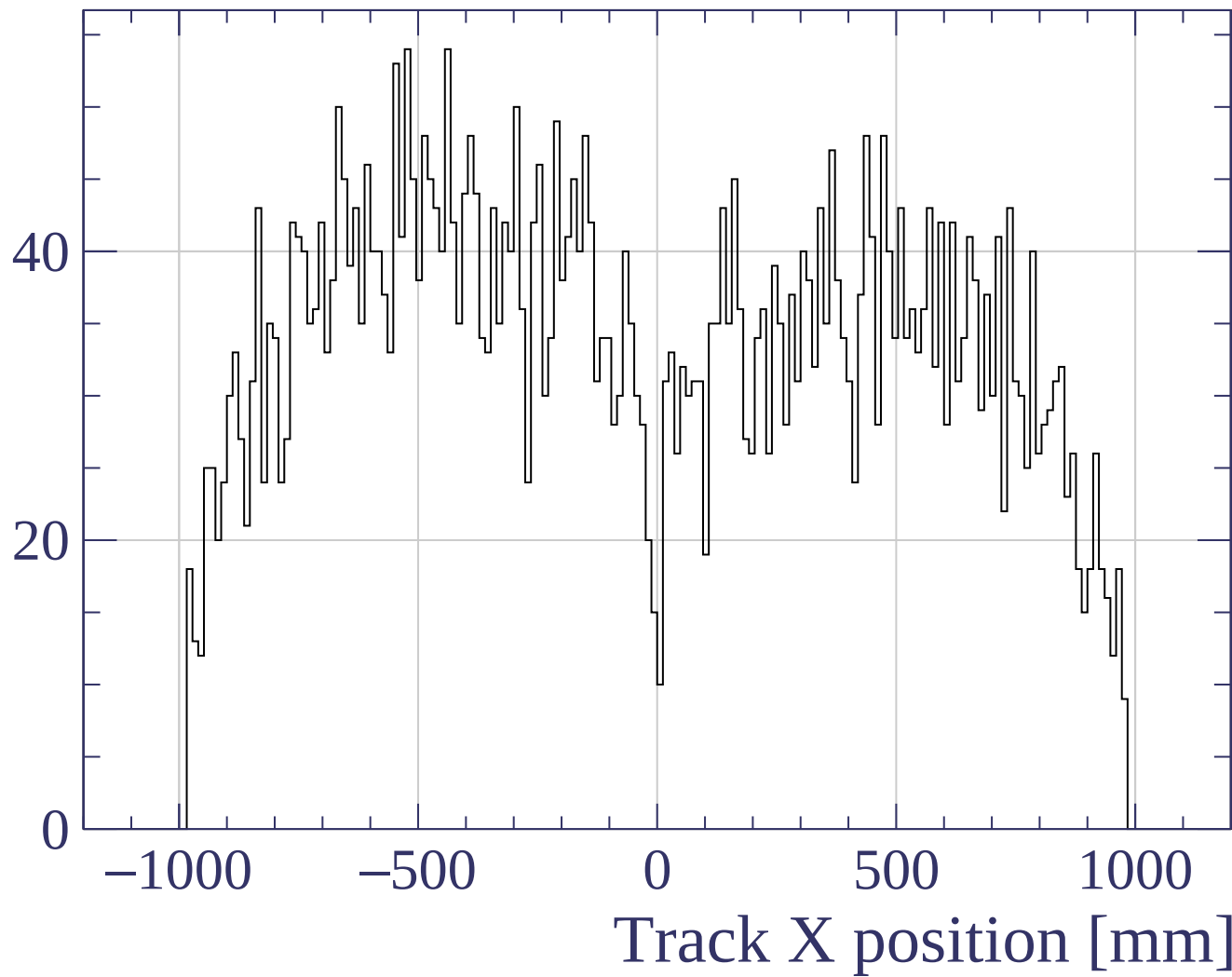


Track length [cm]

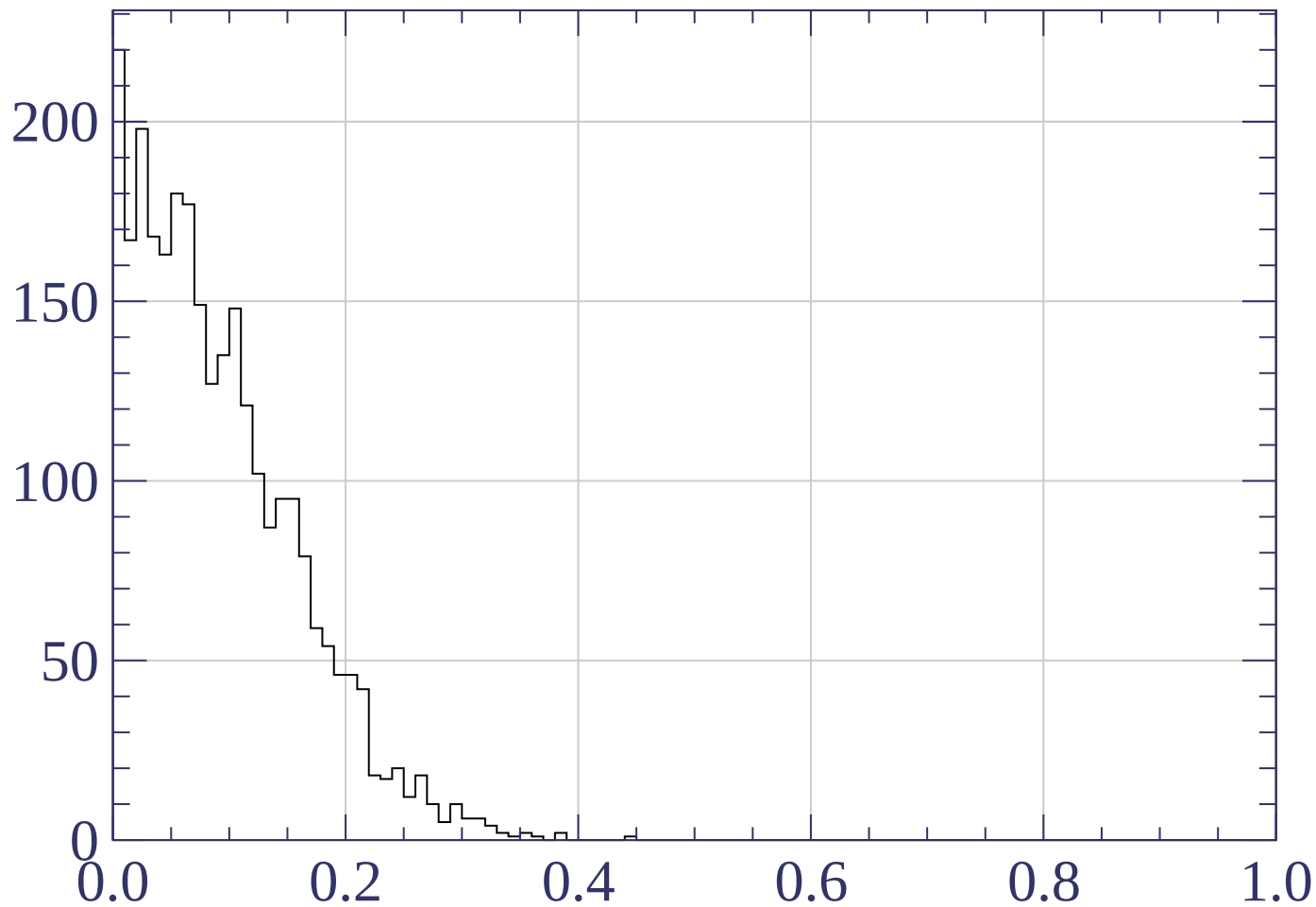
Count



Count

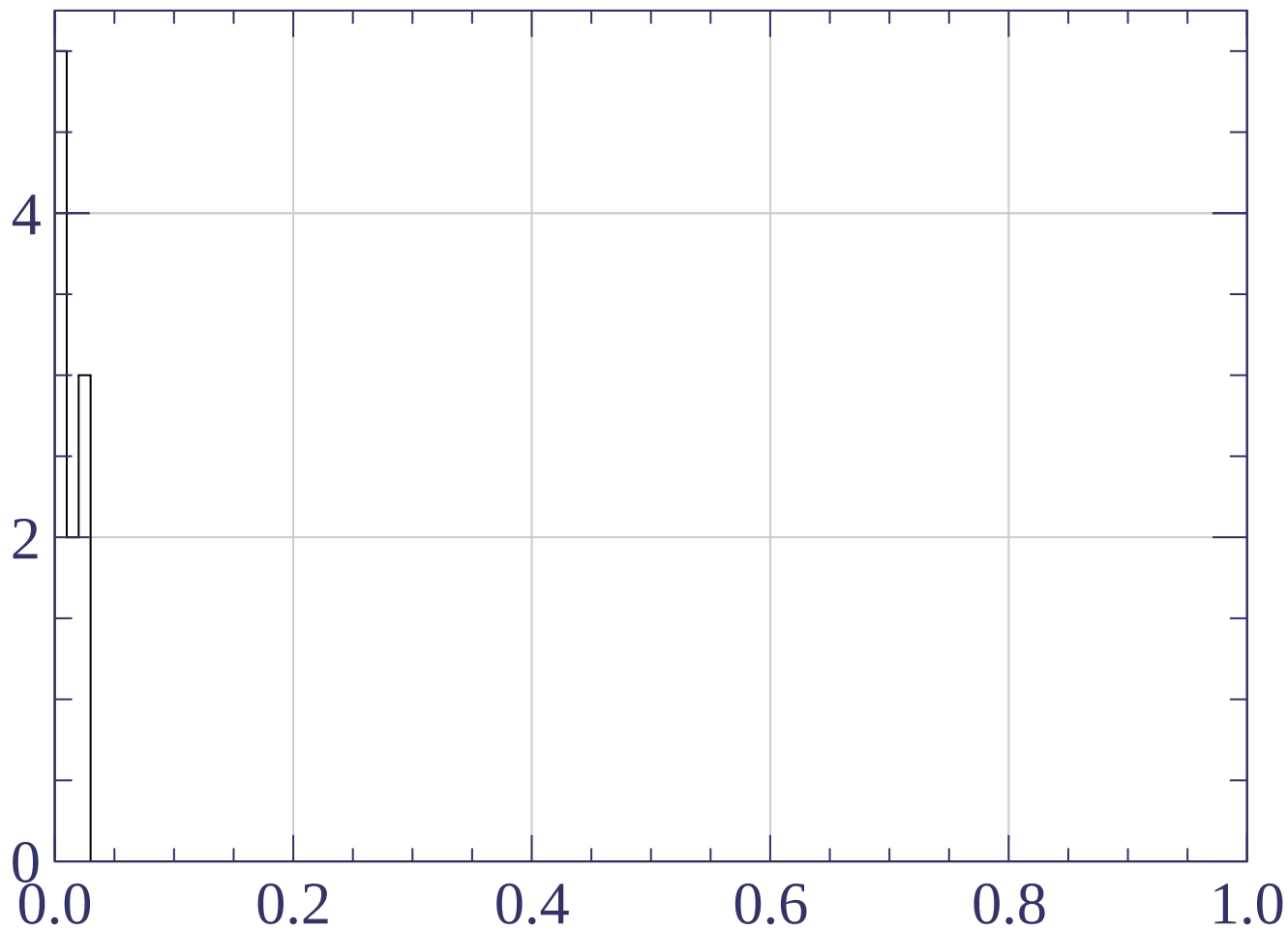


Counts



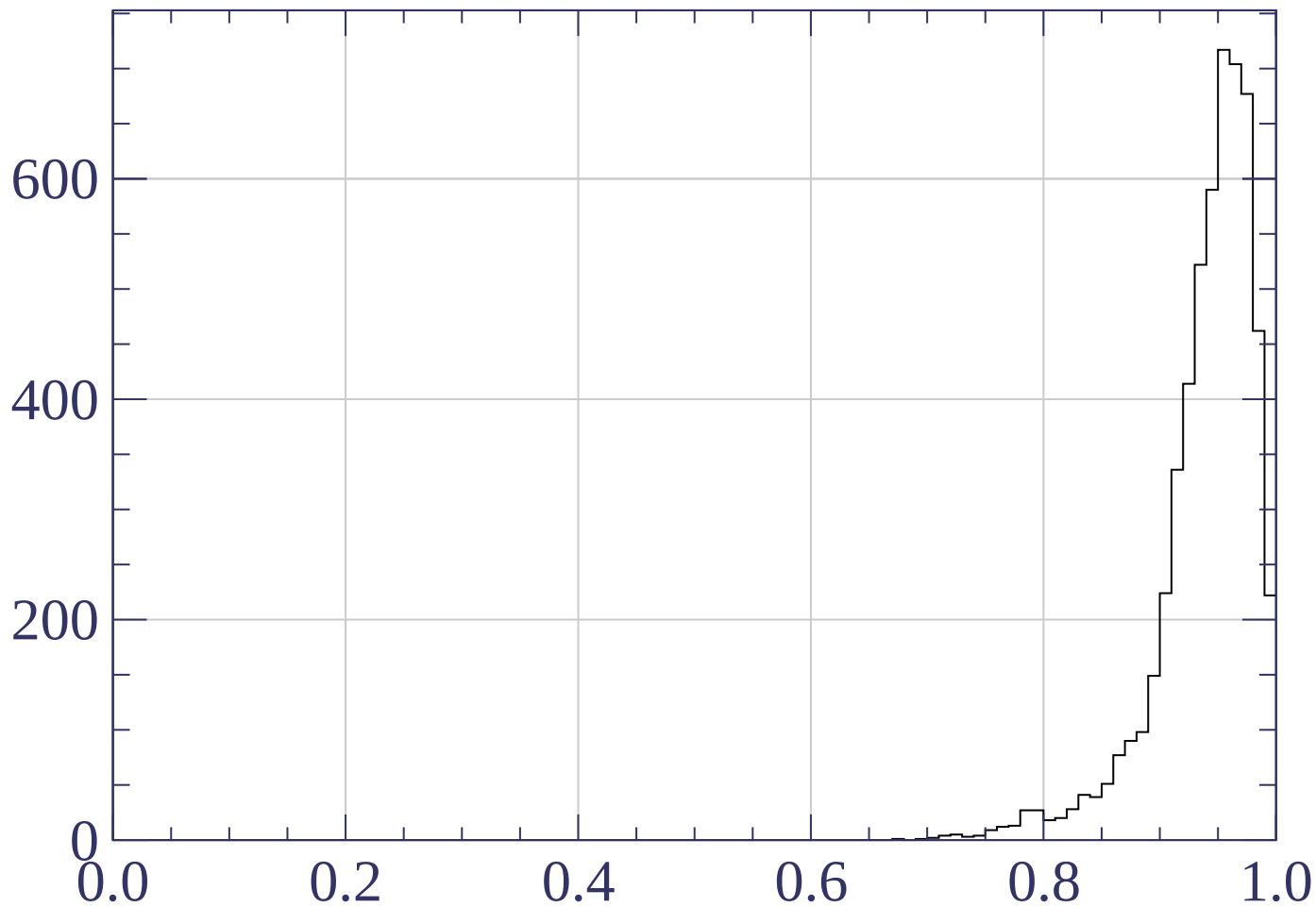
X direction

Counts

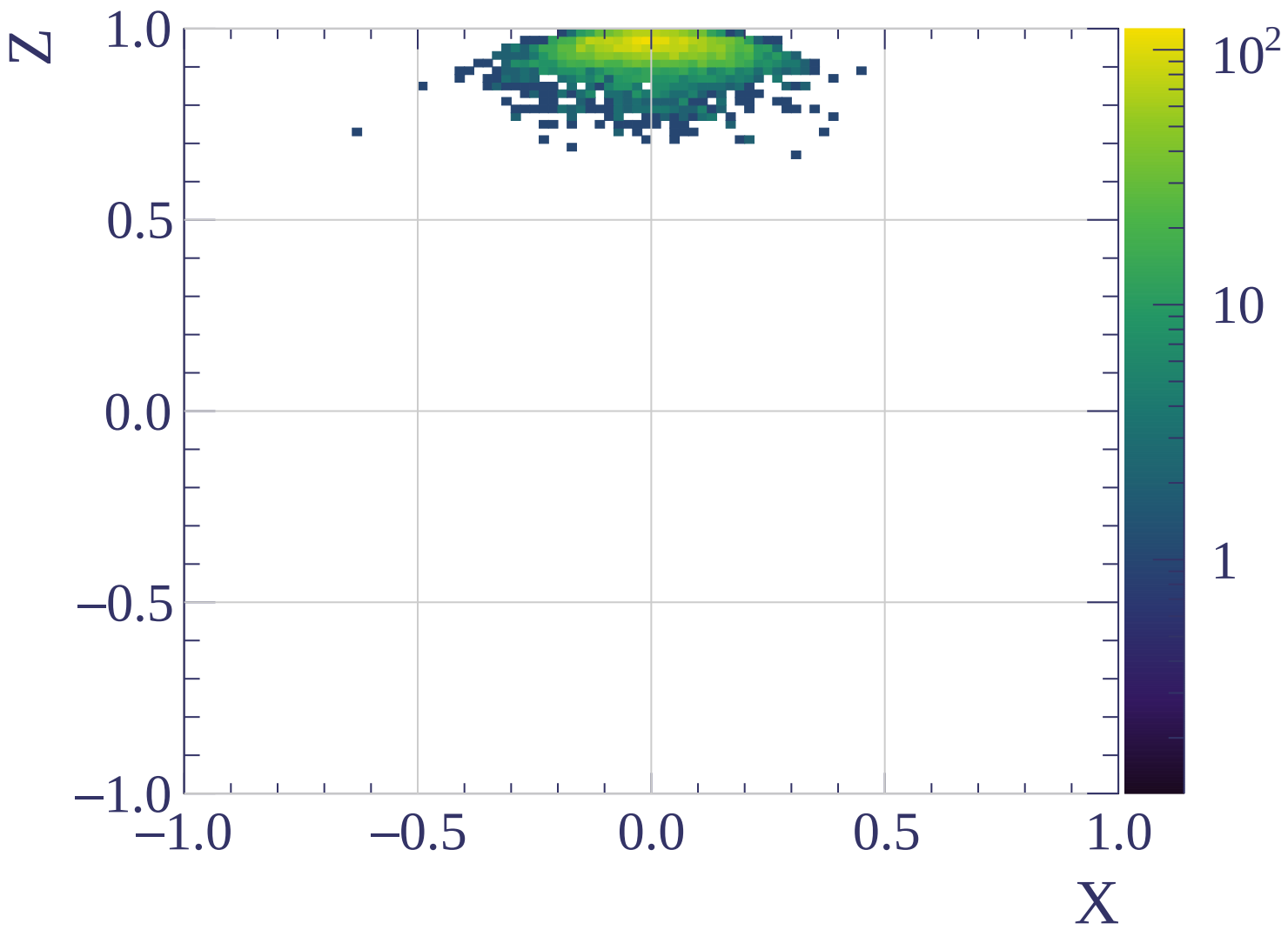


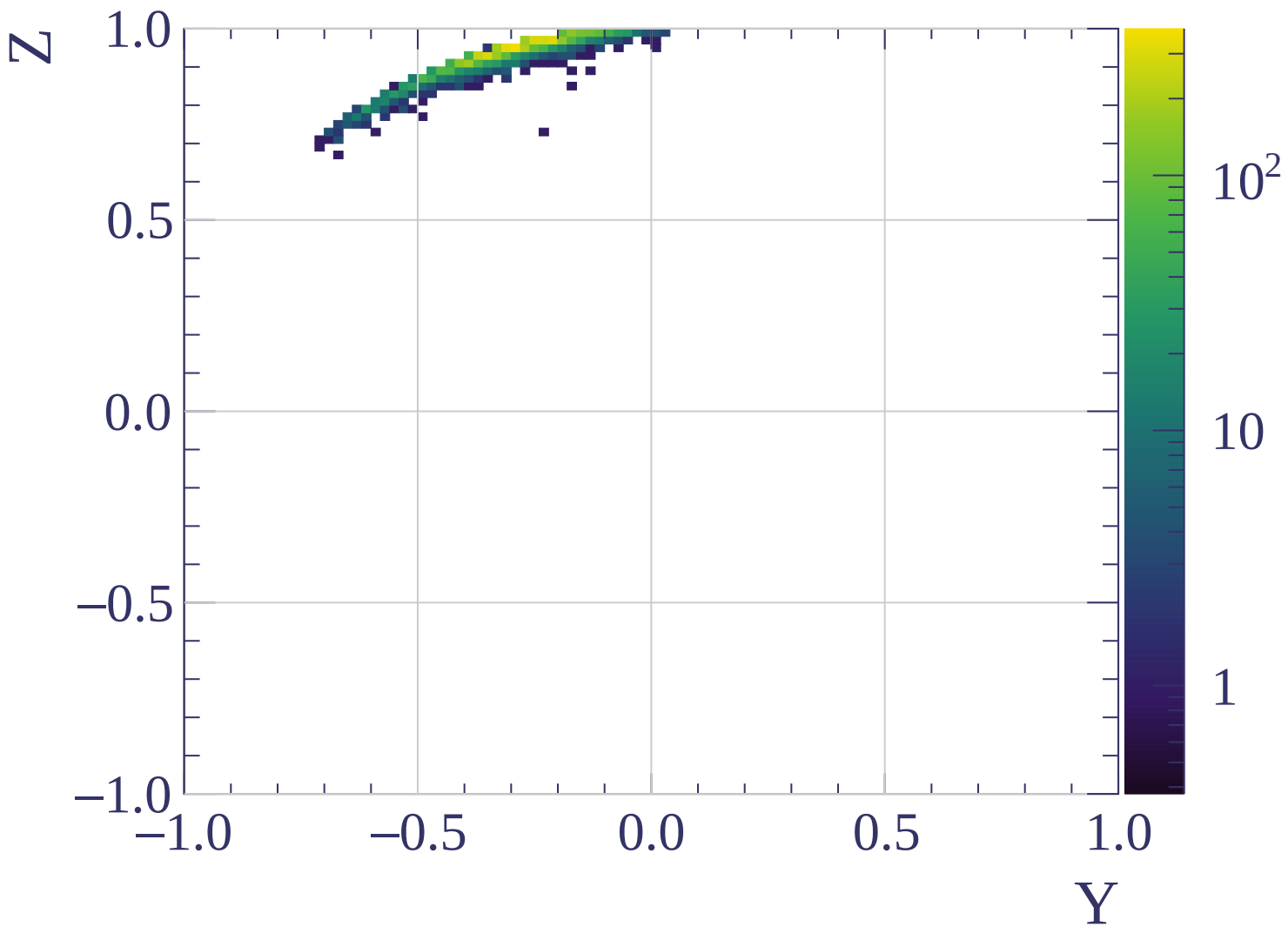
Y direction

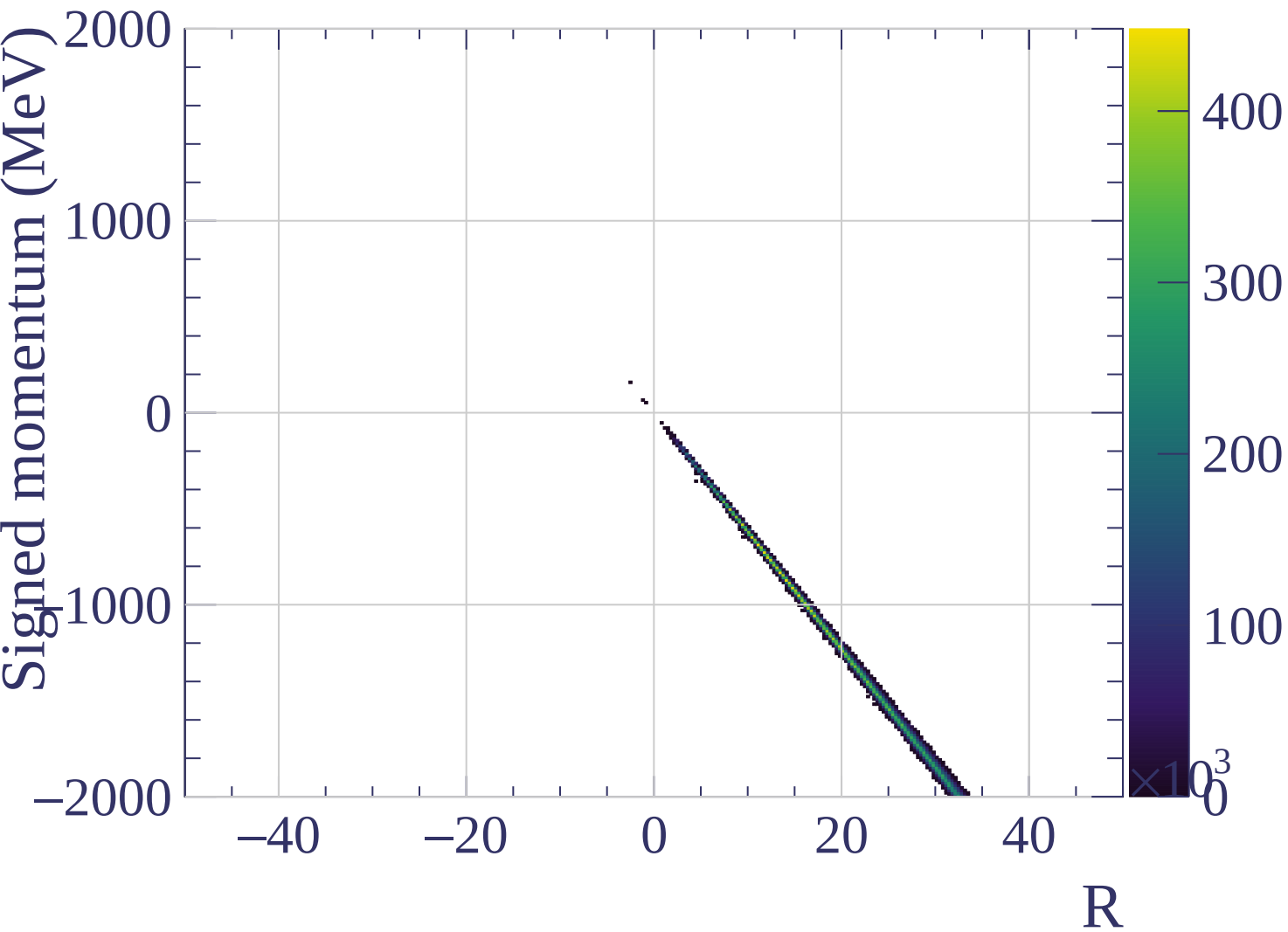
Counts

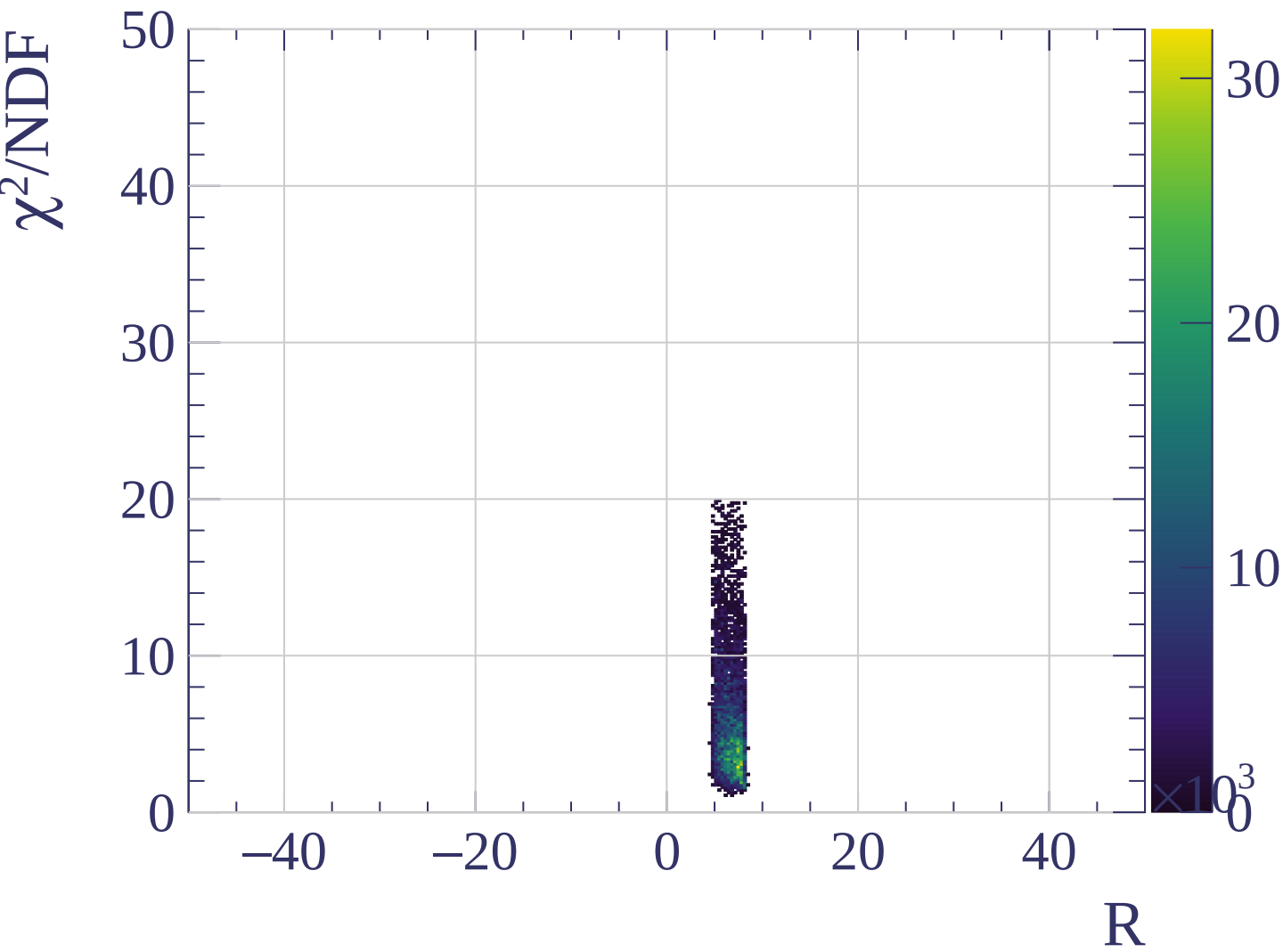


Z direction



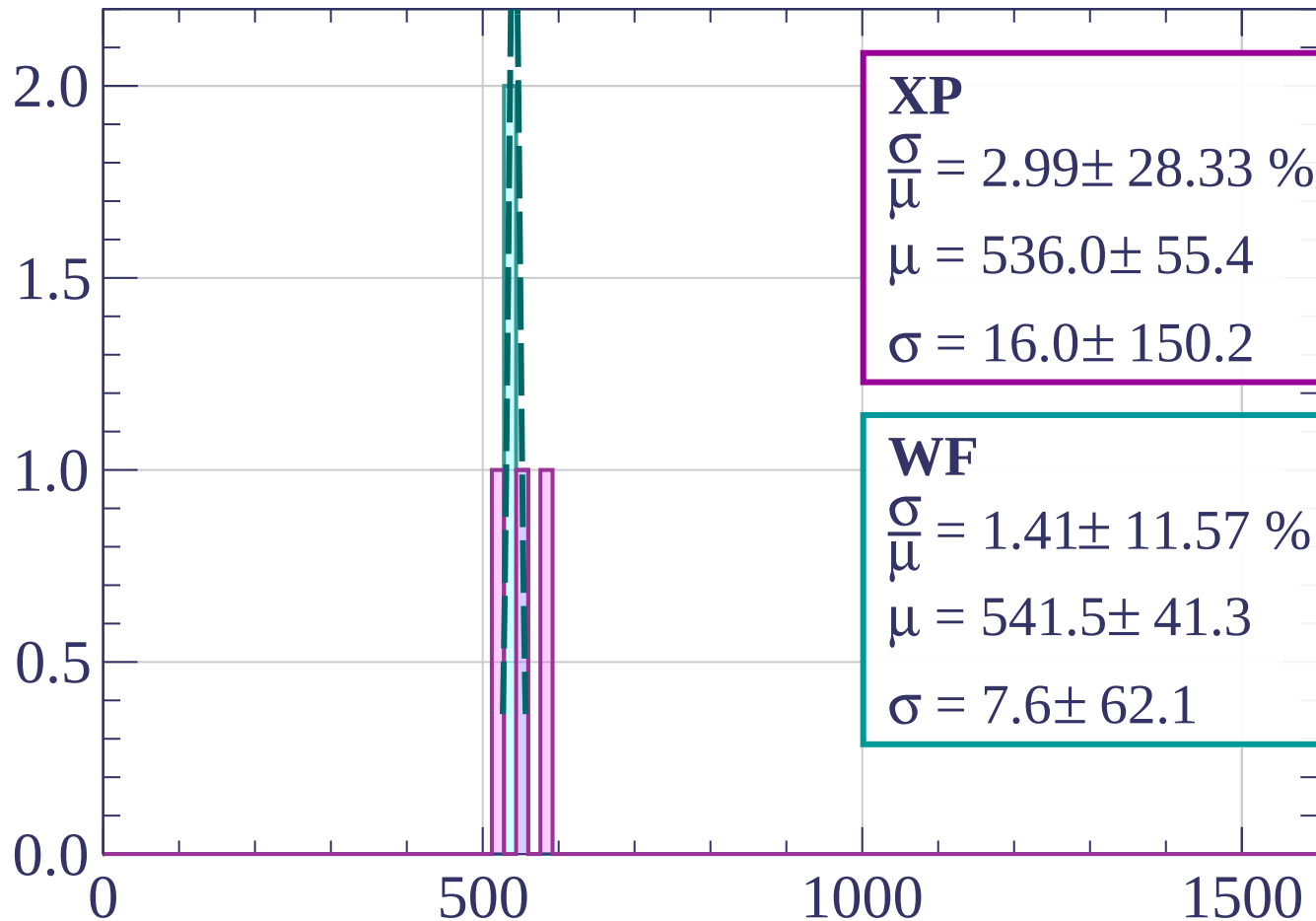






Count

Energy loss | $0 < p < 80$



XP

$$\frac{\sigma}{\mu} = 2.99 \pm 28.33 \%$$

$$\mu = 536.0 \pm 55.4$$

$$\sigma = 16.0 \pm 150.2$$

WF

$$\frac{\sigma}{\mu} = 1.41 \pm 11.57 \%$$

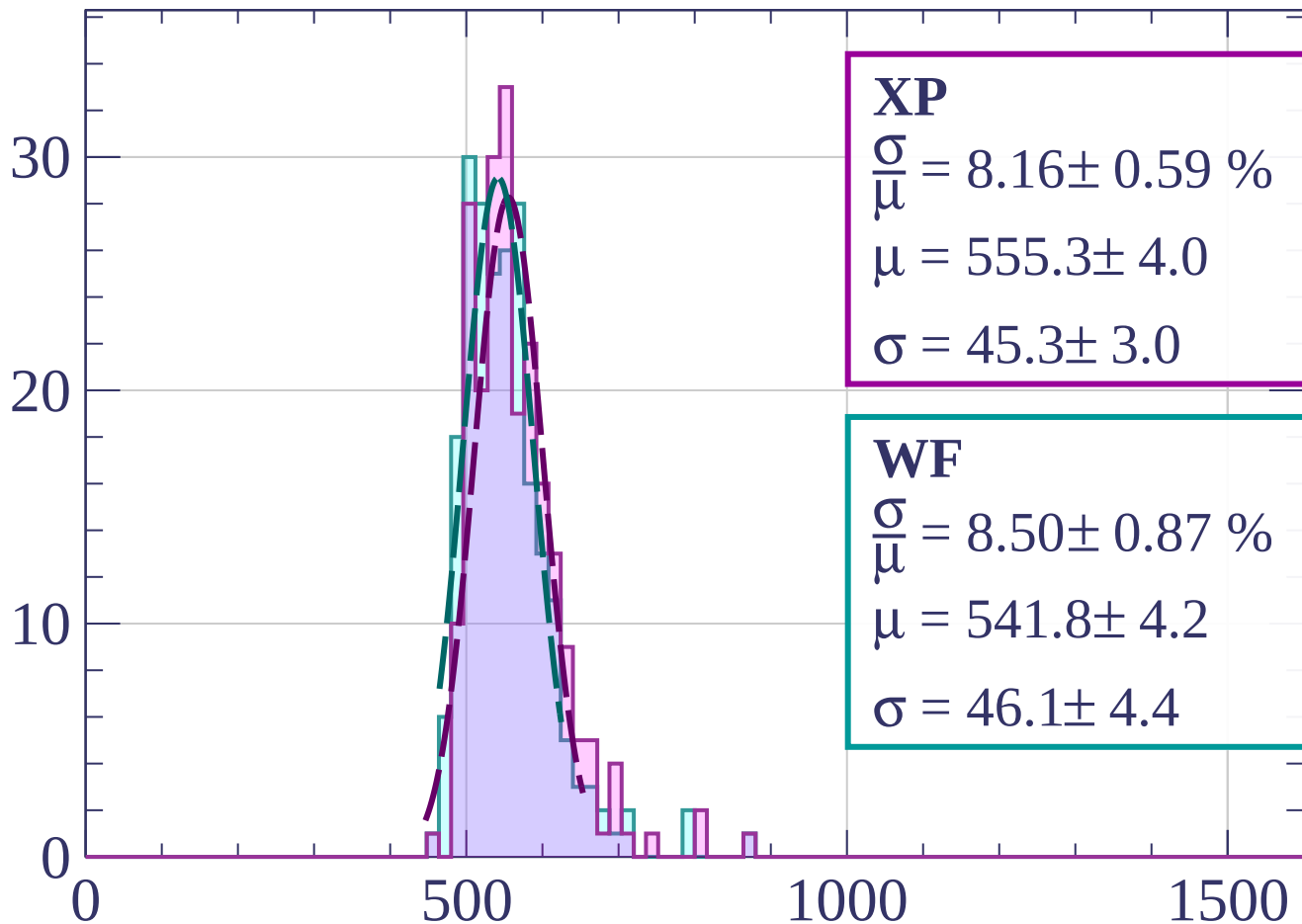
$$\mu = 541.5 \pm 41.3$$

$$\sigma = 7.6 \pm 62.1$$

dE/dx [ADC counts/cm]

Energy loss | $80 < p < 160$

Count



XP

$$\frac{\sigma}{\mu} = 8.16 \pm 0.59 \%$$

$$\mu = 555.3 \pm 4.0$$

$$\sigma = 45.3 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 8.50 \pm 0.87 \%$$

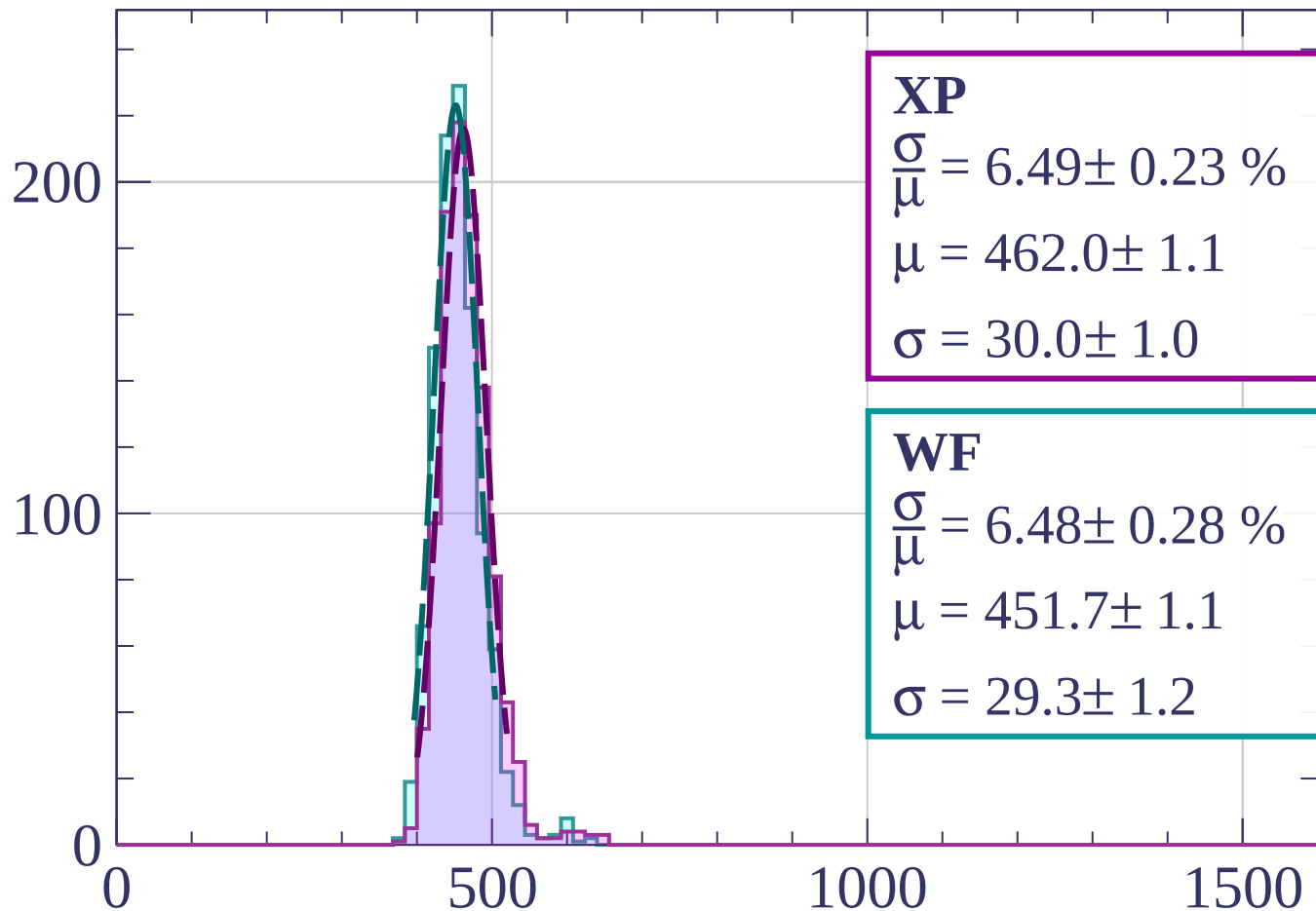
$$\mu = 541.8 \pm 4.2$$

$$\sigma = 46.1 \pm 4.4$$

dE/dx [ADC counts/cm]

Energy loss | $160 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 6.49 \pm 0.23 \%$$

$$\mu = 462.0 \pm 1.1$$

$$\sigma = 30.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.48 \pm 0.28 \%$$

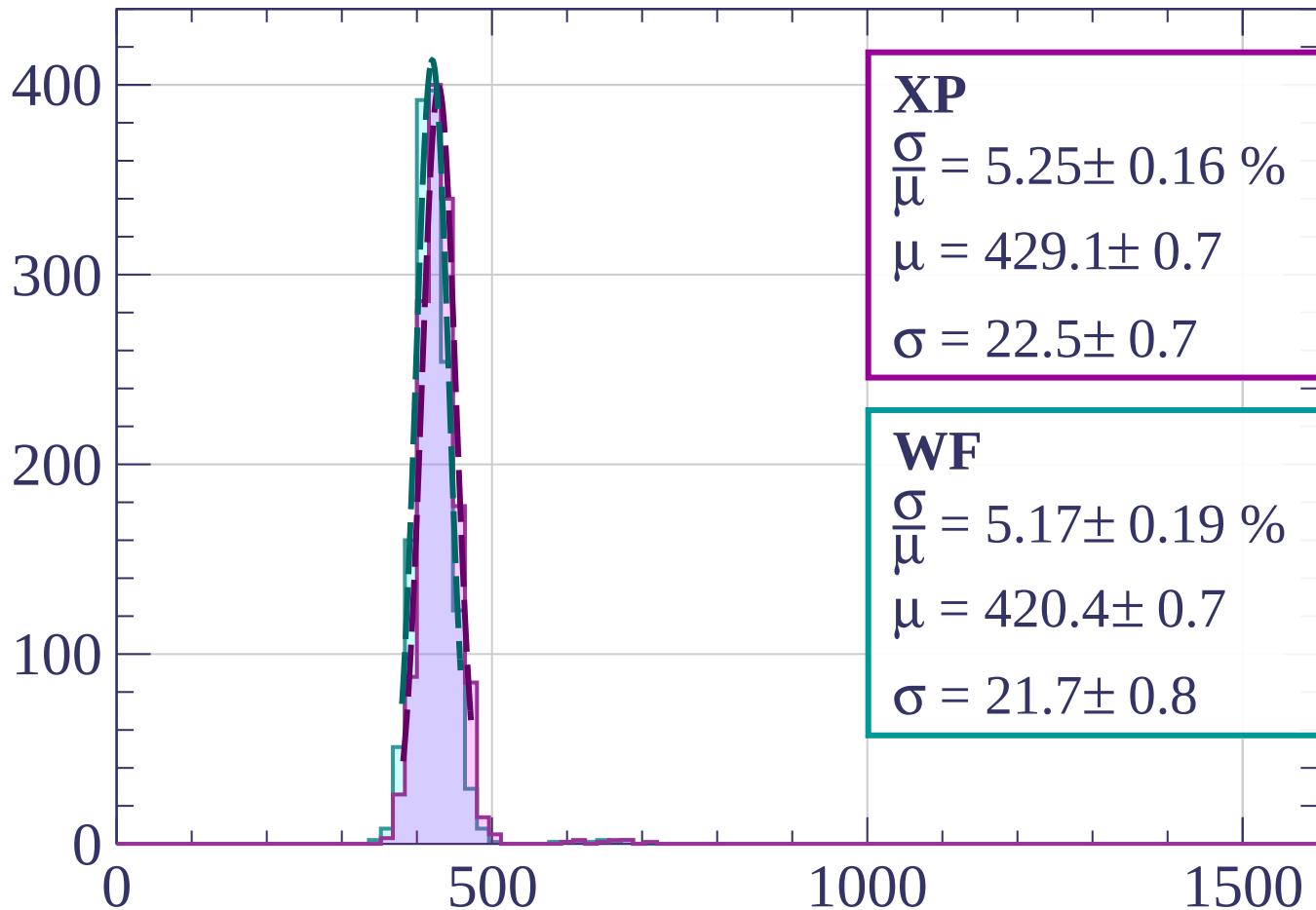
$$\mu = 451.7 \pm 1.1$$

$$\sigma = 29.3 \pm 1.2$$

dE/dx [ADC counts/cm]

Energy loss | $240 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 5.25 \pm 0.16 \%$$

$$\mu = 429.1 \pm 0.7$$

$$\sigma = 22.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.17 \pm 0.19 \%$$

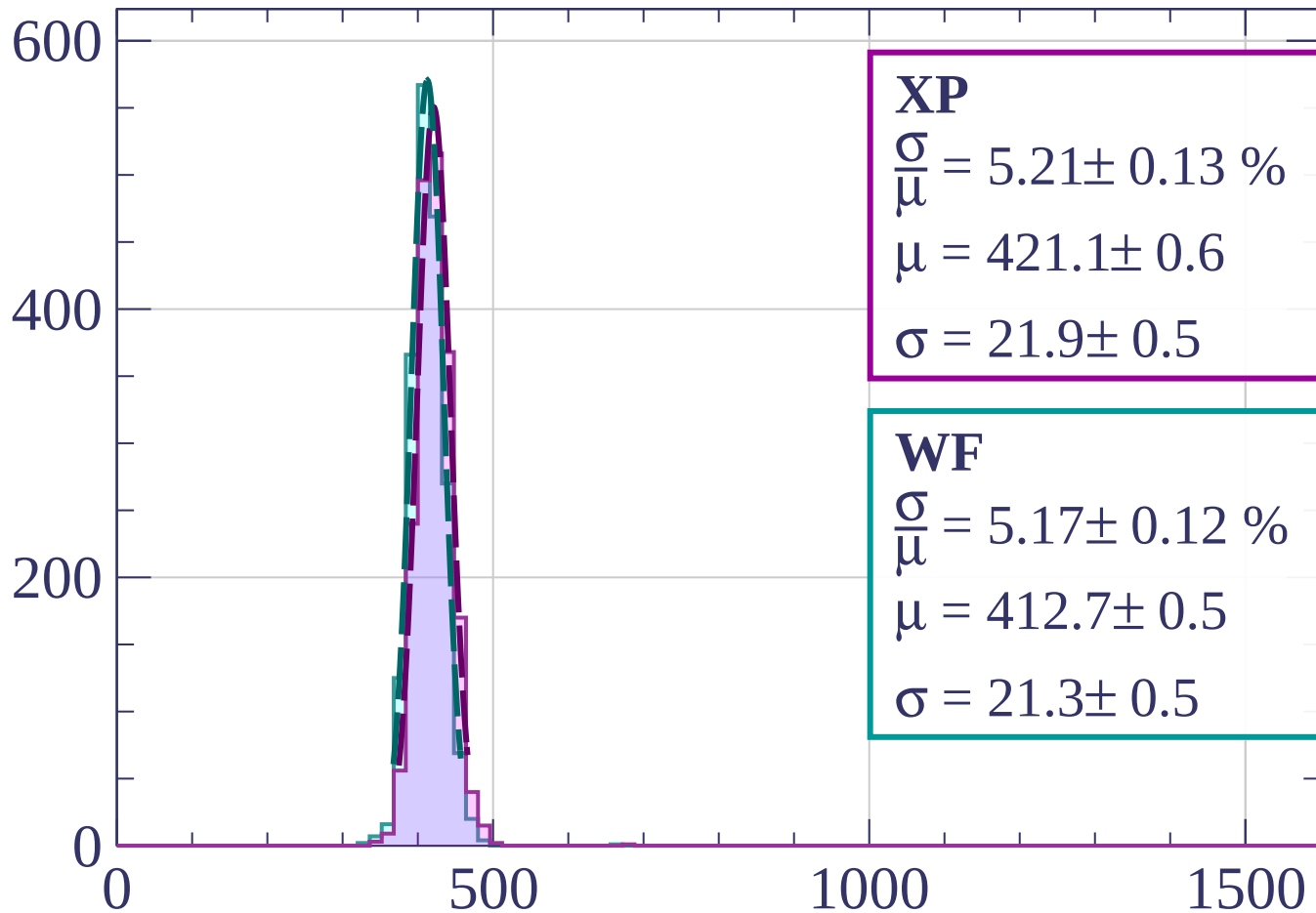
$$\mu = 420.4 \pm 0.7$$

$$\sigma = 21.7 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $320 < p < 400$

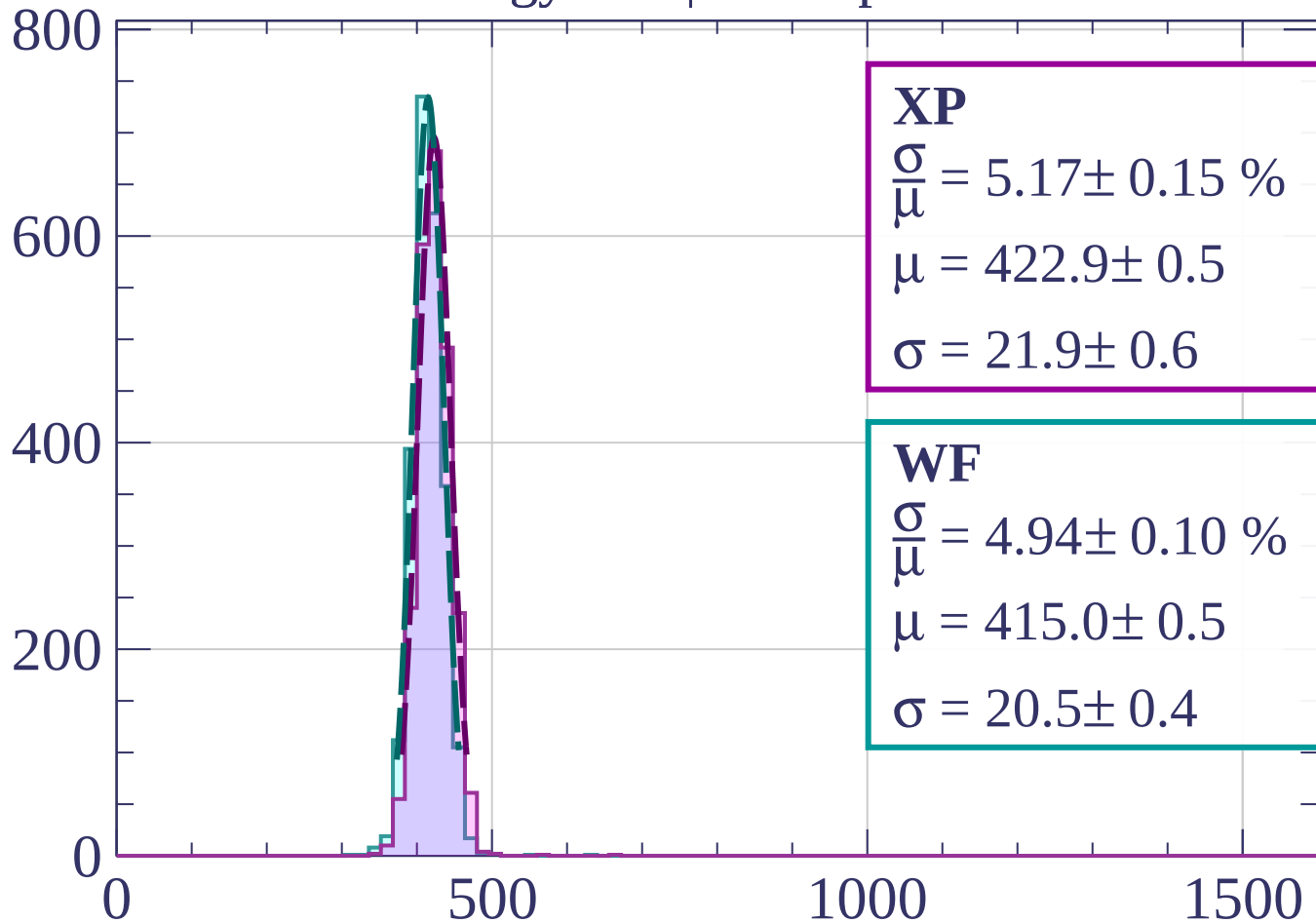
Count



dE/dx [ADC counts/cm]

Energy loss | $400 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 5.17 \pm 0.15 \%$$

$$\mu = 422.9 \pm 0.5$$

$$\sigma = 21.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 4.94 \pm 0.10 \%$$

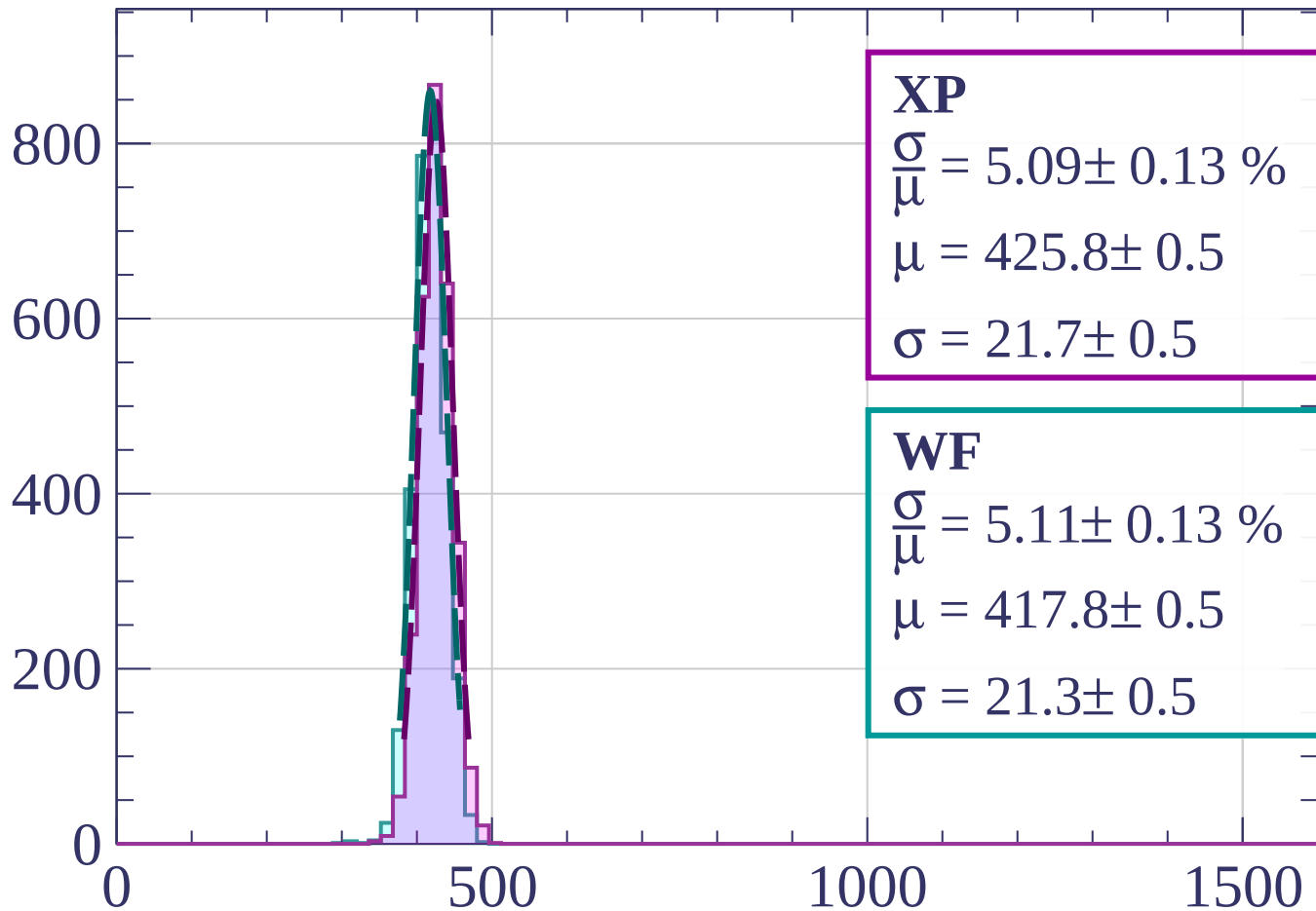
$$\mu = 415.0 \pm 0.5$$

$$\sigma = 20.5 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $480 < p < 560$

Count



dE/dx [ADC counts/cm]

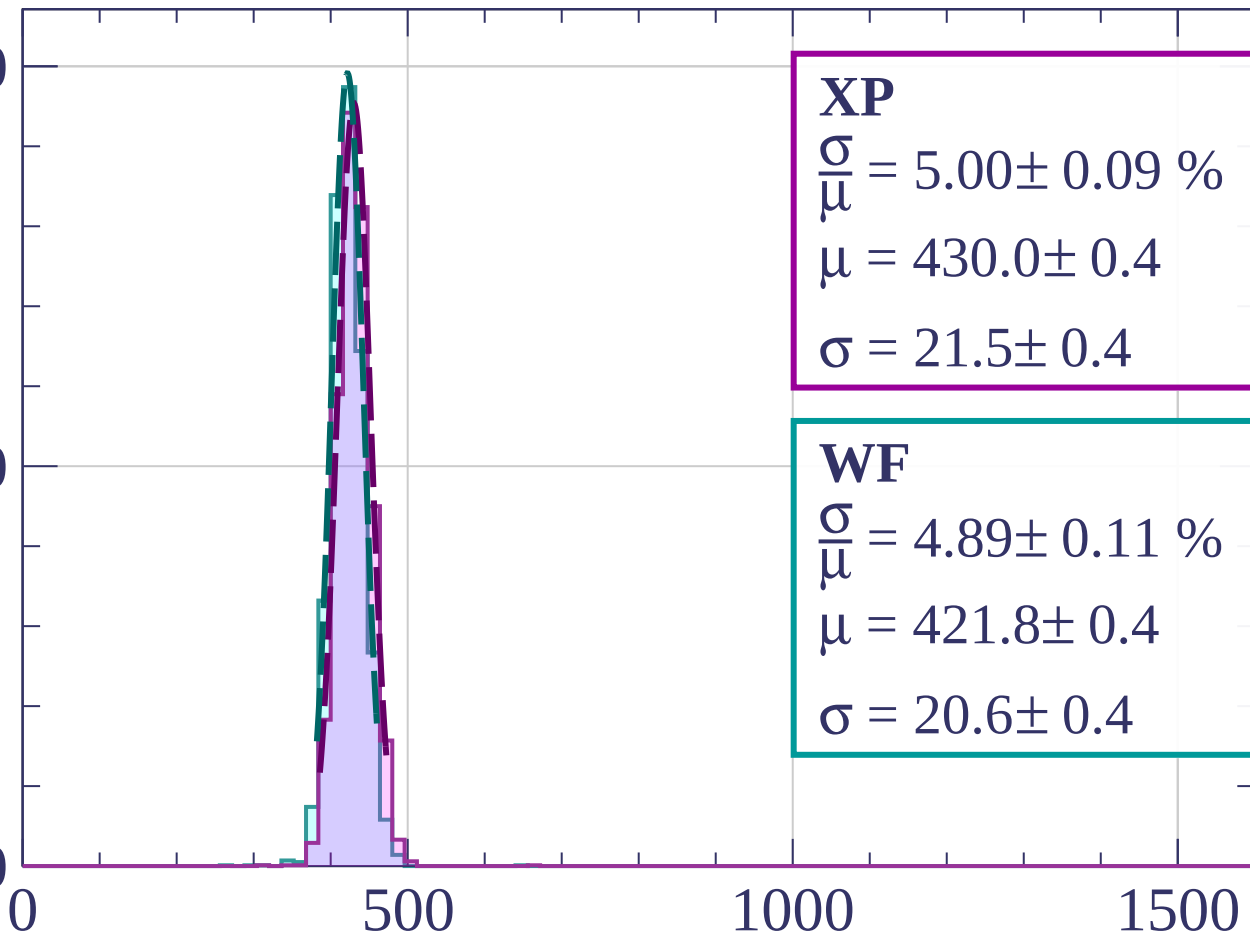
Energy loss | $560 < p < 640$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 5.00 \pm 0.09 \%$$

$$\mu = 430.0 \pm 0.4$$

$$\sigma = 21.5 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.89 \pm 0.11 \%$$

$$\mu = 421.8 \pm 0.4$$

$$\sigma = 20.6 \pm 0.4$$

dE/dx [ADC counts/cm]

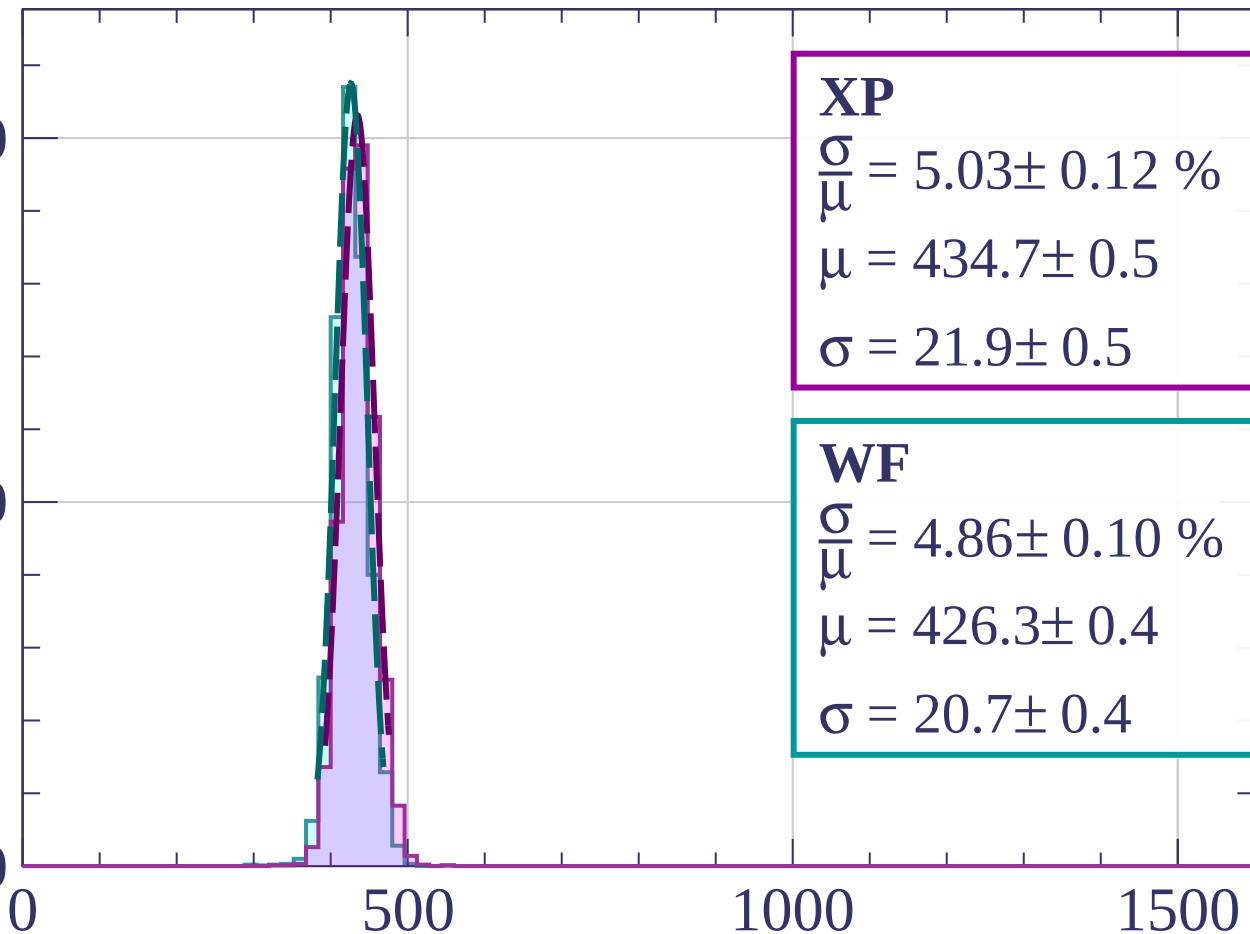
Energy loss | $640 < p < 720$

Count

1000

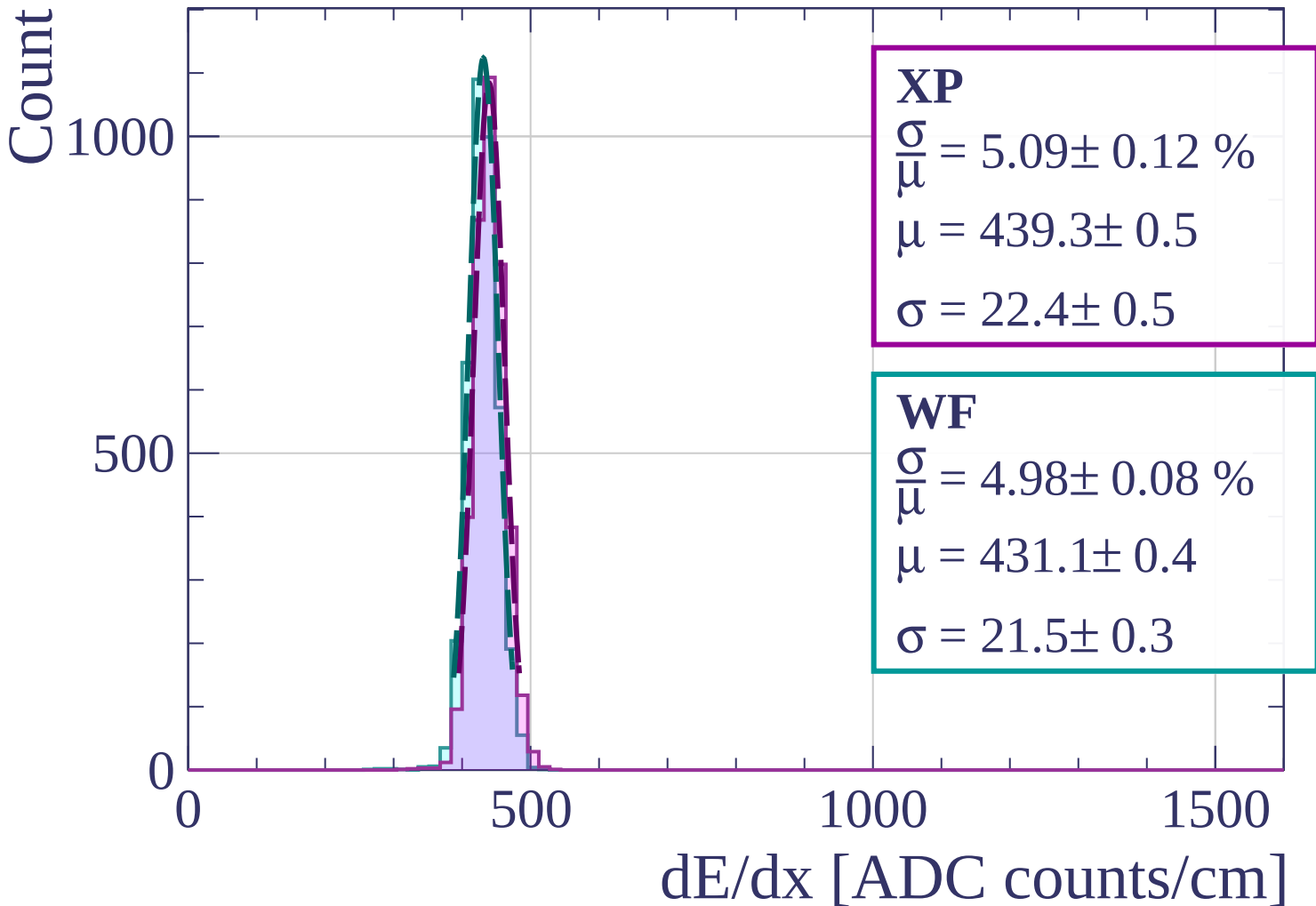
500

0



dE/dx [ADC counts/cm]

Energy loss | $720 < p < 800$



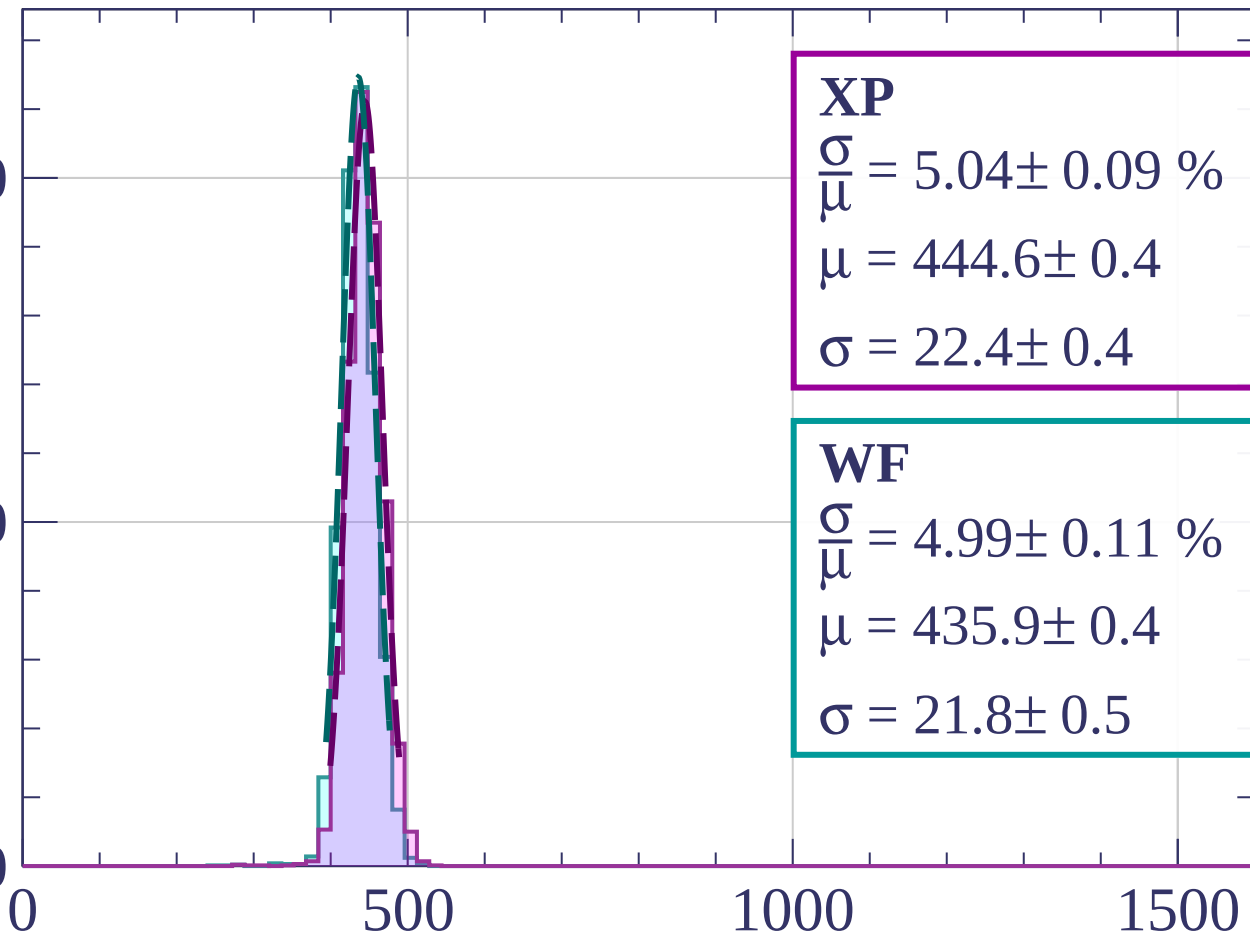
Energy loss | $800 < p < 880$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 5.04 \pm 0.09 \%$$

$$\mu = 444.6 \pm 0.4$$

$$\sigma = 22.4 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.99 \pm 0.11 \%$$

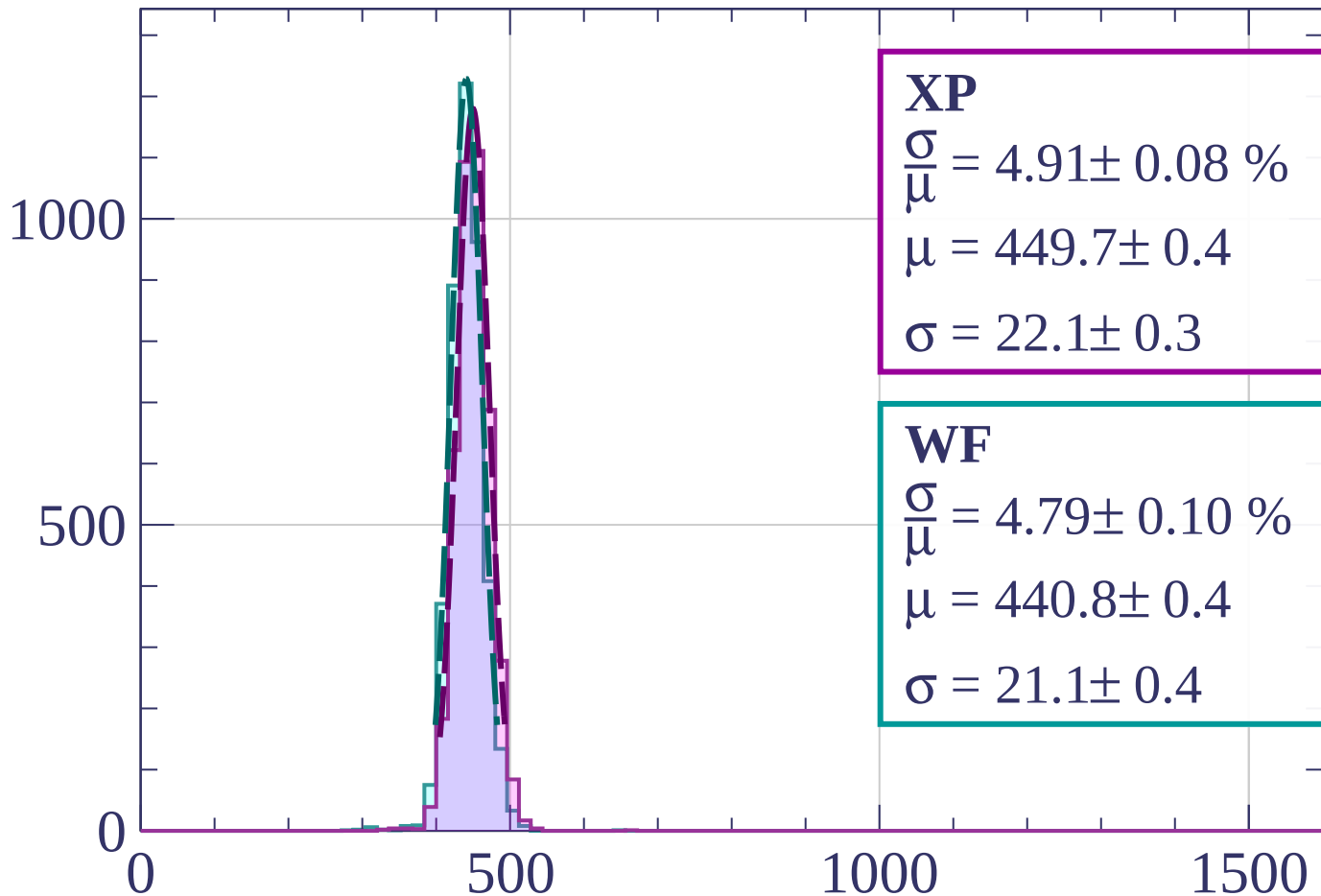
$$\mu = 435.9 \pm 0.4$$

$$\sigma = 21.8 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 4.91 \pm 0.08 \%$$

$$\mu = 449.7 \pm 0.4$$

$$\sigma = 22.1 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 4.79 \pm 0.10 \%$$

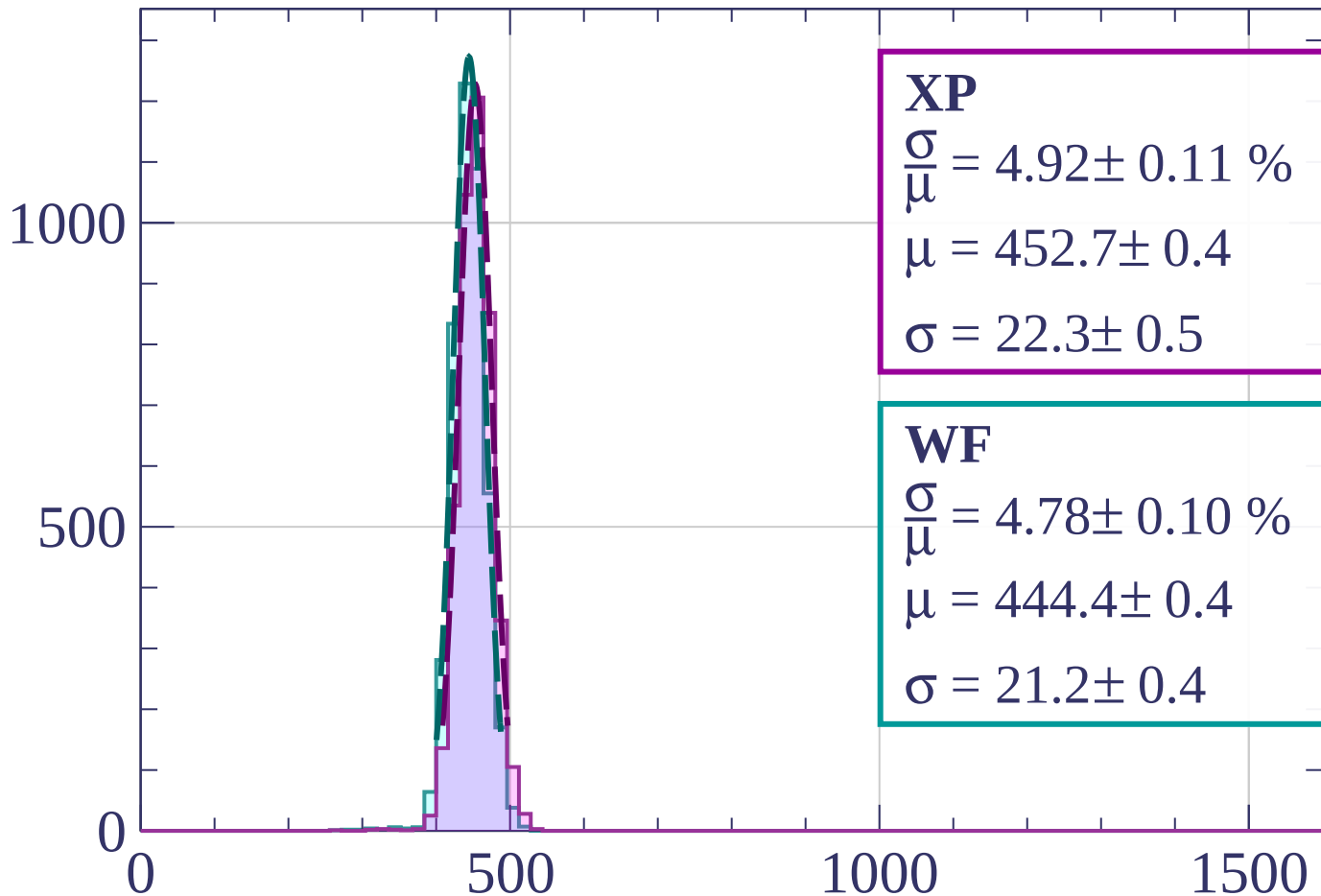
$$\mu = 440.8 \pm 0.4$$

$$\sigma = 21.1 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $960 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 4.92 \pm 0.11 \%$$

$$\mu = 452.7 \pm 0.4$$

$$\sigma = 22.3 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 4.78 \pm 0.10 \%$$

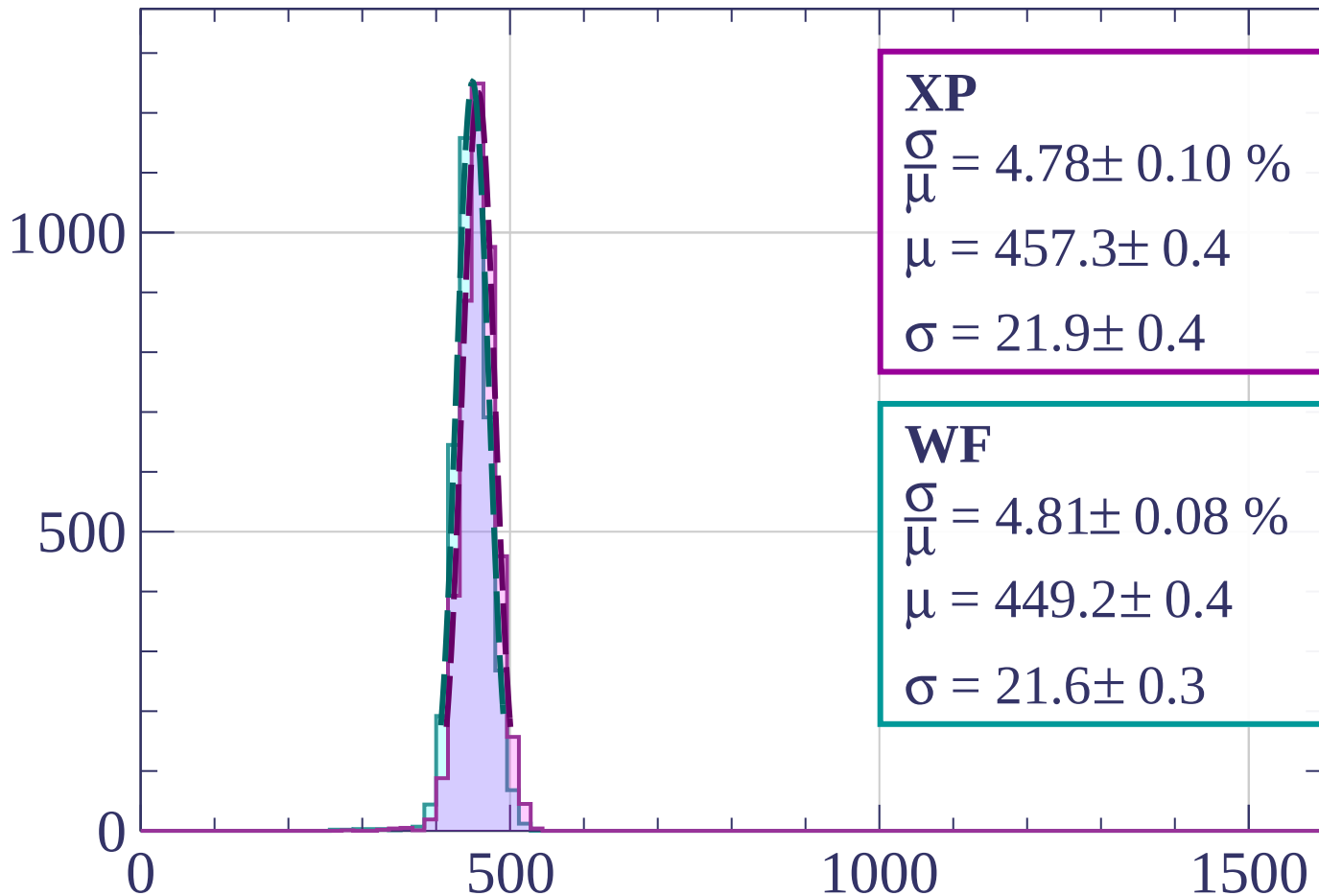
$$\mu = 444.4 \pm 0.4$$

$$\sigma = 21.2 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 4.78 \pm 0.10 \%$$

$$\mu = 457.3 \pm 0.4$$

$$\sigma = 21.9 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.81 \pm 0.08 \%$$

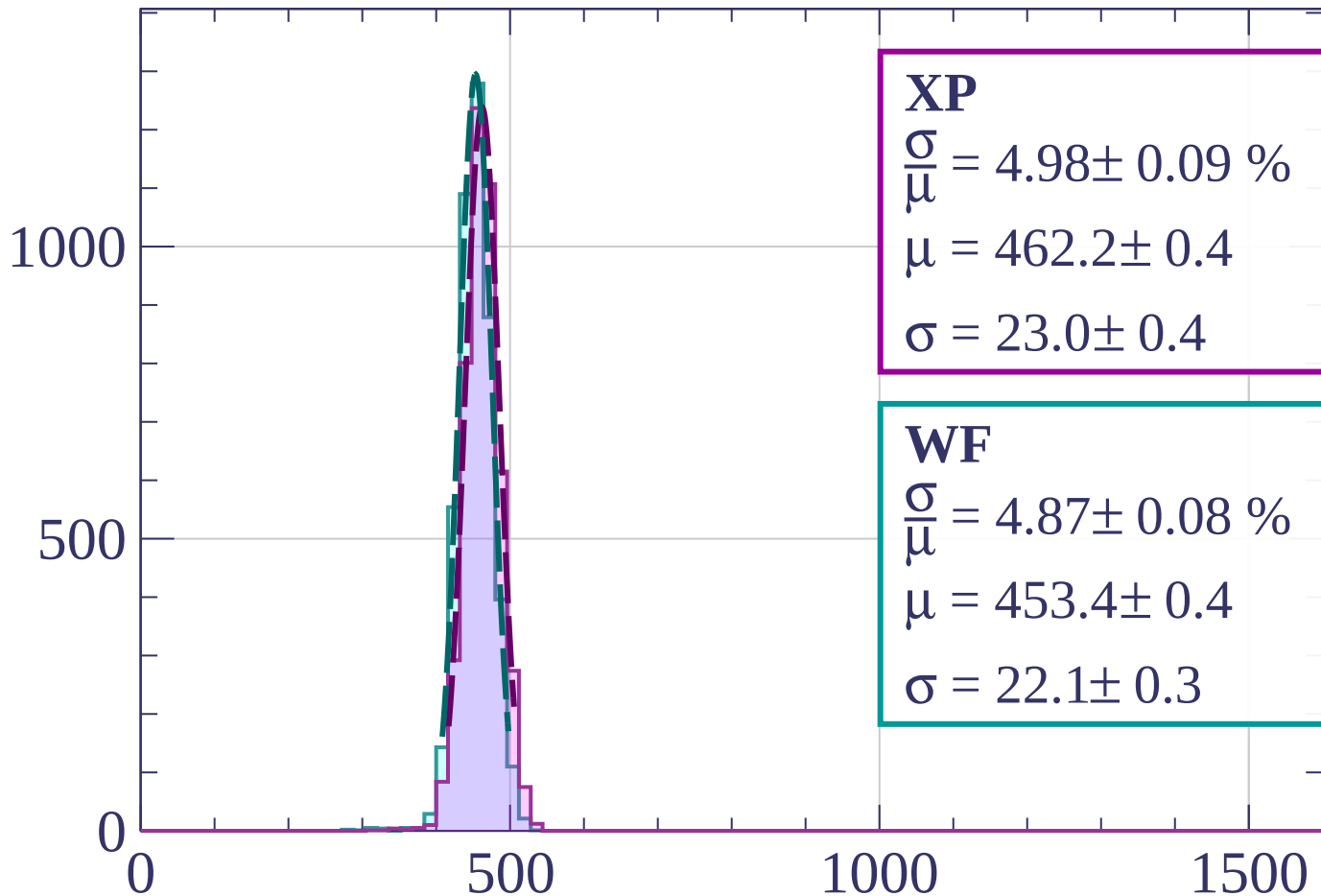
$$\mu = 449.2 \pm 0.4$$

$$\sigma = 21.6 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 4.98 \pm 0.09 \%$$

$$\mu = 462.2 \pm 0.4$$

$$\sigma = 23.0 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.87 \pm 0.08 \%$$

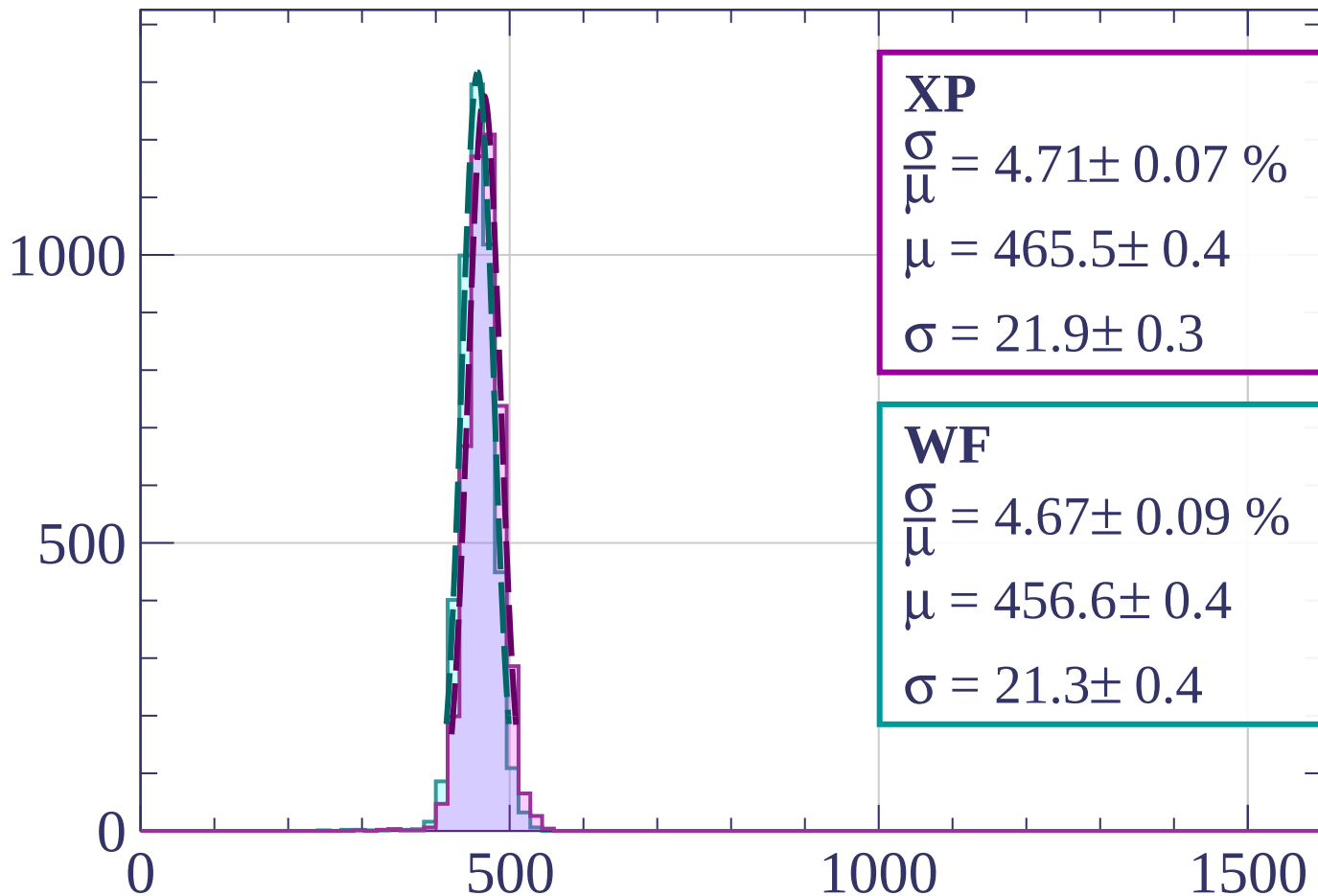
$$\mu = 453.4 \pm 0.4$$

$$\sigma = 22.1 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 4.71 \pm 0.07 \%$$

$$\mu = 465.5 \pm 0.4$$

$$\sigma = 21.9 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 4.67 \pm 0.09 \%$$

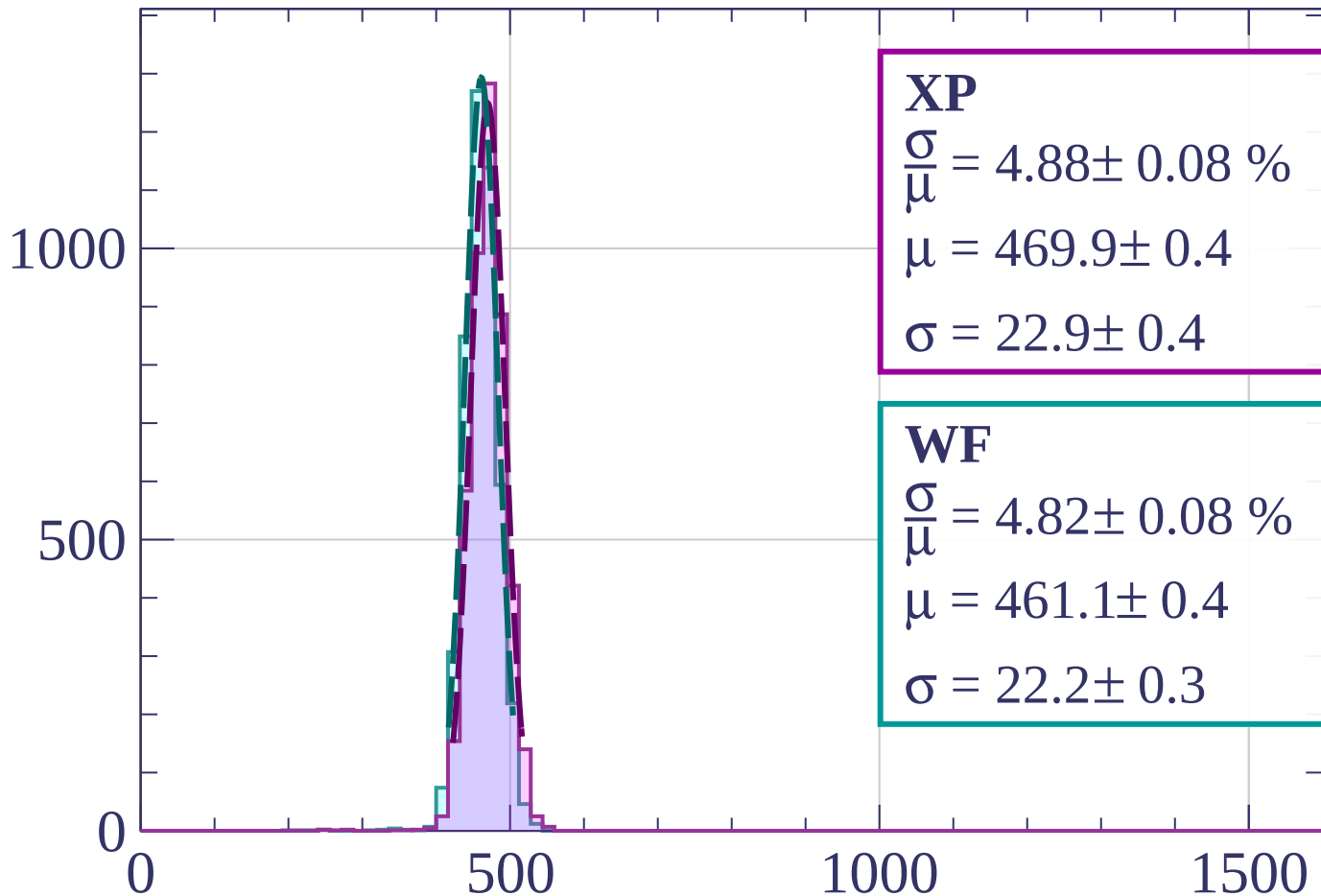
$$\mu = 456.6 \pm 0.4$$

$$\sigma = 21.3 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $1280 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 4.88 \pm 0.08 \%$$

$$\mu = 469.9 \pm 0.4$$

$$\sigma = 22.9 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.82 \pm 0.08 \%$$

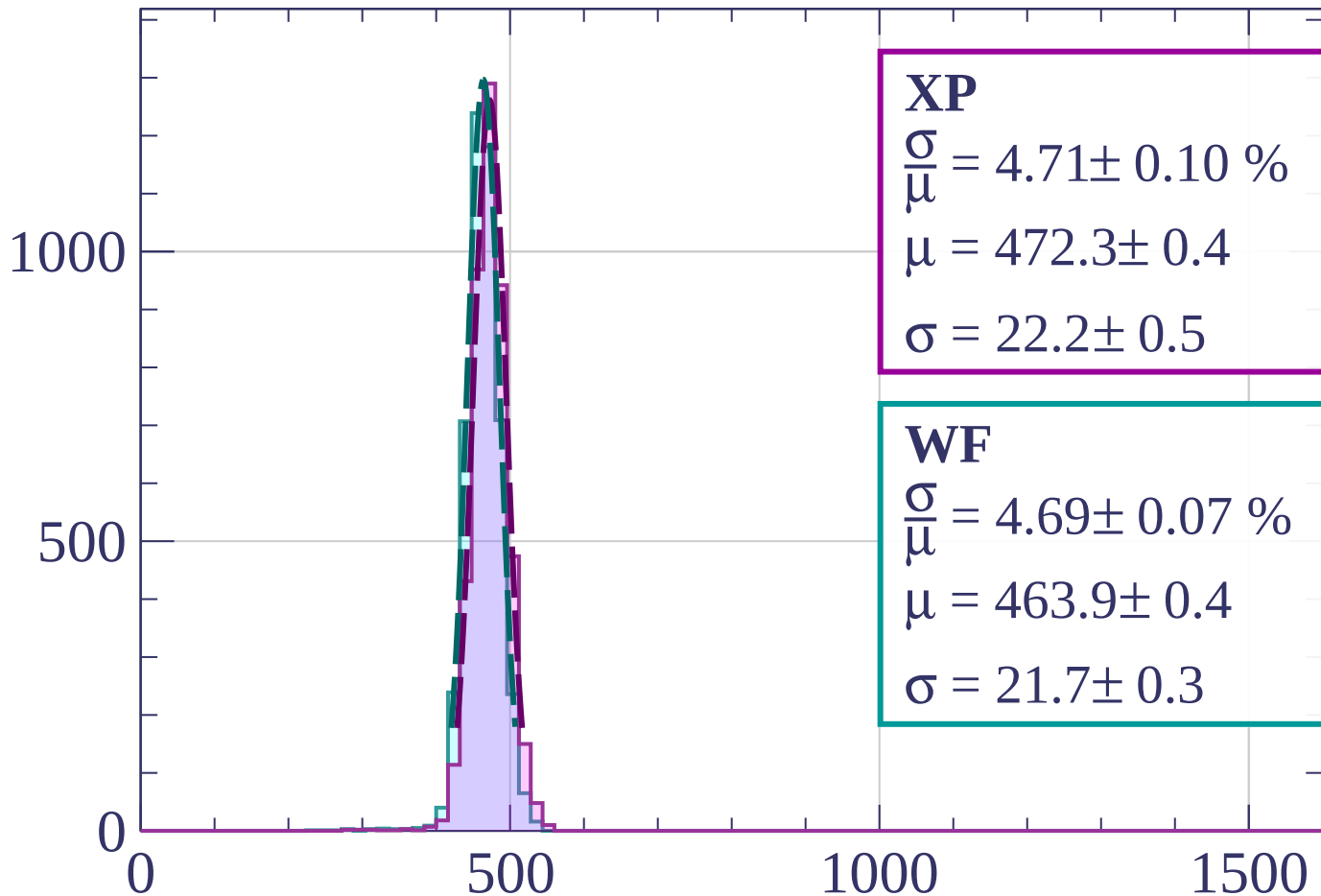
$$\mu = 461.1 \pm 0.4$$

$$\sigma = 22.2 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 4.71 \pm 0.10 \%$$

$$\mu = 472.3 \pm 0.4$$

$$\sigma = 22.2 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 4.69 \pm 0.07 \%$$

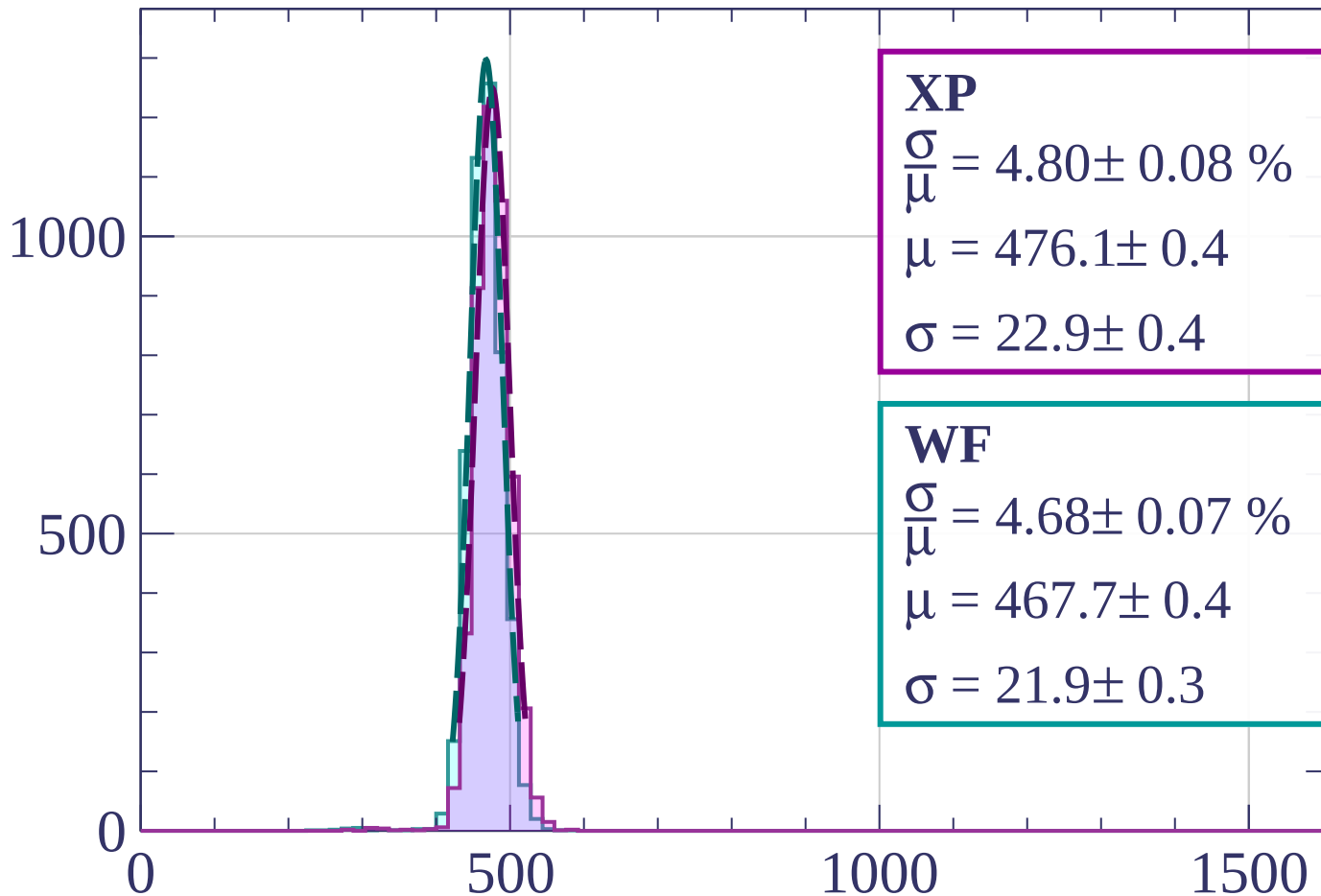
$$\mu = 463.9 \pm 0.4$$

$$\sigma = 21.7 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 4.80 \pm 0.08 \%$$

$$\mu = 476.1 \pm 0.4$$

$$\sigma = 22.9 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.68 \pm 0.07 \%$$

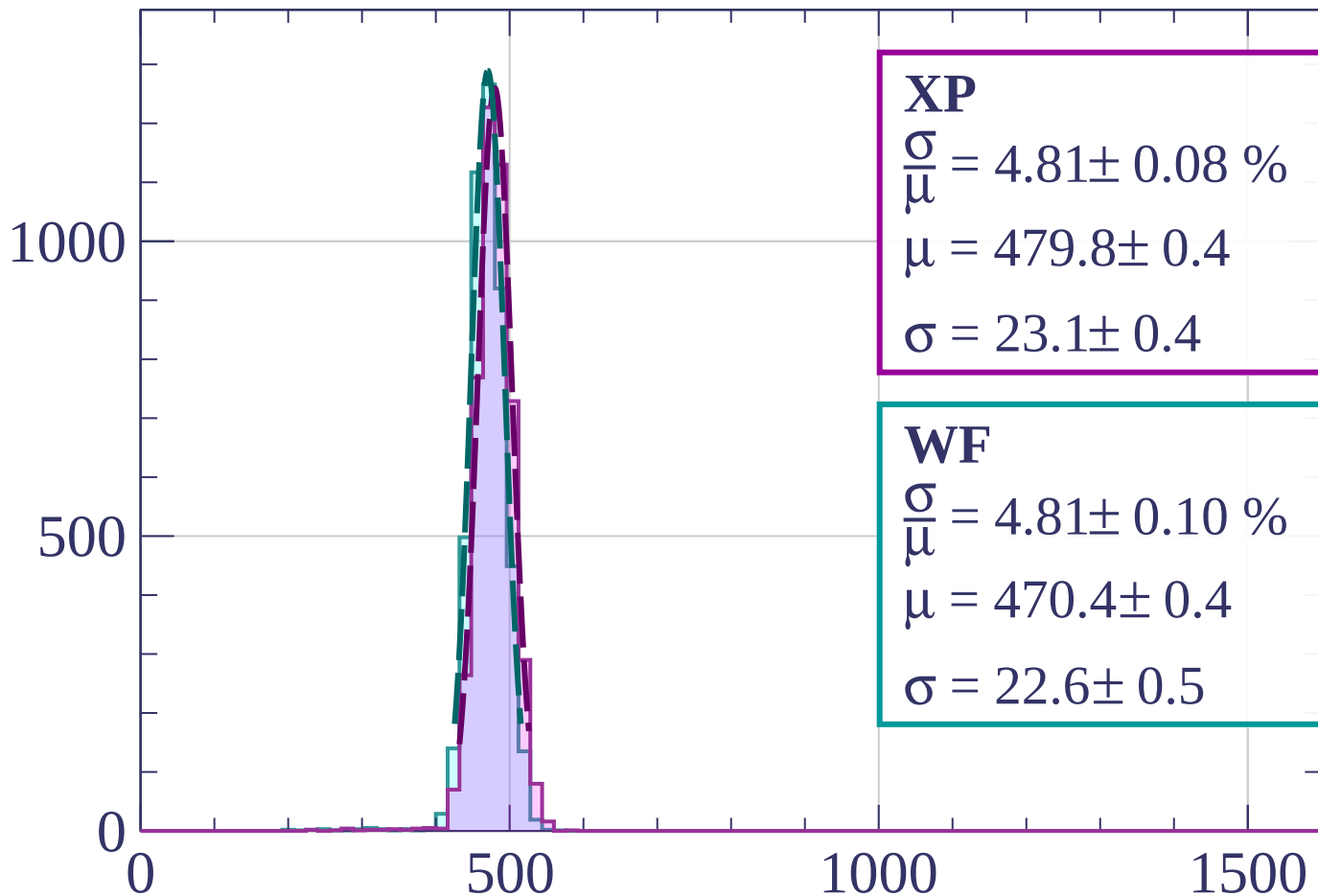
$$\mu = 467.7 \pm 0.4$$

$$\sigma = 21.9 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 4.81 \pm 0.08 \%$$

$$\mu = 479.8 \pm 0.4$$

$$\sigma = 23.1 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.81 \pm 0.10 \%$$

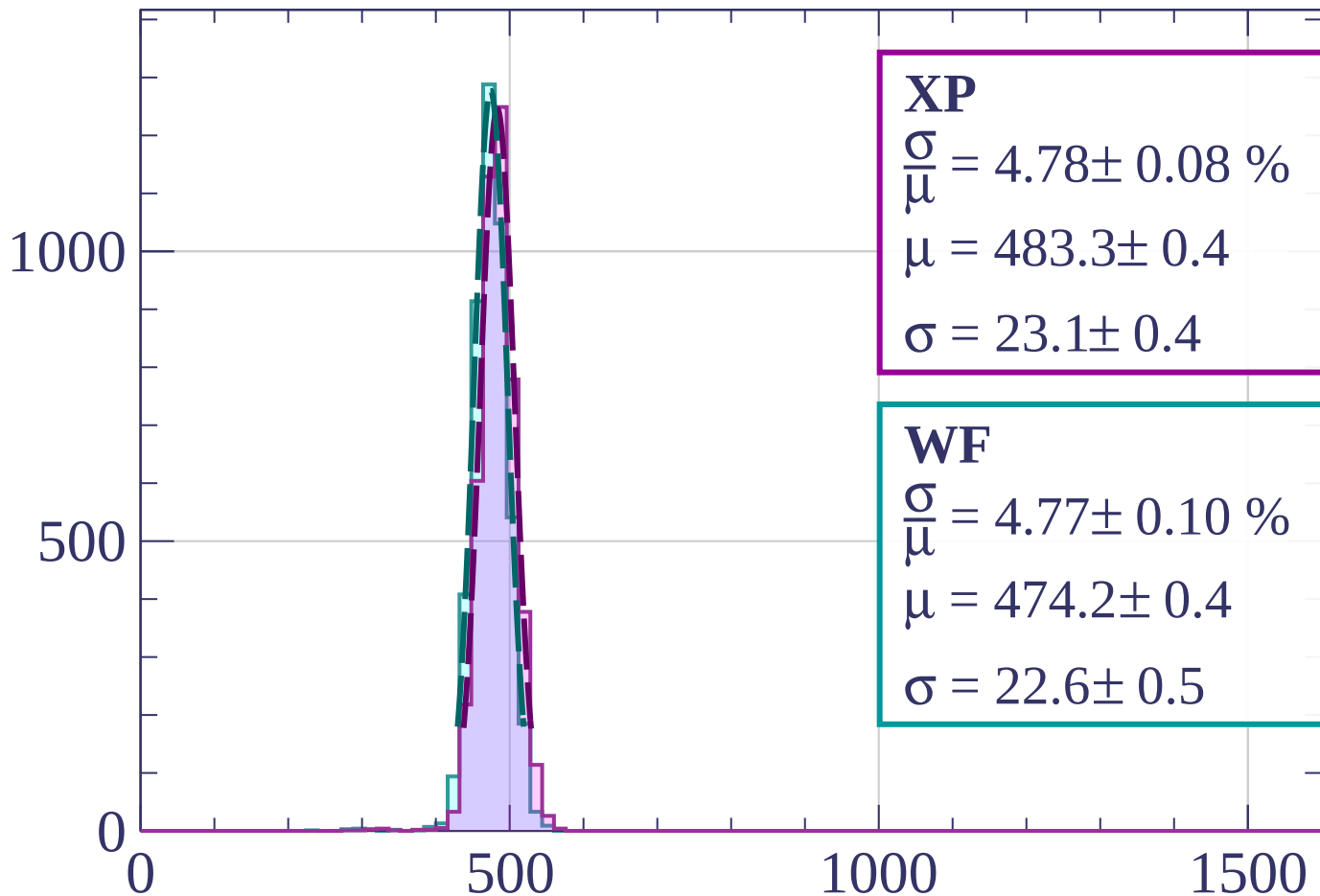
$$\mu = 470.4 \pm 0.4$$

$$\sigma = 22.6 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 4.78 \pm 0.08 \%$$

$$\mu = 483.3 \pm 0.4$$

$$\sigma = 23.1 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.77 \pm 0.10 \%$$

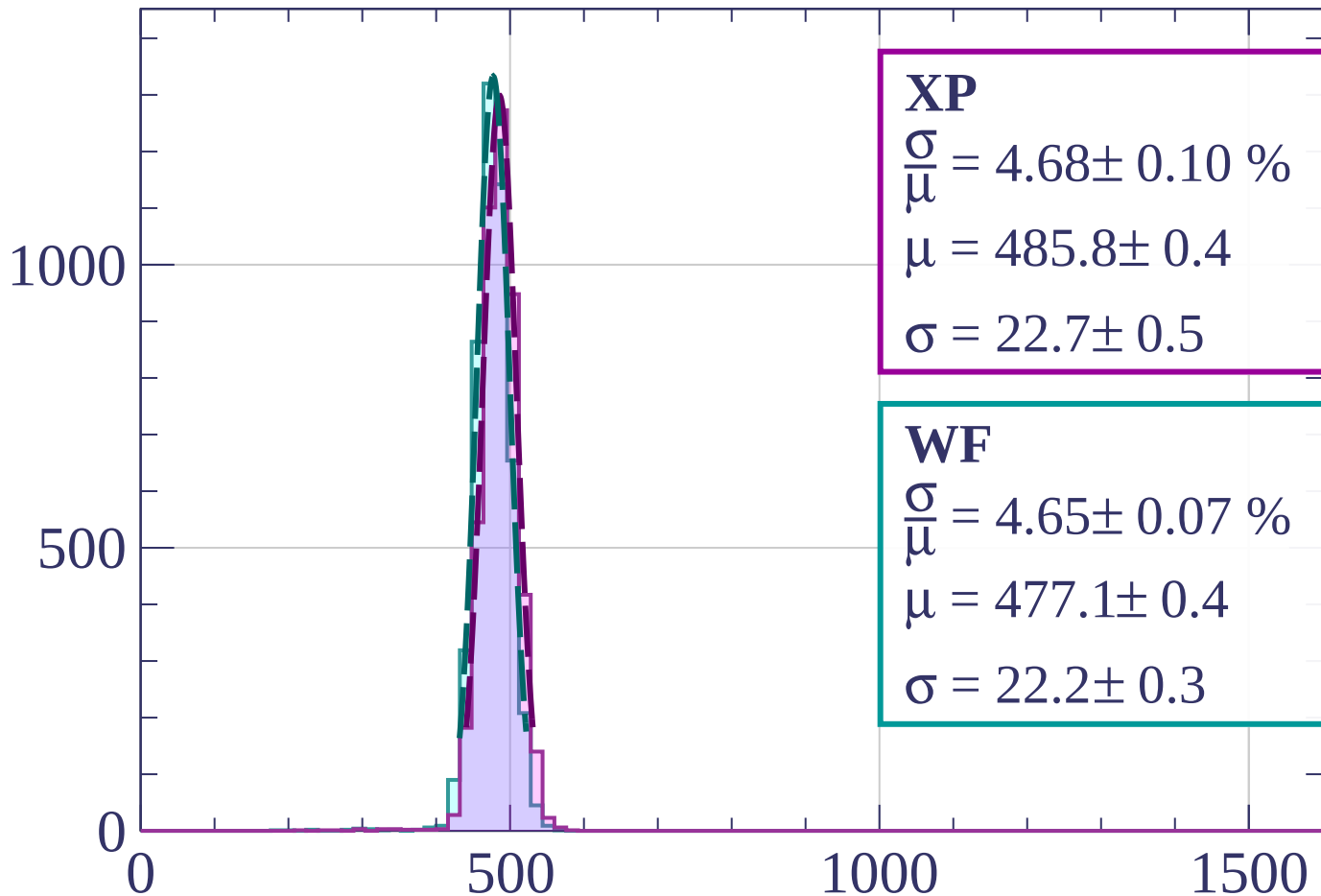
$$\mu = 474.2 \pm 0.4$$

$$\sigma = 22.6 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $1680 < p < 1760$

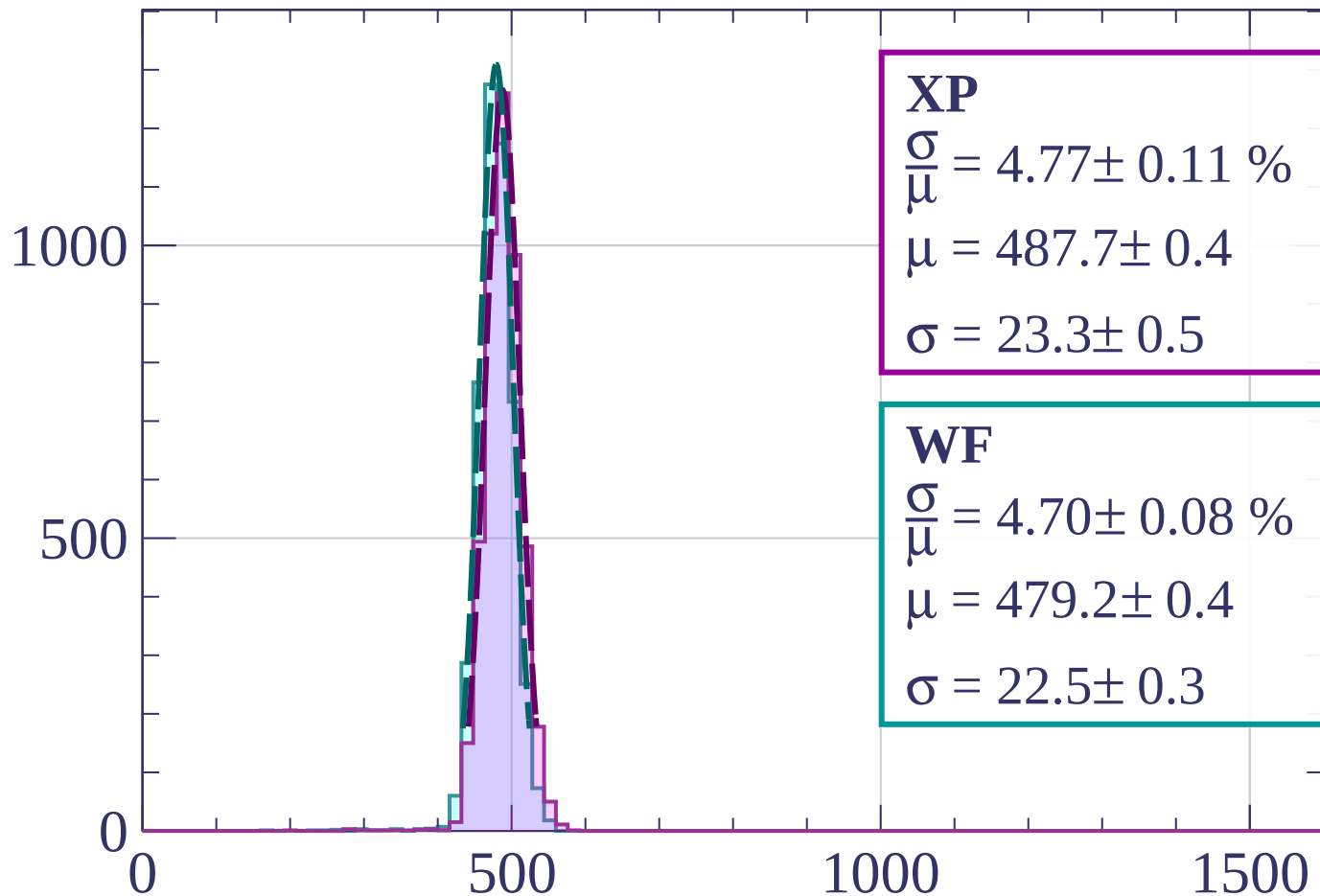
Count



dE/dx [ADC counts/cm]

Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 4.77 \pm 0.11 \%$$

$$\mu = 487.7 \pm 0.4$$

$$\sigma = 23.3 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 4.70 \pm 0.08 \%$$

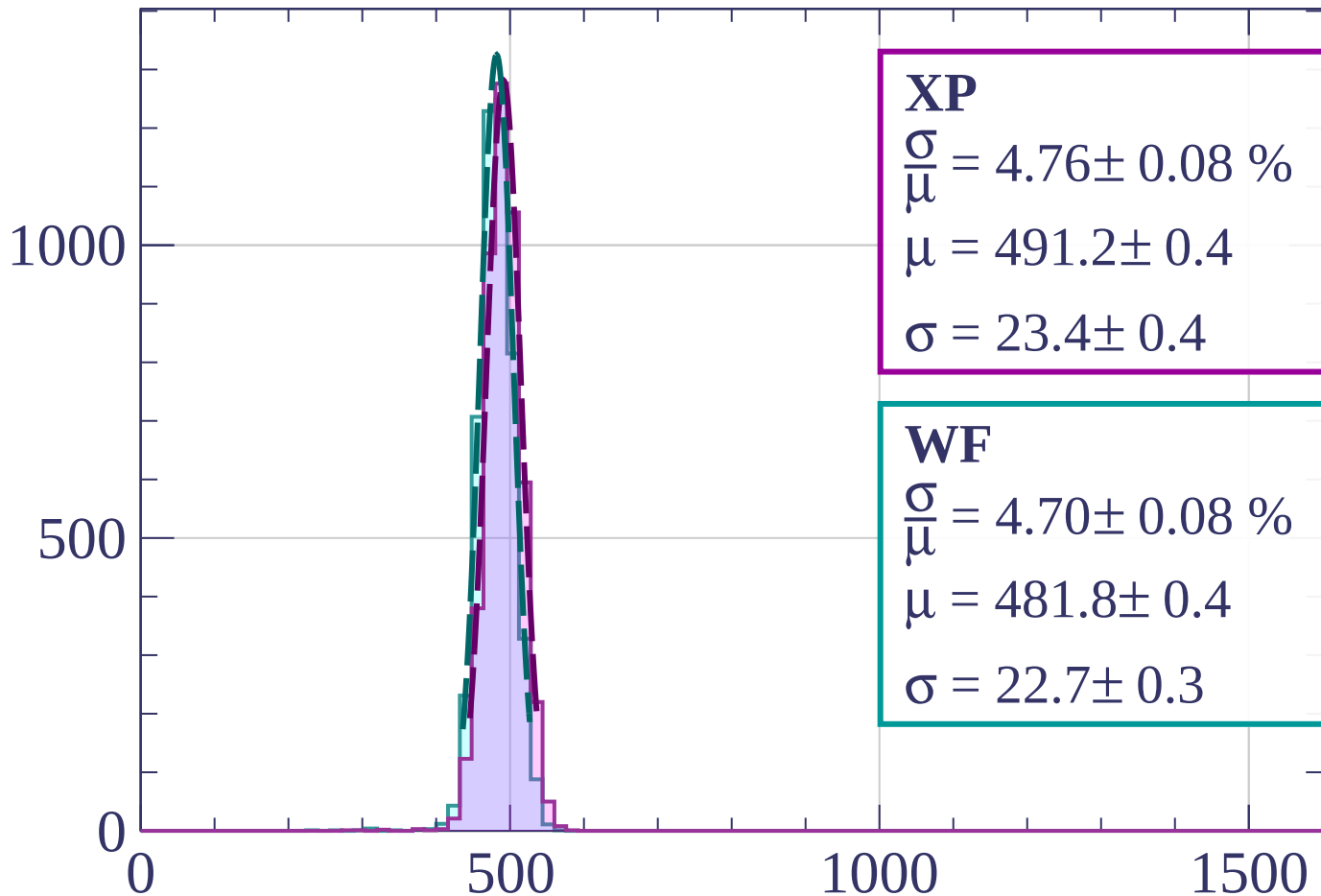
$$\mu = 479.2 \pm 0.4$$

$$\sigma = 22.5 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $1840 < p < 1920$

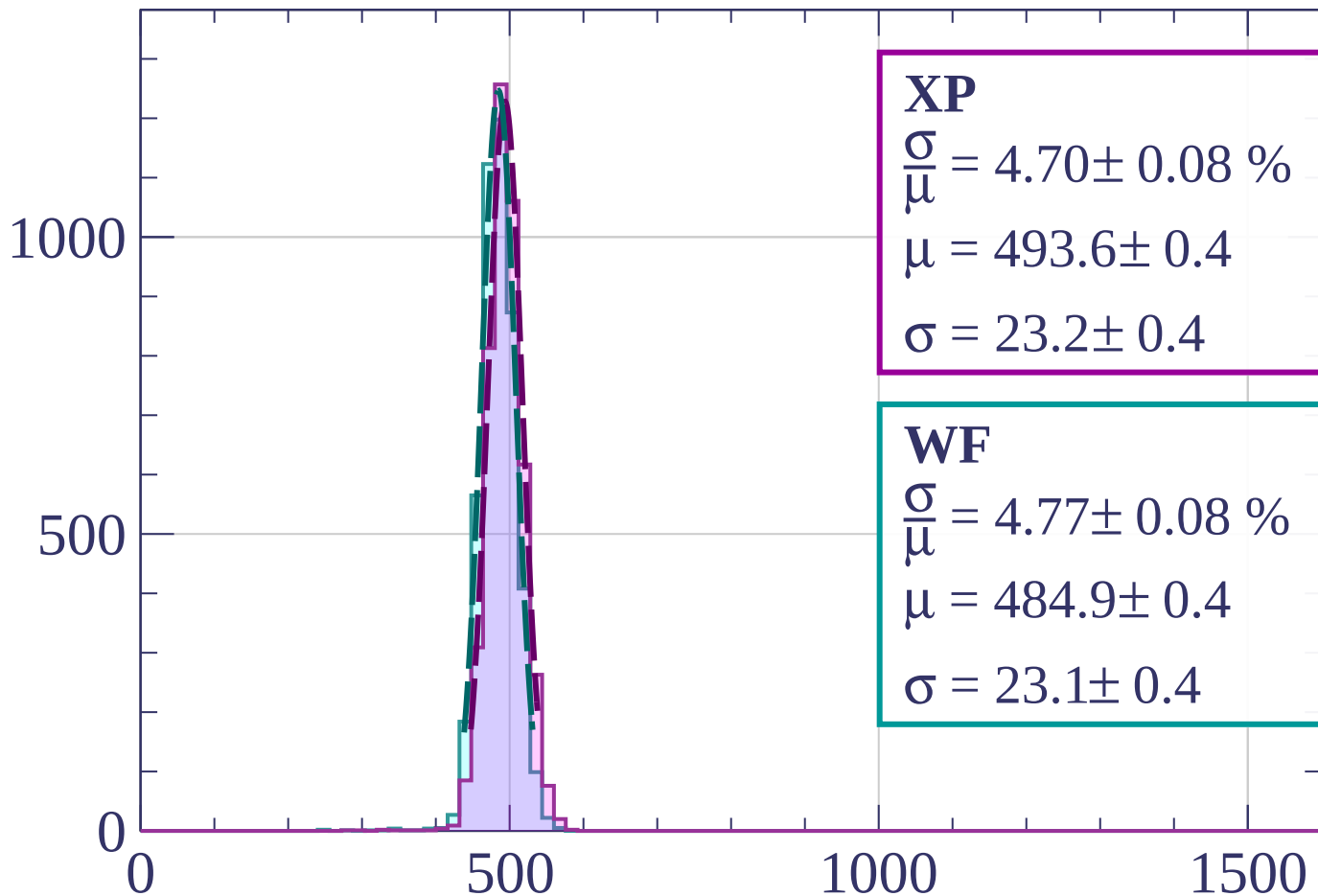
Count



dE/dx [ADC counts/cm]

Energy loss | $1920 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 4.70 \pm 0.08 \%$$

$$\mu = 493.6 \pm 0.4$$

$$\sigma = 23.2 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 4.77 \pm 0.08 \%$$

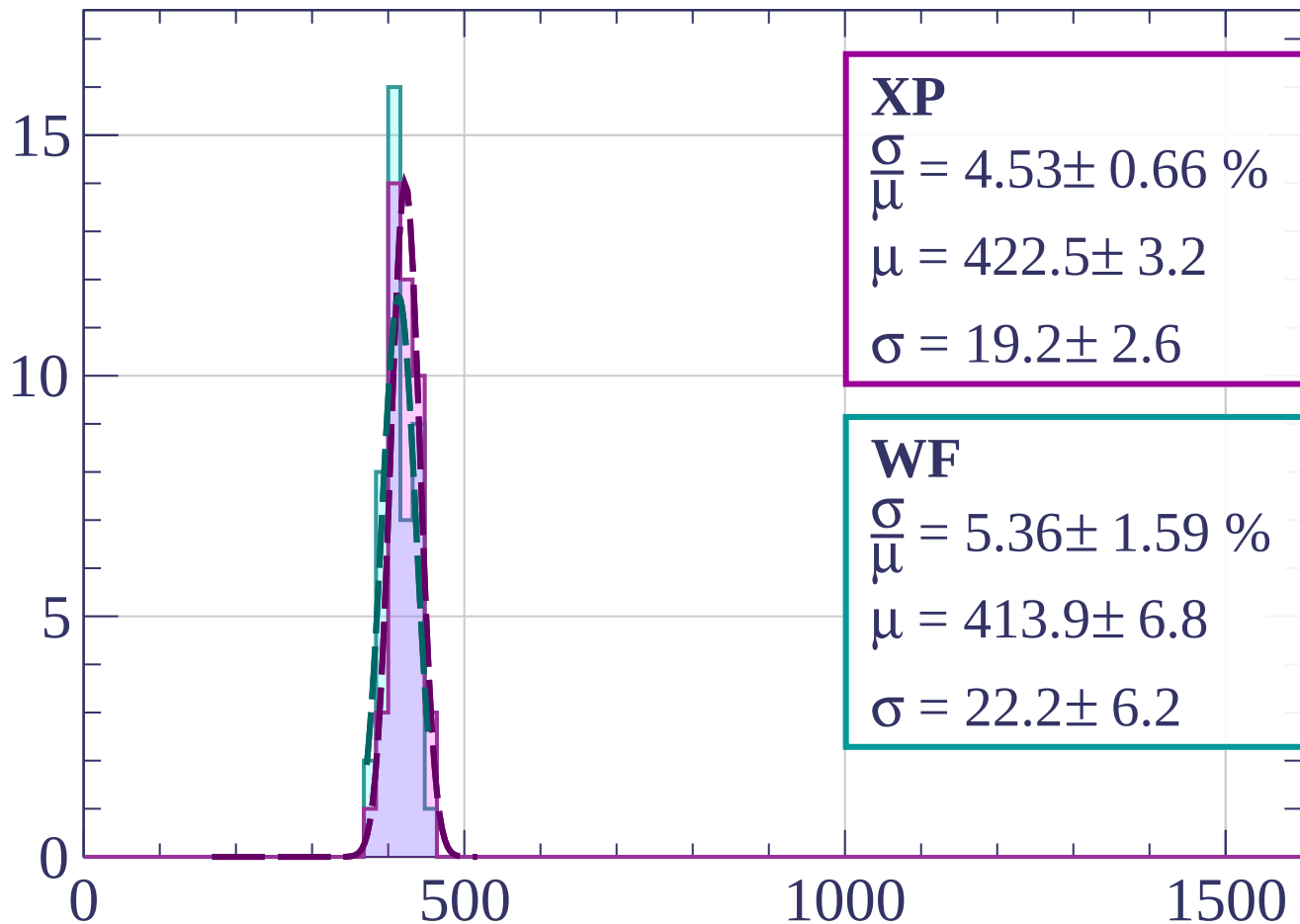
$$\mu = 484.9 \pm 0.4$$

$$\sigma = 23.1 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $0 < \phi < 3$

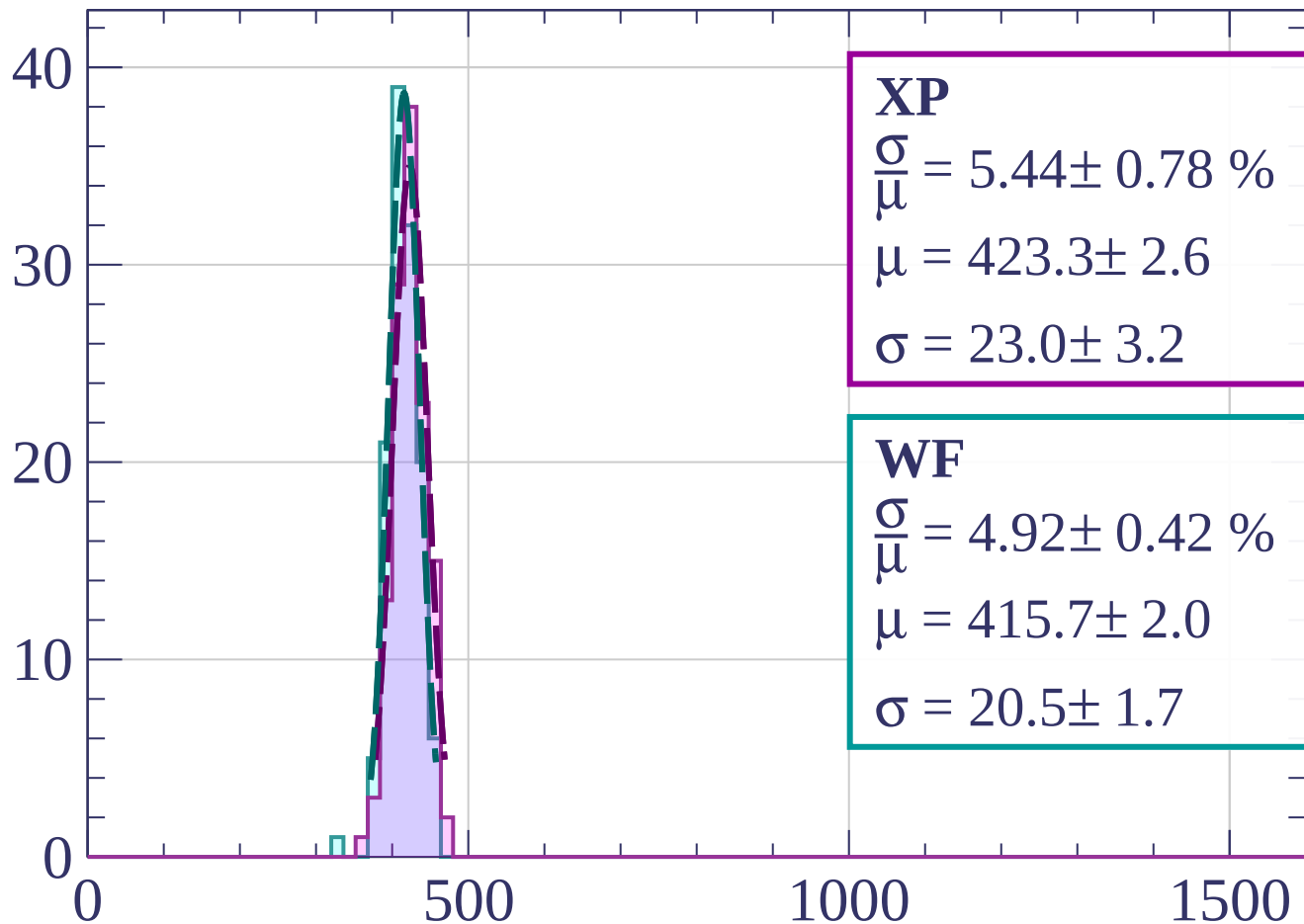
Count



dE/dx [ADC counts/cm]

Energy loss | $3 < \phi < 6$

Count



XP

$$\frac{\sigma}{\mu} = 5.44 \pm 0.78 \%$$

$$\mu = 423.3 \pm 2.6$$

$$\sigma = 23.0 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 4.92 \pm 0.42 \%$$

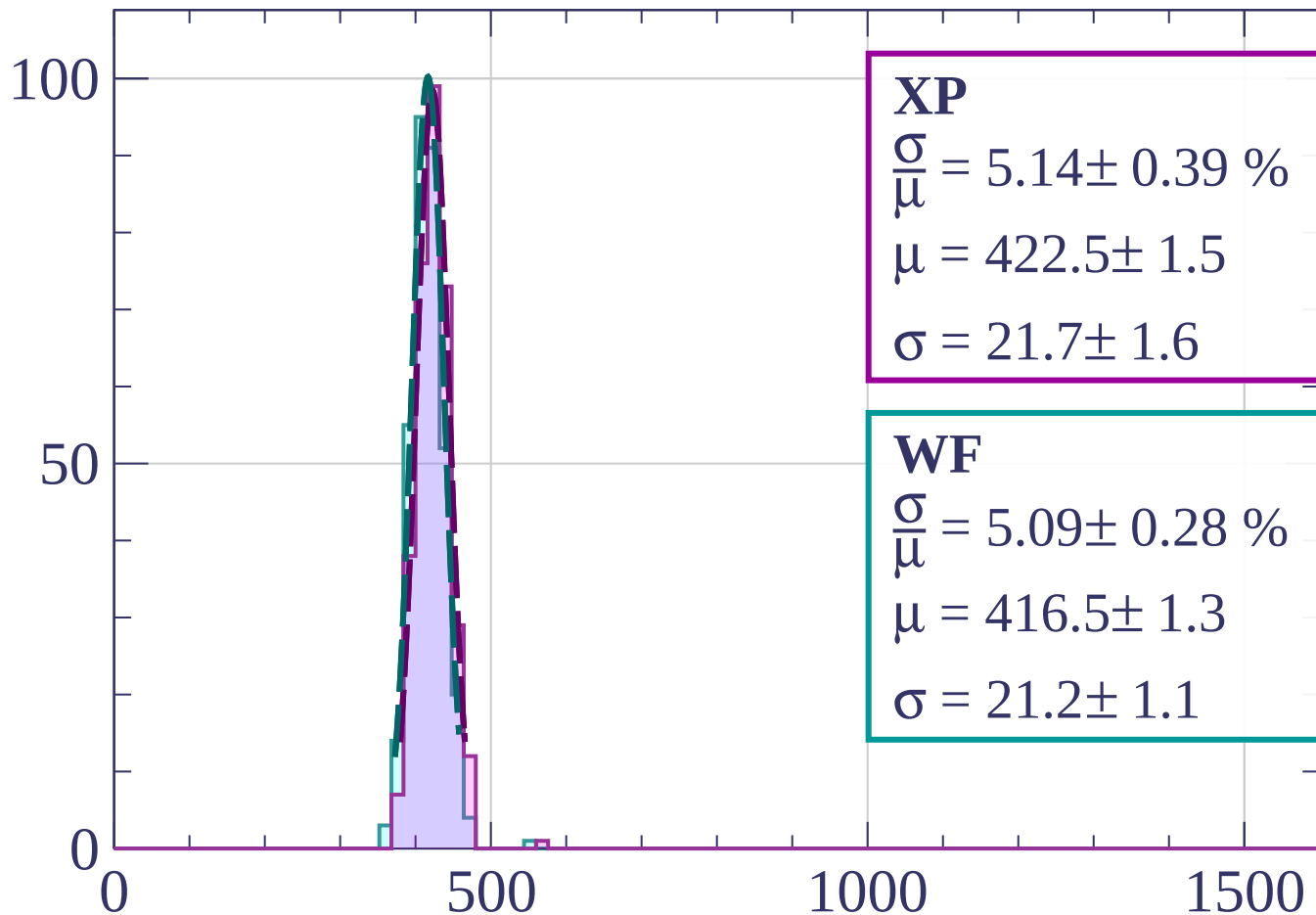
$$\mu = 415.7 \pm 2.0$$

$$\sigma = 20.5 \pm 1.7$$

dE/dx [ADC counts/cm]

Energy loss | $6 < \phi < 9$

Count



XP

$$\frac{\sigma}{\mu} = 5.14 \pm 0.39 \%$$

$$\mu = 422.5 \pm 1.5$$

$$\sigma = 21.7 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 5.09 \pm 0.28 \%$$

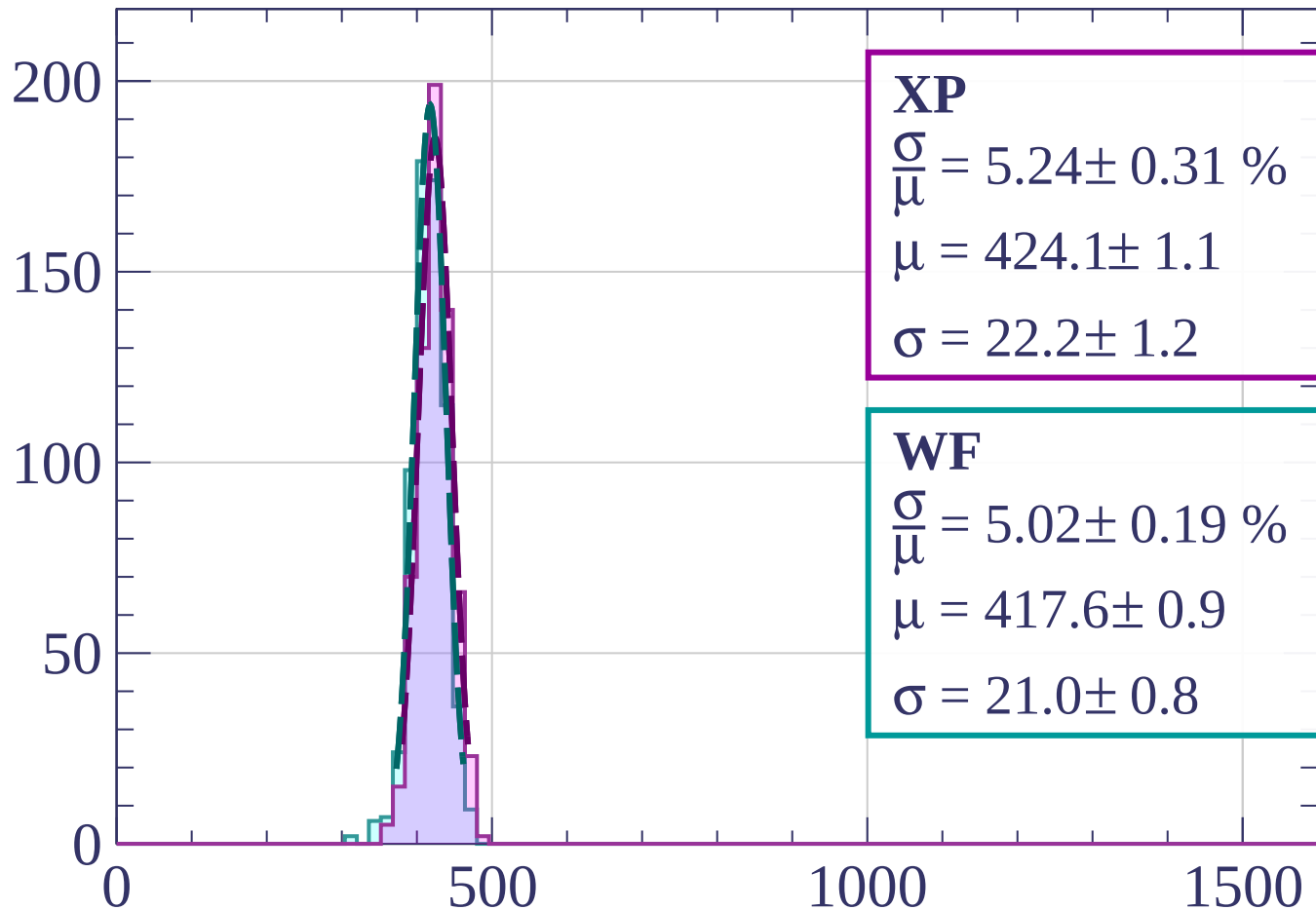
$$\mu = 416.5 \pm 1.3$$

$$\sigma = 21.2 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $9 < \phi < 12$

Count



XP

$$\frac{\sigma}{\mu} = 5.24 \pm 0.31 \%$$

$$\mu = 424.1 \pm 1.1$$

$$\sigma = 22.2 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 5.02 \pm 0.19 \%$$

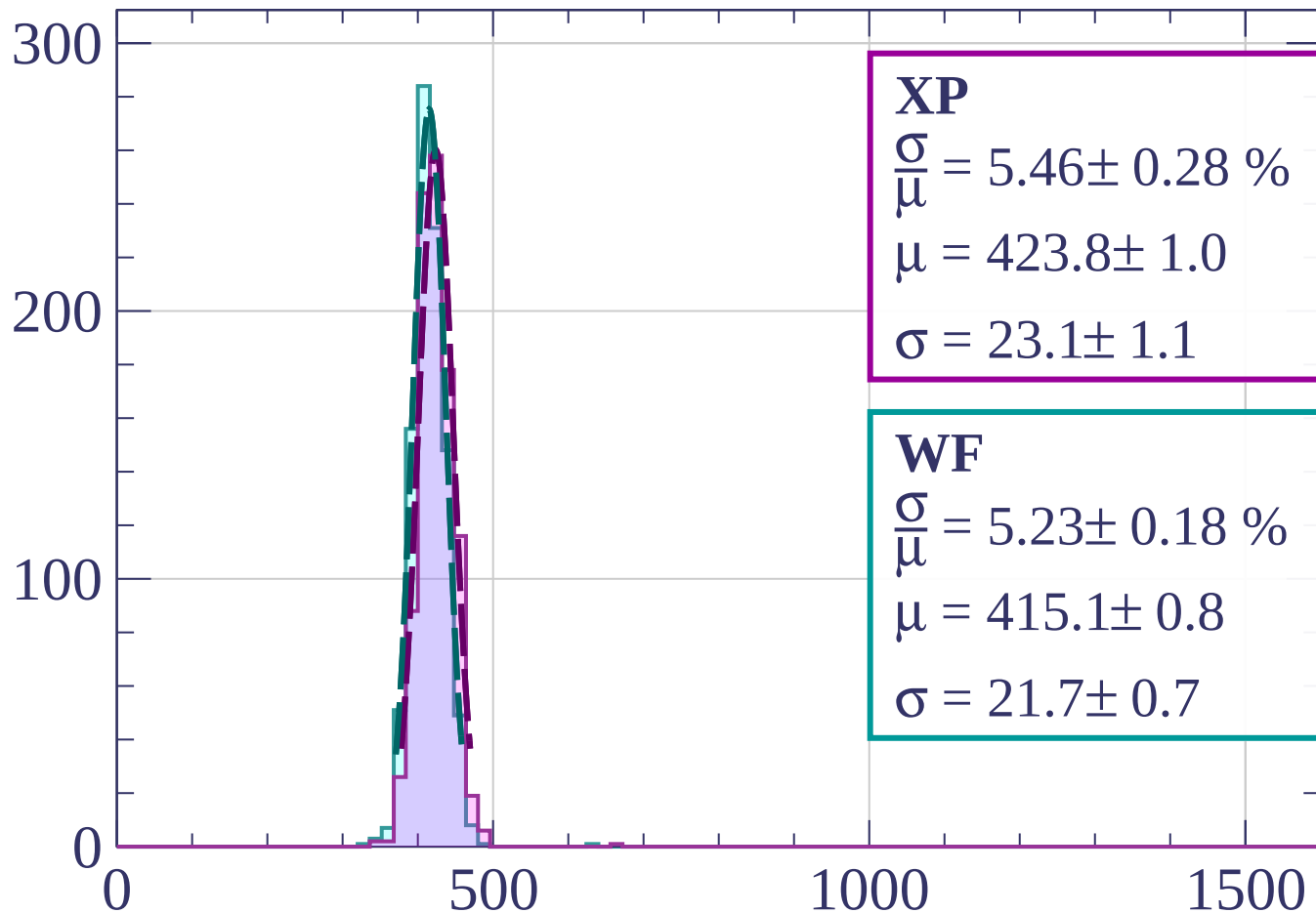
$$\mu = 417.6 \pm 0.9$$

$$\sigma = 21.0 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $12 < \phi < 15$

Count



XP

$$\frac{\sigma}{\mu} = 5.46 \pm 0.28 \%$$

$$\mu = 423.8 \pm 1.0$$

$$\sigma = 23.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 5.23 \pm 0.18 \%$$

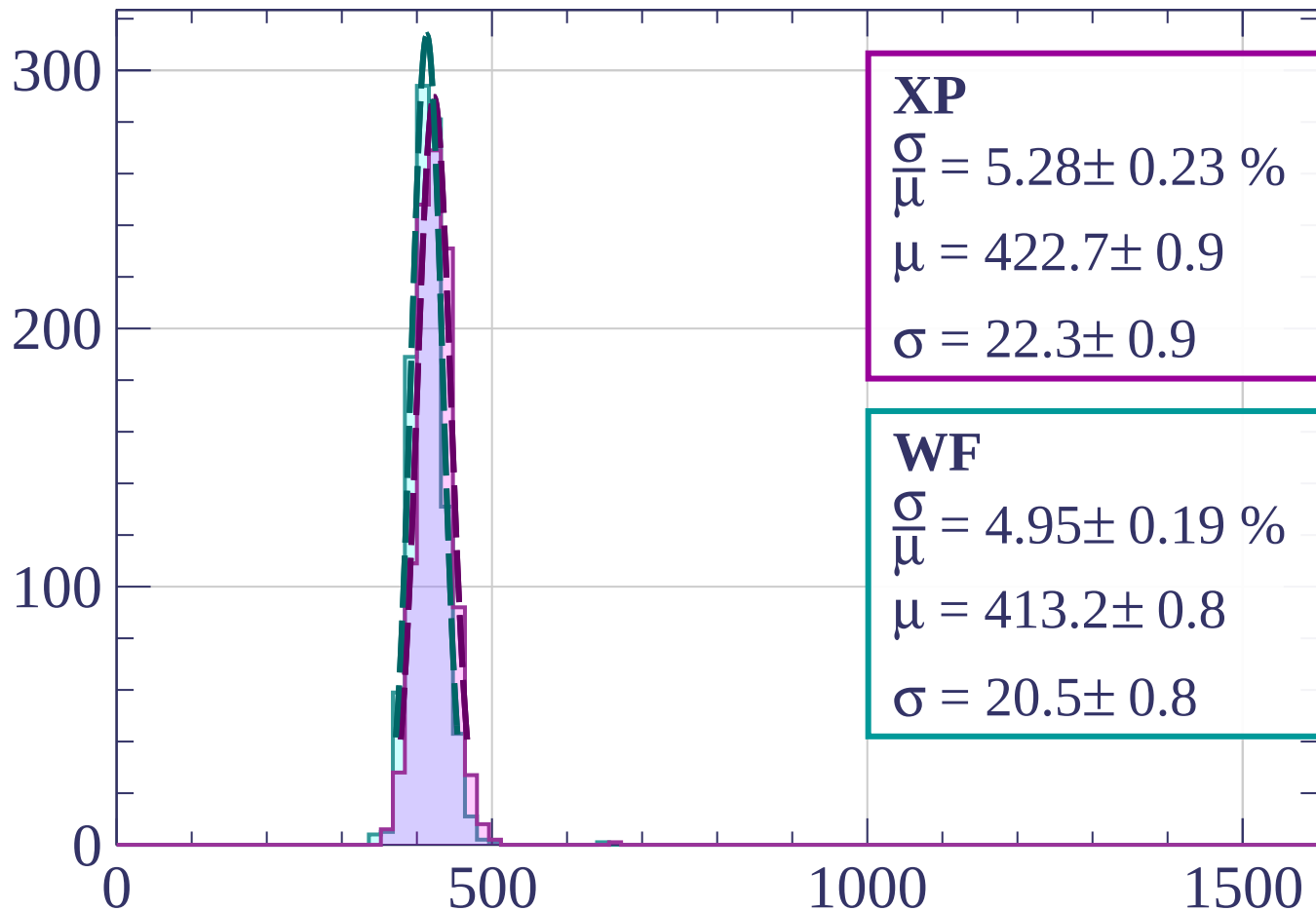
$$\mu = 415.1 \pm 0.8$$

$$\sigma = 21.7 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $15 < \phi < 18$

Count



XP

$$\frac{\sigma}{\mu} = 5.28 \pm 0.23 \%$$

$$\mu = 422.7 \pm 0.9$$

$$\sigma = 22.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 4.95 \pm 0.19 \%$$

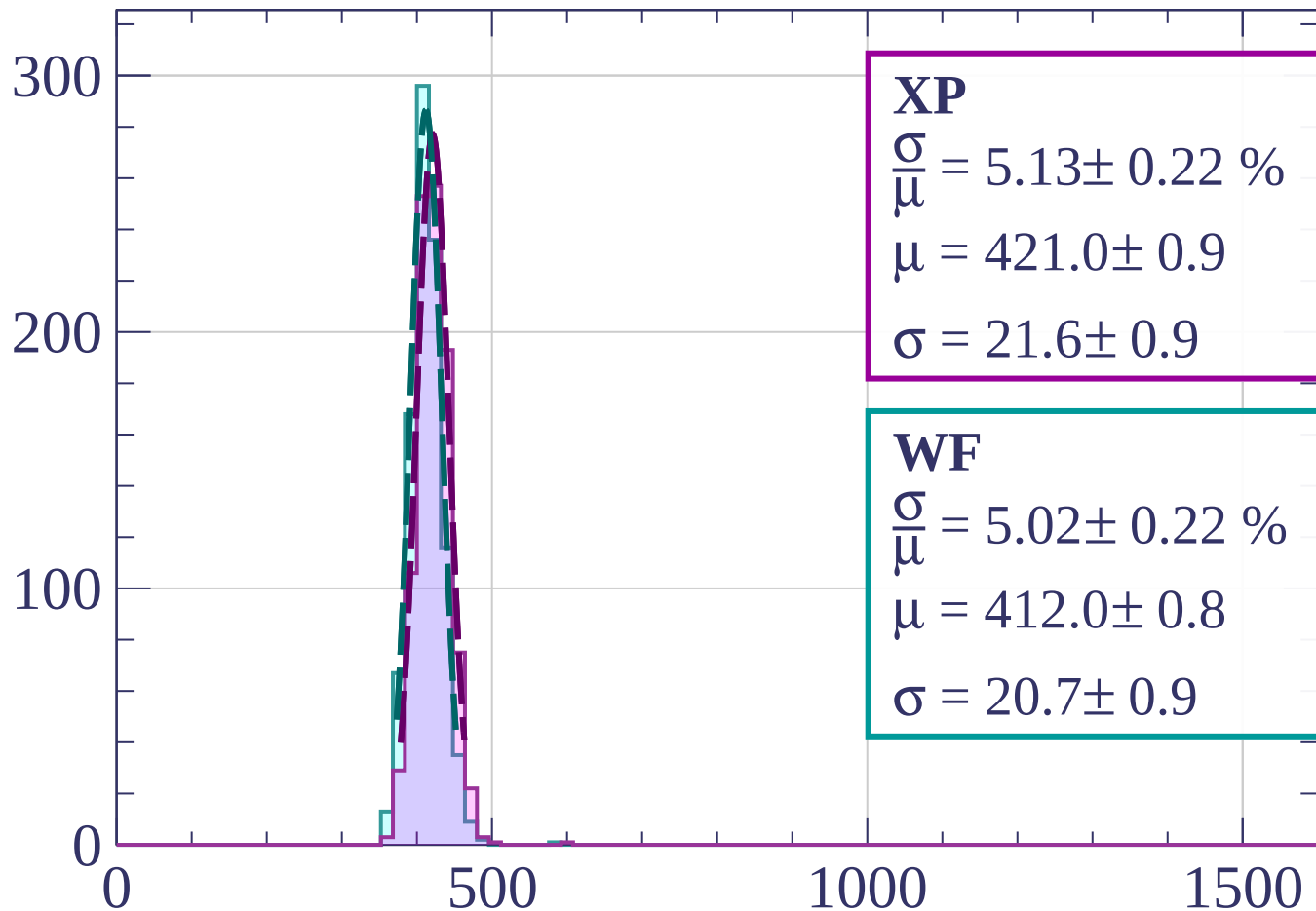
$$\mu = 413.2 \pm 0.8$$

$$\sigma = 20.5 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $18 < \phi < 21$

Count



XP

$$\frac{\sigma}{\mu} = 5.13 \pm 0.22 \%$$

$$\mu = 421.0 \pm 0.9$$

$$\sigma = 21.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 5.02 \pm 0.22 \%$$

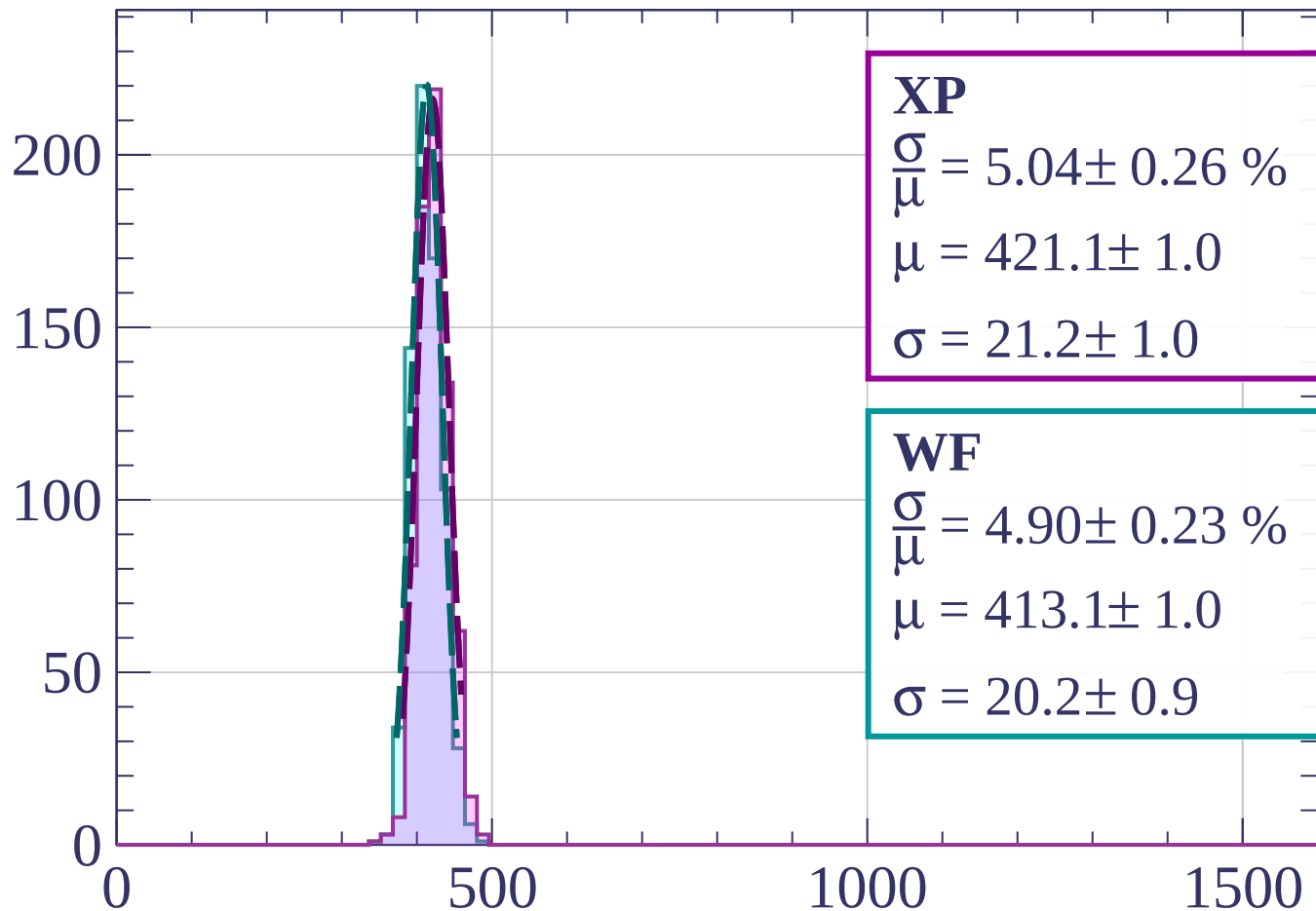
$$\mu = 412.0 \pm 0.8$$

$$\sigma = 20.7 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $21 < \phi < 24$

Count



XP

$$\frac{\sigma}{\mu} = 5.04 \pm 0.26 \%$$

$$\mu = 421.1 \pm 1.0$$

$$\sigma = 21.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 4.90 \pm 0.23 \%$$

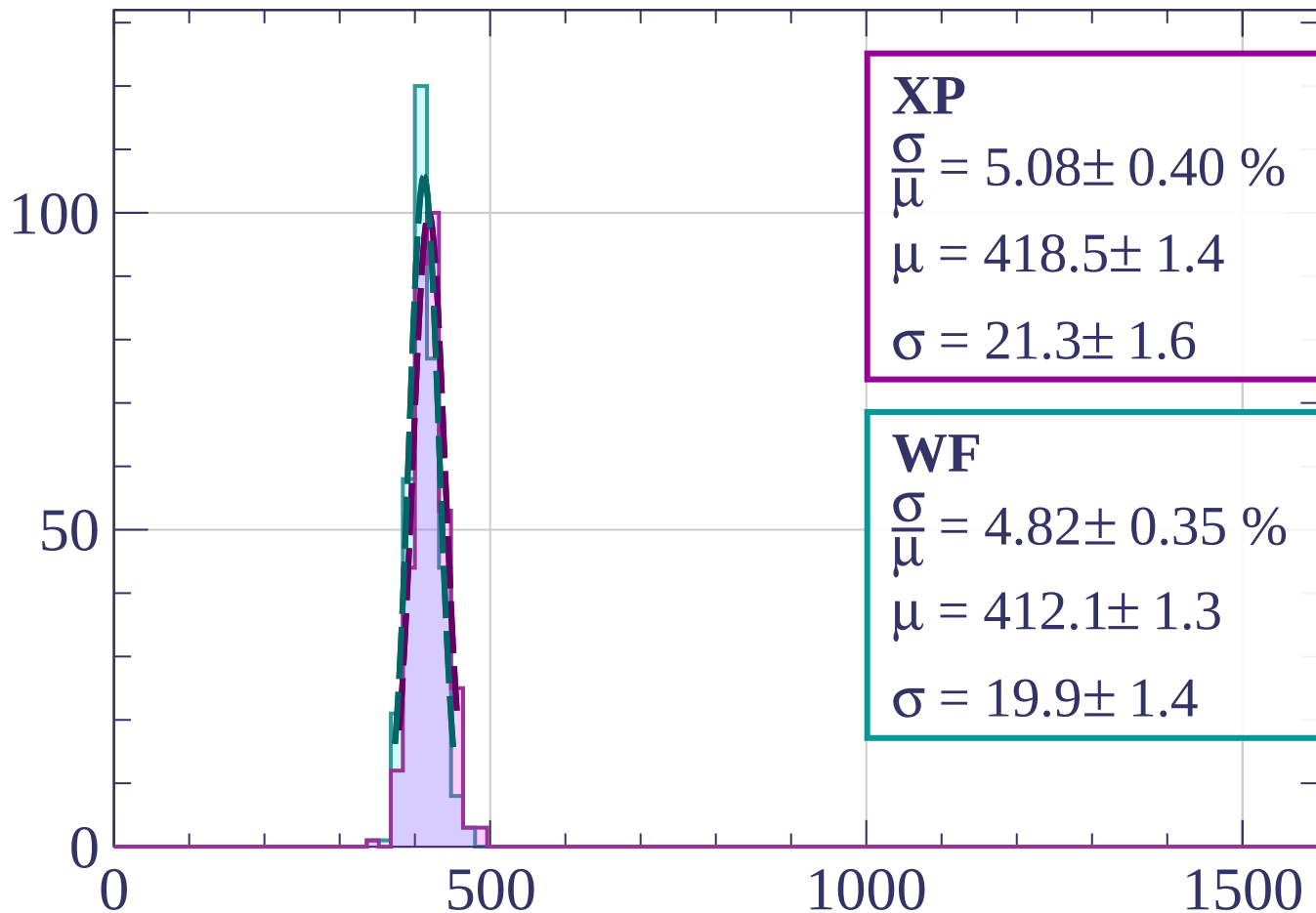
$$\mu = 413.1 \pm 1.0$$

$$\sigma = 20.2 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $24 < \phi < 27$

Count



XP

$$\frac{\sigma}{\mu} = 5.08 \pm 0.40 \%$$

$$\mu = 418.5 \pm 1.4$$

$$\sigma = 21.3 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 4.82 \pm 0.35 \%$$

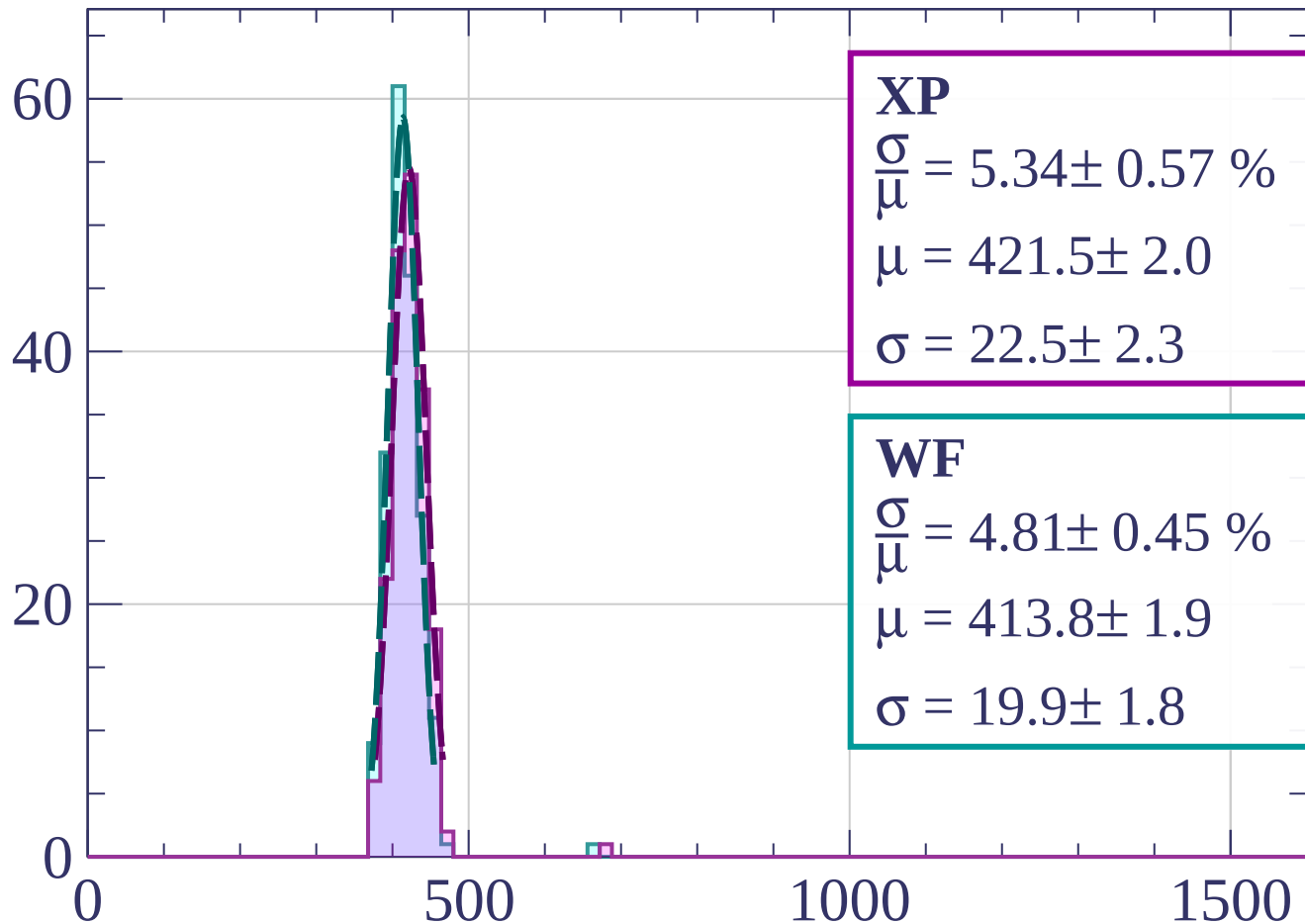
$$\mu = 412.1 \pm 1.3$$

$$\sigma = 19.9 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $27 < \phi < 30$

Count



XP

$$\frac{\sigma}{\mu} = 5.34 \pm 0.57 \%$$

$$\mu = 421.5 \pm 2.0$$

$$\sigma = 22.5 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 4.81 \pm 0.45 \%$$

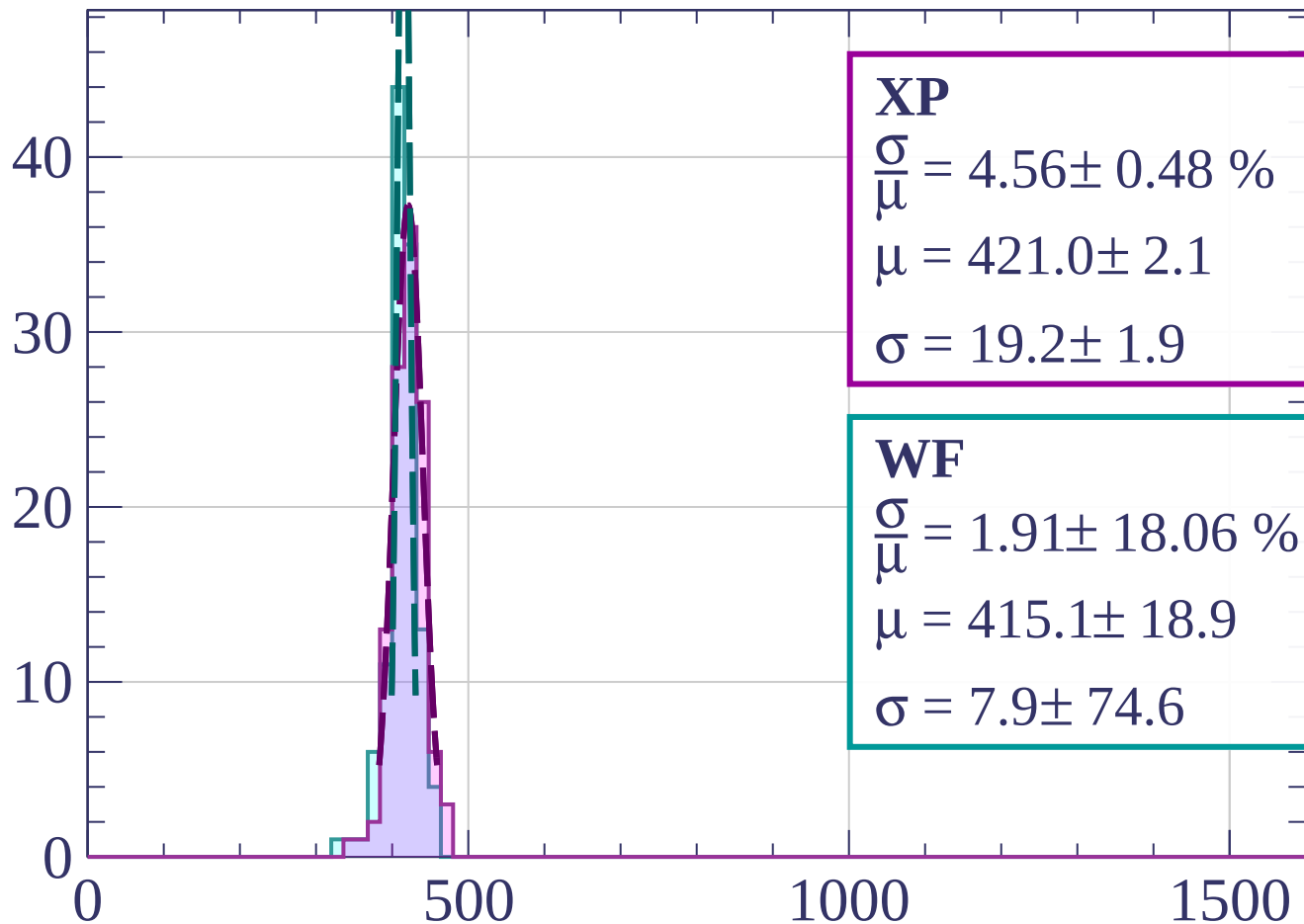
$$\mu = 413.8 \pm 1.9$$

$$\sigma = 19.9 \pm 1.8$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \phi < 33$

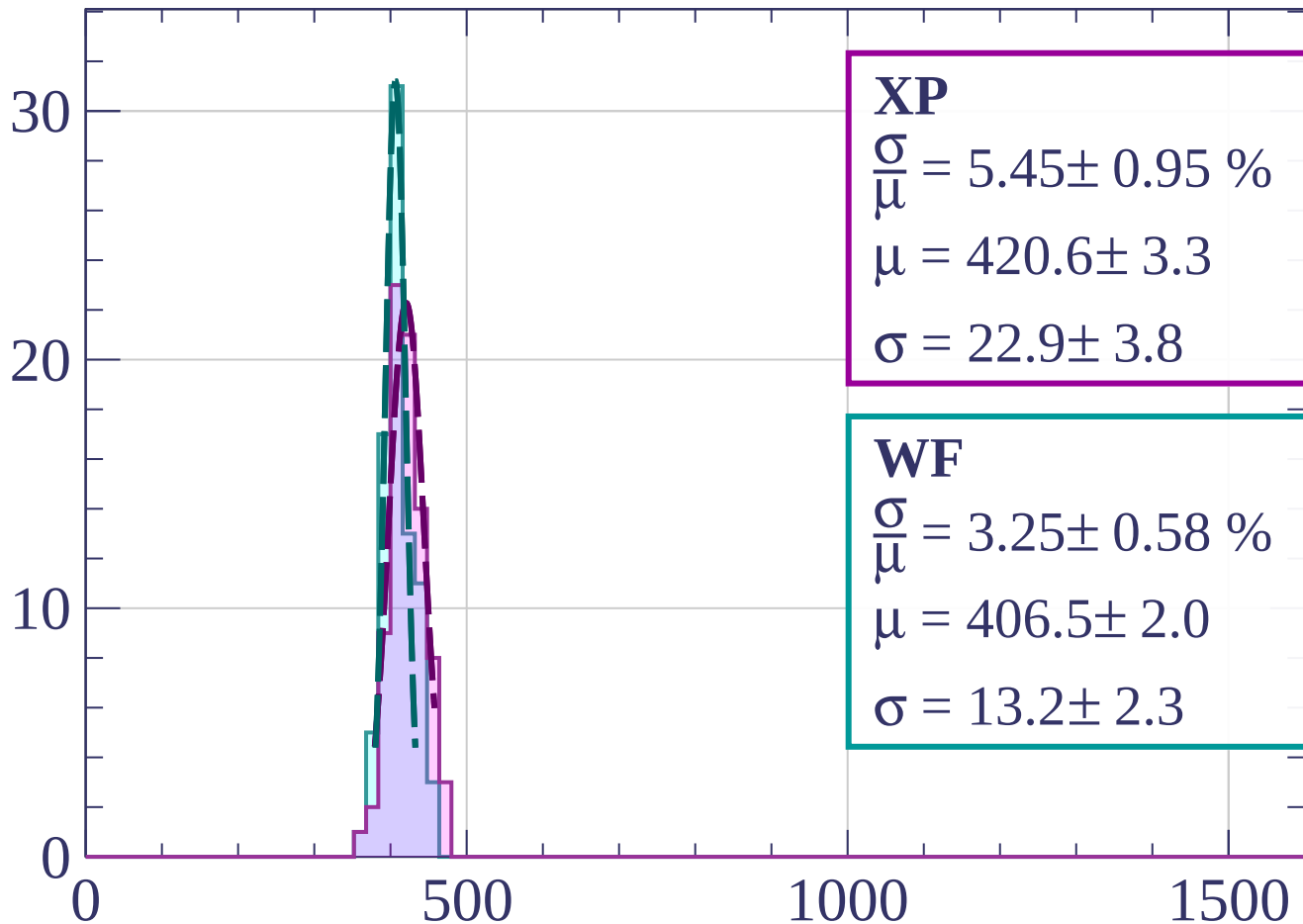
Count



dE/dx [ADC counts/cm]

Energy loss | $33 < \phi < 36$

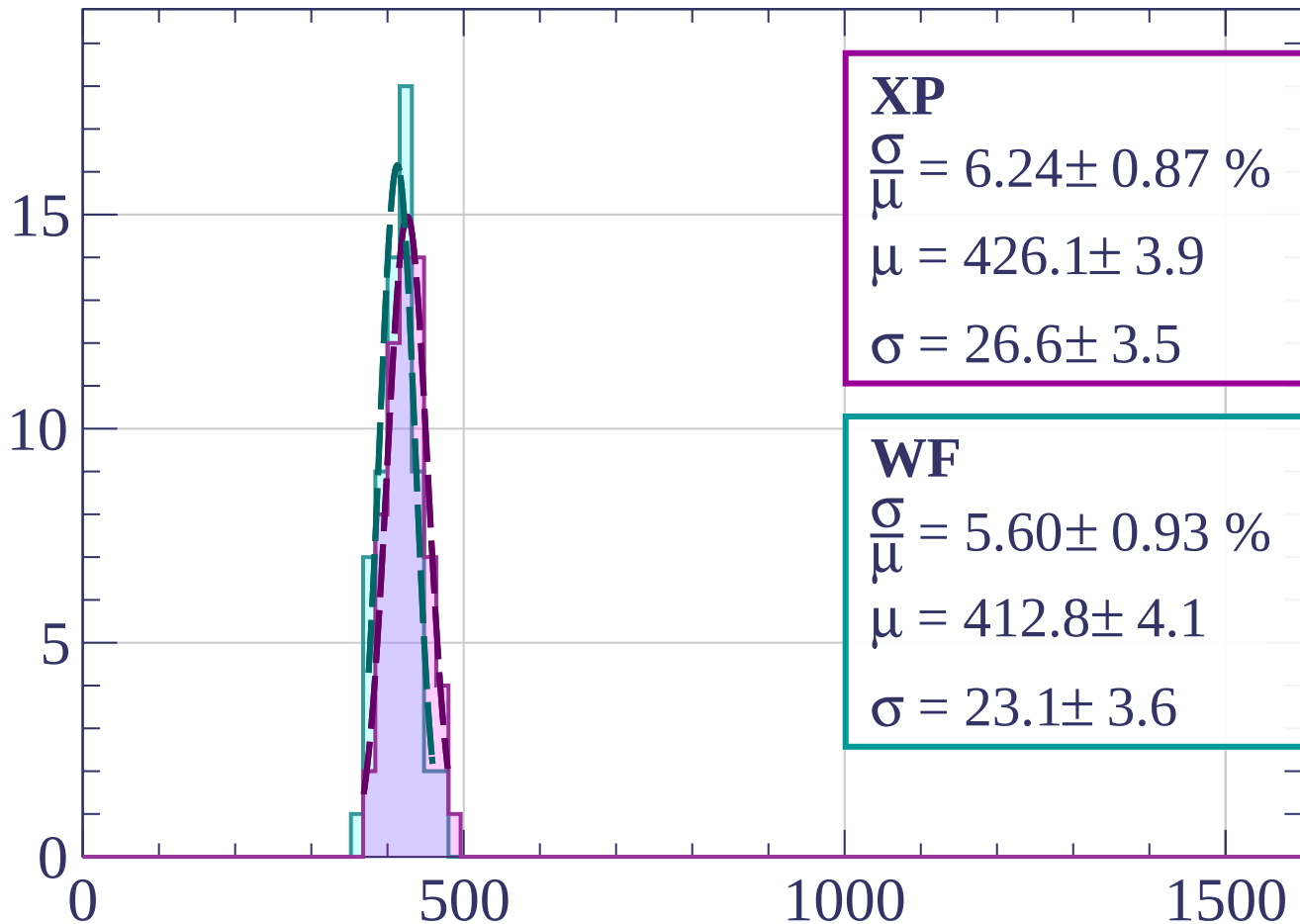
Count



dE/dx [ADC counts/cm]

Energy loss | $36 < \phi < 39$

Count



XP

$$\frac{\sigma}{\mu} = 6.24 \pm 0.87 \%$$

$$\mu = 426.1 \pm 3.9$$

$$\sigma = 26.6 \pm 3.5$$

WF

$$\frac{\sigma}{\mu} = 5.60 \pm 0.93 \%$$

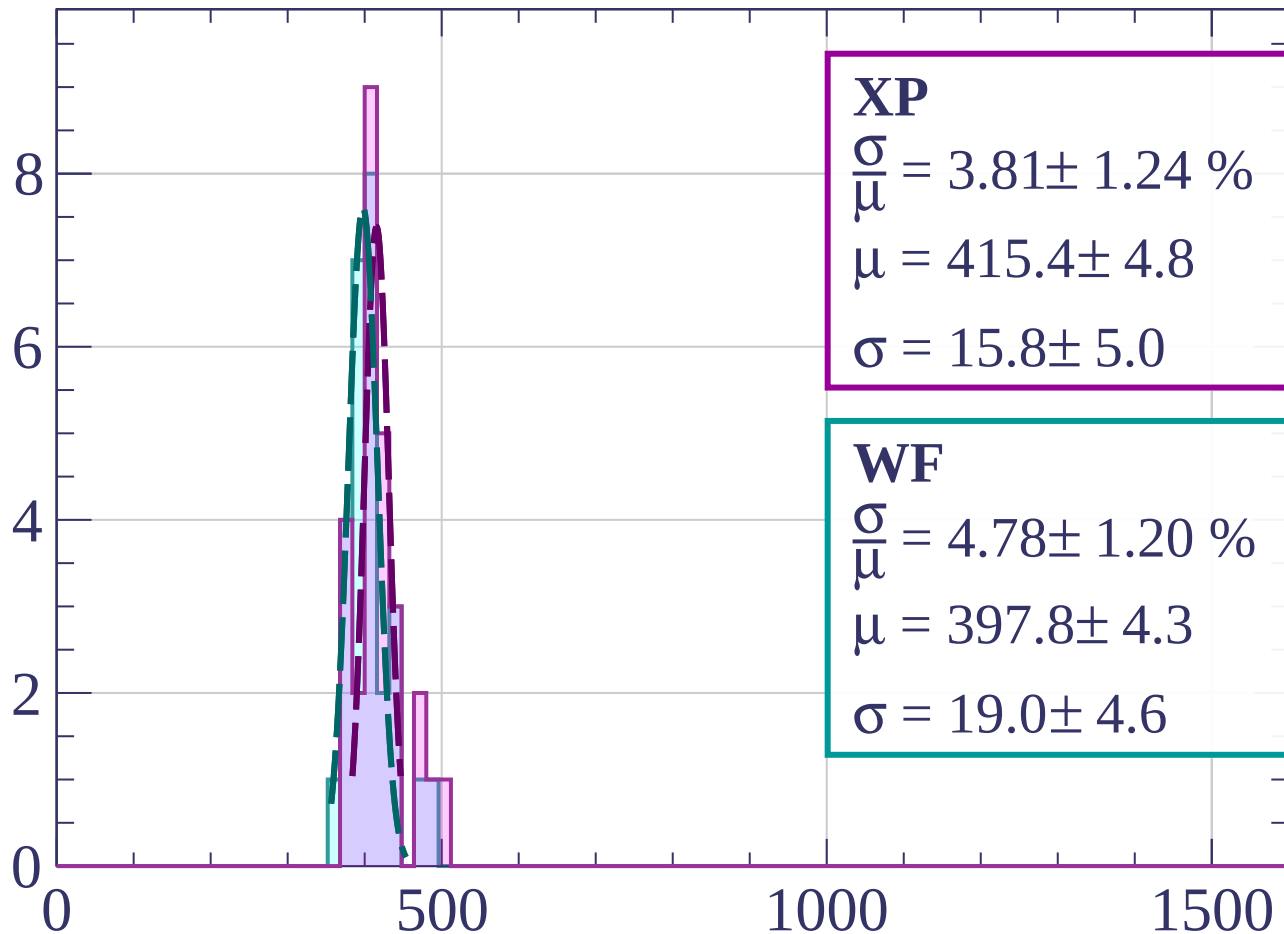
$$\mu = 412.8 \pm 4.1$$

$$\sigma = 23.1 \pm 3.6$$

dE/dx [ADC counts/cm]

Energy loss | $39 < \phi < 42$

Count



XP

$$\frac{\sigma}{\mu} = 3.81 \pm 1.24 \%$$

$$\mu = 415.4 \pm 4.8$$

$$\sigma = 15.8 \pm 5.0$$

WF

$$\frac{\sigma}{\mu} = 4.78 \pm 1.20 \%$$

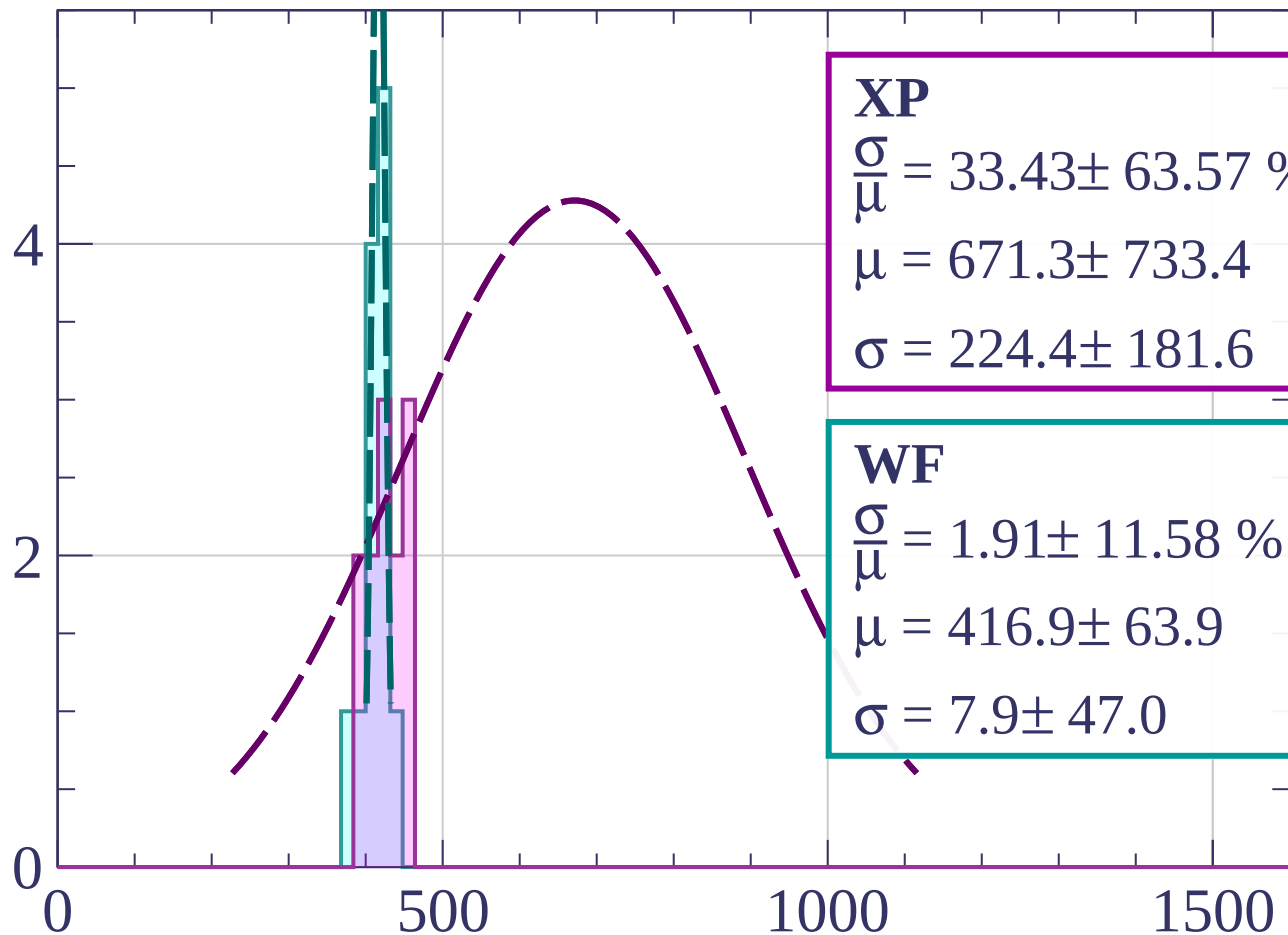
$$\mu = 397.8 \pm 4.3$$

$$\sigma = 19.0 \pm 4.6$$

dE/dx [ADC counts/cm]

Energy loss | $42 < \phi < 45$

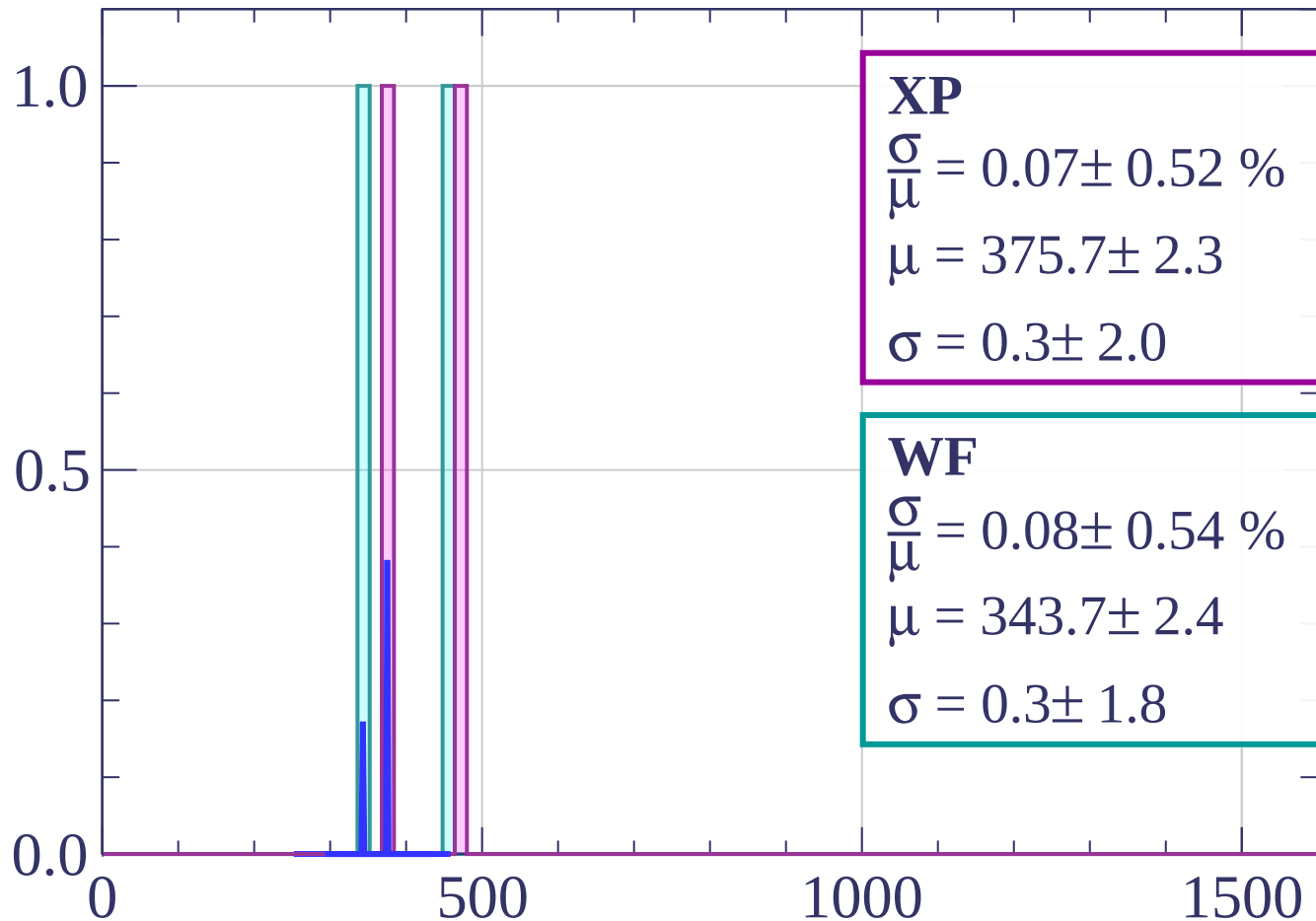
Count



dE/dx [ADC counts/cm]

Energy loss | $45 < \phi < 48$

Count



XP

$$\frac{\sigma}{\mu} = 0.07 \pm 0.52 \%$$

$$\mu = 375.7 \pm 2.3$$

$$\sigma = 0.3 \pm 2.0$$

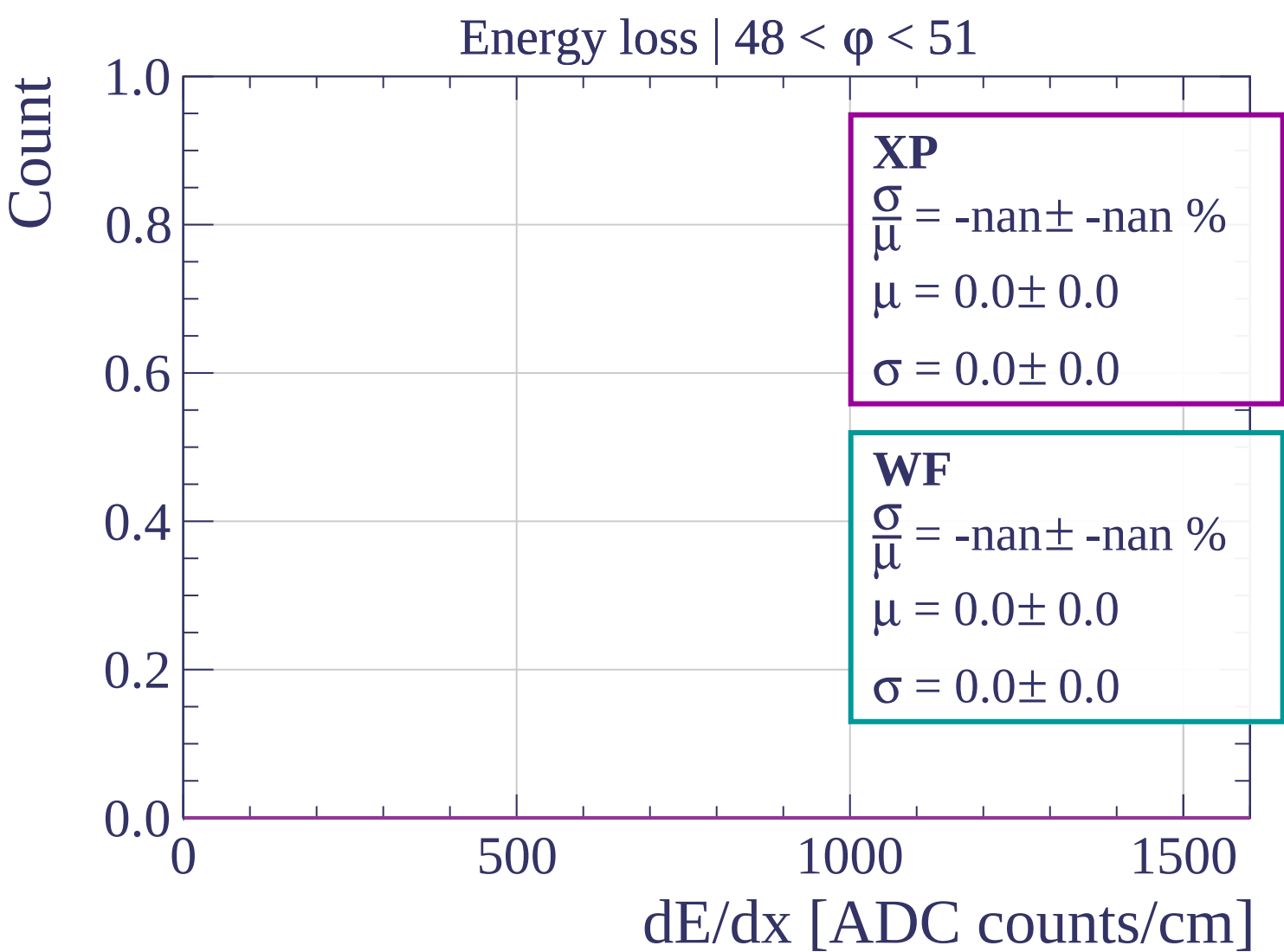
WF

$$\frac{\sigma}{\mu} = 0.08 \pm 0.54 \%$$

$$\mu = 343.7 \pm 2.4$$

$$\sigma = 0.3 \pm 1.8$$

dE/dx [ADC counts/cm]



Count

Energy loss | $51 < \phi < 54$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

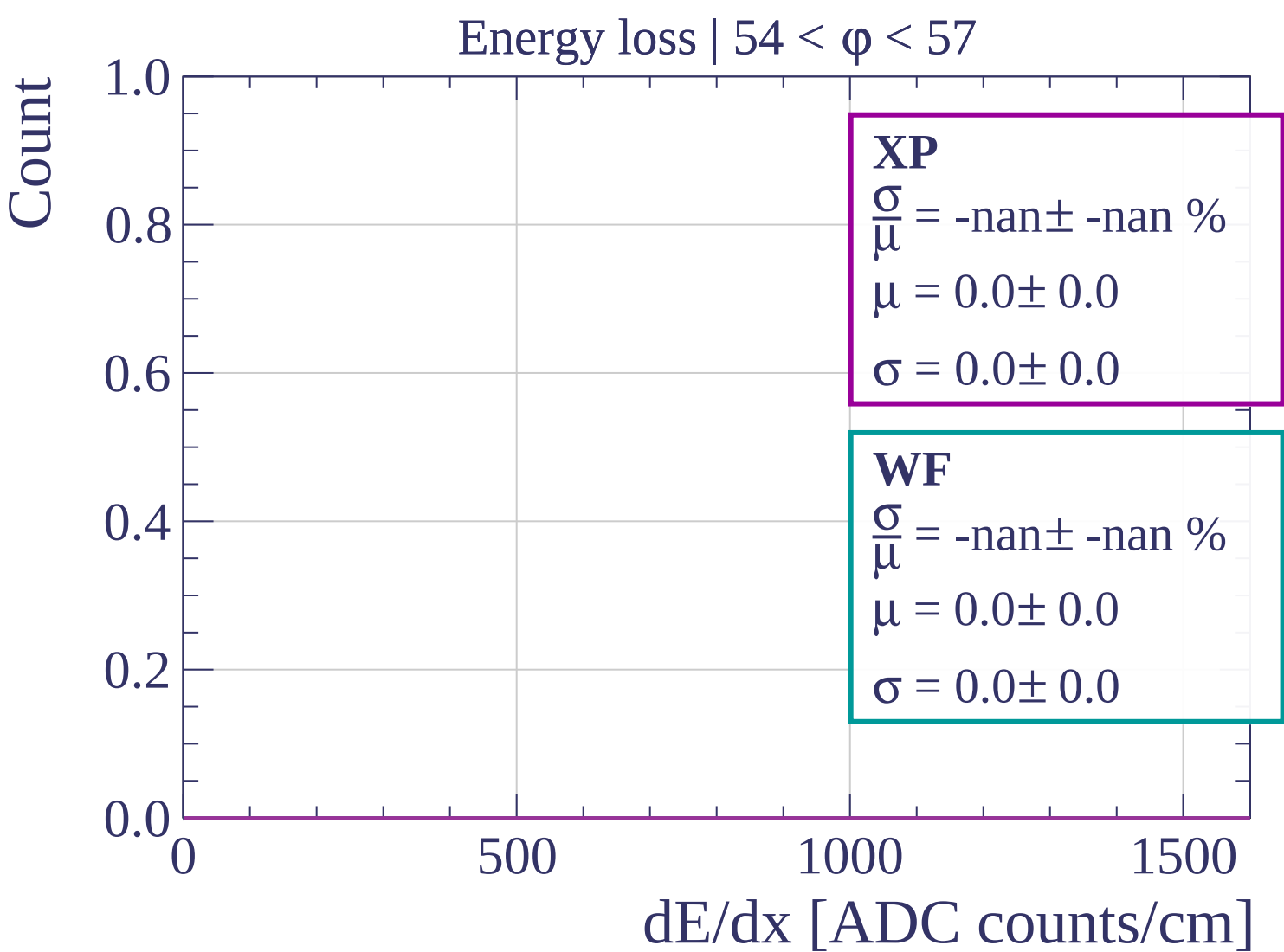
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

Energy loss | $57 < \phi < 60$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

0

500

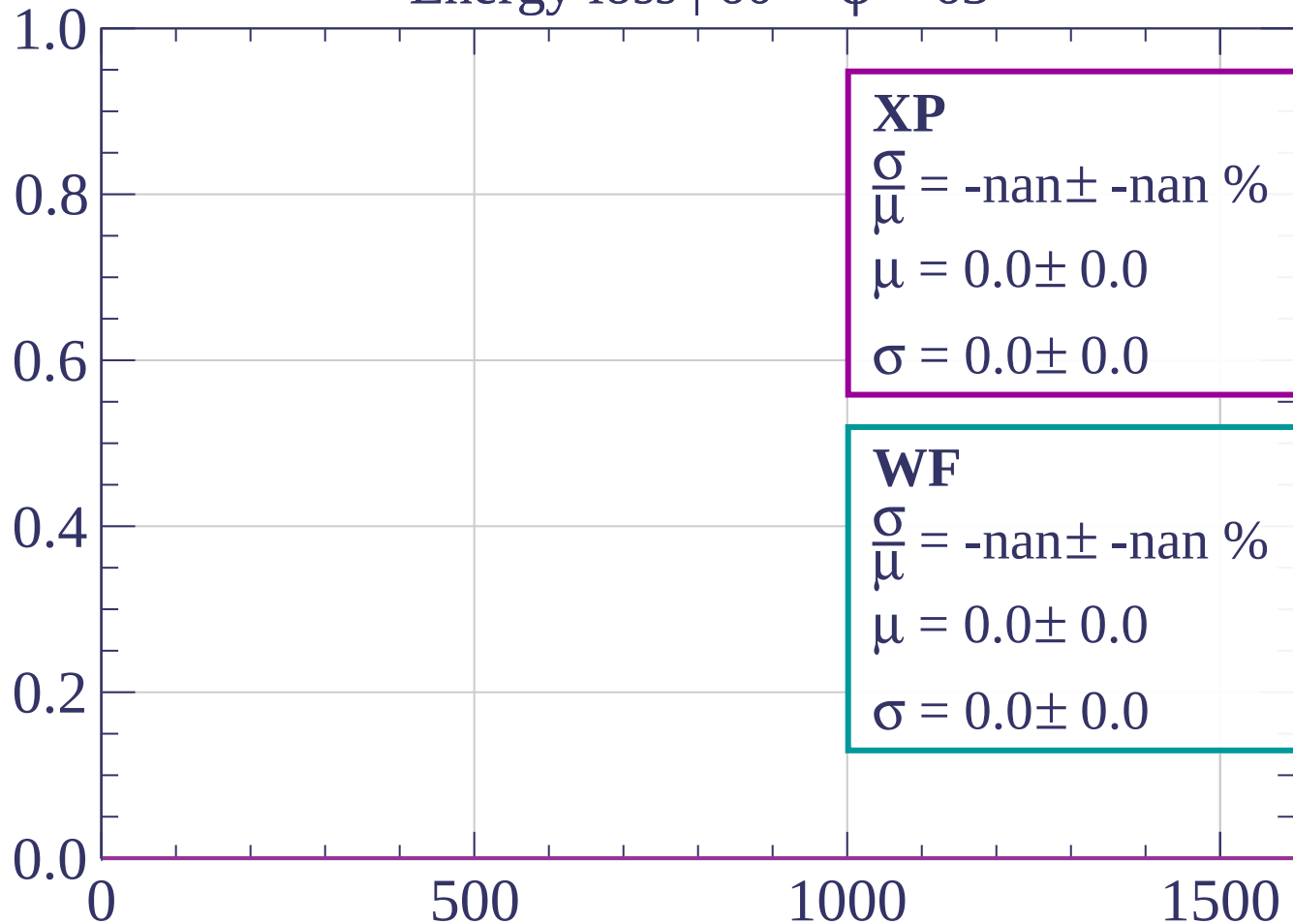
1000

1500

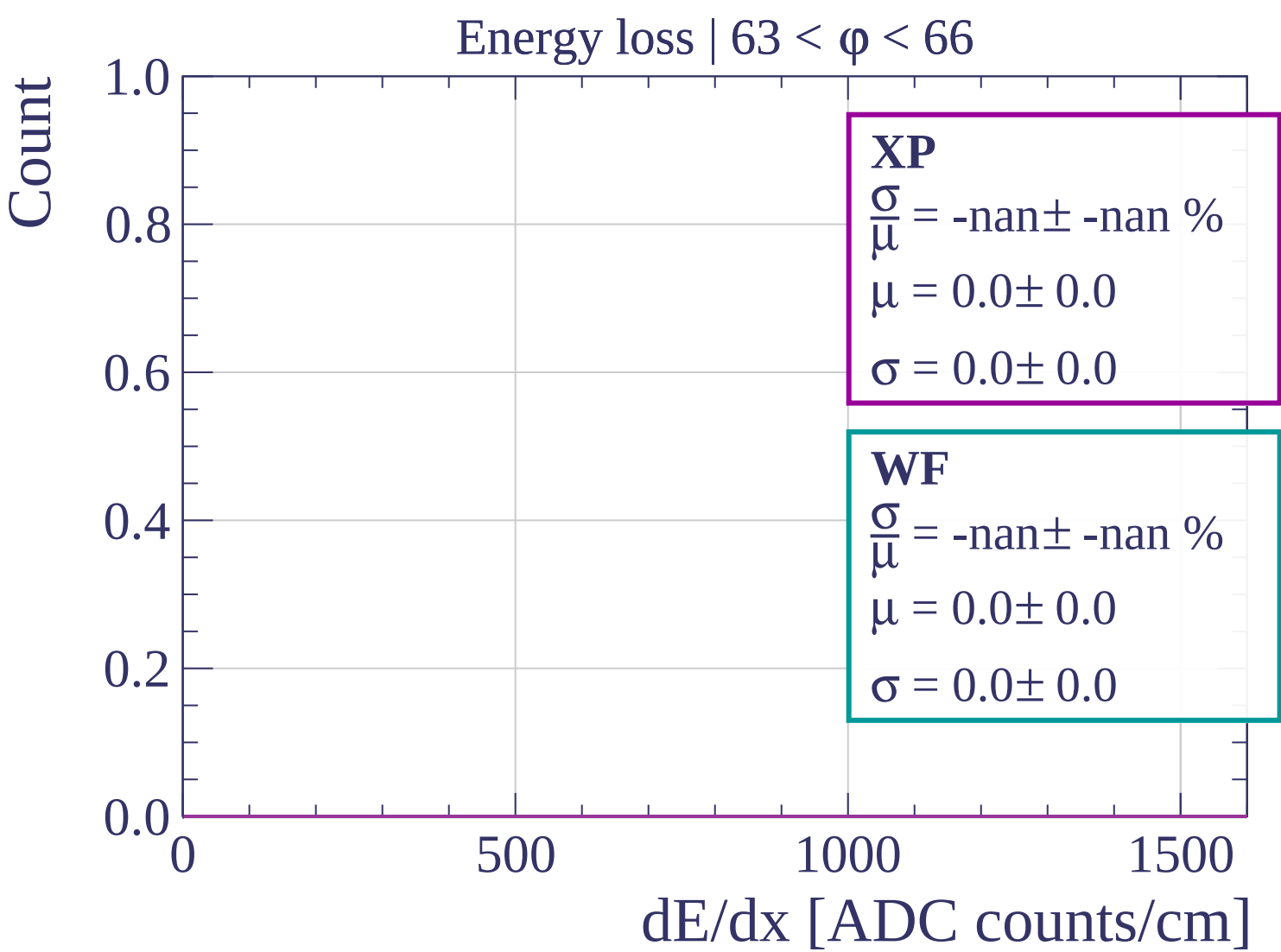
dE/dx [ADC counts/cm]

Count

Energy loss | $60 < \phi < 63$



dE/dx [ADC counts/cm]



Count

Energy loss | $66 < \phi < 69$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $69 < \phi < 72$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $72 < \phi < 75$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $75 < \phi < 78$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

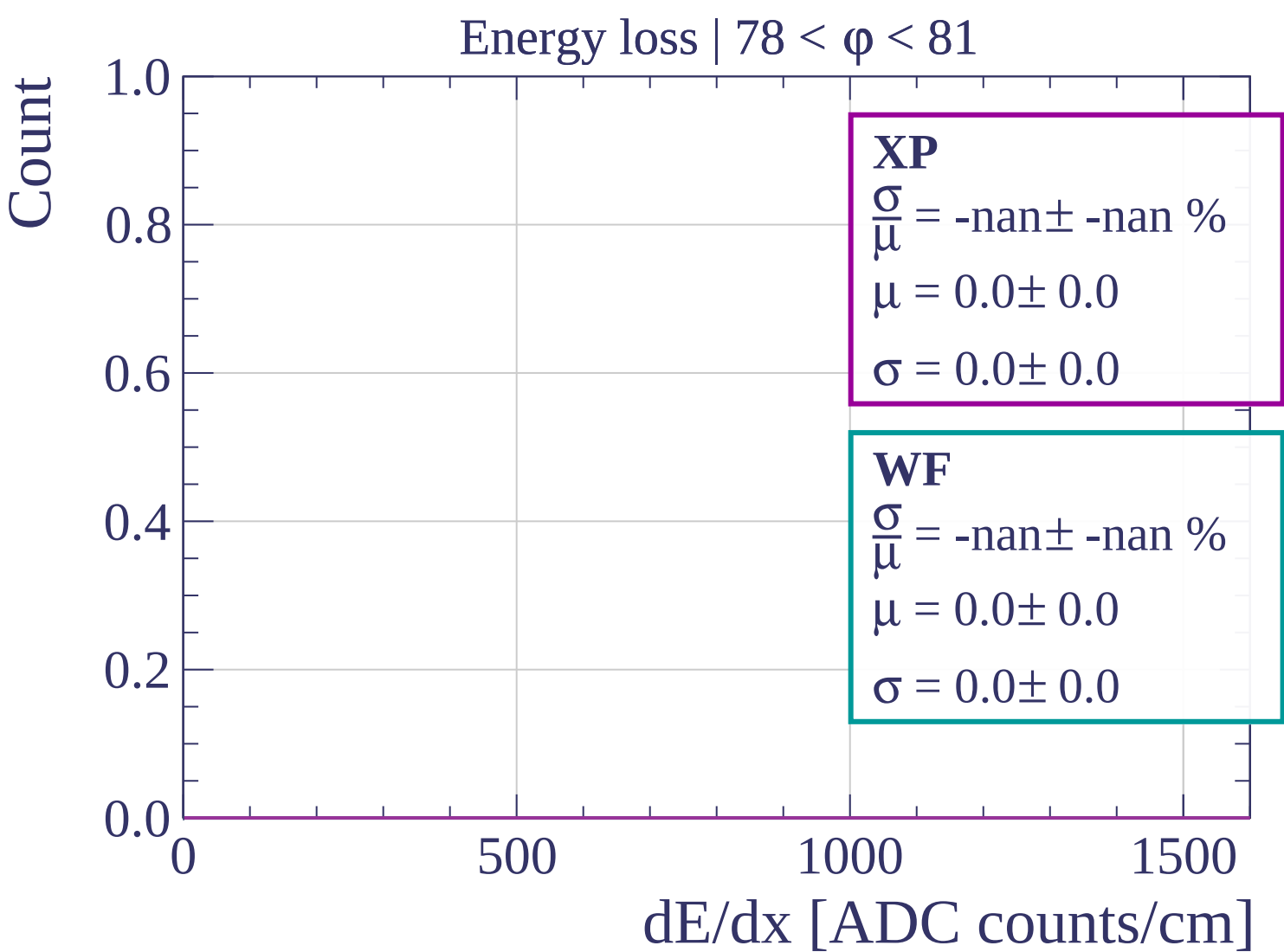
0

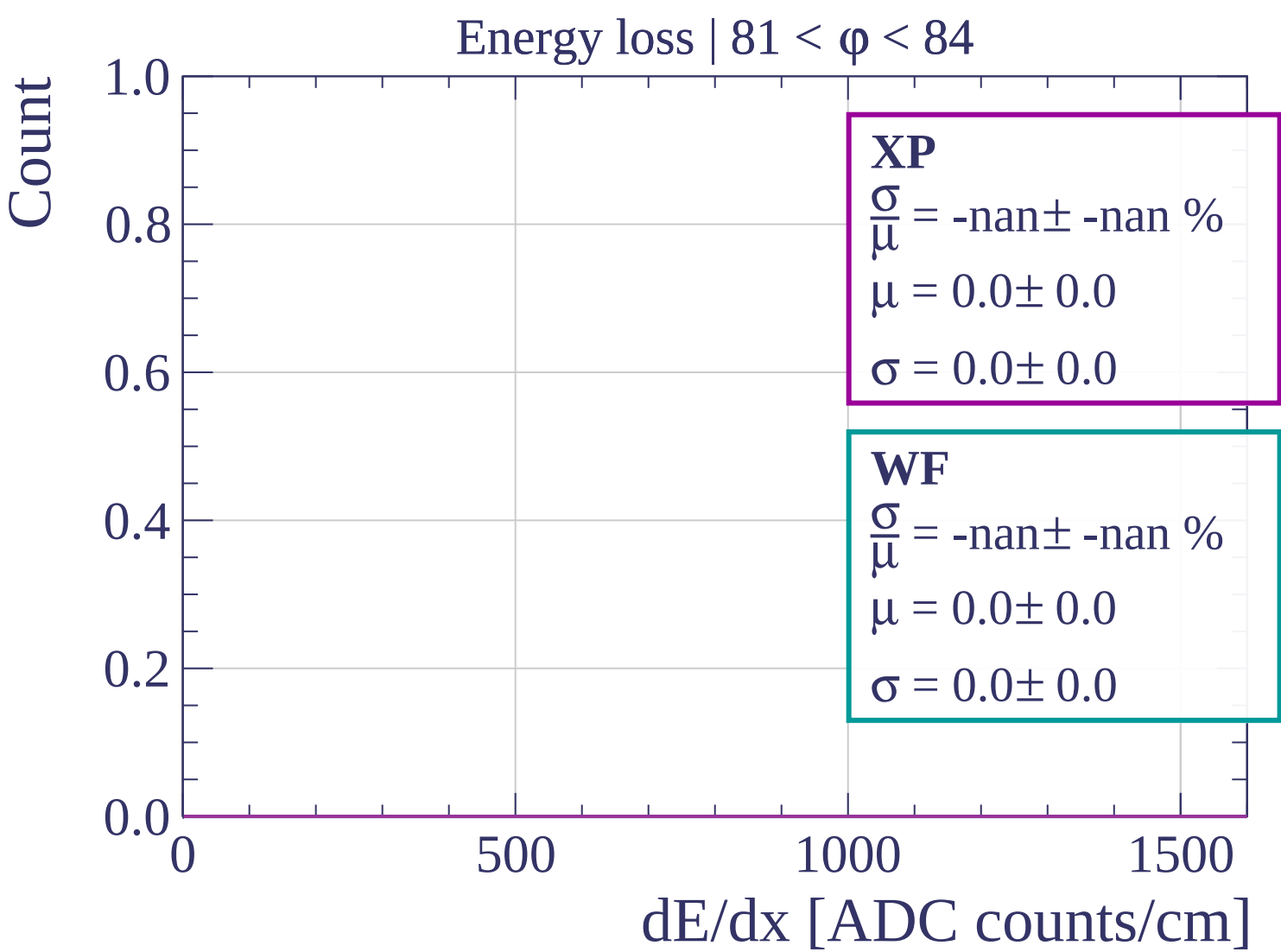
500

1000

1500

dE/dx [ADC counts/cm]





Count

Energy loss | $84 < \phi < 87$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $87 < \phi < 90$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

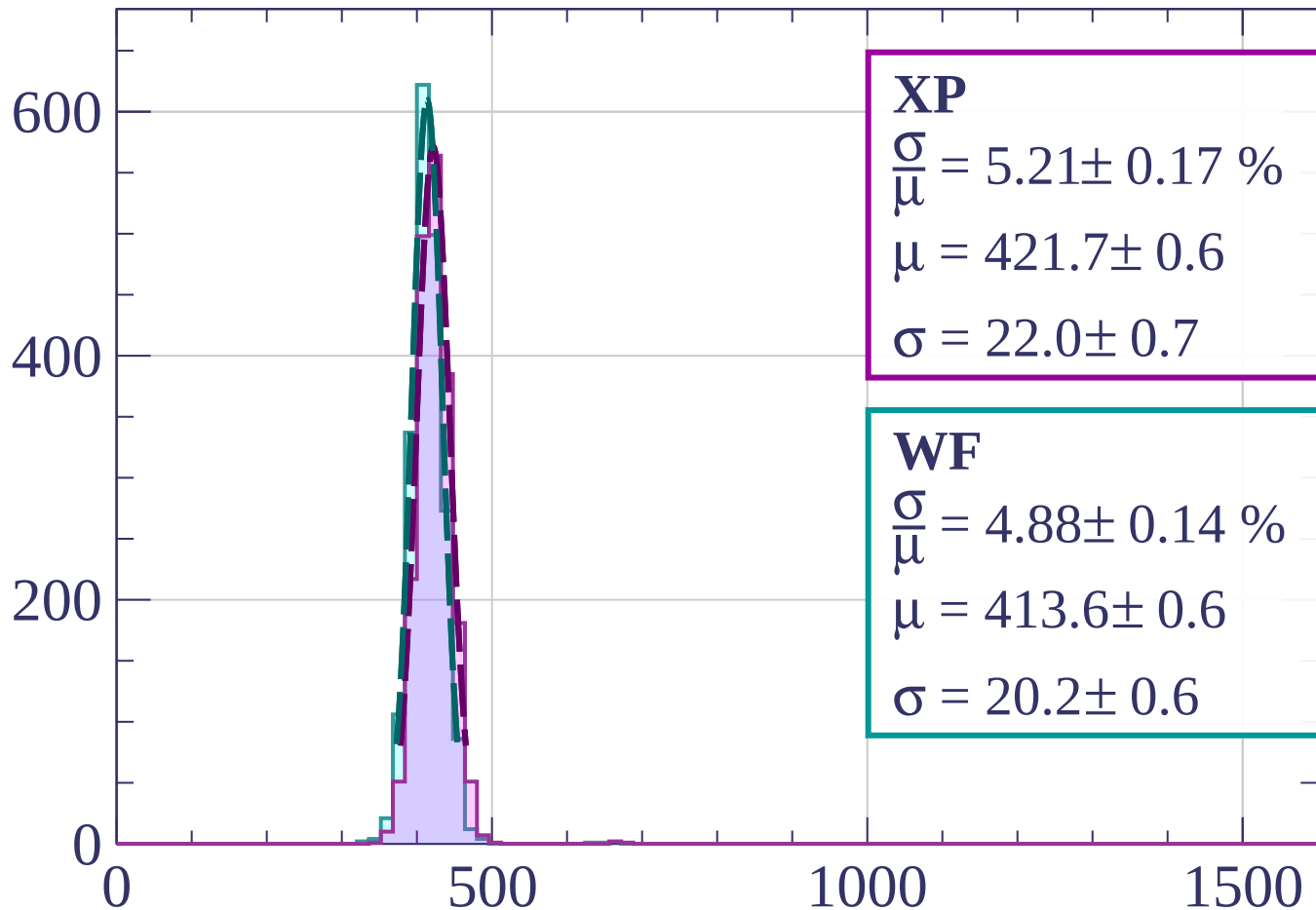
1000

1500

dE/dx [ADC counts/cm]

Energy loss | $0 < \theta < 3$

Count



XP

$$\frac{\sigma}{\mu} = 5.21 \pm 0.17 \%$$

$$\mu = 421.7 \pm 0.6$$

$$\sigma = 22.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 4.88 \pm 0.14 \%$$

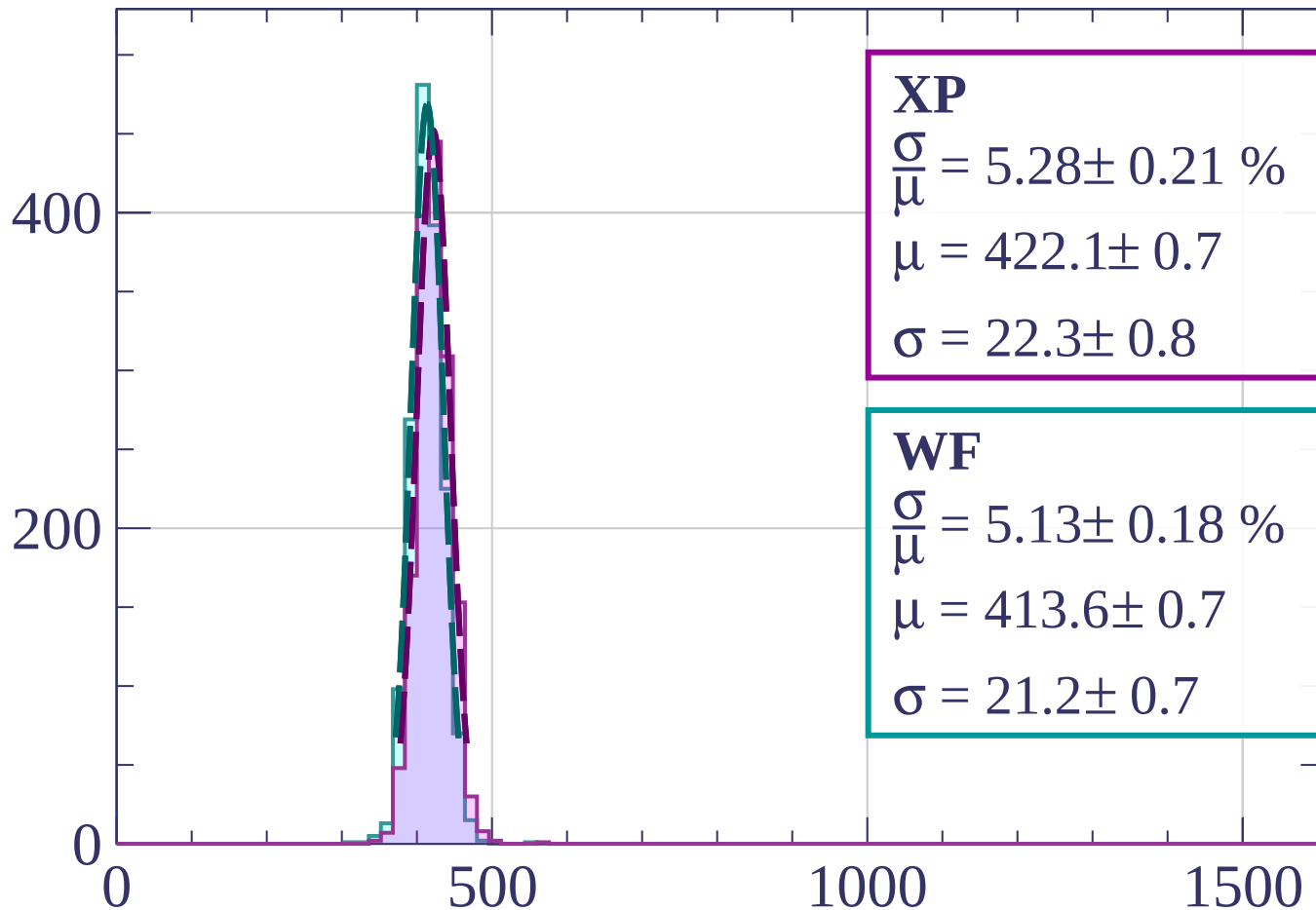
$$\mu = 413.6 \pm 0.6$$

$$\sigma = 20.2 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $3 < \theta < 6$

Count



XP

$$\frac{\sigma}{\mu} = 5.28 \pm 0.21 \%$$

$$\mu = 422.1 \pm 0.7$$

$$\sigma = 22.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 5.13 \pm 0.18 \%$$

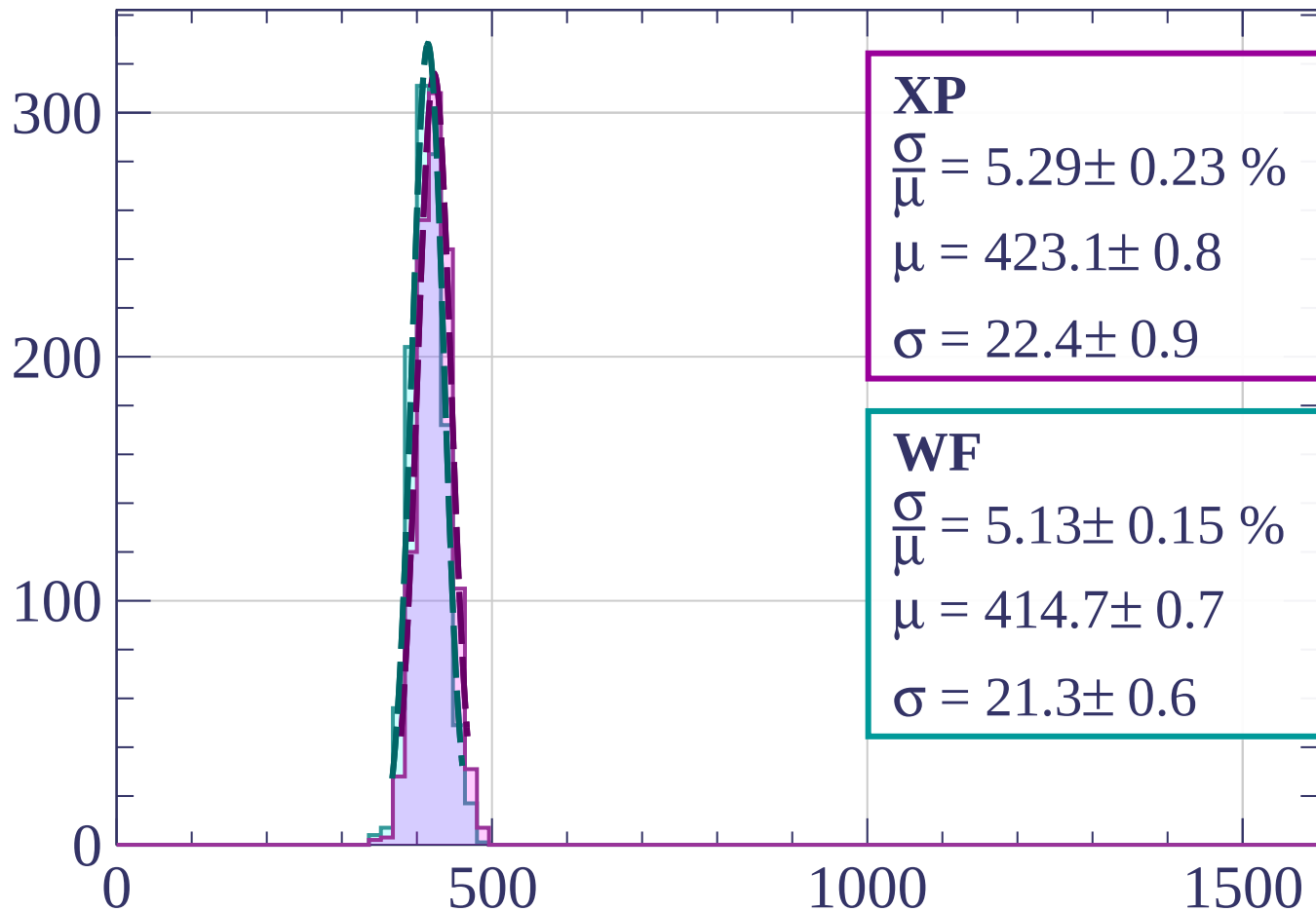
$$\mu = 413.6 \pm 0.7$$

$$\sigma = 21.2 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $6 < \theta < 9$

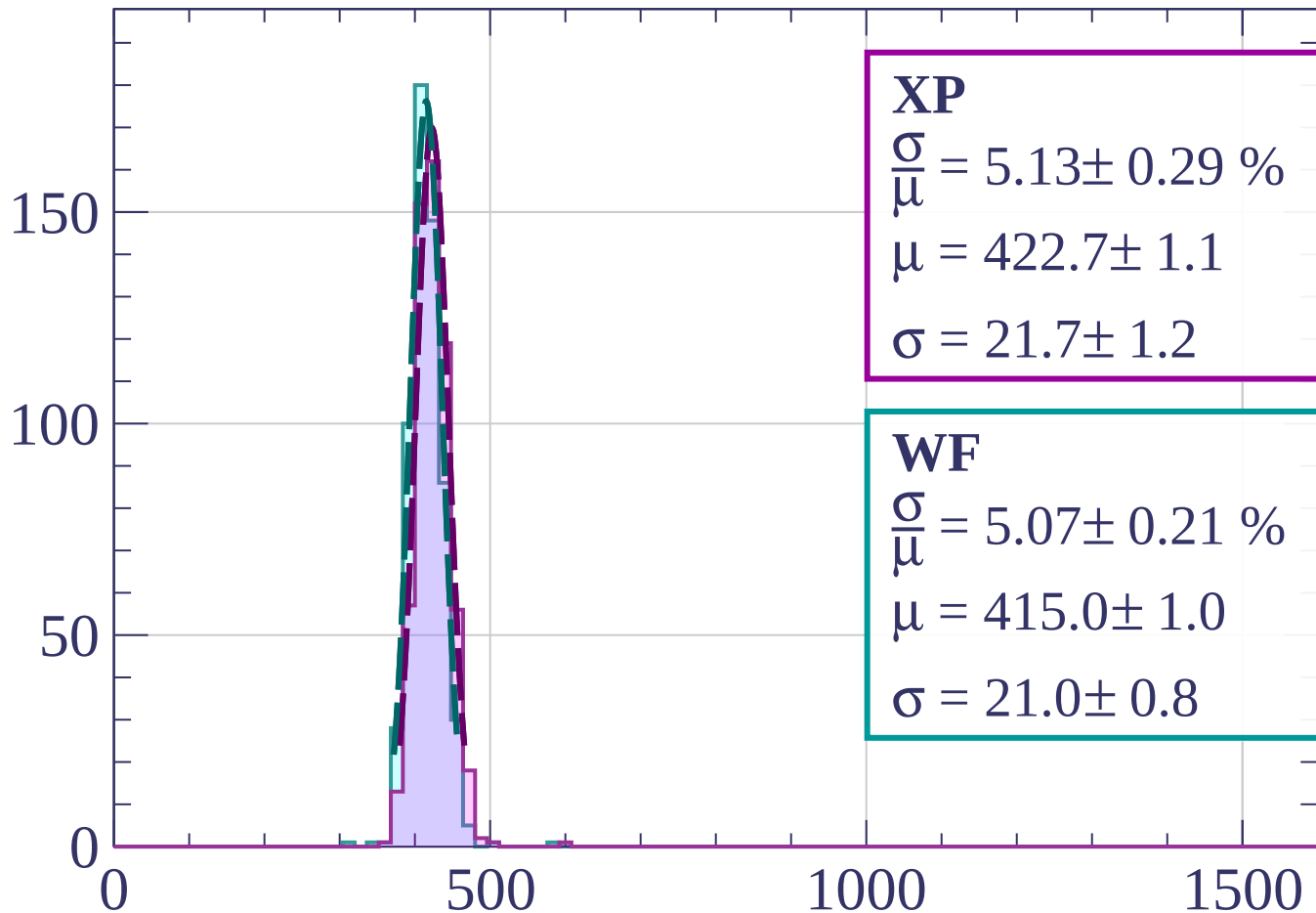
Count



dE/dx [ADC counts/cm]

Energy loss | $9 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 5.13 \pm 0.29 \%$$

$$\mu = 422.7 \pm 1.1$$

$$\sigma = 21.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 5.07 \pm 0.21 \%$$

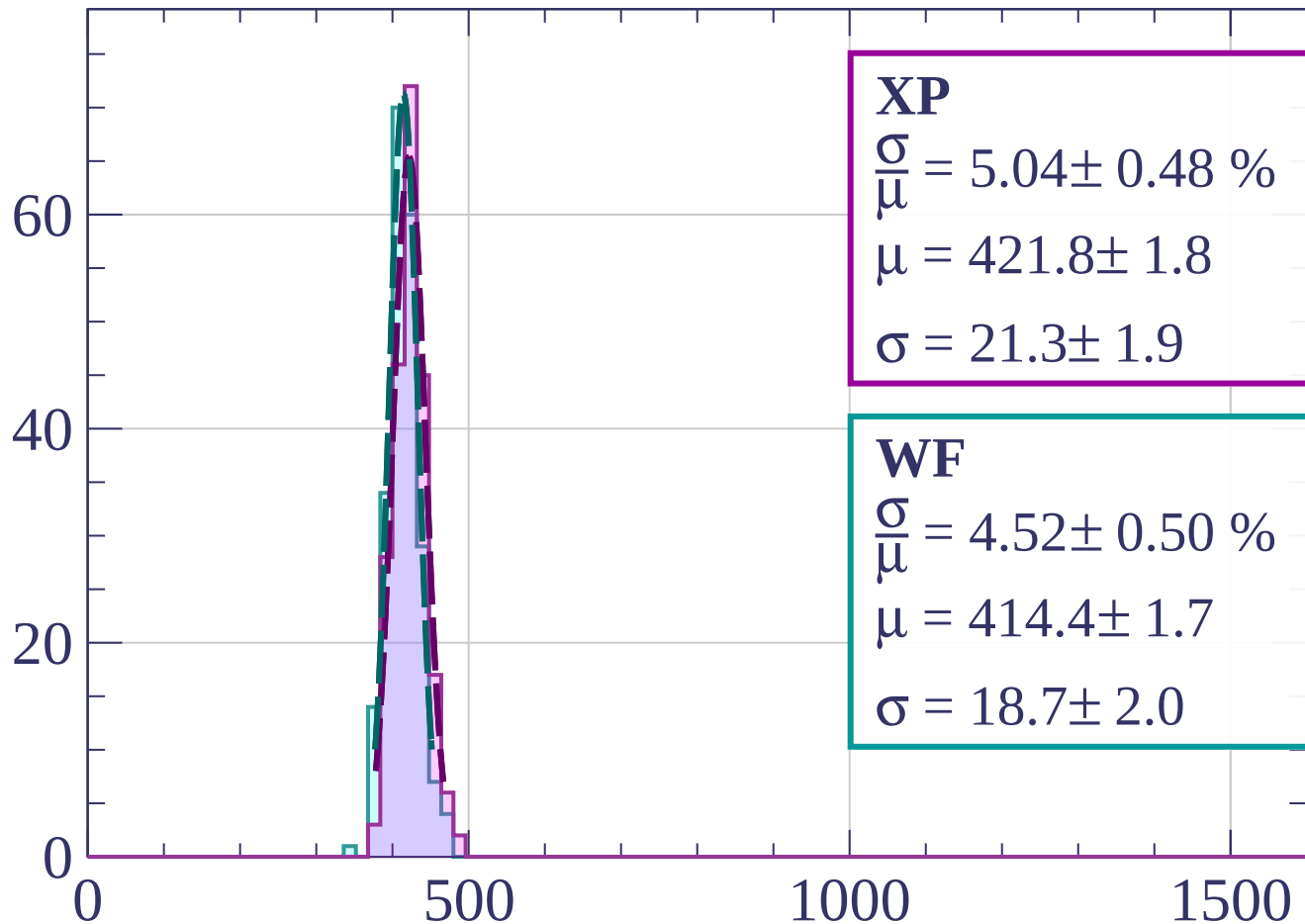
$$\mu = 415.0 \pm 1.0$$

$$\sigma = 21.0 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $12 < \theta < 15$

Count



XP

$$\frac{\sigma}{\mu} = 5.04 \pm 0.48 \%$$

$$\mu = 421.8 \pm 1.8$$

$$\sigma = 21.3 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 4.52 \pm 0.50 \%$$

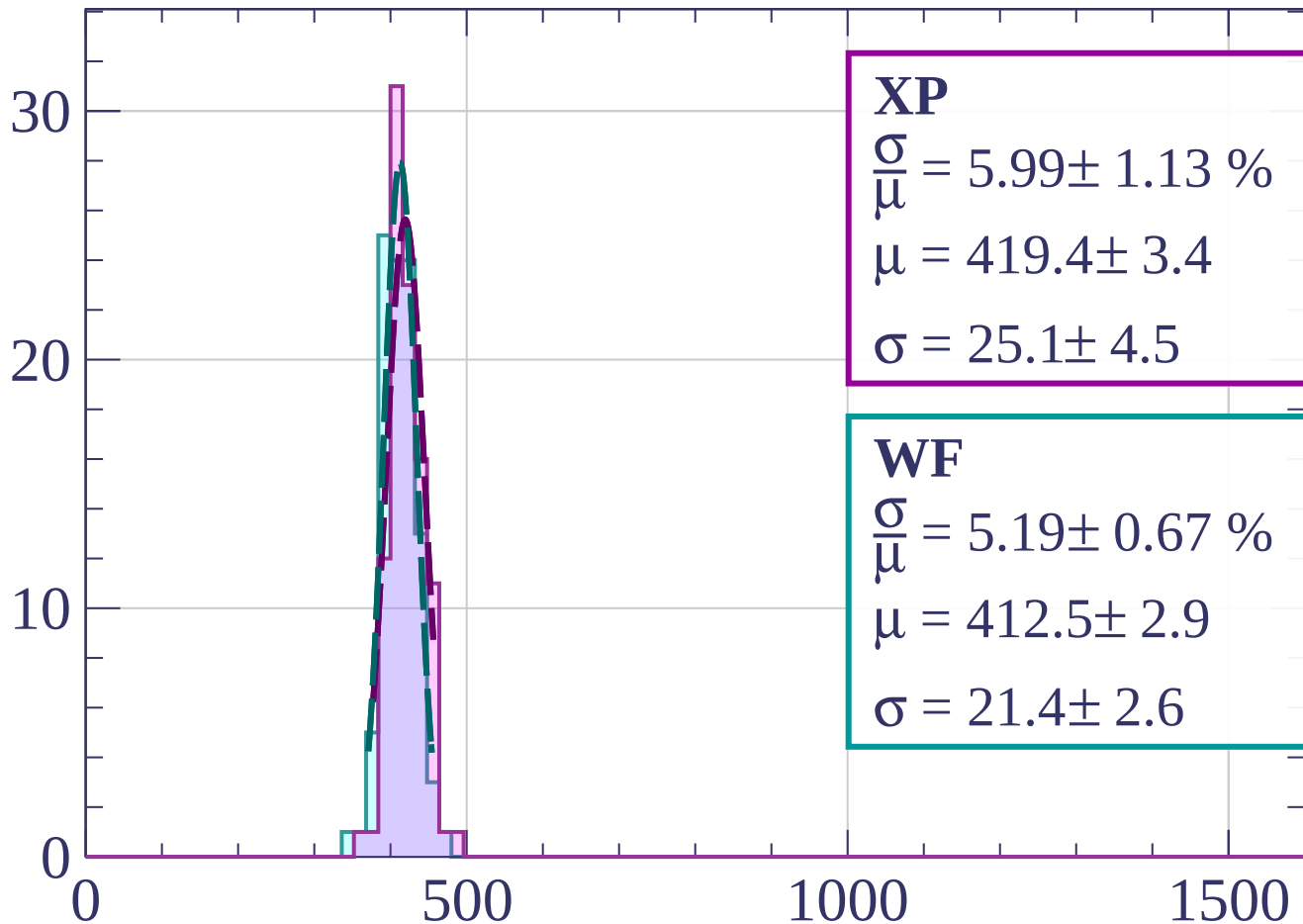
$$\mu = 414.4 \pm 1.7$$

$$\sigma = 18.7 \pm 2.0$$

dE/dx [ADC counts/cm]

Energy loss | $15 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 5.99 \pm 1.13 \%$$

$$\mu = 419.4 \pm 3.4$$

$$\sigma = 25.1 \pm 4.5$$

WF

$$\frac{\sigma}{\mu} = 5.19 \pm 0.67 \%$$

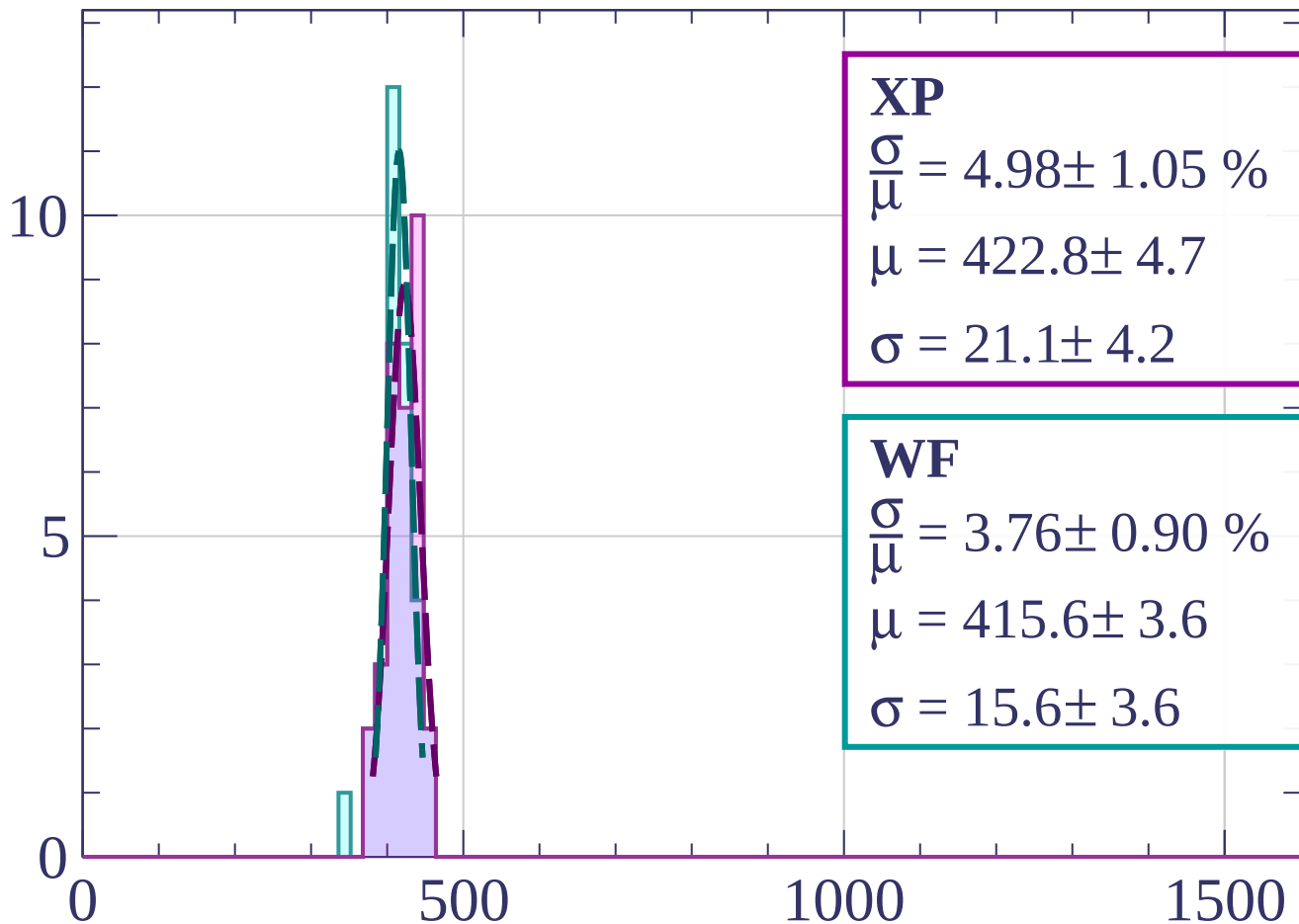
$$\mu = 412.5 \pm 2.9$$

$$\sigma = 21.4 \pm 2.6$$

dE/dx [ADC counts/cm]

Energy loss | $18 < \theta < 21$

Count



XP

$$\frac{\sigma}{\mu} = 4.98 \pm 1.05 \%$$

$$\mu = 422.8 \pm 4.7$$

$$\sigma = 21.1 \pm 4.2$$

WF

$$\frac{\sigma}{\mu} = 3.76 \pm 0.90 \%$$

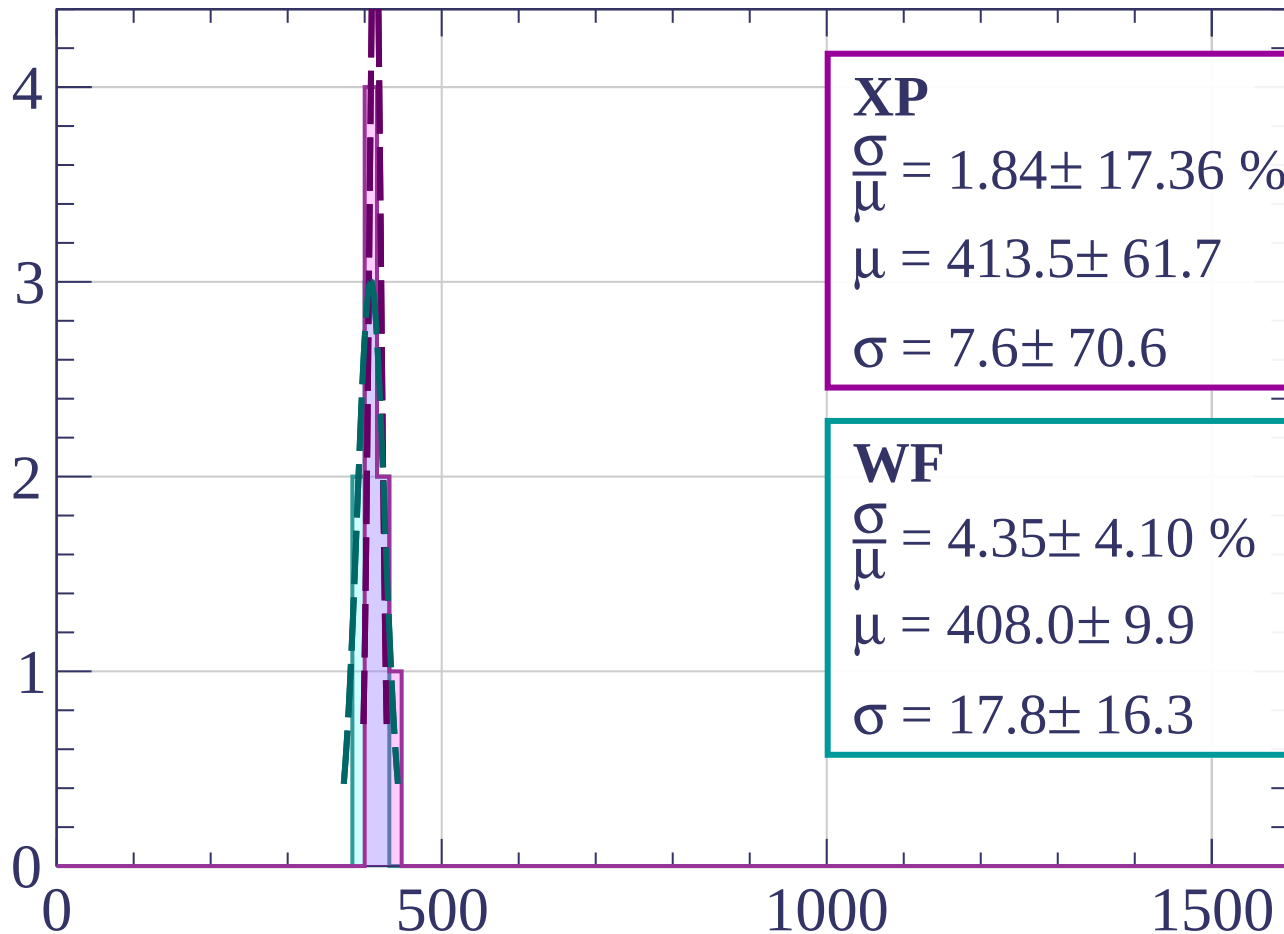
$$\mu = 415.6 \pm 3.6$$

$$\sigma = 15.6 \pm 3.6$$

dE/dx [ADC counts/cm]

Energy loss | $21 < \theta < 24$

Count



XP

$$\frac{\sigma}{\mu} = 1.84 \pm 17.36 \%$$

$$\mu = 413.5 \pm 61.7$$

$$\sigma = 7.6 \pm 70.6$$

WF

$$\frac{\sigma}{\mu} = 4.35 \pm 4.10 \%$$

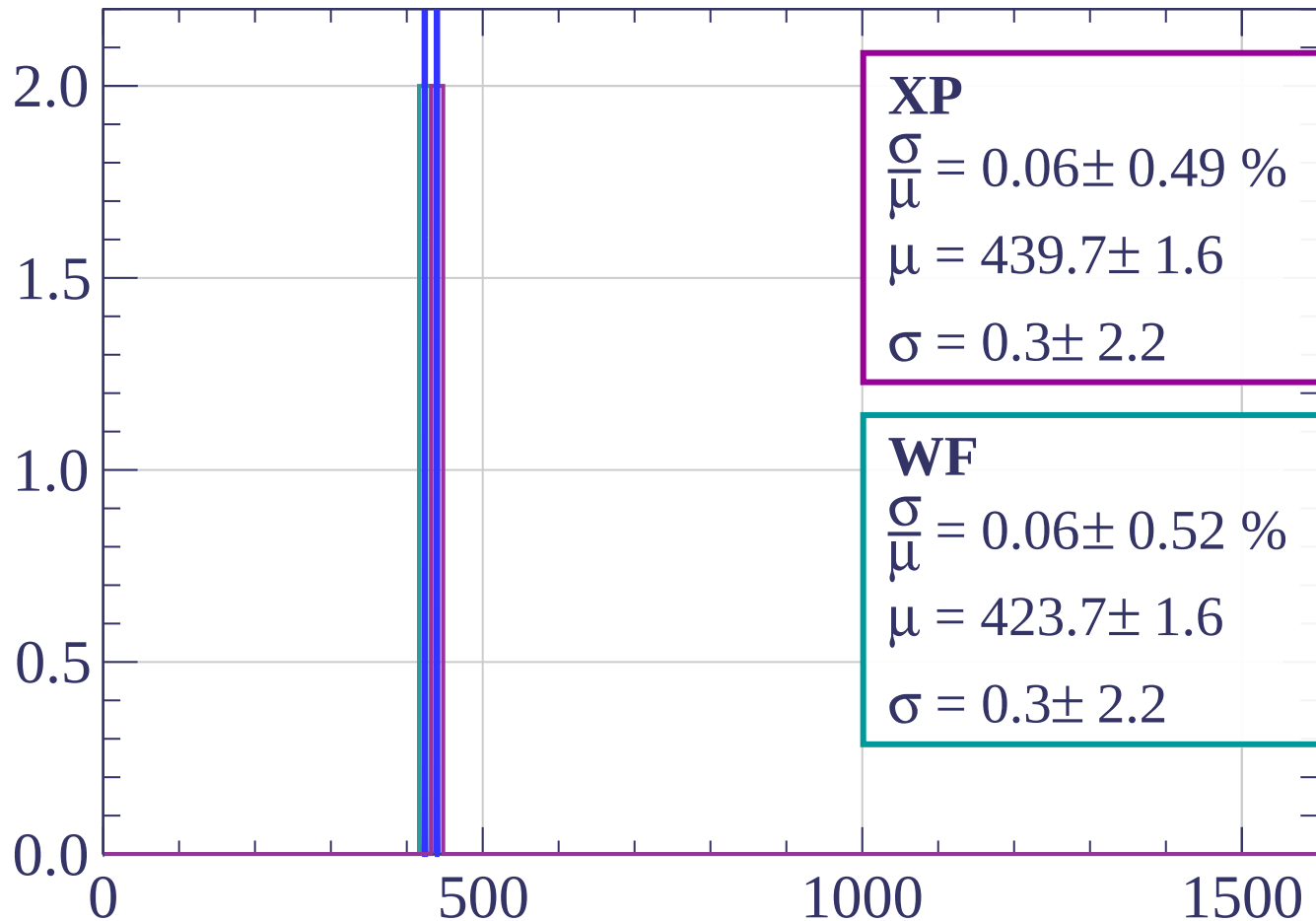
$$\mu = 408.0 \pm 9.9$$

$$\sigma = 17.8 \pm 16.3$$

dE/dx [ADC counts/cm]

Energy loss | $24 < \theta < 27$

Count



XP

$$\frac{\sigma}{\mu} = 0.06 \pm 0.49 \%$$

$$\mu = 439.7 \pm 1.6$$

$$\sigma = 0.3 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 0.06 \pm 0.52 \%$$

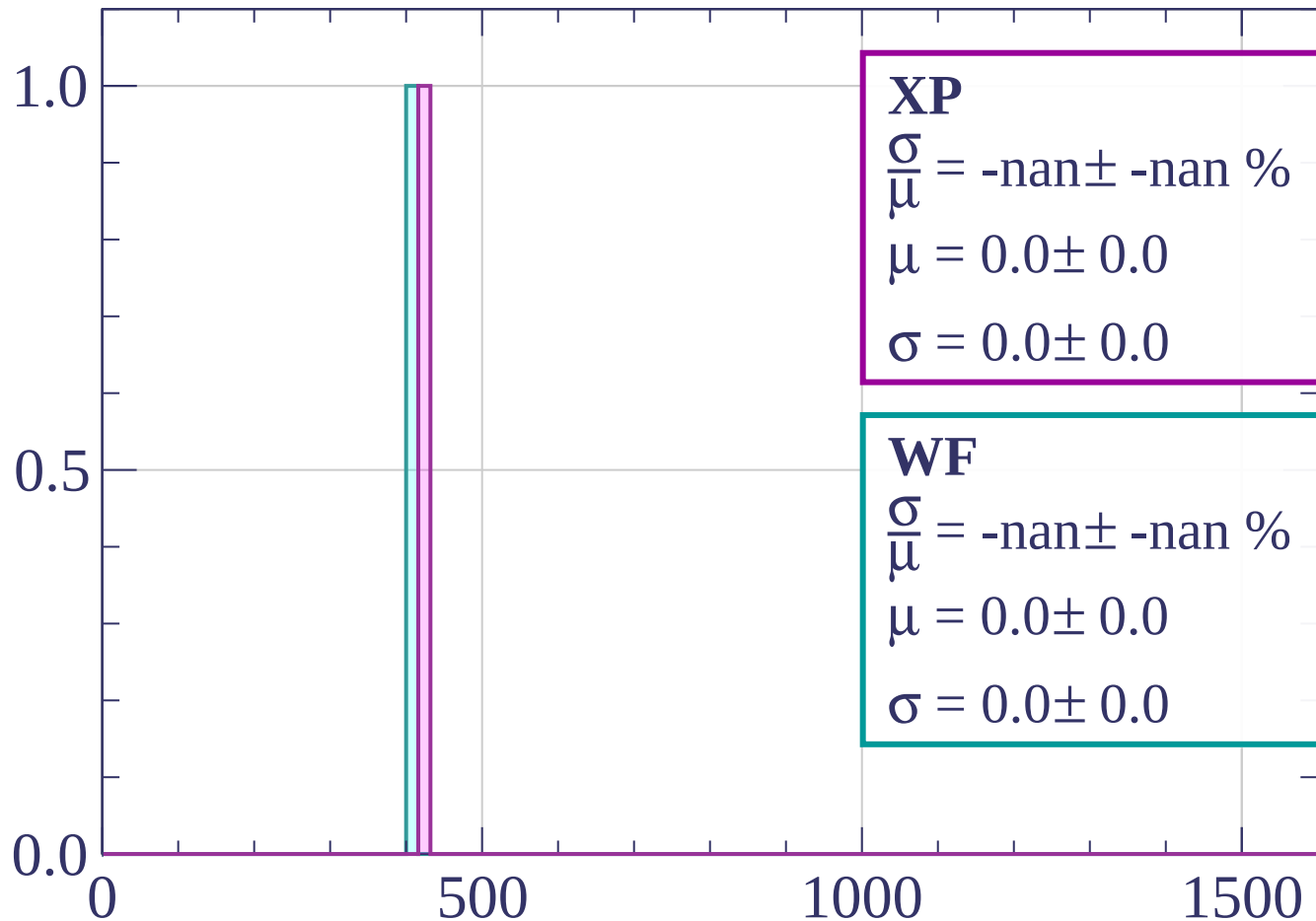
$$\mu = 423.7 \pm 1.6$$

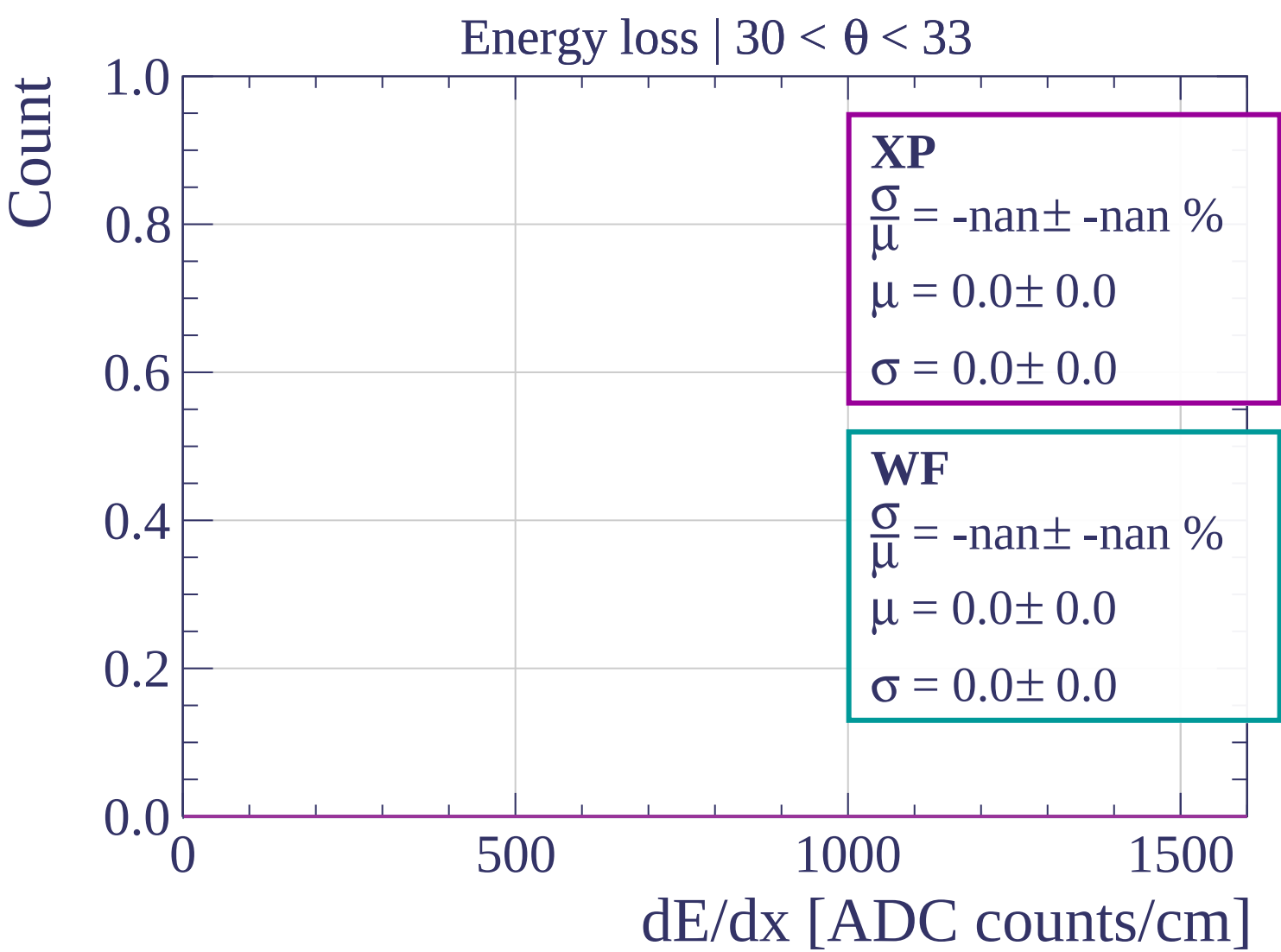
$$\sigma = 0.3 \pm 2.2$$

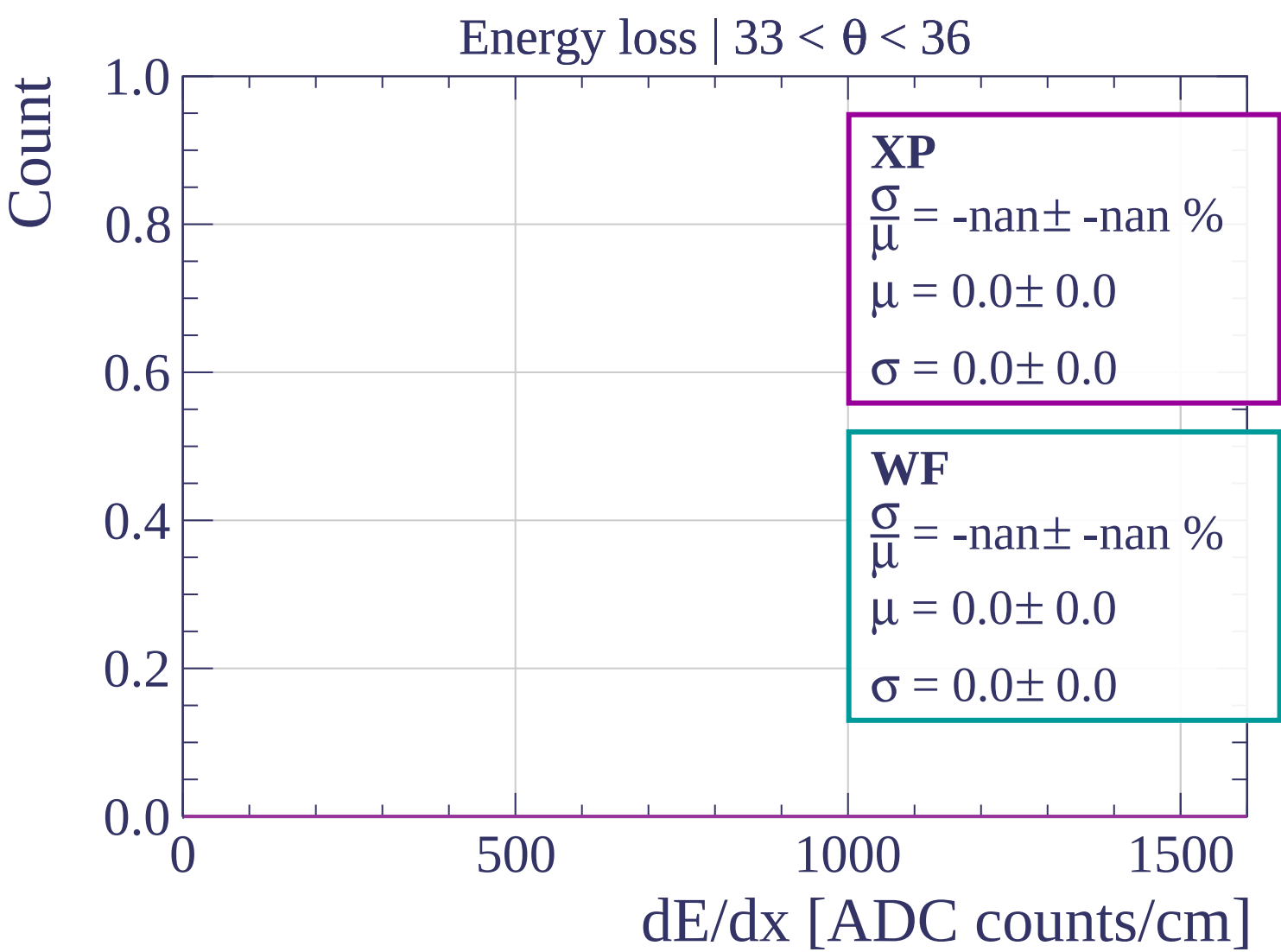
dE/dx [ADC counts/cm]

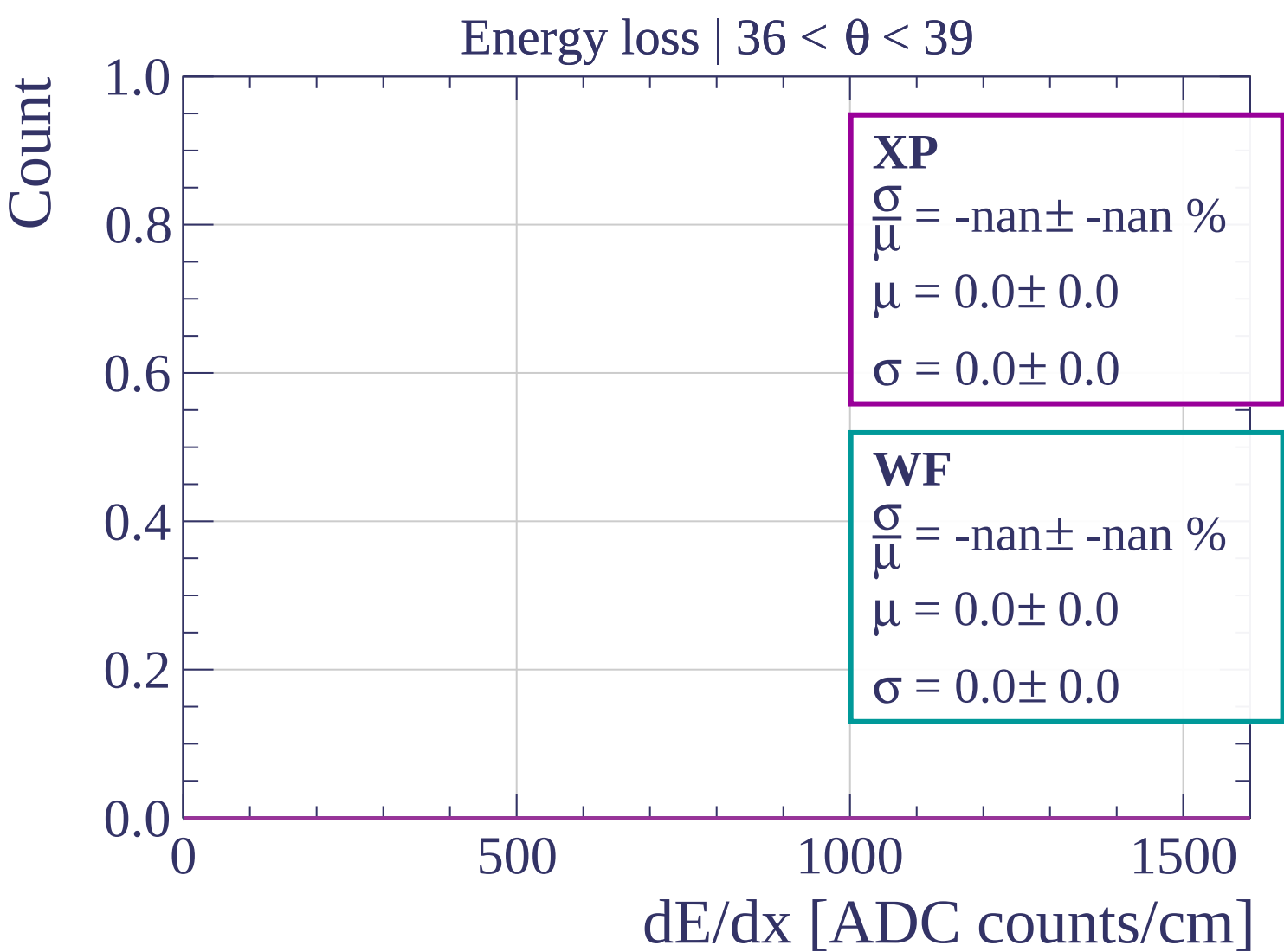
Energy loss | $27 < \theta < 30$

Count



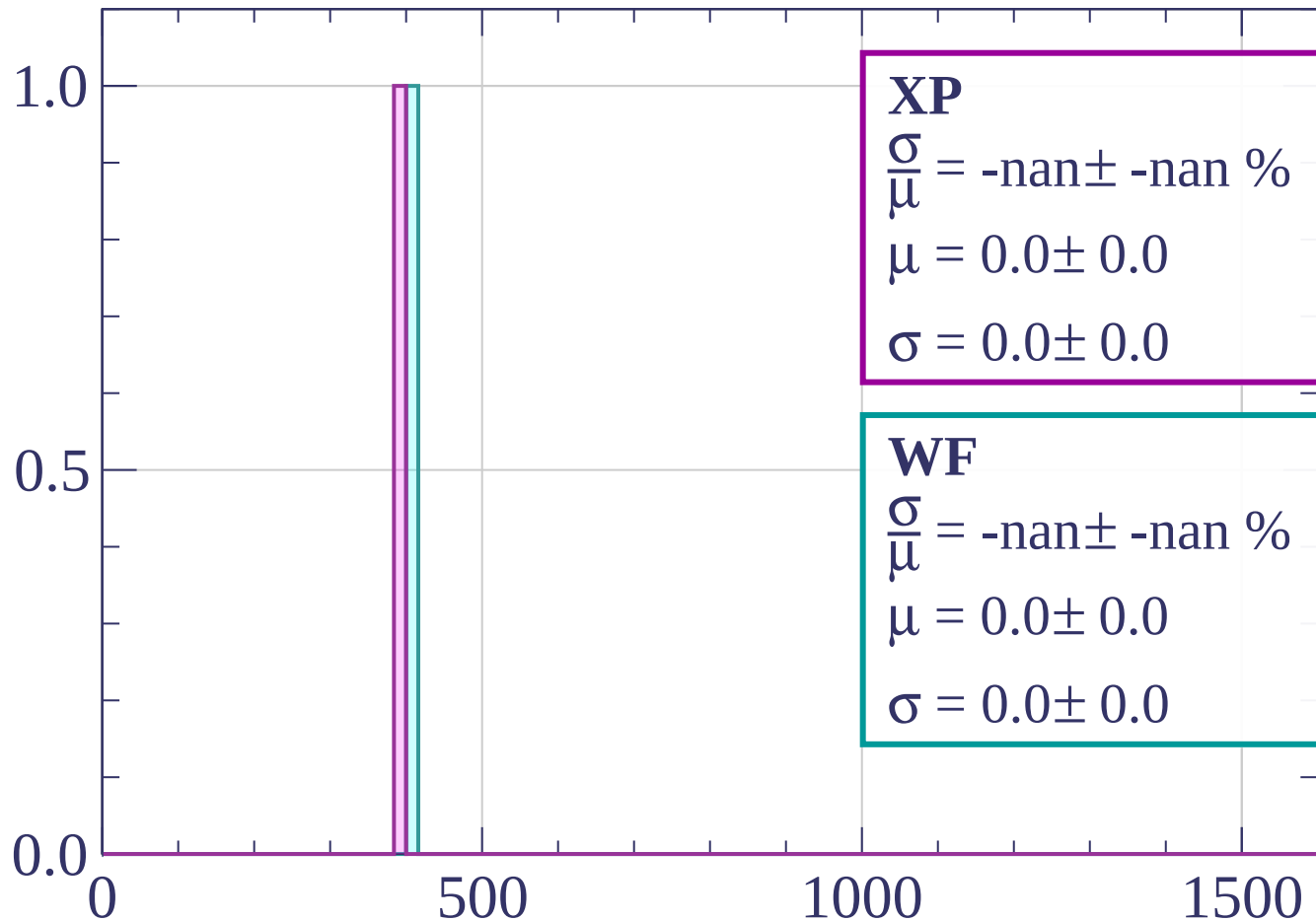






Energy loss | $39 < \theta < 42$

Count



XP

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx [ADC counts/cm]

Count

Energy loss | $42 < \theta < 45$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss | $45 < \theta < 48$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

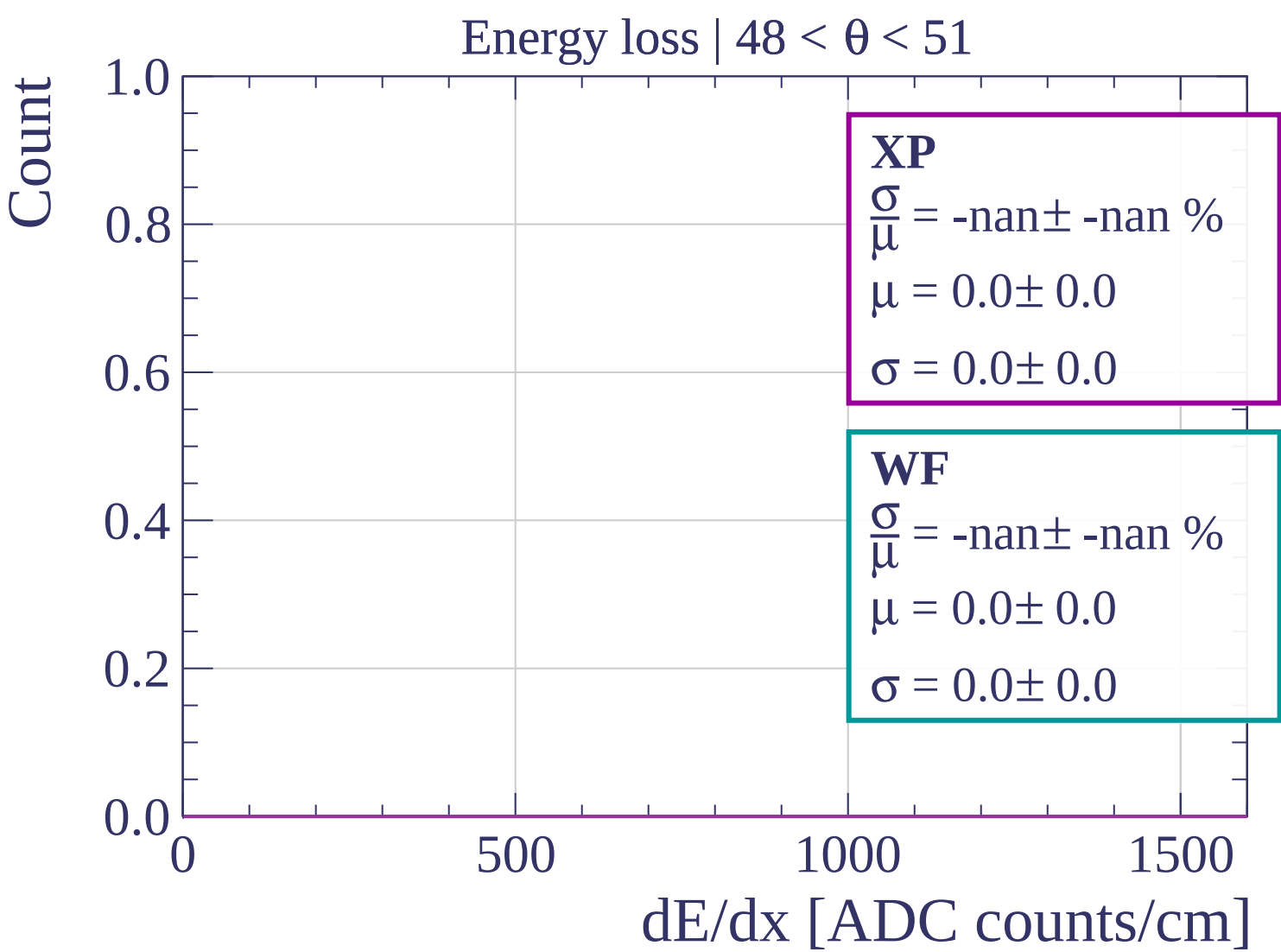
0

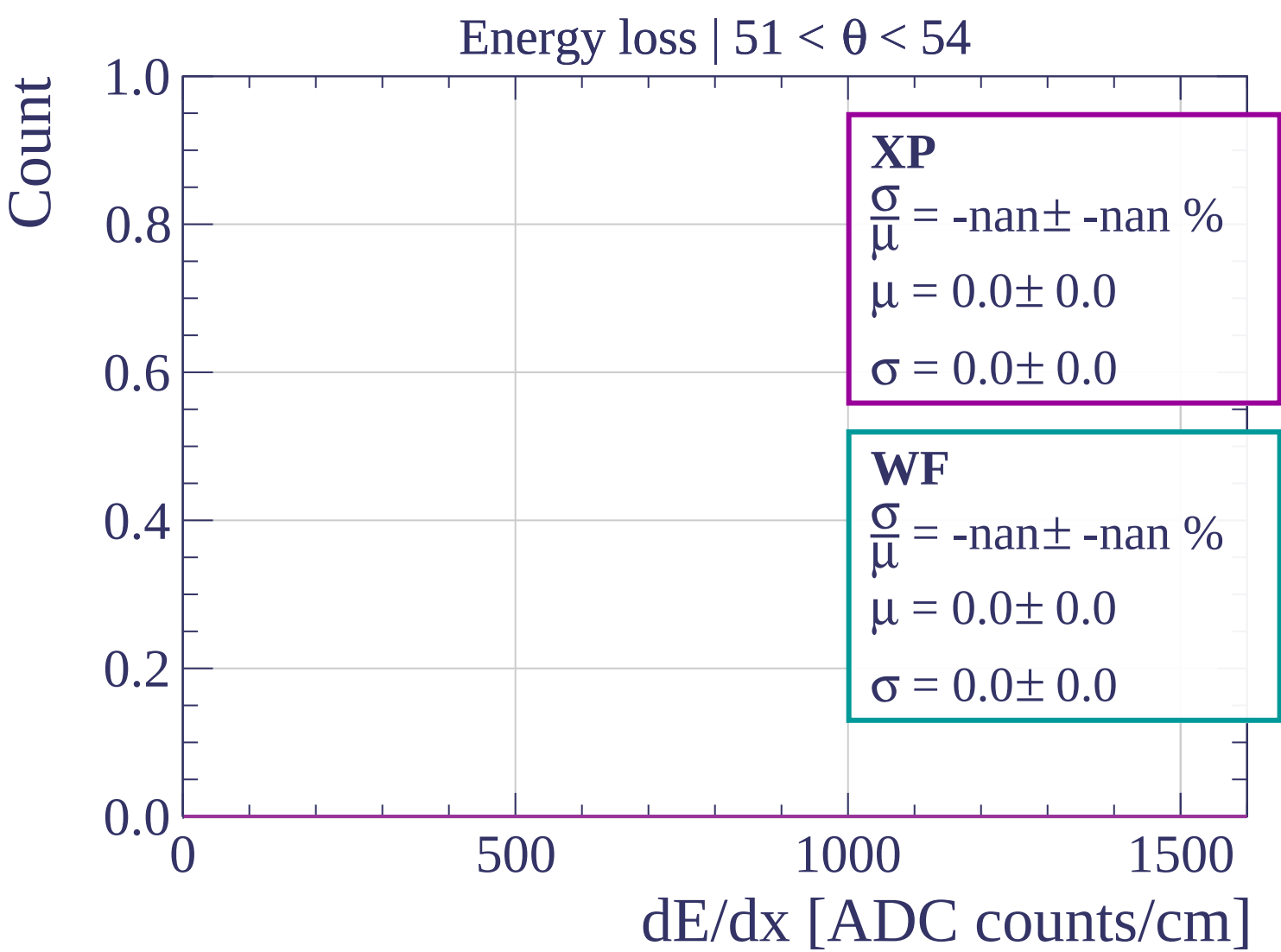
500

1000

1500

dE/dx [ADC counts/cm]





Count

Energy loss | $54 < \theta < 57$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

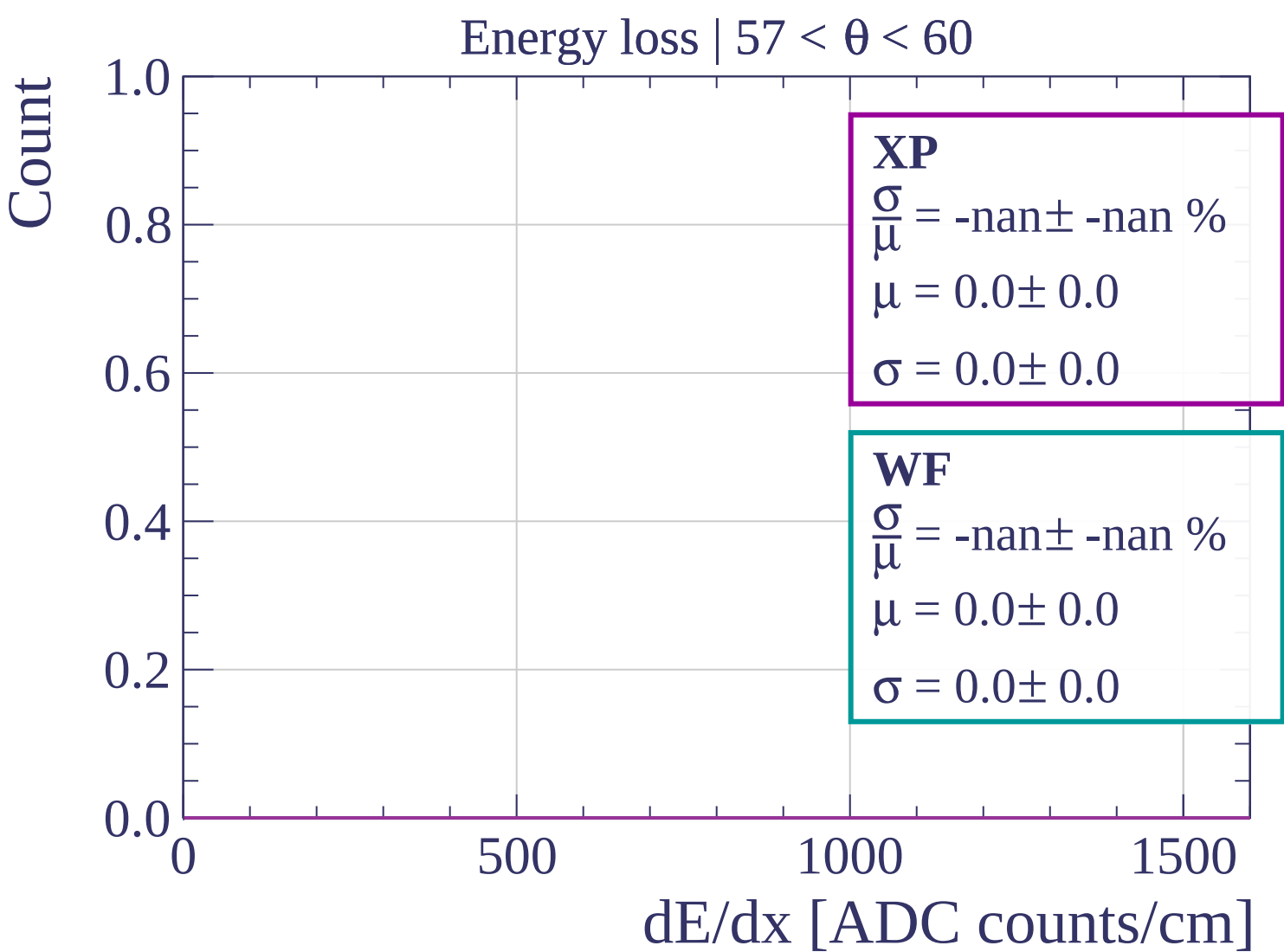
0

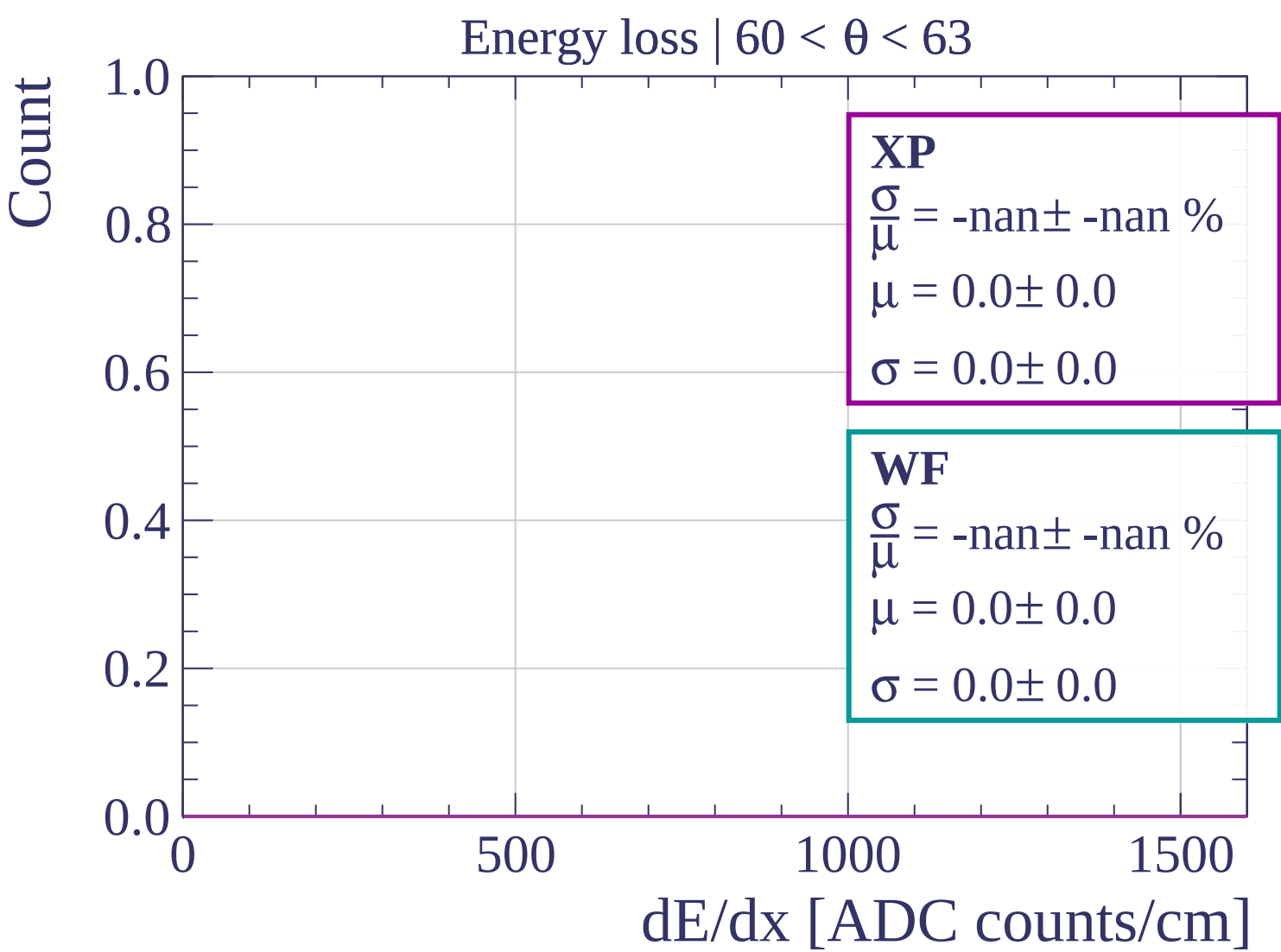
500

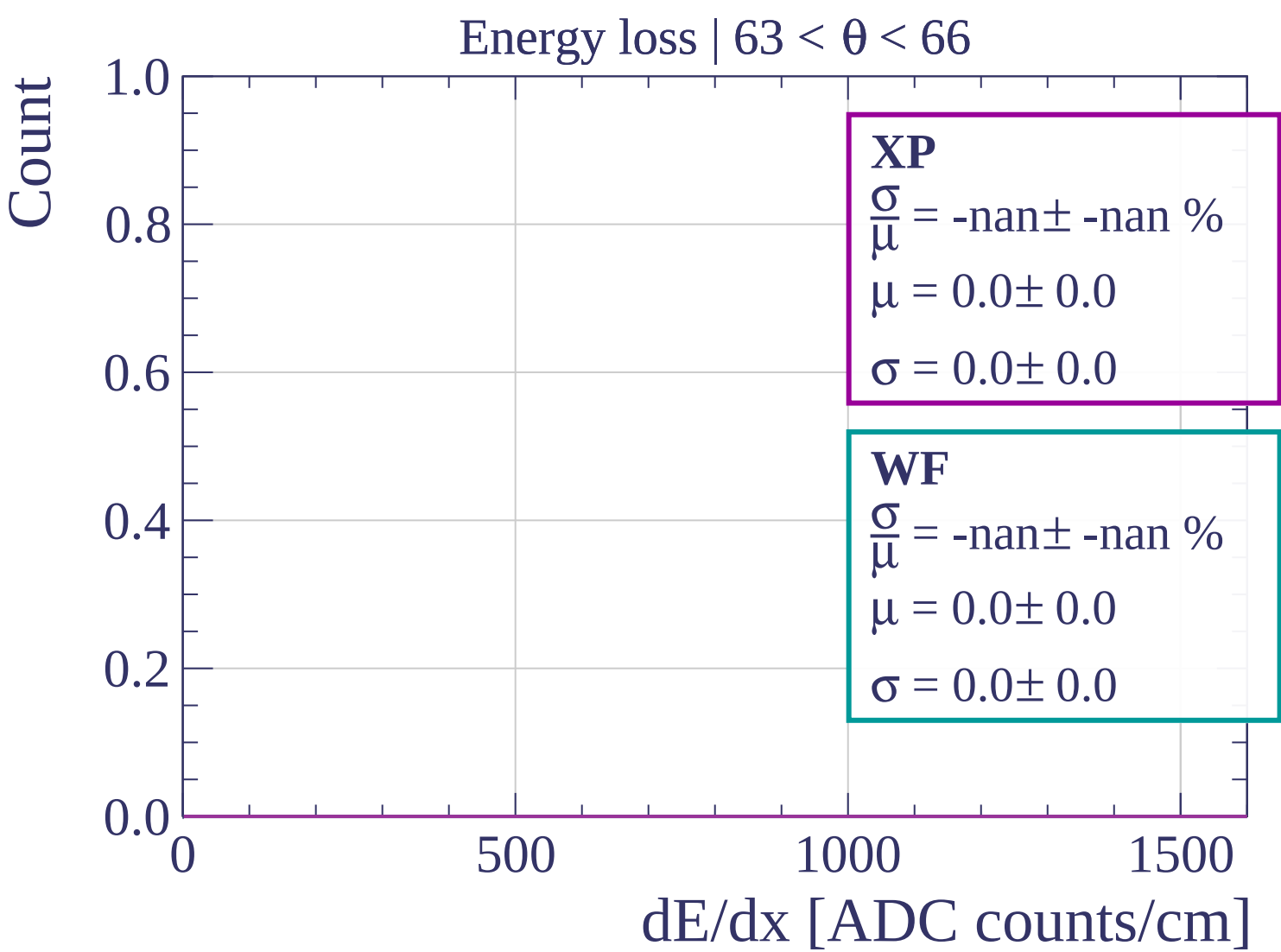
1000

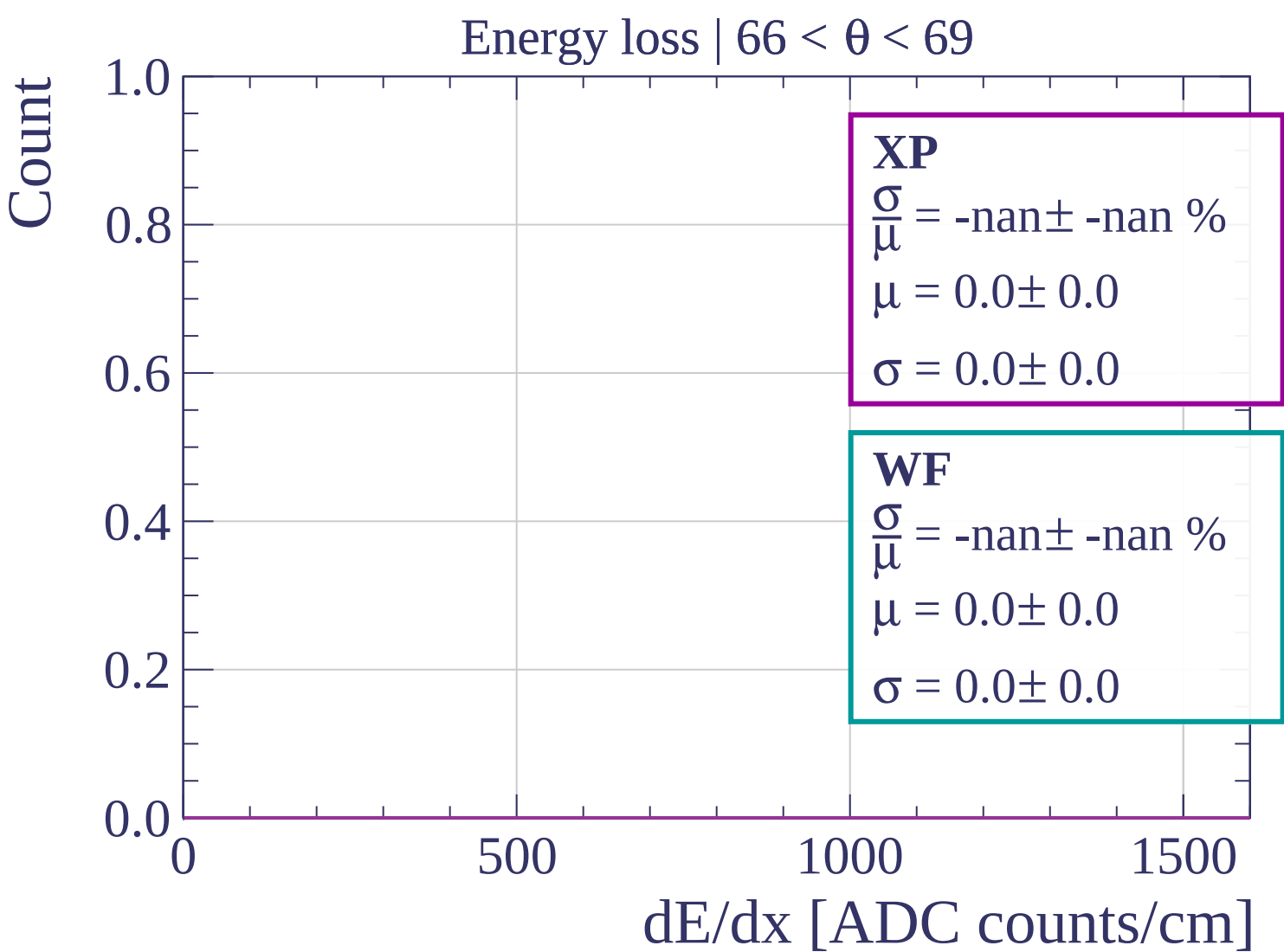
1500

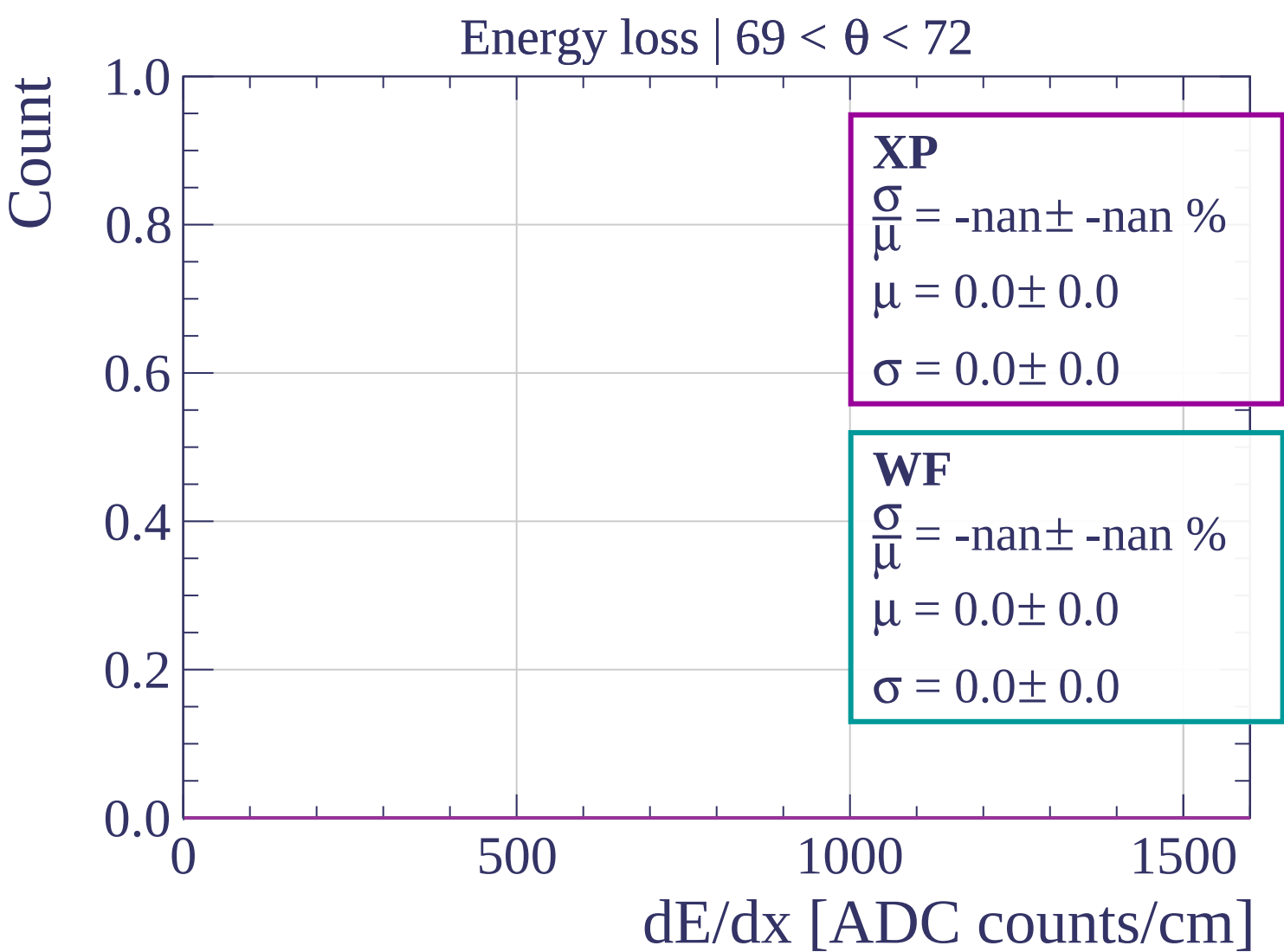
dE/dx [ADC counts/cm]











Count

Energy loss | $72 < \theta < 75$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

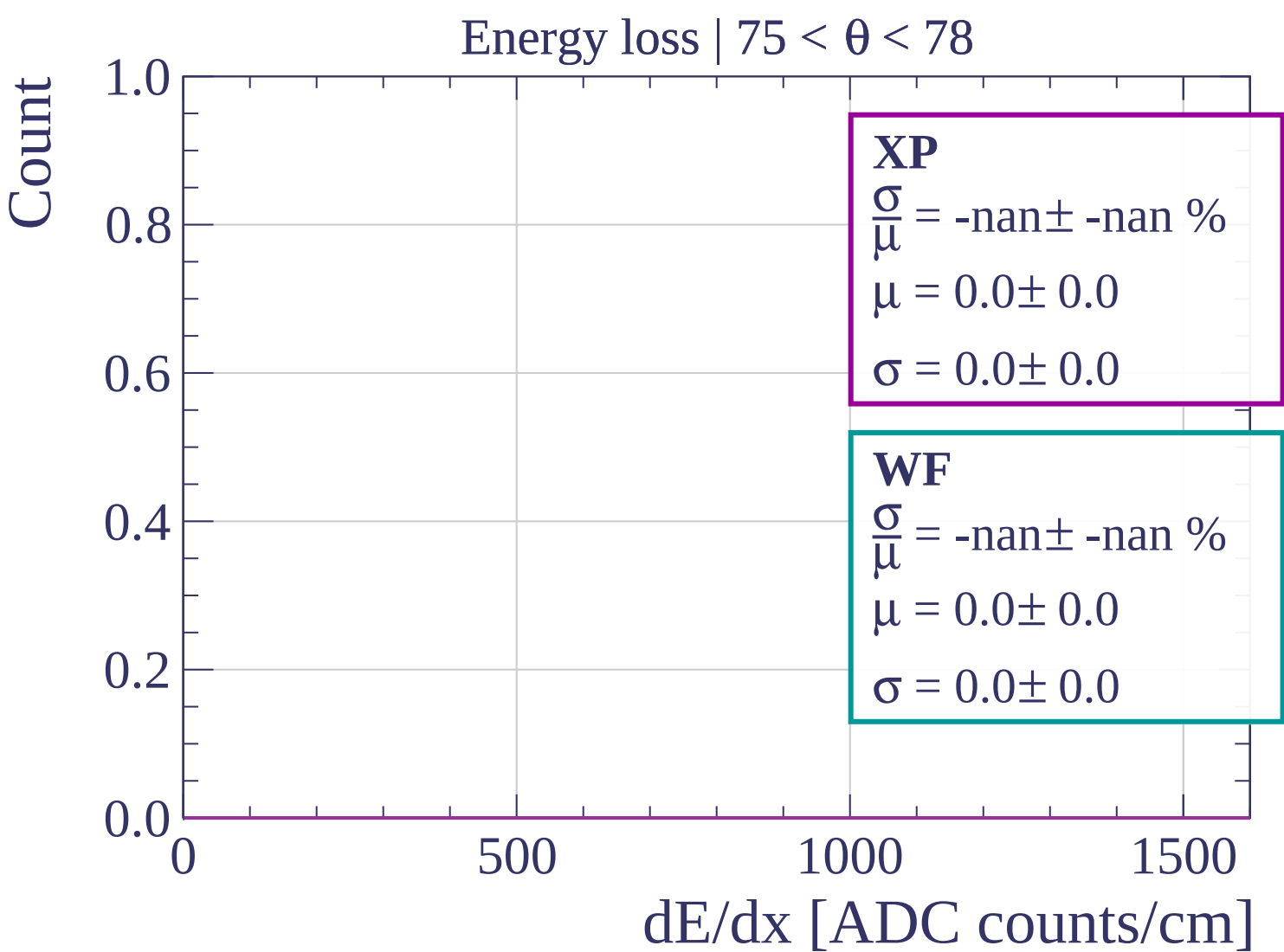
0

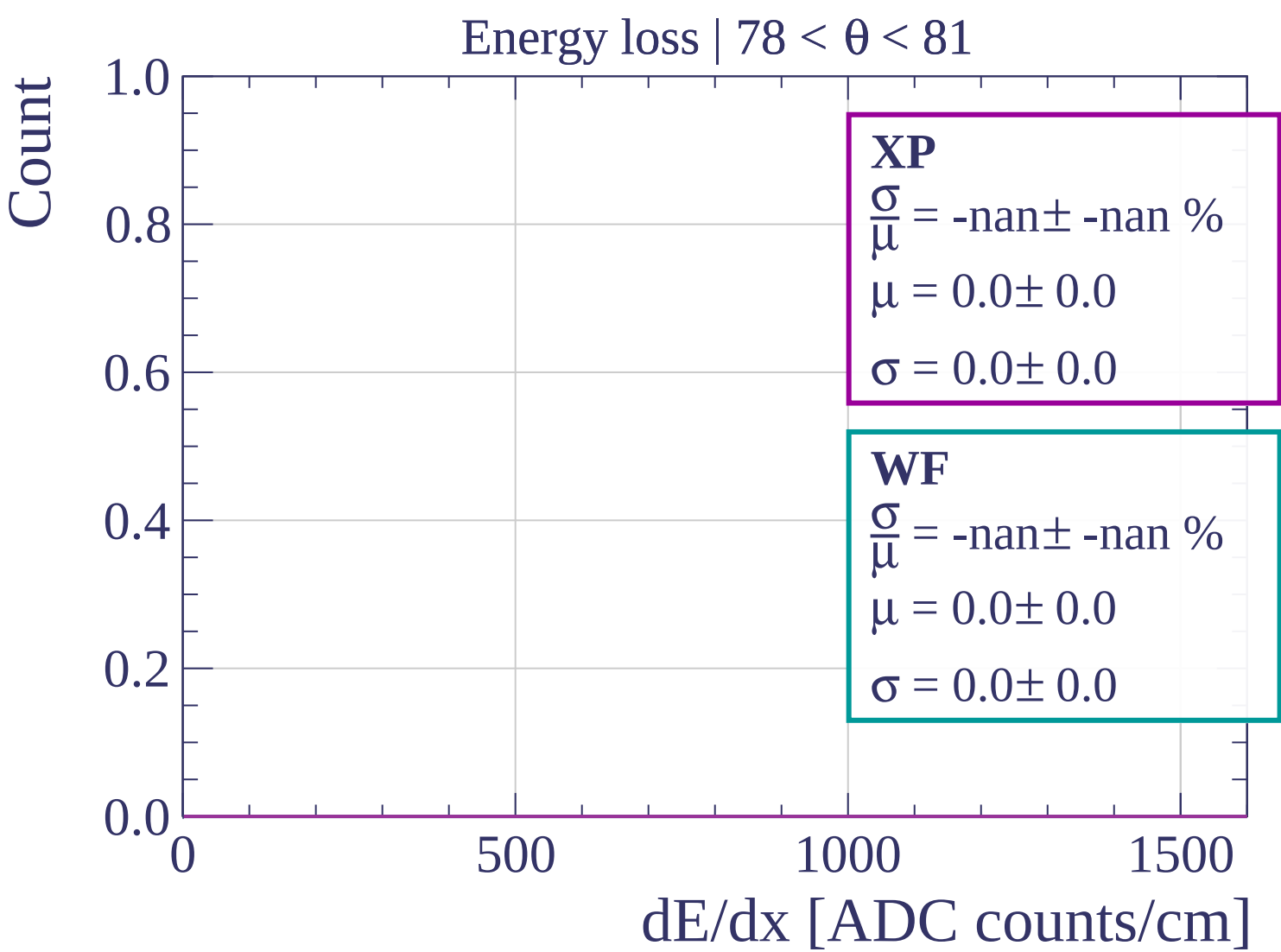
500

1000

1500

dE/dx [ADC counts/cm]





Count

Energy loss | $81 < \theta < 84$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

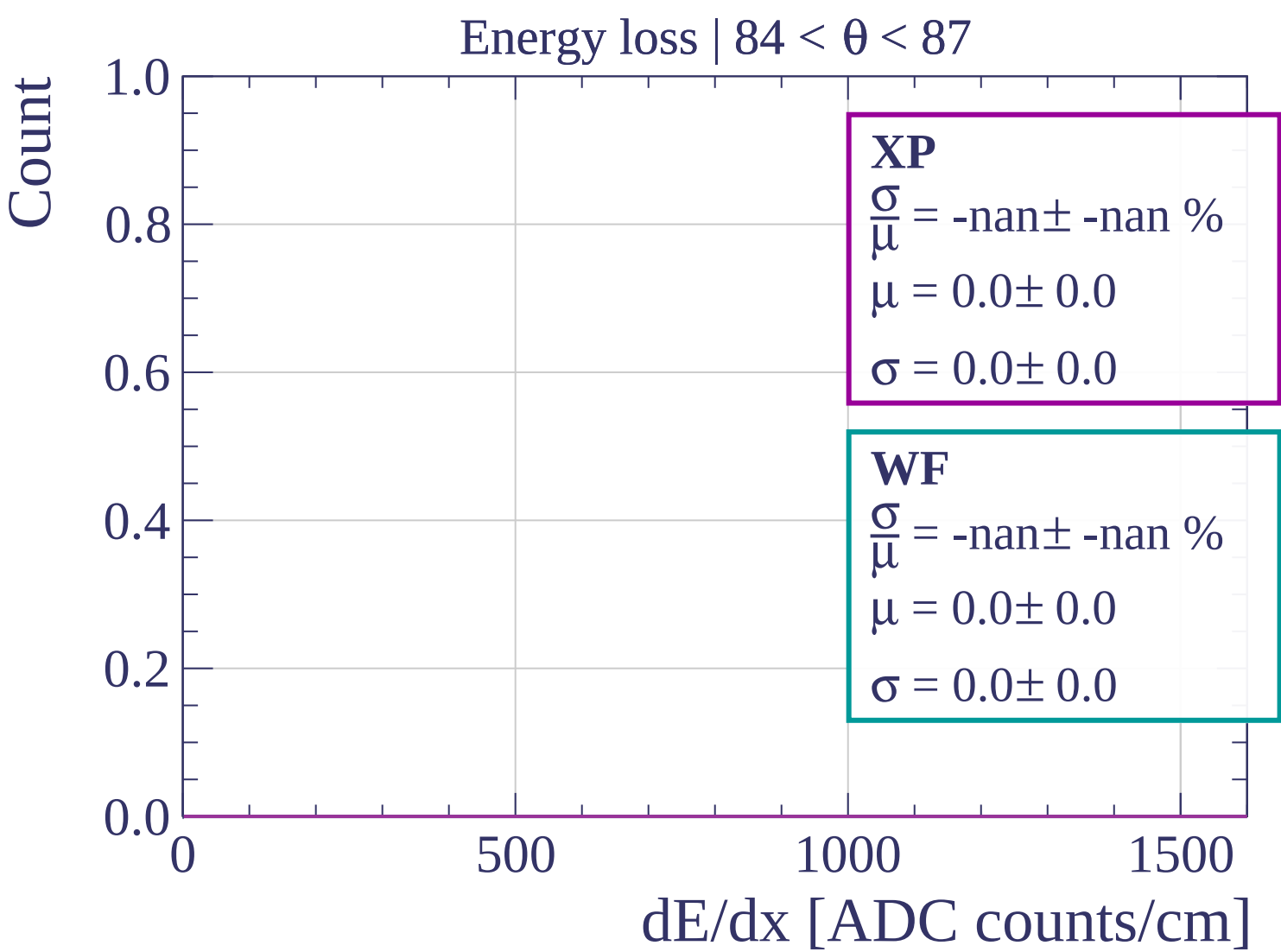
0

500

1000

1500

dE/dx [ADC counts/cm]



Count

Energy loss | $87 < \theta < 90$

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0
0.8
0.6
0.4
0.2
0.0

0

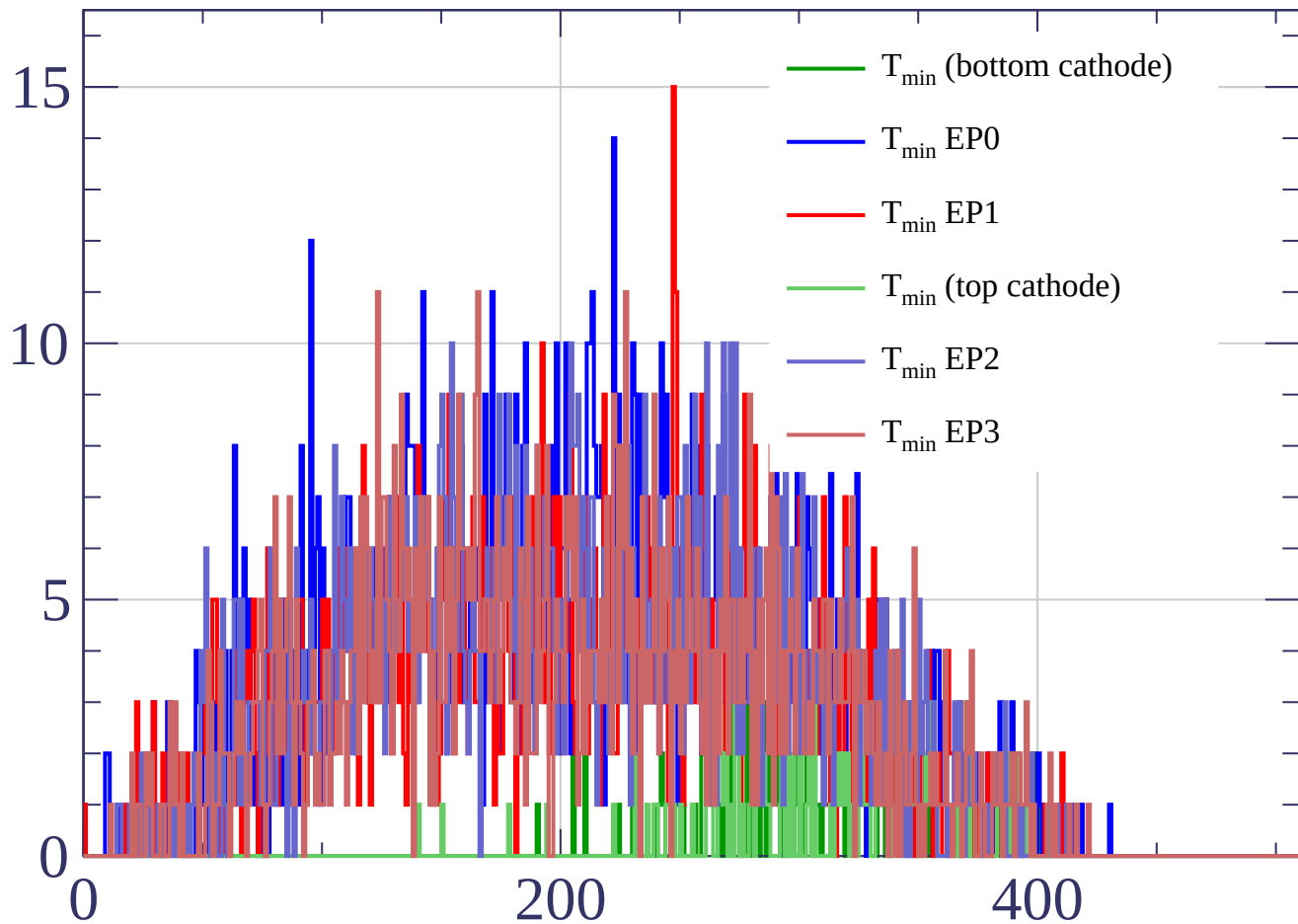
500

1000

1500

dE/dx [ADC counts/cm]

Count



Count

